

Handbook of Industrial Drying

Fourth Edition

**Edited by
Arun S. Mujumdar**

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Foreword to the Third Edition

The *Handbook of Industrial Drying*, as a result of the great success of its first and second editions, has gained high reputation among readers interested in the process of drying. In the last three decades, we have observed a growing interest in the multidisciplinary subject of drying, which had resulted in a major increase of research activity, publication of several monographs, book series, technical papers, international journals, several drying conference series in almost all continents, etc. Today drying R&D continues worldwide at a pace unmatched in any earlier period. To keep abreast with all these scattered sources of information in a broad area like drying is extremely difficult for most readers in academia and industry alike.

So, the third edition of the handbook, nearly a decade after the second edition, will play a very important role in providing comprehensive, updated information and a view of the current state of the art in industrial drying as a more cohesive whole. This third edition continues the style of the two previous ones; the authors are international leaders and generally recognized world authorities from academia, industry, and R&D laboratories from many countries. It maintains the essential interdisciplinary character addressing a broad academic and industrial readership. This book gives the possibility for self-study and of finding a clear overview of the fundamentals and practical information in broad aspects and problems of drying technology. It is like having one's own private *consultant on the desk*.

The topics chosen are constructed to give a quick and clear overview of the fundamental principles and many

practical data referring to the selection of industrial dryers, description of drying equipment, industrial drying technologies, recent developments in R&D in drying, as well as future trends. Over 60% of the chapters are new and some 40% revised. A few chapters from the second edition have been deleted due to space limitations. New sections have been added to encompass the latest data on drying of several materials (foods, wood, herbal medicines, sludge, grain, nanosize products, fish and seafood, etc.); also included are some dryer types (rotary, indirect, drum, fluidized, flash and pneumatic, etc.) with a strong general approach to energy, environmental safety, control and quality aspects. So practically, this edition can be treated as a truly new *Handbook of Industrial Drying* based on the latest achievements in the drying area.

Finally, having in mind the international background of the authors, this handbook gives readers a chance to get acquainted in considerable detail with the literature sources published not only in English but also in other languages. Key relevant references are included at the end of each chapter.

I am confident that this third edition of the handbook will be of great help to the broad audience from academia and in the application, progress, and future trends in drying R&D on a global scale.

Czesław Strumiłło
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Foreword to the Second Edition

The second edition of the *Handbook of Industrial Drying* continues the tradition of the editor and the publisher as international leaders in providing information in the field of industrial drying. The authors are knowledgeable of the subjects and have been chosen from among the world's authorities in industry, academia, government, and consulting. Some 50 authors from 15 countries have written 43 chapters plus 3 appendices. There are 21 new chapters plus 2 new appendices. All chapters have been updated or revised. There is over 60% new material, making this edition practically a new volume.

The mark of an outstanding handbook is that it provides current information on a subject—in this case multidisciplinary in nature—understandable to a broad audience. A balanced approach of covering principles and practices provides a sound basis for the presentations. Students, academics, consultants, and industry people can find information to meet their needs. Researchers, designers, manufacturers, and salespeople can benefit from the book as they consider elements or components related to drying as well as the system itself.

New material has been added to provide the latest information on minimizing environmental impacts, increasing energy efficiency, maintaining quality control, improving safety of operation, and improving the control of drying systems. New sections or chapters have been added to cover in detail microwave drying; infrared drying; impinging stream dryers; use of superheated steam and osmotic dehydration; and drying of biotechnological materials, tissue and towels, peat, coal, and fibrous materials.

The information in this book can be categorized as product related, equipment related, and the relationship between the two—the system of drying. For products not specifically covered, or for the design of dryers not detailed, users can select closely related applicable information to meet many needs. The user may want to pursue a subject in considerably more detail. Pertinent references, but not voluminous overwhelming bibliographies, are included at the end of each chapter. An appendix devoted to an annotated bibliography is also included.

Carl W. Hall

Foreword to the First Edition

The *Handbook of Industrial Drying* fills an important need and is of immeasurable value in the field of drying. Academics, students, and industry people—from sales to research—can learn much from the combination of principles and practices used throughout. The presentation of principles does not overwhelm the coverage of equipment and systems. More appropriate theories will develop as a result of the description of equipment and systems. For example, a description of dryers, particularly industrial dryers, is lacking in many research articles; this handbook provides such information.

The authors have distilled much information from extensive literature to provide generic information as contrasted with details of a specific drying system of a particular manufacturer. The users can extrapolate the use of drying systems, by design and management, to a variety of products. As a special feature, a complete listing of books written on the subject of drying is included.

The authors, a blend of students, faculty, and those in industry, represent experience with different kinds of drying

systems, different applications of principles, and different products. The book provides excellent coverage of the cross-disciplinary nature of drying by inviting well-known authors from many countries of the world. Dr. Mujumdar and his associates have assembled an excellent up-to-date handbook.

The common thread throughout the book is the movement of heat and moisture as well as the movement and handling of products. Also included are instrumentation, sensors, and controls that are important for quality control of products and efficiency of operation. The emphasis on the design of equipment to expedite these processes in an economical manner is appropriate and useful.

The word *handbook* is sometimes used disparagingly to describe a reference for quick answers to limited questions or problems. In that sense, this book is more than a handbook—the knowledge base provided permits the user to build different systems for products other than those covered.

Carl W. Hall

Preface to the Fourth Edition

I am pleased to present this fourth enhanced edition of the *Handbook of Industrial Drying*, which is a reflection on the continuing industrial and academic interest in this interdisciplinary field of thermal dehydration and closely allied operations such as filtration and crystallization. As by far the most commonly encountered and also the most energy-intensive unit operation in almost all industrial sectors, this is not surprising. The awareness of drying as an important unit operation has already led to significant enhancement in energy efficiency in most industrial drying technologies. The biennial International Drying Symposium series (IDS) as well as the only premier journal devoted exclusively to drying technologies, *Drying Technology—An International Journal* (Taylor & Francis Group), have provided a strong impetus to drying R&D around the globe over the past four decades. Their impact on energy savings in industrial drying is truly immeasurably high. With escalating concerns over global warming due to combustion of fossil fuels, the interest in innovation in this field will increase further in the coming decade.

As noted in my earlier prefaces, drying is an applied field where we try to combine basic sciences with engineering applications. Design and scale-up are key issues faced by those associated with selection, design, and operation of dryers. The focus of this book is on industrial applications. Despite the thousands of publications on all aspects of drying science, engineering, and technology, we still suffer from lack of a generalized theory or ability to model all types of dryers. Empirical information and know-how remain the hallmarks of dryer design and scale-up. The tremendous diversity of dryers found in practice is a direct result of the multitude of physical characteristics of the wet materials to be dried and of the diverse specifications of the dried product. Even the production requirements affect the selection of dryer type. This book covers most of the commonly encountered dryers as well as the less common innovative dryers that have potential in coming years. There is yet no truly disruptive drying technology, but numerous incremental advances do lead to significant improvements in the technology.

Because of the long half-life of most drying technologies, the changes in dryer designs are relatively slow. No drastic design changes are imminent. However, better design and operation can improve energy efficiency and cost-effectiveness. Much of what appeared in the earlier editions is still relevant and current. While some chapters have been revised when it was found to be necessary, we have added 12 new chapters adding to the content of the handbook to include some topics or enhance certain themes to widen its usefulness. The need for this edition arose out of the demand for the handbook as well as the need to include additional material that is useful for our readers.

As was the case with the earlier editions, we have selected authors with expertise in the field and have covered a wide range of industry-related and academic topics. This is a truly international effort. I wish to thank all the authors for their support and valuable contribution. Writing for a handbook such as this is a highly demanding task that requires a high degree of professionalism and desire to share knowledge. As one can readily understand, much information on drying and dryers is often proprietary and hence not available in the public domain. While we have made every attempt to provide information from public sources, we cannot take responsibility for accuracy of all the data and information. Readers must use their good judgment and must be diligent before making any decision on selection, design, scale-up, and operation of actual dryers in practice. I believe that general enhancement of knowledge about various aspects of industrial dryers is of valuable assistance in making industrial operations more efficient and effective.

Editing of a massive book such as this is a long and arduous effort for the editor as well as those who assist in one way or the other to complete the editorial process. I want to acknowledge the invaluable assistance of Dr. Sachin Jangam in speeding up the global communications needed to accomplish the goal with a rather tight schedule as I had to complete the processing before moving from the National University of Singapore to the Hong Kong University of Science and Technology. He also coauthored a number of new chapters within the remarkably

short time available for completion. As usual, my better half, Purnima Mujumdar, had to bear through the disruptions in home affairs due to the editorial and writing effort I had to undertake over a whole year. My sincere thanks and appreciation goes to her for being so understanding—something she is used to for nearly four decades of my work on IDSs, the journal, and some 60 books on drying! I do want to record my appreciation of the effort of scores of my graduate students, research fellows, research collaborators, and industry participants for their direct and indirect contribution to this huge effort. I do hope this enhanced edition will perform the role, as one reviewer put it, of drying consultant on the desk.

Arun S. Mujumdar

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Preface to the Third Edition

From the success of the second edition of the *Handbook of Industrial Drying*, the need for an updated and enhanced edition is realized at this time. Interest in industrial drying operations has been growing continuously over the last three decades and still shows no signs of abatement. This unit operation is central to almost all industrial sectors while exposure to its fundamentals and applications is minimal in most engineering and applied science curricula around the world. The escalating interest in drying is evidenced by the large number of international, regional, and national conferences being held regularly around the world, which are devoted exclusively to thermal and nonthermal dehydration and drying. Although deceptively simple, the processes involved are still too complex to be described confidently in mathematical terms. This means that the design and analyses of industrial dryers remain a combination of science, engineering, and art. It is necessary to have both know-how and know-why of the processes involved to improve the design and operation of dryers. This book represents a comprehensive compendium of collected knowledge of experts from around the world. We are grateful to them for contributing to this effort.

As in the earlier editions, we have a blend of academic and industry-based authors. The academics were carefully selected to ensure they also have industrial background so that readers can reliably utilize the knowledge embedded in this book. Nevertheless, we need to include information and resources available in the public domain; despite our best intentions and high degree of selectivity, we cannot assume responsibility for validity of all the data and information given in this book. Readers must exercise due diligence before using the data in an industrial design or operation.

About two thirds of this book contains new material written by new authors using recent literature. A few topics from the second chapter are deleted. Numerous chapters are totally rewritten with new authorship. At least ten new chapters have been added to make the coverage encyclopedic. I believe that individuals and libraries who have the second edition in their collection should keep that as an independent reference. The material in it is still relevant since the shelf-life of drying technologies is rather long—several decades!

As some 50,000 materials are estimated to require drying on varying scales, it is obvious that it is impossible to pretend to cover all possible dryer types and products in any single resource. However, I believe we have covered most of the commonly used drying equipment and ancillaries, as well as addressed industrial sectors where drying is a key operation. In this edition, for the first time we have covered several new topics relevant to drying, for example, risk analysis,

crystallization, and frying. We have also covered new and emerging drying technologies in adequate detail.

This book is organized in much the same way as the earlier editions. The main difference is the wider coverage of topics. Once again, a deliberate attempt is made to cover most industrial sectors and make the content useful to industry as well as academia. Students and instructors in many disciplines will find the content useful for teaching, design, and research. It is particularly useful for researchers who wish to make their findings relevant to real-world needs.

As energy costs escalate and environmental impact becomes a serious issue in the coming decade, it is clear that the significance of drying for industry will rise. It is hoped that industry will encourage academia to include the study of drying, both as a basic and as an applied subject, as an essential part of engineering and technical curricula. Industry–university cooperation and active collaboration is essential to gaining in-depth knowledge of drying and dryers. I believe that rising energy costs and demand for enhanced product quality will drive drying R&D. Although no truly disruptive drying technology appears on the horizon today, it is likely to happen within the next decade. This book addresses some of the new technologies that have the potential to be disruptive.

Production of a massive handbook such as this one is a collective effort of scores of dedicated and enthusiastic individuals from around the globe. Indeed, this book embodies a result of globalization. Aside from the authors and referees, numerous staff members initially at Marcel Dekker, New York, and then at CRC Press, Boca Raton, have helped move this project along over a period of nearly five years. Purnima Mujumdar, as usual, played a pivotal part in bringing this project to a successful closure. Without her enthusiastic volunteer effort, it is highly unlikely this book would have seen the proverbial end of the tunnel. A number of my postgraduate students at McGill, National University of Singapore, and indeed many overseas institutions also assisted in various ways for which I want to express my gratitude. The encouragement I received regularly from Dr. Carl Hall was instrumental in keeping the project alive and kicking over very long periods, especially since it competed for my leisure time used to edit *Drying Technology—An International Journal* and several other books, as well as organizational effort for many drying-related conferences such as IDS, ADC, NDC, and IWSID. I thank the authors for their patience and effort in making this third edition a valuable reference work.

Arun S. Mujumdar
Singapore

Preface to the Second Edition

The second edition of the *Handbook of Industrial Drying* is a testimonial to the success of the first edition published in 1987. Interest in the drying operation has continued to increase on a truly global scale over the past decade. For example, over 1500 papers have been presented at the biennial International Drying Symposia (IDS) since its inception in 1978. *Drying Technology—An International Journal* published some 2000 pages in seven issues in 1993 compared with just over 300 only a decade earlier. The growth in drying R&D is stimulated by the need to design and operate dryers more efficiently and produce products of higher quality.

A handbook is expected to provide the reader with critical information and advice on appropriate use of such information compiled in a readily accessible form. It is intended to bring together widely scattered information and know-how in a coherent format. Since drying of solids is a multidisciplinary field—indeed, a discipline by itself—it is necessary to call on the expertise of individuals from different disciplines, different industrial sectors, and several countries. A quick perusal of the list of contributors will indicate a balanced blend of authorship from industry as well as academia. An attempt has been made to provide the key elements of fundamentals along with details of industrial dryers and special aspects of drying in specific industries, e.g., foods, pulp and paper, and pharmaceuticals.

The first edition contained 29 chapters and 2 appendices; this one contains 43 chapters and 3 appendices. Aside from the addition of new chapters to cover topics missing from the first one, a majority of earlier chapters have been updated—some fully rewritten with new authorship. This edition contains over 60% new updated material. Thus, this book will be a valuable addition even to the bookshelves that already hold the first edition.

This revised and expanded edition follows the same general organization as the first with additions made to each of

the four parts to eliminate some of the weaknesses of the first edition. For example, an extensive chapter is added to Part I on transport properties needed for dryer calculations. Chapters on infrared drying and the novel impinging stream dryers are added to II. III contains the largest enhancement with ten new chapters, while IV is completely new except for the chapter on humidity measurements.

A two-volume set of this magnitude must depend on the direct and indirect contributions of a large number of individuals and organizations. Clearly, it is impossible to name them all. I am grateful to all the contributors for the valuable time and effort they devoted to this project. The companies and publishers who have permitted us to reproduce some of their copyrighted artwork are acknowledged for their support. Appropriate credits are given in the text where applicable. Exergex Corporation, Brossard, Quebec, Canada, provided all the secretarial and related assistance over a three-year period. Without it, this revision would have been nearly impossible.

Over the past two years, most of my graduate students and postdoctoral fellows of McGill University have provided me with very enthusiastic assistance in various forms in connection with this project. In particular, I wish to express my thanks to Dr. T. Kudra for his continued help in various ways. Purnima, Anita, and Amit Mujumdar kindly word-processed numerous chapters and letters and helped me keep track of the incredible paperwork involved. The encouragement I received from Dr. Carl W. Hall was singularly valuable in keeping me going on this project while handling concurrently the editorial responsibilities for *Drying Technology—An International Journal* and a host of other books. Finally, the staff at Marcel Dekker, Inc., have been marvelous; I sincerely appreciate their patience and faith in this project.

Arun S. Mujumdar

Preface to the First Edition

Drying of solids is one of the oldest and most common unit operations found in diverse processes such as those used in the agricultural, ceramic, chemical, food, pharmaceutical, pulp and paper, mineral, polymer, and textile industries. It is also one of the most complex and least understood operations because of the difficulties and deficiencies in mathematical descriptions of the phenomena of simultaneous—and often coupled and multiphase—transport of heat, mass, and momentum in solid media. Drying is therefore an amalgam of science, technology, and art (or know-how based on extensive experimental observations and operating experience) and is likely to remain so, at least for the foreseeable future.

Industrial as well as academic interest in drying of solids has been on the rise for over a decade, as evidenced by the continuing success of the biennial Industrial Drying Symposia (IDS) series. The emergence of several book series and an international journal devoted exclusively to drying and related areas also demonstrates the growing interest in this field. The significant growth in research and development activity in the Western world related to drying and dewatering was no doubt triggered by the energy crunch of the early 1970s, which increased the cost of drying severalfold within only a few years. However, it is worth noting that continued efforts in this area will be driven not only by the need to conserve energy, but also by needs related to increased productivity, better product quality, quality control, new products and new processes, safer and environmentally superior operation, etc.

This book is intended to serve both the practicing engineer involved in the selection or design of drying systems and the researcher as a reference work that covers the wide field of drying principles, various commonly used drying equipment, and aspects of drying in important industries. Since industrial dryers can be finely categorized into over 200 variants and, furthermore, since they are found in practically all major industrial sectors, it is impossible within limited space to cover all aspects of drying and dryers. We have had to make choices. In view of the availability of such publications as *Advances in Drying* and the *Proceedings of the International Drying Symposia*, which emphasize research and development in solids drying, we decided to concentrate on various practical aspects of commonly used industrial dryers following a brief introduction to the basic principles, classification and selection of dryers, process calculation schemes, and basic experimental techniques in drying. For detailed information on the fundamentals of drying, the reader is referred to various textbooks in this area.

The volume is divided into four major parts. Part I covers the basic principles, definitions, and process calculation methods in a general but concise fashion. The second part is

devoted to a series of chapters that describe and discuss the more commonly used industrial dryers. Novel and less prevalent dryers have been excluded from coverage; the reader will find the necessary references in Appendix B, which lists books devoted to drying and related areas in English as well as in other languages. Part III is devoted to the discussion of current drying practices in key industrial sectors in which drying is a significant if not necessarily dominant operation. Some degree of repetition was unavoidable since various dryers are discussed under two possible categories. Most readers will, however, find such information complementary as it is derived from different sources and generally presented in different contexts.

Because of the importance of gas humidity measurement techniques, which can be used to monitor and control the convective drying operation, Part IV includes a chapter that discusses such techniques. Energy savings in drying via the application of energy recovery techniques, and process and design modifications, optimization and control, and new drying techniques and nonconventional energy sources are also covered in some depth in the final part of the book.

Finally, it is my pleasant duty to express my sincerest gratitude to the contributors from industry and academia, from various parts of the world, for their continued enthusiasm and interest in completing this major project. The comments and criticisms received from over 25 reviewers were very valuable in improving the contents within the limitations of space. Many dryer manufacturers assisted me and the contributors directly or indirectly by providing nonproprietary information about their equipment. Dr. Maurits Dekker, chairman of the board, Marcel Dekker, Inc., was instrumental in elevating the level of my interest in drying so that I was able to undertake the major task of compiling and editing a handbook in a truly multidisciplinary area whose advancement depends on closer industry-academia interaction and cooperation. My heartfelt thanks go to Chairman Mau for his kindness, continuous encouragement, and contagious enthusiasm throughout this project.

Over the past four years, many of my graduate students provided me with enthusiastic assistance in connection with this project. In particular, I wish to thank Mainul Hasan and Victor Jariwala for their help and support. In addition, Purnima and Anita Mujumdar kindly word-processed countless drafts of numerous chapters. Without the assistance of my coauthors, it would have been impossible to achieve the degree of coverage attained in this book. I wish to record my appreciation of their efforts. Indeed, this book is a result of the combined and sustained efforts of everyone involved.

Arun S. Mujumdar

Editor

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of over 50 books, including *Mujumdar's Practical Guide to Industrial Drying*, which has already been translated into several languages, including Chinese, Indonesian, French, Vietnamese, and Hungarian. Dr. Mujumdar has lectured in 38 countries across 4 continents. He has also given professional development courses to industrial and academic audiences in the United States, Canada, Japan, China, and India. Details of his research activities and interests in drying can be found at www.arunmujumdar.com.

Dr. Mujumdar has been instrumental in developing the then-neglected field of drying into a major multi- and interdisciplinary field on a truly global scale. Thanks to his missionary efforts, often carried out single-handedly before the field received worldwide recognition, engineers and scientists around the world have been able to pursue their interests in this exciting field, which provides a kaleidoscope of challenging research opportunities for innovation. He is aptly called the drying guru—a label he was first given during the presentation of the esteemed Joseph Janus Medal of the Czech Academy of Sciences in Prague in 1990 to honor his countless contributions to chemical engineering and drying technologies.

Professor Mujumdar has been recognized for his exceptional contributions and service to global drying R&D with honorary doctorates awarded by Lodz Technical University, Poland, and University of Lyon1, France. Most recently, he received the coveted National Award for International Cooperation from the Government of China for his outstanding contribution to the development of food drying technology in cooperation with Jiangnan University, Wuxi, China.

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Section I

Fundamental Aspects

1 Principles, Classification, and Selection of Dryers

Arun S. Mujumdar

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1.1 INTRODUCTION

Drying commonly describes the process of thermally removing volatile substances (moisture) to yield a solid product. Moisture held in loose chemical combination, present as a liquid solution within the solid or even trapped in the microstructure of the solid, which exerts a vapor pressure less than that of pure liquid, is called *bound* moisture. Moisture in excess of bound moisture is called *unbound* moisture.

When a wet solid is subjected to thermal drying, two processes occur simultaneously:

1. Transfer of energy (mostly as heat) from the surrounding environment to evaporate the surface moisture
2. Transfer of internal moisture to the surface of the solid and its subsequent evaporation due to process 1

The rate at which drying is accomplished is governed by the rate at which the two processes proceed. Energy transfer as heat from the surrounding environment to the wet solid can occur as a result of convection, conduction, or radiation and in some cases as a result of a combination of these effects. Industrial dryers differ in type and design, depending on the principal method of heat transfer employed. In most cases heat is transferred to the surface of the wet solid and then to the interior. However, in dielectric, radio frequency (RF), or microwave freeze drying, energy is supplied to generate heat internally within the solid and flows to the exterior surfaces.

Process 1, the removal of water as vapor from the material surface, depends on the external conditions of temperature, air humidity and flow, area of exposed surface, and pressure.

Process 2, the movement of moisture internally within the solid, is a function of the physical nature of the solid, the temperature, and its moisture content. In a drying operation any one of these processes may be the limiting factor governing the rate of drying, although they both proceed simultaneously throughout the drying cycle. In the following sections we shall discuss the terminology and some of the basic concepts behind the two processes involved in drying.

The separation operation of drying converts a solid, semi-solid, or liquid feedstock into a solid product by evaporation of the liquid into a vapor phase through application of heat. In the special case of freeze drying, which takes place below the triple point of the liquid that is removed, drying occurs by sublimation of the solid phase directly into the vapor phase. This definition thus excludes conversion of a liquid phase into a concentrated liquid phase (evaporation), mechanical dewatering operations such as filtration, centrifugation, sedimentation, supercritical extraction of water from gels to produce extremely high porosity aerogels (extraction) or so-called drying of liquids and gases by the use of molecular sieves (adsorption). Phase change and production of a solid phase as end product are essential features of the drying process. Drying is an essential operation in the chemical,

agricultural, biotechnology, food, polymer, ceramics, pharmaceutical, pulp and paper, mineral processing, and wood processing industries.

Drying is perhaps the oldest, most common and most diverse of chemical engineering unit operations. Over 400 types of dryers have been reported whereas over 100 distinct types are commonly available. It competes with distillation as the most energy-intensive unit operation due to the high latent heat of vaporization and the inherent inefficiency of using hot air as the (most common) drying medium. Several studies report national energy consumption for industrial drying operations ranging from 10% to 15% for United States, Canada, France, and U.K. to 20%–25% for Denmark and Germany. The latter figures have been obtained recently based on mandatory energy audit data supplied by industry and hence are more reliable.

Energy consumption in drying ranges from a low value of under 5% for the chemical process industries to 35% for the papermaking operations. In the United States, for example, capital expenditures for dryers are estimated to be in the order of only \$800 million per annum. Thus, the major costs for dryers are in their operation rather than in their initial investment costs.

Drying of various feedstocks is needed for one or several of the following reasons: need for easy-to-handle free-flowing solids, preservation and storage, reduction in cost of transportation, achieving desired quality of product, etc. In many processes, improper drying may lead to irreversible damage to product quality and hence a nonsalable product.

Before proceeding to the basic principles, it is useful to note the following unique features of drying, which make it a fascinating and challenging area for research and development (R&D):

- Product size may range from microns to tens of centimeters (in thickness or depth)
- Product porosity may range from 0% to 99.9%
- Drying times range from 0.25 s (drying of tissue paper) to 5 months (for certain hardwood species)
- Production capacities may range from 0.10 kg/h to 100 tons/h
- Product speeds range from 0 (stationary) to 2000 m/min (tissue paper)
- Drying temperatures range from below the triple point to above the critical point of the liquid
- Operating pressure may range from fraction of a millibar to 25 atm
- Heat may be transferred continuously or intermittently by convection, conduction, radiation, or electromagnetic fields

Clearly, no single design procedure that can apply to all or even several of the dryer variants is possible. It is therefore essential to revert to the fundamentals of heat, mass and momentum transfer coupled with knowledge of the material properties (quality) when attempting design of a dryer or analysis of an existing dryer. Mathematically speaking,

all processes involved, even in the simplest dryer, are highly nonlinear and hence scale-up of dryers is generally very difficult. Experimentation at laboratory and pilot scales coupled with field experience and know how for it is essential to the development of a new dryer application. Dryer vendors are necessarily specialized and normally offer only a narrow range of drying equipment. The buyer must therefore be reasonably conversant with the basic knowledge of the wide assortment of dryers and be able to come up with an informal preliminary selection before going to the vendors with notable exceptions. In general, several different dryers may be able to handle a given application.

Drying is a complex operation involving transient transfer of heat and mass along with several rate processes, such as physical or chemical transformations, which, in turn, may cause changes in product quality as well as the mechanisms of heat and mass transfer. Physical changes that may occur include shrinkage, puffing, crystallization, and glass transitions. In some cases, desirable or undesirable chemical or biochemical reactions may occur, leading to changes in color, texture, odor, or other properties of the solid product. In the manufacture of catalysts, for example, drying conditions can yield significant differences in the activity of the catalyst by changing the internal surface area.

Drying occurs by effecting vaporization of the liquid by supplying heat to the wet feedstock. As noted earlier, heat may be supplied by convection (direct dryers), by conduction (contact or indirect dryers), radiation or volumetrically by placing the wet material in a microwave or RF electromagnetic field. Over 85% of industrial dryers are of the convective type with hot air or direct combustion gases as the drying medium. Over 99% of the applications involve removal of water. All modes except the dielectric (microwave and RF) supply heat at the boundaries of the drying object so that the heat must diffuse into the solid primarily by conduction. The liquid must travel to the boundary of the material before it is transported away by the carrier gas (or by application of vacuum for nonconvective dryers).

Transport of moisture within the solid may occur by any one or more of the following mechanisms of mass transfer:

- Liquid diffusion, if the wet solid is at a temperature below the boiling point of the liquid
- Vapor diffusion, if the liquid vaporizes within material
- Knudsen diffusion, if drying takes place at very low temperatures and pressures, e.g., in freeze drying
- Surface diffusion (possible although not proven)
- Hydrostatic pressure differences, when internal vaporization rates exceed the rate of vapor transport through the solid to the surroundings
- Combinations of the above mechanisms

Note that since the physical structure of the drying solid is subject to change during drying, the mechanisms of moisture transfer may also change with elapsed time of drying.

1.2 EXTERNAL CONDITIONS (PROCESS 1)

Here the essential external variables are temperature, humidity, rate and direction of airflow, the physical form of the solid, the desirability of agitation, and the method of supporting the solid during the drying operation [1]. External drying conditions are especially important during the initial stages of drying when unbound surface moisture is removed. In certain cases, for example, in materials like ceramics and timber in which considerable shrinkage occurs, excessive surface evaporation after the initial free moisture has been removed sets up high moisture gradients from the interior to the surface. This is liable to cause overdrying and excessive shrinkage and consequently high tension within the material, resulting in cracking and warping. In these cases surface evaporation should be retarded through the employment of high air relative humidities while maintaining the highest safe rate of internal moisture movement by heat transfer.

Surface evaporation is controlled by the diffusion of vapor from the surface of the solid to the surrounding atmosphere through a thin film of air in contact with the surface. Since drying involves the interphase transfer of mass when a gas is brought in contact with a liquid in which it is essentially insoluble, it is necessary to be familiar with the equilibrium characteristics of the wet solid. Also, since the mass transfer is usually accompanied by the simultaneous transfer of heat, due consideration must be given to the enthalpy characteristics.

1.2.1 VAPOR-LIQUID EQUILIBRIUM AND ENTHALPY FOR A PURE SUBSTANCE VAPOR-PRESSURE CURVE

When a liquid is exposed to a dry gas, the liquid evaporates, that is, forms vapor and passes into the gaseous phase. If m_w is the mass of vapor in the gaseous phase, then this vapor exerts a pressure over the liquid, the *partial pressure*, which, assuming ideal gas behavior for the vapor, is given by

$$P_w V = \frac{m_w}{M_w} RT \quad \text{or} \quad P_w V_w = RT \quad (1.1)$$

The maximum value of P_w that can be reached at any temperature is the saturated vapor pressure P_w^0 . If the vapor pressure of a substance is plotted against temperature, a curve such as *TC* of Figure 1.1 is obtained. Also plotted in the figure are the solid-liquid equilibrium curve (melting curve) and the solid-vapor (sublimation) curve. The point *T* in the graph at which all three phases can coexist is called the *triple point*. For all conditions along the curve *TC*, liquid and vapor may coexist, and these points correspond with the *saturated liquid* and the *saturated vapor state*. Point *C* is the critical point at which distinction between the liquid and vapor phases disappears, and all properties of the liquid, such as density, viscosity, and refractive index, are identical with those of the vapor. The substance above the critical temperature is called a gas, the temperature corresponding to a pressure at each point on the curve *TC* is the *boiling point*, and that corresponding to a pressure of 101.3 kPa is the *normal boiling point*.

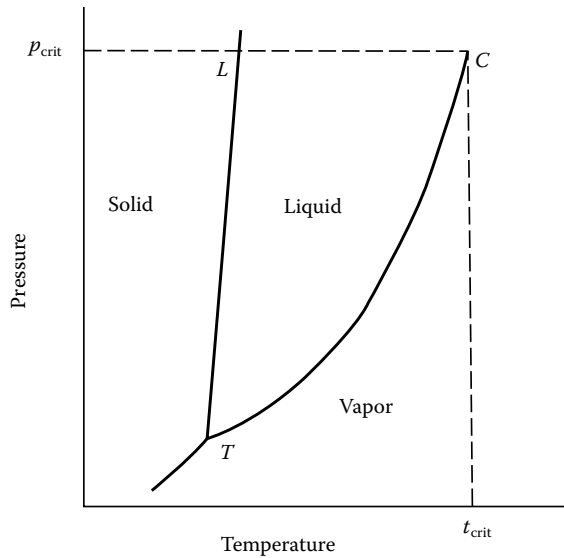


FIGURE 1.1 Vapor pressure of a pure liquid.

1.2.1.1 The Clausius–Clapeyron Equation

Comprehensive tables of vapor-pressure data of common liquids, such as water, common refrigerants, and others, may be found in Refs. [2,3]. For most liquids, the vapor–pressure data are obtained at a few discrete temperatures, and it might frequently be necessary to interpolate between or extrapolate beyond these measurement points. At a constant pressure, the Clausius–Clapeyron equation relates the slope of the vapor pressure–temperature curve to the latent heat of vaporization through the relation

$$\frac{dP_W^0}{dT} = \frac{\Delta H_W}{T(V_W - V_L)} \quad (1.2)$$

where

V_W and V_L are the specific molar volumes of saturated vapor and saturated liquid, respectively

ΔH_W is the molar latent heat of vaporization

Since the molar volume of the liquid is very small compared with that of the vapor, we neglect V_L and substitute for V_W from Equation 1.1 to obtain

$$d \ln P_W^0 = \frac{\Delta H_W}{RT^2} dT \quad (1.3)$$

Since ΔH_W could be assumed to be a constant over short temperature ranges, Equation 1.3 can be integrated to

$$\ln P_W^0 = -\frac{\Delta H_W}{RT} + \text{constant} \quad (1.4)$$

and this equation can be used for interpolation. Alternatively, reference-substance plots [6] may be constructed. For the reference substance,

$$d \ln P_R^0 = \frac{\Delta H_R}{RT^2} dT \quad (1.5)$$

Dividing Equation 1.3 by Equation 1.5 and integrating provides

$$\ln P_W^0 = \frac{M_W \Delta H_W}{M_R \Delta H_R} \ln P_R^0 + \text{constant} \quad (1.6)$$

The reference substance chosen is one whose vapor pressure data are known.

1.2.1.2 Enthalpy

All substances have an internal energy due to the motion and relative position of the constituent atoms and molecules. Absolute values of the internal energy, u , are unknown, but numerical values relative to an arbitrarily defined baseline at a particular temperature can be computed. In any steady flow system there is an additional energy associated with forcing streams into a system against a pressure and in forcing streams out of the system. This flow work per unit mass is PV , where P is the pressure and V is the specific volume. The internal energy and the flow work per unit mass have been conveniently grouped together into a composite energy called the enthalpy H . The enthalpy is defined by the expression

$$H = u + PV \quad (1.7)$$

and has the units of energy per unit mass (J/kg or N m/kg).

Absolute values of enthalpy of a substance like the internal energy are not known. Relative values of enthalpy at other conditions may be calculated by arbitrarily setting the enthalpy to zero at a convenient reference state. One convenient reference state for zero enthalpy is liquid water under its own vapor pressure of 611.2 Pa at the triple-point temperature of 273.16 K (0.01°C).

The isobaric variation of enthalpy with temperature is shown in Figure 1.2. At low pressures in the gaseous state, when the gas behavior is essentially ideal, the enthalpy is almost independent of the pressure, so the isobars nearly superimpose on each other. The curves marked “saturated liquid” and “saturated vapor,” however, cut across the constant pressure lines and show the enthalpies for these conditions at temperatures and pressures corresponding to the equilibrium vapor pressure relationship for the substance. The distance between the saturated vapor and saturated liquid curves, such as the distance $V - L$ corresponds to the latent heat of vaporization at a temperature T . Both T and $V - L$ are dependent on pressure, the distance $V - L$ decreases and becomes zero at the critical temperature T_C . Except near the critical temperature, the enthalpy of the liquid is almost independent of pressure until exceedingly high pressures are reached.

1.2.1.3 Heat Capacity

The *heat capacity* is defined as the heat required to raise the temperature of a unit mass of substance by a unit

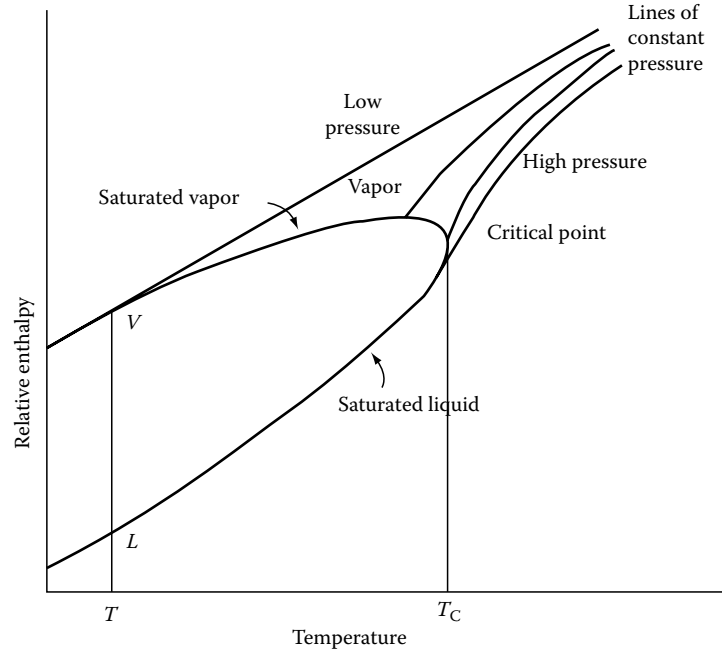


FIGURE 1.2 Typical enthalpy-temperature diagram for a pure substance.

temperature. For a constant pressure process, the heat capacity C_p is given by

$$C_p = \left(\frac{\partial Q}{\partial T} \right)_p \quad (1.8)$$

where the heat flow Q is the sum of the internal energy change ∂u and the work done against pressure $P \partial V$. Equation 1.8 may be expanded as follows:

$$C_p = \left(\frac{\partial u}{\partial T} \right)_p + P \left(\frac{\partial V}{\partial T} \right)_p = \left(\frac{\partial H}{\partial T} \right)_p \quad (1.9)$$

The slope of the isobars of Figure 1.2 yields the heat capacities.

In drying calculation, it is more convenient to use the mean values of heat capacity over a finite temperature step:

$$\bar{C}_p = \left(\frac{\Delta Q}{\Delta T} \right)_p = \frac{1}{(T_2 - T_1)} \int_{T_1}^{T_2} C_p dT \quad (1.10)$$

Second-order polynomials in temperature have been found to adequately describe the variation of C_p with temperature in the temperature range 300–1500 K [4], but for the temperature changes normally occurring in drying the quadratic term can be neglected.

Thus if

$$C_p = a + bT \quad (1.11)$$

then from Equation 1.10,

$$\bar{C}_p = a + \frac{1}{2}b(T_1 + T_2) = C_p(T_{av}) \quad (1.12)$$

The mean heat capacity is the heat capacity evaluated at the arithmetic mean temperature T_{av} .

From Equations 1.9 and 1.10, the enthalpy of the pure substance can be estimated from its heat capacity by

$$H = \bar{C}_p \theta \quad (1.13)$$

where θ denotes the temperature difference or excess over the zero enthalpy reference state. Heat capacity data for a large number of liquids and vapors are found in Ref. [5].

1.2.2 VAPOR-GAS MIXTURES

When a gas or gaseous mixture remains in contact with a liquid surface, it will acquire vapor from the liquid until the partial pressure of the vapor in the gas mixture equals the vapor pressure of the liquid at the existing temperature. In drying applications, the gas frequently used is air and the liquid used is water. Although common concentration units (partial pressure, mole fraction, and others) based on total quantity of gas and vapor are useful, for operations that involve changes in vapor content of a vapor-gas mixture without changes in the amount of gas, it is more convenient to use a unit based on the unchanging amount of gas.

Humid air is a mixture of water vapor and gas, composed of a mass m_w of water vapor and a mass m_G of gas (air). The *moisture content* or *absolute humidity* can be expressed as

$$Y = \frac{m_w}{m_G} \quad (1.14)$$

The total mass can be written in terms of Y and m_G as

$$m_G + m_w = m_G(1 + Y) \quad (1.15)$$

Using the gas law for vapor and air fractions at constant total volume V and temperature T ,

$$m_G = \frac{P_G V}{RT} M_G \quad \text{and} \quad m_w = \frac{P_w V}{RT} M_w \quad (1.16)$$

Thus,

$$Y = \frac{P_w}{P_G} \frac{M_w}{M_G} \quad (1.17)$$

Using Dalton's law of partial pressures,

$$P = P_w + P_G \quad (1.18)$$

and

$$Y = \frac{P_w}{P - P_w} \frac{M_w}{M_G} \quad (1.19)$$

When the partial pressure of the vapor in the gas equals the vapor pressure of the liquid, an equilibrium is reached and the gas is said to be *saturated* with vapor. The ideal saturated absolute humidity is then

$$Y_s = \frac{P_w}{P - P_w^0} \frac{M_w}{M_G} \quad (1.20)$$

The *relative humidity* ψ of a vapor–gas mixture is a measure of its fractional saturation with moisture and is defined as the ratio of the partial pressure of the vapor P_w to the saturated pressure P_w^0 at the same temperature. Thus ψ is given by

$$\psi = \frac{P_w}{P_w^0} \quad (1.21)$$

Equation 1.19 may now be written as

$$Y = \frac{M_w}{M_G} \frac{\psi P_w^0}{P - \psi P_w^0} \quad (1.22)$$

For water vapor and air when $M_w = 18.01$ kg/kmol and $M_G = 28.96$ kg/kmol, respectively, Equation 1.22 becomes

$$Y = 0.622 \frac{\psi P_w^0}{P - \psi P_w^0} \quad (1.23)$$

1.2.3 UNSATURATED VAPOR–GAS MIXTURES: PSYCHROMETRY IN RELATION TO DRYING

If the partial pressure of the vapor in the vapor–gas mixture is for any reason less than the vapor pressure of the liquid at the same temperature, the vapor–gas mixture is said to be unsaturated. As mentioned earlier, two processes occur simultaneously during the thermal process of drying a wet solid, namely, heat transfer to change the temperature of the wet solid and to evaporate its surface moisture and the mass transfer of moisture to the surface of the solid and its subsequent evaporation from the surface to the surrounding atmosphere. Frequently, the surrounding medium is the drying medium, usually heated air or combustion gases. Consideration of the actual quantities of air required to remove the moisture liberated by evaporation is based on psychrometry and the use of humidity charts. The following are definitions of expressions used in psychrometry [6].

1.2.3.1 Dry Bulb Temperature

This is the temperature of a vapor–gas mixture as ordinarily determined by the immersion of a thermometer in the mixture.

1.2.3.2 Dew Point

This is the temperature at which a vapor–gas mixture becomes saturated when cooled at a constant total pressure out of contact with a liquid (i.e., at constant absolute humidity). The concept of the dew point is best illustrated by referring to Figure 1.3, a plot of the absolute humidity versus temperature for a fixed pressure and the same gas. If an unsaturated mixture initially at point F is cooled at constant pressure out of contact of liquid, the gas saturation increases until the point G is reached, when the gas is fully saturated. The temperature at which the gas is fully saturated is called the dew point T_D . If the temperature is reduced to an infinitesimal amount below T_D , the vapor will condense and the process follows the saturation curve.

While condensation occurs the gas always remains saturated. Except under specially controlled circumstances, supersaturation will not occur and no vapor–gas mixture whose coordinates lie to the left of the saturation curve will result.

1.2.3.3 Humid Volume

The humid volume V_H of a vapor–gas mixture is the volume in cubic meters of 1 kg of dry gas and its accompanying vapor at the prevailing temperature and pressure. The volume of an ideal gas or vapor at 273 K and 1 atm (101.3 kPa) is 22.4 m³/kg mol. For a mixture with an absolute humidity Y at T_G (K) and P (atm), the ideal gas law gives the humid volume as

$$V_H = \left(\frac{1}{M_G} + \frac{Y}{M_w} \right) 22.4 \frac{T}{273.14 P} \quad (1.24)$$

$$V_H = 0.082 \left(\frac{1}{M_G} + \frac{Y}{M_w} \right) \frac{T}{P}$$

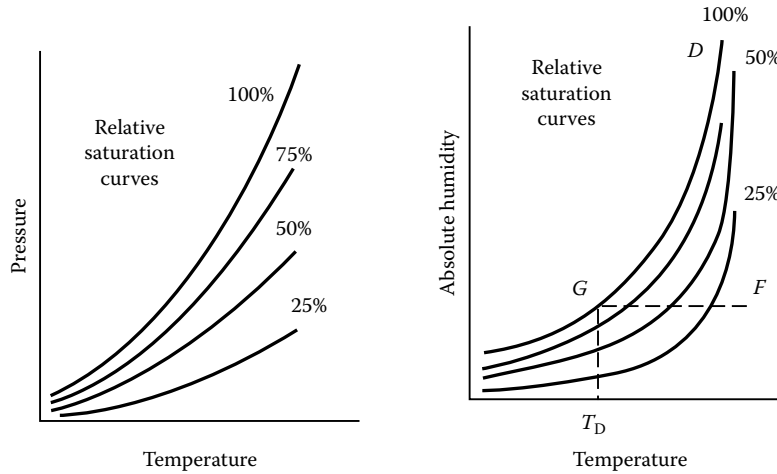


FIGURE 1.3 Two forms of psychrometric charts.

When the mass of dry gas in the vapor–gas mixture is multiplied by the humid volume, the volume of the vapor–gas mixture is obtained. The humid volume at saturation is computed with $Y = Y_s$, and the specific volume of the dry gas can be obtained by substituting $Y = 0$. For partially saturated mixtures, V_H may be interpolated between values for 0% and 100% saturation at the same temperature and pressure.

1.2.3.4 Enthalpy

Since the enthalpy is an extensive property, it could be expected that the enthalpy of a humid gas is the sum of the partial enthalpies of the constituents and a term to take into account the heat of mixing and other effects. The humid enthalpy I_G is defined as the enthalpy of a unit mass of dry gas and its associated moisture. With this definition of enthalpy,

$$I_G = H_{GG} + YH_{GW} + \Delta H_{GM} \quad (1.25)$$

where

H_{GG} is the enthalpy of dry gas

H_{GW} is the enthalpy of moisture

ΔH_{GM} is the residual enthalpy of mixing and other effects

In air saturated with water vapor, this residual enthalpy is only -0.63 kJ/kg at 60°C (333.14 K) [3] and is only 1% of H_{GG} ; thus it is customary to neglect the influences of this residual enthalpy.

It is sometimes convenient to express the enthalpy in terms of specific heat. Analogous to Equation 1.13, we could express the enthalpy of the vapor–gas mixture by

$$I_G = \bar{C}_{PY}\theta + \Delta H_{V0}Y \quad (1.26)$$

$\bar{C}_{PY}$ is called the *humid heat*, defined as the heat required to raise the temperature of 1 kg of gas and its associated moisture by 1 K at constant pressure. For a mixture with absolute humidity Y ,

$$\bar{C}_{PY} = \bar{C}_{PG} + \bar{C}_{PW}Y \quad (1.27)$$

where $\bar{C}_{PG}$ and $\bar{C}_{PW}$ are the mean heat capacities of the dry gas and moisture, respectively.

The path followed from the liquid to the vapor state is described as follows. The liquid is heated up to the dew point T_D , vaporized at this temperature, and superheated to the dry bulb temperature T_G . Thus

$$H_{GW} = \bar{C}_{LW}(T_D - T_0) + \Delta H_{VD} + \bar{C}_{PW}(T_G - T_D) \quad (1.28)$$

However, since the isothermal pressure gradient $(\Delta H/\Delta P)_T$ is negligibly small, it could be assumed that the final enthalpy is independent of the vaporization path followed. For the sake of convenience it could be assumed that vaporization occurs at 0°C (273.14 K), at which the enthalpy is zero, and then directly superheated to the final temperature T_G . The enthalpy of the vapor can now be written as

$$H_{GW} = \bar{C}_{PW}(T_G - T_0) + \Delta H_{V0} \quad (1.29)$$

and the humid enthalpy given by

$$I_G = \bar{C}_{PG}(T_G - T_0) + Y(\bar{C}_{PW}(T_G - T_0) + \Delta H_{V0}) \quad (1.30)$$

Using the definition for the humid heat capacity, Equation 1.30 reduces to

$$I_G = \bar{C}_{PY}(T_G - T_0) + \Delta H_{V0}Y \quad (1.31)$$

In Equation 1.31 the humid heat is evaluated at $(T_G + T_0)/2$ and ΔH_{V0} , the latent heat of vaporization at 0°C (273.14 K). Despite its handiness, the use of Equation 1.31 is not recommended above a humidity of 0.05 kg/kg. For more accurate work, it is necessary to resort to the use of Equation 1.28 in conjunction with Equation 1.25. In Equation 1.28 it should be noted that $\bar{C}_{LW}$ is the mean capacity of liquid moisture between T_0 and T_D , $\bar{C}_{PW}$ is the mean capacity of the moisture vapor evaluated between T_D and T_G , and ΔH_{VD} is the latent heat of vaporization at the dew point T_D .

The value of ΔH_{VD} can be approximately calculated from a known latent heat value at temperature T_0 by

$$\frac{\Delta H_{VD}}{\Delta H_{V0}} = \left(\frac{T_D - T_C}{T_0 - T_C} \right)^{1/3} \quad (1.32)$$

where T_C is the critical temperature. Better and more accurate methods of estimating ΔH_{VD} are available in Refs. [5,7].

1.2.4 ENTHALPY-HUMIDITY CHARTS

Using Equations 1.23, 1.25 and 1.28, the enthalpy-humidity diagram for unsaturated air ($\psi < 1$) can be constructed using the parameters ψ and θ . In order to follow the drying process we need access to enthalpy-humidity values. There seems to be no better, convenient, and cheaper way to store these data than in graphic form. The first of these enthalpy-humidity charts is attributed to Mollier. Mollier's original enthalpy-humidity chart was drawn with standard rectangular coordinates (Figure 1.4), but in order to extend the area over which it can be read, an oblique-angle system of coordinates is chosen for $I_G = f(Y)$.

In the unsaturated region, it can be seen from Equation 1.30 that I_G varies linearly with the humidity Y and the temperature T_G . If zero temperature (0°C) is taken as the datum for zero enthalpy, then

$$I_G = \bar{C}_{PG}\theta + Y(\bar{C}_{PW}\theta + \Delta H_{V0}) \quad (1.33)$$

where θ is the temperature in degree Celsius.

The isotherms ($\theta = \text{constant}$) cut the ordinate ($Y = 0$) at a value $\bar{C}_{PG}\theta$ (the dry gas enthalpy). If the isenthalpic lines ($I_G = \text{constant}$) are so inclined that they fall with a slope $-\Delta H_{V0}$, and if only $\Delta H_{V0}Y$ were taken into account in the contribution of vapor to the vapor-gas enthalpy, then the isotherms would run horizontally, but because of the contribution of $\bar{C}_{PW}\theta Y$, they increase with Y for $\theta < 0^\circ\text{C}$ and decrease with Y for $\theta > 0^\circ\text{C}$. Contours of relative humidity ψ are also plotted.

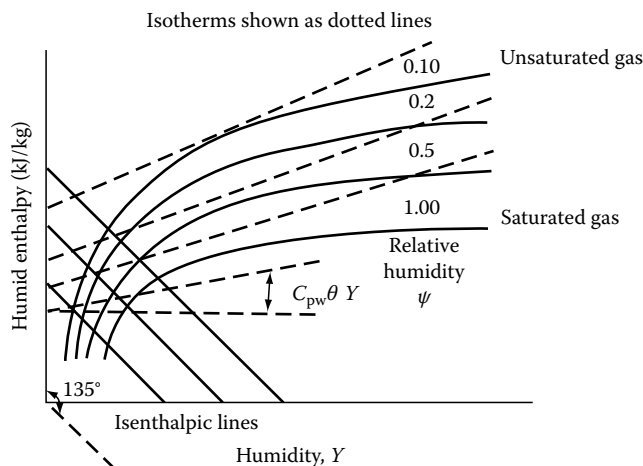


FIGURE 1.4 An enthalpy-humidity diagram for a moist gas.

The region above the curve $\psi = 1$ at which air is saturated corresponds to an unsaturated moist gas; the region below the curve corresponds to fogging conditions. At a fixed temperature air cannot take up more than a certain amount of vapor. Liquid droplets then precipitate due to oversaturation, and this is called the *cloud* or *fog state*. Detailed enthalpy-humidity diagrams are available elsewhere in this handbook and in Ref. [10].

A humidity chart is not only limited to a specific system of gas and vapor but is also limited to a particular total pressure. The thermophysical properties of air may be generally used with reasonable accuracy for diatomic gases [3], so that charts developed for mixtures in air can be used to describe the properties of the same moisture vapor in a gas such as nitrogen. Charts other than those of moist air are often required in the drying of fine chemicals and pharmaceutical products. These are available in Refs. [3,8,9].

1.2.4.1 Adiabatic Saturation Curves

Also plotted on the psychrometric chart are a family of adiabatic saturation curves. The operation of adiabatic saturation is indicated schematically in Figure 1.5. The entering gas is contacted with a liquid and as a result of mass and heat transfer between the gas and liquid the gas leaves at conditions of humidity and temperature different from those at the entrance. The operation is adiabatic as no heat is gained or lost by the surroundings. Doing a mass balance on the vapor results in

$$G_V = G_G(Y_{\text{out}} - Y_{\text{in}}) \quad (1.34)$$

The enthalpy balance yields

$$I_{G_{\text{in}}} + (Y_{\text{out}} - Y_{\text{in}})I_{LW} = I_{G_{\text{out}}} \quad (1.35)$$

Substituting for I_G from Equation 1.31, we have

$$\begin{aligned} \bar{C}_{PY_{\text{in}}}(T_{\text{in}} - T_0) + \Delta H_{V0}Y_{\text{in}} + (Y_{\text{out}} - Y_{\text{in}})\bar{C}_{LW}(T_L - T_0) \\ = \bar{C}_{PY_{\text{out}}}(T_{\text{out}} - T_0) + \Delta H_{V0}Y_{\text{out}} \end{aligned} \quad (1.36)$$

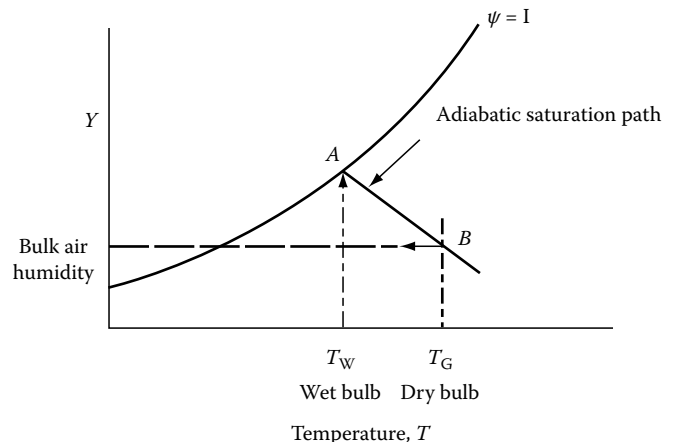


FIGURE 1.5 A temperature-humidity diagram for moist air.

Now, if a further restriction is made that the gas and the liquid phases reach equilibrium when they leave the system (i.e., the gas–vapor mixture leaving the system is saturated with liquid), then $T_{\text{out}} = T_{\text{GS}}$, $I_{\text{Gout}} = I_{\text{GS}}$, and $Y_{\text{out}} = Y_{\text{GS}}$ where T_{GS} is the adiabatic saturation temperature and Y_{GS} is the absolute humidity saturated at T_{GS} . Still further, if the liquid enters at the adiabatic saturation temperature T_{GS} , that is, $T_{\text{L}} = T_{\text{GS}}$, Equation 1.36 becomes

$$\begin{aligned} \bar{C}_{\text{PYin}}(T_{\text{in}} - T_0) + \Delta H_{\text{V0}}Y_{\text{in}} + (Y_{\text{GS}} - Y_{\text{in}})\bar{C}_{\text{LW}}(T_{\text{GS}} - T_0) \\ = \bar{C}_{\text{PYGS}}(T_{\text{GS}} - T_0) + \Delta H_{\text{V0}}Y_{\text{GS}} \end{aligned} \quad (1.37)$$

or substituting for $\bar{C}_{\text{PG}}$ from Equation 1.27

$$\begin{aligned} \bar{C}_{\text{PYin}}(T_{\text{in}} - T_0) + Y_{\text{in}}\bar{C}_{\text{PWin}}(T_{\text{in}} - T_0) + \Delta H_{\text{V0}}Y_{\text{in}} \\ + (Y_{\text{GS}} - Y_{\text{in}})\bar{C}_{\text{LW}}(T_{\text{GS}} - T_0) \\ = \bar{C}_{\text{PGGS}}(T_{\text{GS}} - T_0) + \bar{C}_{\text{PWGS}}Y_{\text{GS}}(T_{\text{GS}} - T_0) + \Delta H_{\text{V0}}Y_{\text{GS}} \end{aligned} \quad (1.38)$$

Assuming that the heat capacities are essentially constant over the temperature range involved, $\bar{C}_{\text{PGin}} = \bar{C}_{\text{PGGS}} = \bar{C}_{\text{PG}}$ and $\bar{C}_{\text{PWin}} = \bar{C}_{\text{PWGS}} = \bar{C}_{\text{PW}}$. Further subtracting $Y_{\text{in}}\bar{C}_{\text{PW}}T_{\text{GS}}$ from both sides of Equation 1.38 and simplifying, we have

$$\begin{aligned} \bar{C}_{\text{PY}}(T_{\text{in}} - T_{\text{GS}}) = (Y_{\text{GS}} - Y_{\text{in}}) \\ \times [(\bar{C}_{\text{PW}}(T_{\text{GS}} - T_0) + \Delta H_{\text{V0}} - \bar{C}_{\text{LW}}(T_{\text{GS}} - T_0))] \end{aligned} \quad (1.39)$$

From Figure 1.2 the quantity in square brackets is equal to ΔH_{VS} , and thus,

$$\bar{C}_{\text{PY}}(T_{\text{in}} - T_{\text{GS}}) = (Y_{\text{GS}} - Y_{\text{in}})\Delta H_{\text{VS}} \quad (1.40)$$

or

$$T_{\text{in}} - T_{\text{GS}} = (Y_{\text{GS}} - Y_{\text{in}}) \frac{\Delta H_{\text{VS}}}{\bar{C}_{\text{PY}}} \quad (1.41)$$

Equation 1.41 represents the “adiabatic saturation curve” on the psychrometric chart, which passes through the points $A(Y_{\text{GS}}, T_{\text{GS}})$ on the 100% saturation curve ($\psi = 1$) and $B(Y_{\text{in}}, T_{\text{in}})$, the initial condition. Since the humid heat contains the term Y_{in} , the curve is not straight but is curved slightly concave upward. Knowing the adiabatic saturation temperature and the actual gas temperature, the actual gas humidity can be easily obtained as the absolute humidity from the saturation locus. Equation 1.40 indicates that the sensible heat given up by the gas in cooling equals the latent heat required to evaporate the added vapor. It is important to note that, since Equation 1.41 is derived from the overall mass and energy balances between the initial gas conditions and the adiabatic

saturation conditions, it is applicable only at these points and may not describe the path followed by the gas as it becomes saturated. A family of these adiabatic saturation curves for the air–water system are contained in the psychrometric charts [10].

1.2.4.2 Wet Bulb Temperature

One of the oldest and best-known methods of determining the humidity of a gas is to measure its “wet bulb temperature” and its dry bulb temperature. The wet bulb temperature is the steady temperature reached by a small amount of liquid evaporating into a large amount of rapidly moving unsaturated vapor–gas mixture. It is measured by passing the gas rapidly past a thermometer bulb kept wet by a saturated wick and shielded from the effects of radiation. If the gas is unsaturated, some liquid is evaporated from the wick into the gas stream, carrying with it the associated latent heat. This latent heat is taken from within the liquid in the wick, and the wick is cooled. As the temperature of the wick is lowered, sensible heat is transferred by convection from the gas stream and by radiation from the surroundings. At steady state, the net heat flow to the wick is zero and the temperature is constant.

The heat transfer to the wick can be written as

$$q = (h_{\text{C}} + h_{\text{R}})A(T_{\text{G}} - T_{\text{W}}) \quad (1.42)$$

where

h_{C} and h_{R} are the convective and radiative heat transfer coefficients, respectively

T_{G} is the gas temperature

T_{W} is the temperature indicated by thermometer

By using h_{R} , it is assumed that radiant heat transfer can be approximated:

$$q_{\text{R}} = h_{\text{R}}A(T_{\text{G}} - T_{\text{W}}) \quad (1.43)$$

The rate of mass transfer from the wick is

$$N_{\text{G}} = KA(Y_{\text{W}} - Y_{\text{G}}) \quad (1.44)$$

An amount of heat given by

$$q = N_{\text{G}}\Delta H_{\text{VW}} \quad (1.45)$$

is associated with this mass transfer. Since under steady conditions all the heat transferred to the wick is utilized in mass transfer, from Equations 1.42, 1.44 and 1.45 we have

$$T_{\text{G}} - T_{\text{W}} = \frac{K\Delta H_{\text{VW}}}{h_{\text{C}} + h_{\text{R}}}(Y_{\text{W}} - Y_{\text{G}}) \quad (1.46)$$

The quantity $T_G - T_W$ is called the *wet bulb depression*. In order to determine the humidity Y_G from Equation 1.46, predictable values of $K\Delta H_{vw}/(h_C + h_R)$ must be obtained. This ratio of coefficients depends upon the flow, boundary, and temperature conditions encountered. In measuring the wet bulb temperature, several precautions are taken to ensure reproducible values of $K\Delta H_{vw}/(h_C + h_R)$. The contribution by radiation is minimized by shielding the wick. The convective heat transfer can be enhanced by making the gas movement past the bulb rapid, often by swinging the thermometer through the gas, as in the sling psychrometer, or by inserting the wet bulb thermometer in a constriction in the gas flow path. Under these conditions Equation 1.46 reduces to

$$T_G - T_W = \frac{K\Delta H_{vw}}{h_C} (Y_W - Y_G) \quad (1.47)$$

For turbulent flow past a wet cylinder, such as a wet bulb thermometer, the accumulated experimental data give

$$\frac{h_C}{K} = 35.53 \left(-\frac{\mu}{\rho D} \right)^{0.56} \text{ J/mol } ^\circ\text{C} \quad (1.48)$$

when air is the noncondensable gas and

$$\frac{h_C}{K} = \bar{C}_{PY} \left(\frac{Sc}{Pr} \right)^{0.56} \quad (1.49)$$

for other gases. Equation 1.49 is based on heat and mass transfer experiments with various gases flowing normal to cylinders. For pure air, $Sc \cong Pr \cong 0.70$ and $h_C/K = 29.08 \text{ J/mol } ^\circ\text{C}$ from Equations 1.48 and 1.49. Experimental data for the air–water system yield values of h_C/K ranging between 32.68 and 28.54 $\text{J/mol } ^\circ\text{C}$. The latter figure is recommended [11]. For the air–water system, the h_C/K value can be replaced by $\bar{C}_{PY}$ within moderate ranges of temperature and humidity, provided flow is turbulent. Under these conditions, Equation 1.47 becomes identical to the adiabatic saturation curve Equation 1.41 and thus the adiabatic saturation temperature is the same as the wet bulb temperature for the air–water system. For systems other than air–water, they are not the same, as can be seen from the psychrometric charts given by Perry [7].

It is worthwhile pointing out here that, although the adiabatic saturation curve equation does not reveal anything of the enthalpy–humidity path of either the liquid phase or gas phase at various points in the contacting device (except for the air–water vapor system), each point within the system must conform with the wet bulb relation, which requires that the heat transferred be exactly consumed as latent heat of vaporization of the mass of liquid evaporated. The identity of h_C/K with $\bar{C}_{PY}$ was first found empirically by Lewis and hence is called the Lewis relation. The treatment given here on the wet bulb temperature applies only in the limit of very mild drying conditions when the vapor flux becomes directly

proportional to the humidity potential ΔY . This is the case in most drying operations.

A more detailed treatment using a logarithmic driving force for vapor flux and the concept of the humidity potential coefficient Φ while accounting for the influence of the moisture vapor flux on the transfer of heat to the surface, namely, the Ackermann correction Φ_E , has been given in Ref. [3]. The concept of Luikov number Lu , which is essentially the ratio of the Prandtl number Pr to the Schmidt number Sc , has also been introduced.

1.2.5 TYPES OF PSYCHROMETRIC REPRESENTATION

As stated previously, two processes occur simultaneously during the thermal process of drying a wet solid: heat transfer, to change the temperature of the wet solid, and mass transfer of moisture to the surface of a solid accompanied by its evaporation from the surface to the surrounding atmosphere, which in convection or direct dryers is the drying medium. Consideration of the actual quantities of air required to remove the moisture liberated by evaporation is based on psychrometry and the use of humidity charts. This procedure is extremely important in the design of forced convection, pneumatic, and rotary dryers. The definitions of terms and expressions involved in psychrometry have been discussed in Section 1.2.3.

There are different ways of plotting humidity charts. One procedure involves plotting the absolute humidity against the dry bulb temperature. A series of curves is obtained for different percentage humidity values from saturation downward (Figure 1.3). On this chart, the saturation humidities are plotted from vapor pressure data with the help of Equation 1.23 to give curve GD . The curve for humidities at 50% saturation is plotted at half the ordinate of curve GD . All curves at constant percentage saturation reach infinity at the boiling point of the liquid at the prevailing pressure.

Another alternative is the graphic representation of conditions of constant relative saturation on a vapor pressure–temperature chart (Figure 1.3). The curve for 50% relative saturation shows a partial pressure equal to one-half of the equilibrium vapor pressure at any temperature. A common method of portraying humidity charts is by using the enthalpy–humidity chart indicated earlier [10].

1.3 INTERNAL CONDITIONS (PROCESS 2)

After having discussed the factors and definitions related to the external conditions of air temperature and humidity, attention will now be paid to the solid characteristics.

As a result of heat transfer to a wet solid, a temperature gradient develops within the solid while moisture evaporation occurs from the surface. This produces a migration of moisture from within the solid to the surface, which occurs through one or more mechanisms, namely, diffusion, capillary flow, internal pressures set up by shrinkage during drying, and, in the case of indirect (conduction) dryers, through a repeated and progressive occurring vaporization and

recondensation of moisture to the exposed surface. An appreciation of this internal movement of moisture is important when it is the controlling factor, as it occurs after the critical moisture content, in a drying operation carried to low final moisture contents. Variables such as air velocity and temperature, which normally enhance the rate of surface evaporation, are of decreasing importance except to promote the heat transfer rates. Longer residence times, and, where permissible, higher temperatures become necessary. In the case of such materials as ceramics and timber, in which considerable shrinkage occurs, excessive surface evaporation sets up high moisture gradients from the interior toward the surface, which is liable to cause overdrying, excessive shrinkage, and, consequently, high tension, resulting in cracking or warping. In such cases, it is essential not to incur too high moisture gradients by retarding surface evaporation through the employment of high air relative humidities while maintaining the highest safe rate of internal moisture movement by virtue of heat transfer. The temperature gradient set up in the solid will also create a vapor–pressure gradient, which will in turn result in moisture vapor diffusion to the surface; this will occur simultaneously with liquid moisture movement.

1.3.1 MOISTURE CONTENT OF SOLIDS

The moisture contained in a wet solid or liquid solution exerts a vapor pressure to an extent depending upon the nature of moisture, the nature of solid, and the temperature. A wet solid exposed to a continuous supply of fresh gas continues to lose moisture until the vapor pressure of the moisture in the solid is equal to the partial pressure of the vapor in the gas. The solid and gas are then said to be in *equilibrium*, and the moisture content of the solid is called the *equilibrium moisture content* under the prevailing conditions. Further exposure to this air for indefinitely long periods will not bring about any additional loss of moisture. The moisture content in the solid could be reduced further by exposing it to air of lower relative humidity. Solids can best be classified as follows [12]:

Nonhygroscopic capillary-porous media, such as sand, crushed minerals, nonhygroscopic crystals, polymer particles, and some ceramics. The defining criteria are as follows. (1) There is a clearly recognizable pore space; the pore space is filled with liquid if the capillary-porous medium is completely saturated and is filled with air when the medium is completely dry. (2) The amount of physically bound moisture is negligible; that is, the material is nonhygroscopic. (3) The medium does not shrink during drying.

Hygroscopic-porous media, such as clay, molecular sieves, wood, and textiles. The defining criteria are as follows. (1) There is a clearly recognizable pore space. (2) There is a large amount of physically bound liquid. (3) Shrinkage often occurs in the initial stages of drying. This category was further classified into (a) hygroscopic capillary-porous media (micropores and macropores, including bidisperse

media, such as wood, clays, and textiles) and (b) strictly hygroscopic media (only micropores, such as silica gel, alumina, and zeolites).

Colloidal (nonporous) media, such as soap, glue, some polymers (e.g., nylons), and various food products. The defining criteria are as follows: (1) there is no pore space (evaporation can take place only at the surface); (2) all liquid is physically bound.

It should be noted that such classifications are applicable only to homogeneous media that could be considered as continua for transport.

As a wet solid is usually swollen compared with its condition when free of moisture and its volume changes during the drying process, it is not convenient to express moisture content in terms of volume. The moisture content of a solid is usually expressed as the moisture content by weight of bone-dry material in the solid, X . Sometimes a wet basis moisture content W , which is the moisture–solid ratio based on the total mass of wet material, is used. The two moisture contents are related by the expression

$$X = \frac{W}{1 - W} \quad (1.50)$$

Water may become bound in a solid by retention in capillaries, solution in cellular structures, solution with the solid, or chemical or physical adsorption on the surface of the solid. Unbound moisture in a hygroscopic material is the moisture in excess of the equilibrium moisture content corresponding to saturation humidity. All the moisture content of a nonhygroscopic material is unbound moisture. Free moisture content is the moisture content removable at a given temperature and may include both bound and unbound moisture.

In the immediate vicinity of the interface between free water and vapor, the vapor pressure at equilibrium is the saturated vapor pressure. Very moist products have a vapor pressure at the interface almost equal to the saturation vapor pressure. If the concentration of solids is increased by the removal of water, then the dissolved hygroscopic solids produce a fall in the vapor pressure due to osmotic forces. Further removal of water finally results in the surface of the product dried. Water now exists only in the interior in very small capillaries, between small particles, between large molecules, and bound to the molecules themselves. This binding produces a considerable lowering of vapor pressure. Such a product can therefore be in equilibrium only with an external atmosphere in which the vapor pressure is considerably decreased.

1.3.2 MOISTURE ISOTHERMS [10]

A dry product is called hygroscopic if it is able to bind water with a simultaneous lowering of vapor pressure. Different products vary widely in their hygroscopic properties. The reason for this is their molecular structure, their solubility, and the extent of reactive surface.

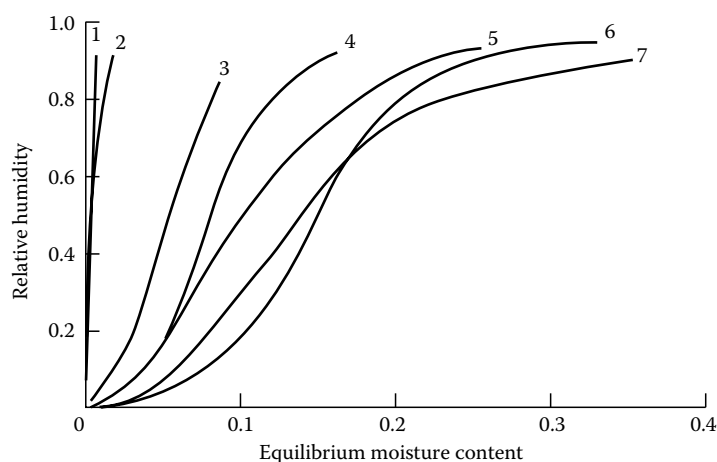


FIGURE 1.6 Typical equilibrium moisture isotherms at room temperature for selected substances: (1) asbestos fiber, (2) PVC (50°C), (3) wood charcoal, (4) Kraft paper, (5) jute, (6) wheat, (7) potatoes.

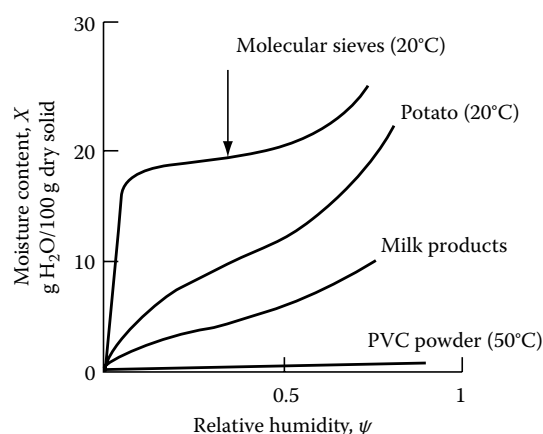


FIGURE 1.7 Shapes of sorption isotherms for materials of varying hygroscopicity.

Sorption isotherms measured experimentally under isothermal conditions are used to describe the hygroscopic properties of a product. A graph is constructed in which the moisture bound by sorption per unit weight is plotted against relative humidity, and vice versa. Such isotherms are shown in Figures 1.6 and 1.7. From Figure 1.7 it is seen that molecular sieves are highly hygroscopic but polyvinyl chloride (PVC) powder is mildly hygroscopic. Potatoes and milk exhibit intermediate hygroscopicity.

Figure 1.8 shows the shape of the sorption isotherm characteristic of many dry food products. If the partial pressure of the external atmosphere P_w is nearly zero, then the equilibrium moisture inside the dry product will also be almost zero. Section A of the curve represents a region in which the monomolecular layers are formed, although there may be multimolecular layers in some places toward the end of A. Section B is a transitional region in which double and multiple layers are mainly formed. Capillary condensation could also have taken place. In section C the slope of the curve increases again, which is attributed mainly to increasing capillary condensation and swelling. The maximum hygroscopicity

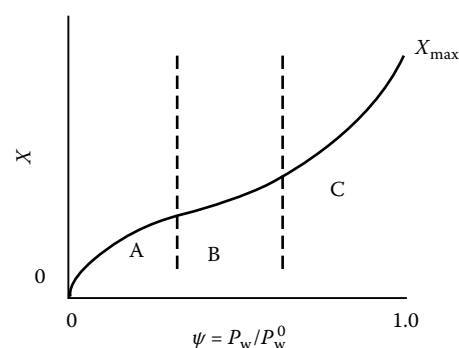


FIGURE 1.8 A typical isotherm (see text for explanation of areas within dashed lines).

$X_{\max}$ is achieved when the solid is in equilibrium with air saturated with moisture ($\psi = 1$).

1.3.2.1 Sorption–Desorption Hysteresis

The equilibrium moisture content of a product may be different depending on whether the product is wetted (sorption or absorption) or dried (desorption) (Figure 1.9). These differences are observed to varying degrees in almost all hygroscopic products.

One of the hypotheses used to explain hysteresis is to consider a pore connected to its surroundings by a small capillary [10]. During absorption, as the relative humidity rises, the capillary begins to fill while the pore is empty. Only when the partial pressure of the vapor in air is greater than the vapor pressure of the liquid in the capillary will the moisture move into the pore. Starting from saturation the pore is full of liquid. This fluid can only escape when the partial pressure of the surrounding air falls below the vapor pressure of the liquid in the capillary. Since the system of pores has generally a large range of capillary diameters, it follows that differences between adsorption and desorption will be observed. This theory assumes that the pore is a rigid structure. This is not true for foods or synthetic materials, although these show hysteresis. The explanation is that contraction and swelling

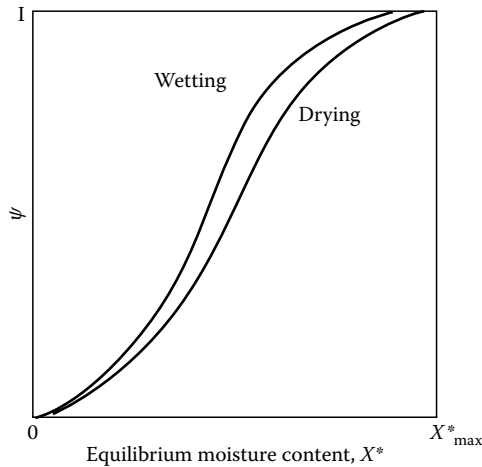


FIGURE 1.9 Wetting and drying isotherms for a typical hygroscopic solid.

are superimposed on the drying and wetting processes, producing states of tension in the interior of the products and leading to varying equilibrium moisture contents depending on whether desorption or absorption is in progress.

1.3.2.2 Temperature Variations and Enthalpy of Binding

Moisture isotherms pertain to a particular temperature. However, the variation in equilibrium moisture content for small changes of temperature ($<10^{\circ}\text{C}$) is neglected [3]. To a first approximation, the temperature coefficient of the equilibrium moisture content is proportional to the moisture content at a given relative humidity:

$$\left(\frac{\partial X^*}{\partial T} \right)_{\psi} = -AX^* \quad (1.51)$$

The coefficient A lies between 0.005 and 0.01 per kelvin for relative humidities between 0.1 and 0.9 for such materials as natural and synthetic fibers, wood, and potatoes. A could be taken to increase linearly with ψ . So for $\psi = 0.5$ there is a 0.75% fall in moisture content for each degree kelvin rise in temperature. The extent of absorption–desorption hysteresis becomes smaller with increasing temperature.

Figure 1.10 shows moisture isotherms at various temperatures. The binding forces decrease with increasing temperature; that is, less moisture is absorbed at higher temperatures at the same relative humidity. Kessler [10] has shown that the slope of a plot of $\ln(P_w/P_w^0)$ versus $1/T$ at constant X (Figure 1.11) gives the enthalpy of binding. The variation of enthalpy of binding versus moisture content is shown in Figure 1.12. From the figure it is seen that in the region where monomolecular layers are formed, enthalpies of binding are very high.

1.3.3 DETERMINATION OF SORPTION ISOTHERMS [10]

The sorption isotherms are established experimentally starting mostly with dry products. The initial humidity of the air with which the product is in equilibrium should be brought

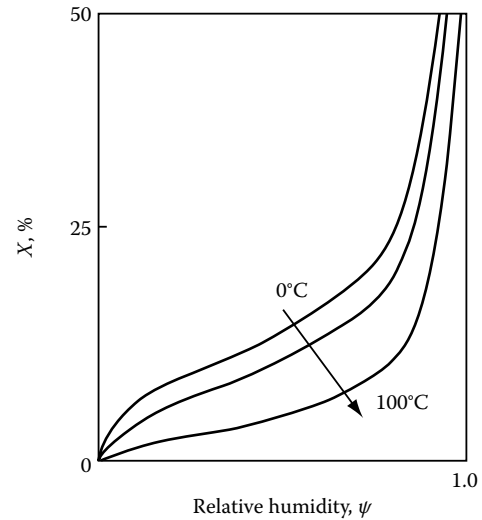


FIGURE 1.10 Sorption isotherms for potatoes.

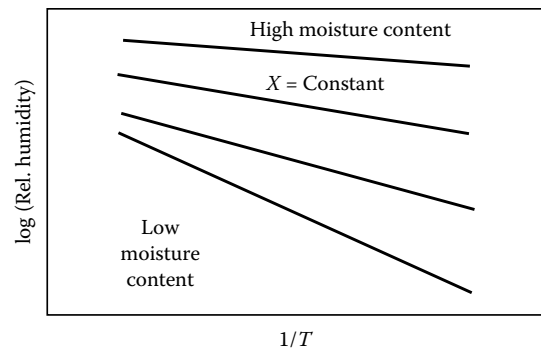


FIGURE 1.11 Determination of the heat of sorption from sorption isotherms.

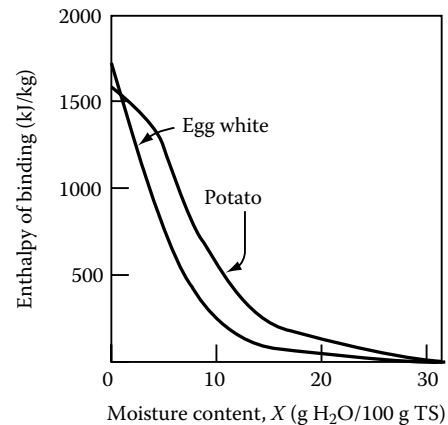


FIGURE 1.12 Enthalpy of sorption as a function of the hygroscopic moisture content. (Egg white data by Nemitz; potato data by Krischer.)

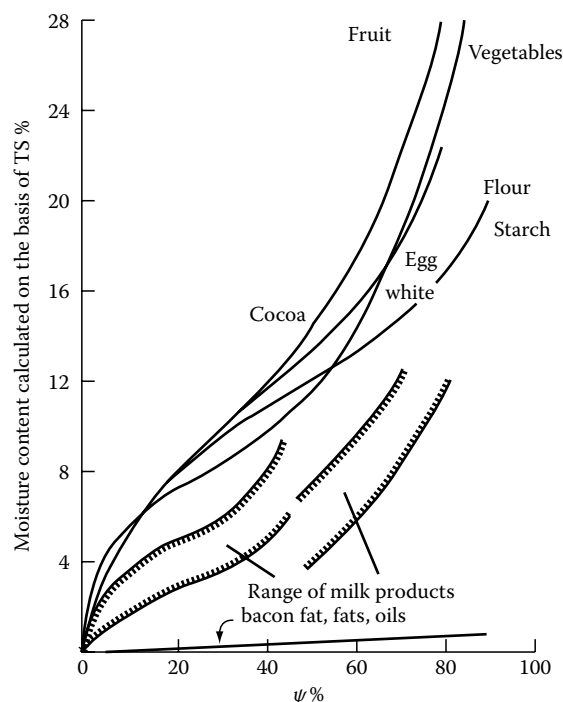


FIGURE 1.13 Range of sorption isotherms of various foods at room temperature.

to extremely low values using either concentrated sulfuric acid or phosphorus pentoxide, so that the moisture content of the product is close to zero at the beginning. The product is then exposed to successively greater humidities in a thermostatically controlled atmosphere. Sufficient time must be allowed for equilibrium between the air and solid to be attained. Using thin slices of the product, moving air and especially vacuum help to establish equilibrium quickly. This is especially important for foodstuffs: there is always the danger of spoilage. There are severe problems associated with the maintenance of constant humidity and temperature. These problems could be alleviated by using sulfuric acid–water mixtures and saturated salt solutions to obtain different relative humidities [10,13]. Figure 1.13 depicts the absorption isotherms of a range of food products. Further information on solid moisture characteristics, enthalpy of wetting, and sorption isotherms are available in Refs. [3,10].

1.4 MECHANISM OF DRYING

As mentioned above, moisture in a solid may be either unbound or bound. There are two methods of removing unbound moisture: evaporation and vaporization. Evaporation occurs when the vapor pressure of the moisture on the solid surface is equal to the atmospheric pressure. This is done by raising the temperature of the moisture to the boiling point. This kind of phenomenon occurs in roller dryers.

If the material dried is heat sensitive, then the temperature at which evaporation occurs, that is, the boiling point, could be lowered by lowering the pressure (vacuum evaporation). If the pressure is lowered below the triple point, then no liquid

phase can exist and the moisture in the product is frozen. The addition of heat causes sublimation of ice directly to water vapor as in the case of freeze drying.

Second, in vaporization, drying is carried out by convection, that is, by passing warm air over the product. The air is cooled by the product, and moisture is transferred to the air by the product and carried away. In this case the saturation vapor pressure of the moisture over the solid is less than the atmospheric pressure.

A preliminary necessity to the selection of a suitable type of dryer and design and sizing thereof is the determination of the drying characteristics. Information also required are the solid-handling characteristics, solid moisture equilibrium, and material sensitivity to temperature, together with the limits of temperature attainable with the particular heat source. These will be considered later and in other sections of this book.

The drying behavior of solids can be characterized by measuring the moisture content loss as a function of time. The methods used are humidity difference, continuous weighing, and intermittent weighing. Descriptions of these methods are available in Refs. [3,13].

Figure 1.14 qualitatively depicts a typical drying rate curve of a hygroscopic product. Products that contain water behave differently on drying according to their moisture content. During the first stage of drying the drying rate is constant. The surface contains free moisture. Vaporization takes place from there, and some shrinkage might occur as the moisture surface is drawn back toward the solid surface. In this stage of drying the rate-controlling step is the diffusion of the water vapor across the air–moisture interface and the rate at which the surface for diffusion is removed. Toward the end of the constant rate period, moisture has to be transported from the inside of the solid to the surface by capillary forces and the drying rate may still be constant. When the average moisture content has reached the critical

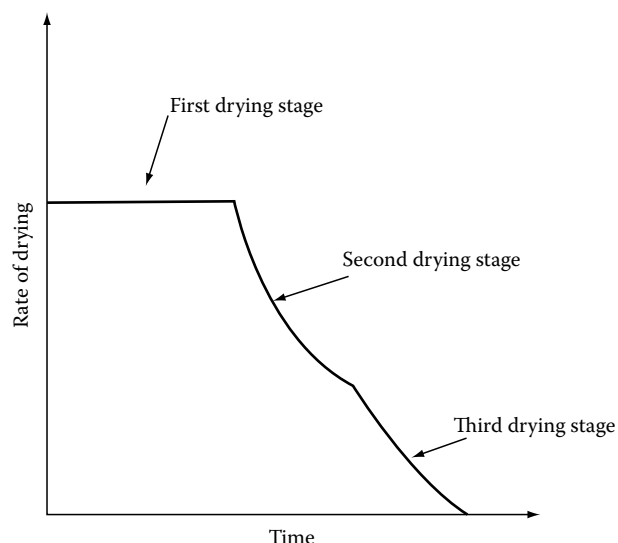


FIGURE 1.14 Typical rate-of-drying curve, constant drying conditions.

moisture content X_{cr} , the surface film of moisture has been so reduced by evaporation that further drying causes dry spots to appear upon the surface. Since, however, the rate is computed with respect to the overall solid surface area, the drying rate falls even though the rate per unit wet solid surface area remains constant. This gives rise to the second drying stage or the first part of the falling rate period, the period of unsaturated surface drying. This stage proceeds until the surface film of liquid is entirely evaporated. This part of the curve may be missing entirely, or it may constitute the whole falling rate period.

On further drying (the second falling rate period or the third drying stage), the rate at which moisture may move through the solid as a result of concentration gradients between the deeper parts and the surface is the controlling step. The heat transmission now consists of heat transfer to the surface and heat conduction in the product. Since the average depth of the moisture level increases progressively and the heat conductivity of the dry external zones is very small, the drying rate is increasingly influenced by the heat conduction. However, if the dry product has a relatively high bulk density and a small cavity volume with very small pores, drying is determined not so much by heat conduction but by a rather high resistance to diffusion within the product. The drying rate is controlled by diffusion of moisture from the inside to the surface and then mass transfer from the surface. During this stage some of the moisture bound by sorption is removed. As the moisture concentration is lowered by drying, the rate of internal movement of moisture decreases. The rate of drying falls even more rapidly than before and continues until the moisture content falls down to the equilibrium value X^* for the prevailing air humidity and then drying stops. The transition from one drying stage to another is not sharp, as indicated in Figure 1.14.

In actual practice, the original feedstock may have a high moisture content and the product may be required to have a high residual moisture content so that all the drying may occur in the constant rate period. In most cases however both phenomena exist, and for slow-drying materials most of the drying may occur in the falling rate period. As mentioned earlier, in the constant rate period the rate of drying is determined by the rate of evaporation. When all the exposed surface of the solid ceases to be wetted, vapor movement by diffusion and capillarity from within the solid to the surface are the rate-controlling steps. Whenever considerable shrinkage occurs, as in the drying of timber, pressure gradients are set up within the solid and these may assume importance. In this case, as in the case of materials that “caseharden,” that is, form a hard impermeable skin, it is essential to retard evaporation and bring it in step with the rate of moisture movement from the interior. This could be achieved by increasing the relative humidity of the drying air. With solids, in which the initial moisture content is relatively low and the final moisture content required is extremely low, the falling rate period becomes important. Dryness times are long. Air velocities will be important only to the extent to which they enhance heat transfer rates. Air temperature, humidity, material

thickness, and bed depth all become important. When the rate of diffusion is the controlling factor, particularly when long drying periods are required to attain low moisture contents, the rate of drying during the falling rate period varies as the square of the material thickness, which indicates the desirability of granulating the feedstock using agitation or using thin layers in case of crossflow tray dryers. Thus the drying characteristics of the solid are extremely important in dryer design.

1.4.1 CHARACTERISTIC DRYING RATE CURVE [14]

When the drying rate curves are determined over a range of conditions for a given solid, the curves appear to be geometrically similar and are simply a function of the extent to which drying has occurred. If these curves were normalized with respect to the initial drying rate and average moisture content, then all the curves could often be approximated to a single curve, “characteristic” of a particular substance. This is the characteristic drying curve. The normalized variables, the characteristic drying rate f and the characteristic moisture content Φ , are defined as follows:

$$f = \frac{N_v}{N_w}$$

and

$$\phi = \frac{\bar{X} - X^*}{X_{cr} - X^*}$$

where

- N_v is the rate of drying for a unit surface
- N_w is the rate when the body is fully saturated or the initial drying rate
- $\bar{X}$ is the average moisture content in the body
- X_{cr} is the corresponding critical point value
- X^* is the equilibrium moisture content

If a solid's drying behavior is to be described by the characteristic curve, then its properties must satisfy the following two criteria:

1. The critical moisture content $\bar{X}_{cr}$ is invariant and independent of initial moisture content and external conditions
2. All drying curves for a specific substance are geometrically similar so that the shape of the curve is unique and independent of external conditions

These criteria are restrictive, and it is quite unlikely that any solid will satisfy them over an exhaustive range of conditions; however, the concept is widely used and often utilized for interpolation and prediction of dryer performance [3,17]. The use of the mean moisture content as an index of the degree of drying contains the implicit assumption that the extent of

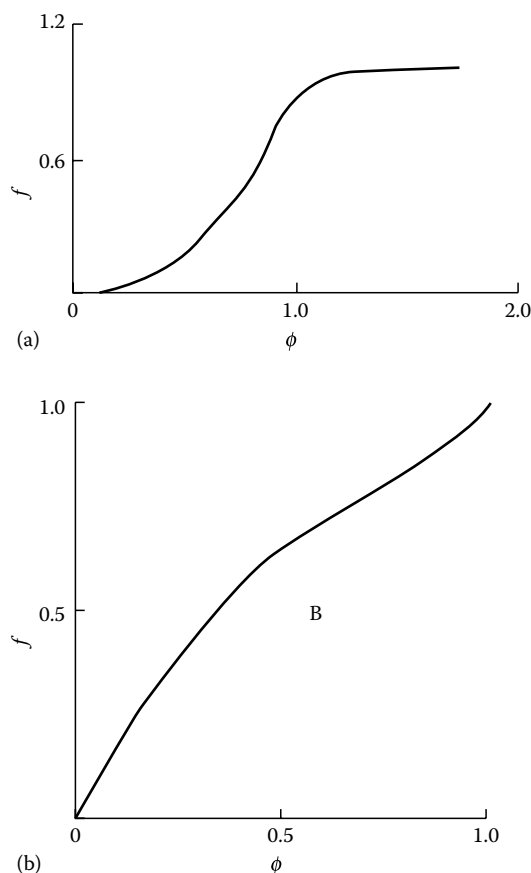


FIGURE 1.15 Experimental drying rates for (a) molecular sieves 13-X particles with $X_0 = 0.27$, diameter = 2.2 mm, air velocity = 4.4 m/s, $T_G \cong 36.5^\circ\text{C}$ – 97°C ; (b) silica-gel particles with diameter = 3.0 mm, air velocity = 1 m/s, $T_G = 54^\circ\text{C}$ – 68°C , $T_w = 25^\circ\text{C}$ – 29°C .

Drying of solids

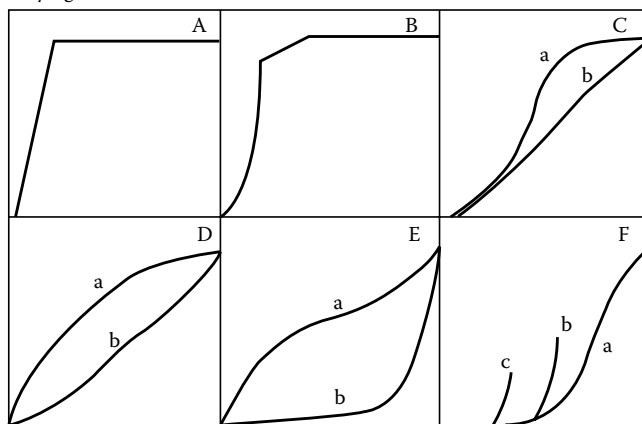


FIGURE 1.16 Examples of normalized drying rate curves for different types of media: (A) sand, clay, silica gel, paper pulp, leather; (B) sand, plastic-clay mix, silica-brick mix, ceramic plate, leather; (C) [a] fir wood and [b] Cyprus wood; (D) [a] paper, wool, and [b] potatoes, tapioca tuber, rice flour; (E) [a] rye bread, yeast, and [b] butter and margarine; (F) [a] wheat corns, [b] and [c] represent curves at lower values of initial moisture.

drying at a mean moisture content will also depend on the relative extensiveness of the exposed surface per unit volume of material. Thus, similar drying behavior may be expected only in the case of materials that are unchanged in form. A typical characteristic drying curve is shown in Figure 1.15.

Further information on the characteristic drying curve, extrapolation procedures used, and the theoretical developments in examining the range of validity of the characteristic drying rate model are available [3,13,25]. The various types of characteristic drying curves have been depicted schematically in Figure 1.16.

1.5 CLASSIFICATION AND SELECTION OF DRYERS

With a very few exceptions, most products from today's industry undergo drying at some stage or another. A product must be suitable for either subsequent processing or sale. Materials need to have a particular moisture content for processing, molding, or pelleting. Powders must be dried to suitable low moisture contents for satisfactory packaging. Whenever products are heated to high temperatures, as in ceramic and metallurgical processes, predrying at lower temperatures ahead of firing kilns is advantageous for energy savings. Cost of transport (as in the case of coal) depends on the moisture content of the product, and a balance must be struck between the cost of conveying and the cost of drying. Excessive drying is wasteful; not only is more heat, that is, expense, involved than is necessary, but often overdrying results in a degraded product, as in the case of paper and timber. Consideration must be given to methods involved in energy savings in dryers [15]. Examples of products and the types of precautions that need to be taken during drying are highlighted. Thermal drying is an essential stage in the manufacture of colors and dyes. Many inorganic colors and most organic dyes are heat sensitive; the time–temperature effect in drying may be critical in arriving at a correct shade of color or in the elimination of thermal degradation. Drying is normally accomplished at low temperatures and in the absence of air. The most widely used dryer is the recirculation type truck and tray compartment dryer. Most pharmaceuticals and fine chemicals require drying before packaging. Large turbo-tray and through circulation dryers are employed. Excessively heat-sensitive products, such as antibiotics and blood plasma, call for special treatment, like freeze drying or high vacuum tray drying. Continuous rotary dryers are usually used to handle large tonnages of natural ores, minerals, and heavy chemicals.

The largest demand for drying equipment is for the continuous drying of paper, which is done on cylinder or “can” dryers. The temperature and humidity conditions are important to the consistency of the paper. Thermal drying is essential in the foodstuffs and agricultural fields. Spray drying and freeze drying are also widely used. In the ceramic industry, drying is a vital operation. Great care must be exercised because of the considerable shrinkage that occurs in drying. Thus control of humidity is important. Drying is also widely

used in the textile industry. The need for product quality puts grave constraints on the dryer chosen and dryer operation [13]. Quality depends on the end use of the product. For many bulk chemicals, handling considerations determine moisture content requirements. For foodstuffs, flavor retention, palatability, and rehydration properties are important. Timber must retain its strength and decorative properties after drying.

The choice of end moisture content is largely dictated by storage and stability requirements. The end moisture content determines the drying time and conditions required for drying. Overdrying should be avoided. Internal moisture gradients within particles and interparticle moisture content variation are important. Temperature restrictions may arise because of degradation, phase changes, discoloration and staining, flammability of dust, and other factors. Thermal sensitivity fixes the maximum temperature to which the substance can be exposed for the drying time. In spray and pneumatic dryers, the retention time of a few seconds permits drying heat-sensitive materials at higher temperatures. Many hygroscopic materials shrink on drying. The extent of shrinkage is linearly related to the moisture content change below the hygroscopic limit. Case hardening and tensile cracking pose problems. Details on how some of these problems are overcome by a compromise between energy efficiency and drying time are available [13].

The first subdivision is based on methods of heat transfer, namely, (1) conduction heating, (2) convection heating, (3) radiant heating, and (4) dielectric heating. Freeze drying is classified as a special case of conduction heating. The next subdivision is the type of drying vessel: tray, rotating drum, fluidized bed, pneumatic, or spray. Dryer classification based on physical form of the feed has also been done. These are indicated in Ref. [16].

Sloan [18] has identified some 20 types of dryers and classified them according to whether they are batch or continuous,

conduct heat exchange through direct contact with gases or by heat exchange through vessel walls, and according to the motion of the equipment. Such a classification, though helpful in distinguishing and describing discrete systems, does not go far in relating the discrete systems to the process problems they are supposed to handle. McCormick [19] has tried to tackle the problem from the user's point of view. A total of 19 types of dryers were classified according to how well they handle different materials.

Schlünder [23] has given a classification of dryers that encompasses the physical state of the product as well as the dwell time of the product in the dryer. For very short drying times (<1 min), flash, spray, or drum dryers are used. For very long drying times (>1 h), only tunnel, truck, or conveyor dryers are appropriate. Most dryers operate in the intermediate range, for which a very wide assortment of dryers is available [21].

Table 1.1 gives a summary of the type of dryer versus the type of feedstock, which may be a slurry, paste, filter cake, powder, granules, crystal, pellet, or fibrous or shaped material. Since thermal sensitivity as well as efficiency and dryer size depend to a major extent on the thermal conditions the product is exposed to within the dryer, Table 1.2 is presented to classify convection and conduction dryers on this basis. Such information is often helpful in narrowing down the choice of dryers.

Keey [3,25] has noted three principal factors that could be utilized in classifying dryers:

1. Manner in which heat is supplied to the material
2. Temperature and pressure of operation (high, medium, or low temperature; atmospheric or vacuum drying)
3. Manner in which the material is handled within the dryer

TABLE 1.1
Dryer Selection versus Feedstock Form

Nature of Feed	Liquids			Cakes		Free-Flowing Solids					Formed Solids
	Solution	Slurry	Pastes	Centrifuge	Filter	Powder	Granule	Fragile Crystal	Pellet	Fiber	
<i>Convection dryers</i>											
Belt conveyer dryer							×	×	×	×	×
Flash dryer				×	×	×	×			×	
Fluid bed dryer	×	×		×	×	×	×		×		
Rotary dryer				×	×	×	×		×	×	
Spray dryer	×	×	×								
Tray dryer (batch)				×	×	×	×	×	×	×	×
Tray dryer (continuous)				×	×	×	×	×	×	×	
<i>Conduction dryers</i>											
Drum dryer	×	×	×								
Steam jacket rotary dryer				×	×	×	×		×	×	
Steam tube rotary dryer				×	×	×	×		×	×	
Tray dryer (batch)				×	×	×	×	×	×	×	×
Tray dryer (continuous)				×	×	×	×	×	×	×	

TABLE 1.2
Solids' Exposures to Heat Conditions

Dryers	Typical Residence Time within Dryer				
	0–10 (s)	10–30 (s)	5–10 (min)	10–60 (min)	1–6 (h)
<i>Convection</i>					
Belt conveyor dryer				×	
Flash dryer	×				
Fluid bed dryer				×	
Rotary dryer				×	
Spray dryer		×			
Tray dryer (batch)					×
Tray dryer (continuous)				×	
<i>Conduction</i>					
Drum dryer		×			
Steam jacket rotary dryer				×	
Steam tube rotary dryer				×	
Tray dryer (batch)					×
Tray dryer (continuous)				×	

Further subclassification is of course possible but generally unnecessary. For example, a given dryer may be operated in batchwise or continuous mode.

1.5.1 HEATING METHODS

1.5.1.1 Convection

Convection is possibly the most common mode of drying particulate or sheet-form or pasty solids. Heat is supplied by heated air or gas flowing over the surface of the solid. Heat for evaporation is supplied by convection to the exposed surface of the material and the evaporated moisture carried away by the drying medium. Air (most common), inert gas (such as N_2 for drying solids wet with organic solvent), direct combustion gases, or superheated steam (or solvent vapor) can be used in convective drying systems.

Such dryers are also called direct dryers. In the initial constant rate drying period (drying in which surface moisture is removed), the solid surface takes on the wet bulb temperature corresponding to the air temperature and humidity conditions at the same location. In the falling rate period the solids' temperature approaches the dry bulb temperature of the medium. These factors must be considered when drying heat-sensitive solids.

When drying with superheated vapors, the solids' temperature corresponds to the saturation temperature at the operating pressure, for example, 100°C for steam at 1 atm. For solids susceptible to oxidation or denaturation, for example, in the presence of oxygen, the heat sensitivity can be quite different in a steam environment. Product quality may differ as well. This can only be ascertained through laboratory tests.

Examples of convective (direct) dryers are air suspension dryers, such as fluid bed, flash, rotary, or spray dryers; air impingement dryers for paper or pulp; packed bed or through dryers; and conveyor-truck-tunnel dryers.

1.5.1.2 Conduction

Conduction or indirect dryers are more appropriate for thin products or for very wet solids. Heat for evaporation is supplied through heated surfaces (stationary or moving) placed within the dryer to support, convey, or confine the solids. The evaporated moisture is carried away by vacuum operation or by a stream of gas that is mainly a carrier of moisture. Vacuum operation is recommended for heat-sensitive solids. Because the enthalpy lost with the drying air in convective dryers is large, their thermal efficiency tends to be low. For conduction dryers the thermal efficiency is higher. Paddle dryers for drying of pastes, rotary dryers with internal steam tubes, and drum dryers for drying thin slurries are examples of indirect dryers.

A more efficient dryer can be designed for some operations that combines advantages of both direct and indirect heating, for example, a fluid bed dryer with immersed heating tubes or coils for drying of heat-sensitive polymer or resin pellets. Such a dryer can be only one third the size of a purely convective fluid bed dryer for the same duty.

It is noteworthy that sometimes one can operate the same apparatus in direct, indirect, or combined modes. For example, a vibrated fluid bed dryer can be purely convective (e.g., drying of tea), purely conductive (e.g., vacuum drying of pharmaceutical granules), or combined direct–indirect (e.g., drying of pulverized coal with immersed heating tubes). The drying medium could be steam for such products as coal.

1.5.1.3 Radiation

Various sources of electromagnetic radiation with wavelengths ranging from the solar spectrum to microwave (0.2 m to 0.2 μ m). Solar radiation barely penetrates beyond the skin of the material, which absorbs only a part of the incident radiation depending on its wavelength. Infrared radiation is often used in drying coatings, thin sheets, and films, for example (4–8 μ m band). Although most moist materials are poor conductors of 50–60 Hz current, the impedance falls dramatically at RF; such radiation can be used to heat the solid volumetrically, thus reducing internal resistance to heat transfer. Energy is absorbed selectively by the water molecules: as the product gets drier less energy is used. The capital as well as operating costs are high, and so these techniques are useful for drying high unit value products or for final correction of moisture profile wherein only small quantities of hard-to-get moisture are removed, as in moisture profiling of paper using RF heating. Combined mode drying with convection (e.g., infrared plus air jets or microwave with impingement for drying of sheet-form foodstuffs) is also commercially feasible.

1.5.2 TEMPERATURE AND PRESSURE OF OPERATION

Most dryers are operated at near atmospheric pressures. A slight positive pressure avoids in-leakage from outside, which may not be permissible in some cases. If no leakage is permitted to the outside, then a slight negative pressure is used.

Vacuum operation is expensive and is recommended only if the product must be dried at low temperatures or in

the absence of oxygen or has flavors that are generated at medium- or high-temperature operation. High-temperature operation tends to be more efficient since lower gas flow rates and smaller equipment may be used for a given evaporation duty. Availability of low-temperature waste heat or energy from solar collectors may dictate the choice of a lower temperature operation. These dryers will then be large in size.

Freeze drying is a special case of drying under vacuum at a temperature below the triple point of water; here water (ice) sublimates directly into water vapor. Although the heat required for sublimation is severalfold lower than for evaporation, vacuum operation is expensive. Freeze drying of coffee, for example, costs two to three times more than spray drying. On the other hand, the product quality and flavor retention are better.

1.5.3 CONVEYING OF MATERIAL IN DRYER

Handling the material to be dried is of course one of the key considerations in dryer selection. This is best illustrated in Table 1.3. In some cases the material may be treated or preformed to make it suitable for handling in a particular dryer, for example, reslurrying of filter cake to make it pumpable for atomization and spray drying or pelletizing pasty materials. This, of course, costs extra and must be considered in an overall evaluation. The final product characteristics and quality requirements also govern the choice of dryer. In some cases a combination of two or more different types of dryers may be the optimal strategy if the product-handling properties change significantly as it dries or if its heat sensitivity changes during the process of drying.

It should be pointed out that many new dryers cannot conveniently fit the classification suggested earlier. For example, a pulsed combustion dryer for pasty solids or waste sludge, the Remaflam process for drying textiles by controlled combustion of solvent (alcohol) on the wet fabric itself, and vibrated bed drying of pastes cannot be placed under any one single category of dryers. Such a “coarse” classification is still of interest in that it allows one to “home in” on a limited number of possible dryers, which can then be evaluated in depth.

One must look very carefully into some of the newer and novel dryers although they are not even mentioned in most textbooks or handbooks [22]. Many of them have the potential to supplant some of the age-old drying technologies, at least in some industrial applications.

Dittman [20] has proposed a structured classification of dryers according to two general classes and five subclasses. The general classes are adiabatic or nonadiabatic dryers. Adiabatic dryers are further subclassified according to whether drying gases pass through the material (for permeable solids or beds of solids) or across the surface. Nonadiabatic dryers are categorized according to the mode of heat supply, such as heat applied through a heat exchange surface or direct radiation, and according to the mode of moisture carryover, for example, moisture removal by vacuum or by a carrier gas.

Many difficulties are encountered in the selection of dryers. These mainly arise because there is no standard set of systematized laboratory tests using standardized apparatus to provide key data on the drying characteristics of materials. The real mechanics of liquid removal from the solid is not really understood, nor is the operation of many dryers. A systematic comprehensive classification of existing dryers has yet to be agreed upon. There is also a lack of a reliable procedure for scaling up laboratory data and even pilot-plant data for some types of dryers.

In spite of the above-mentioned lacunae, dryers have still to be selected and some prior information is required to facilitate this job. This includes (1) flow sheet quantities, such as dry solid quantity, total liquid to be removed, and the source of the wet material; (2) batch or continuous feed physical characteristics, such as source of feed, presence of any previous dewatering stage, like filtration, mechanical pressing, or centrifuging, method of supplying material to the dryer, particle size distribution in the wet feed, physical characteristics and handleability, and abrasive properties of wet and dry materials; (3) feed chemical properties, such as toxicity, odor problems, whether the material can be dried with hot combustion gases containing carbon dioxide, sulfur dioxide, some nitrogen oxides, and traces of partially burnt

TABLE 1.3
Capacity and Energy Consumption for Selected Dryers

Method	Typical Dryer	Typical Materials
Material not conveyed	Tray dryer	Wide range of pastes, granules
Material falls by gravity	Rotary dryer	Free-flowing granules
Material conveyed mechanically	Screw-conveyor, paddle	Wet sludges, pastes
Transported on trucks	Tunnel dryer	Wide range of materials
Sheet-form materials, supported on rolls	Cylinder dryers	Paper, textiles, pulp
Conveyed on bands	Band, conveyor dryer	Wide range of solids (pellets, grains)
Material suspended in air	Fluid bed, flash	Free-flowing granules
Slurries or solutions atomized in air	Spray dryer	Milk, coffee, etc.

Note: Most dryers may operate continuously, semicontinuously, or batchwise. Labor costs are high for tray and tunnel dryers.

hydrocarbon, fire and explosion hazards, temperature limitations, temperatures of relevant phase changes, and corrosive properties; (4) dry product specification and properties, such as moisture content, removal of solvent odor, particle size distribution, bulk density, maximum percentage of impurities, desired granular or crystalline form, flow properties, and temperature to which the dried product must be cooled before storage; and (5) drying data obtained from a pilot plant or laboratory as well as previous experience of the drying performance of similar materials in a full-scale plant. Information on solvent recovery, product loss, and site conditions would be an added bonus.

The best method of selection involves using past experience. One of the preliminary ways of selecting dryers is based on the nature of the feed [16]. There is little difficulty in handling liquid feeds, and the choice of the equipment is normally limited to (1) spray dryer, (2) drum dryer, atmospheric or vacuum, and (3) agitated batch vacuum dryer. Other considerations that might influence the final choice are the need for small product losses and a clean plant, solvent recovery or the need to use an inert atmosphere in which an agitated vacuum dryer is preferred, and the temperature sensitivity of the material. The agitated vacuum dryer has a long residence time, the through circulation dryer, a moderate temperature and moderate residence time. The drum dryer can have a high mean temperature with a short contact time; the spray dryer has a short contact time with a wide range of operating temperatures. The above-mentioned selection is applicable to pumpable suspensions of fine solids, excluding pastes.

For the continuous drying of pastes and sludges, in which the solids are in a finely divided state, dust problems are a major consideration. However, the choice between batch and continuous operation is difficult. The batch dryers normally used are tray atmospheric or vacuum, agitated batch atmospheric or vacuum, and rotary atmospheric or vacuum. The vacuum operation is preferable in cases of solvent recovery, fire, or toxic hazards or when temperature limitations are necessary. Dryers used in continuous operation are (1) spray, where atomization itself poses a considerable problem; (2) fluidized bed, where dispersion of the feed in a deep bed is difficult; (3) continuous band circulation, suitable if dust-free product is required; (4) pneumatic, requires mixing of feed with dry product to facilitate dispersion of wet solid in the gas entering the dryer; and (5) continuous rotary, direct or indirect; here too blending of wet feed with dry product is necessary to facilitate handling. If the feed contains fine particles, the indirect mode of heat transfer is normally preferred. In case of free-blowing wet powders (particle size less than 300 μm) all the dryers used for pastes and sludges could be used with the inclusion of the vertical rotating shelf dryer. For granular crystalline solids with particle sizes greater than 300 μm , the direct rotary dryer is commonly used. With this type of dryer crystal breakage is a problem that can be overcome by proper flight design. For particles larger than 25 mesh, a through circulation dryer using a moving band or a vibrating screen could be used. Fibrous solids hold a considerable amount of water but dry quite easily. These materials

are often temperature sensitive due to their high specific surface, and care should be taken to keep the air temperature down. Through circulation tests at various temperatures should establish the need, if any, to avoid overheating. Apart from this, fibrous materials could be treated in a similar way to any of the solids mentioned above.

The final selection of the dryer will usually represent a compromise between the total cost, operating cost, quality of the product, safety consideration, and convenience of installation. It is always wise, in case of uncertainty, to run preliminary tests to ascertain both design and operating data and also the suitability of the dryer for the particular operation. On certain types of dryers, full-scale tests are the only way of establishing reliable design and operating data, but in some cases, such as through circulation dryers, there are reliable techniques for establishing reliable data from laboratory tests. Details on test techniques and as to when tests are needed are available in Refs. [13,16]. It should be noted that specialized drying techniques, such as vacuum freeze drying, microwave freeze drying, and dielectric or infrared heating or drying, have not been considered here. Details of some of these specialized techniques are given elsewhere in this handbook.

1.6 EFFECT OF ENERGY COSTS, SAFETY, AND ENVIRONMENTAL FACTORS ON DRYER SELECTION

Escalating energy costs and increasingly stringent legislation on pollution, working conditions, and safety have a direct bearing on the design as well as selection of industrial dryers. Lang [27] has discussed the effects of these factors on design of suspension dryers for particulates, such as spray, flash, and fluidized bed dryers. These factors must be considered during the phase of selection of dryers. In some cases a choice between competing drying systems exists; in others one must incorporate these factors at the design stage.

For a given dryer system (including preprocessing, such as mechanical dewatering, centrifugation, evaporation, and pressing, and postprocessing, such as product collection, cooling, agglomeration, granulation, and scrubbing), in general several energy-saving flow sheets may be devised, including gas recycle, closed cycle operation, self-inertization, multistage drying, and exhaust incineration. Areas of conflict may exist between legal requirements, hygienic operation, and energy efficiency. Lang gives the following possible scenarios of conflict:

1. Explosion vents could be a hygiene problem.
2. Dust in recycling streams fouls heat exchanger surfaces or causes difficulties in direct combustion systems.
3. Thermal expansion joints or fire-extinguishing equipment can cause product buildup and hence a fire hazard.
4. High product collection efficiency for particulate dryers means high pressure drop and increased fan noise.

Note that unnecessarily stringent product specifications can cause significant increase in dryer costs, both capital and operating.

In selecting energy-saving drying systems, it is important to note the following (mainly for particulate drying, but some factors are of general applicability):

1. When handling a thermally sensitive product, recycled exhaust must be totally free of product if the stream is to pass through or near a burner.
2. Recycling increases humidity level in drying, which may increase the equilibrium moisture content to unacceptable levels in some cases.
3. To avoid passing dust in recycled gas through air heaters, if fresh makeup air is heated and mixed with recycled gas, to obtain high mixture temperature (say, 400°C), the fresh gas must be heated to a temperature too high for simple materials of construction. Exotic metals or refractories are needed, which can cause product contamination or a source of ignition if it reaches high enough temperatures.
4. In multiple-stage drying, heat economy requires that the first-stage drying give a partially dried product, which is sometimes too sticky to handle.

A drying installation may cause air pollution by emission of dust and gases. Even plumes of clean water vapor are unacceptable in some areas. Particulates below the range 20–50 mg/nm³ of exhaust air are a common requirement. High-efficiency dust collection is essential. It is important to operate the dryer under conditions conducive to production of coarse product. On the other hand, larger products take longer to dry. Cyclones, bag filters, scrubbers, and electrostatic precipitators are commonly used for particle collection and gas cleaning on particulate material drying of materials in other forms, such as pulp sheets. For the removal of noxious gaseous pollutants, one may resort to absorption, adsorption, or incineration; the last operation is becoming more common.

Although rare, care must be taken in drying of airborne materials that can catch fire. Reduction in oxygen content (by recycle) can suppress explosion hazard. Should explosion occur, suitable explosion vents must be included to avoid buildup of excessive pressure in the system. Elimination of sources of ignition is not acceptable as adequate assurance against fire or explosion hazard. When an explosion risk exists, buildup of product in the dryer or collector must be avoided. For example, venting doors on the roof of large-volume spray dryers must be flush with the interior surface to prevent any buildup of product. It is often cheaper to design a partially or fully inertized drying system than to design larger dryer chambers to withstand high internal pressures (5–6 psig).

Finally, local legislation about noise levels must be considered at the selection and design state. Depending on the stringency of noise requirements, the cost of acoustic treatment can be as much as 20% of the total system cost. For air suspension dryers, the fan is the main noise generator. Other sources, such as pumps, gearboxes, compressors, atomization

equipment, burners, and mixers, also contribute to noise. Low fan noise requires a low-pressure drop in the system, which is in conflict with the high-pressure drop required for higher collection efficiency.

A simple direct-fired dryer with once through airflow heating, especially with gas, can be achieved with little noise generation. On the other hand, a more efficient air recirculation unit often requires high-noise burners and ancillaries.

The aforementioned discussion is intended to provide the specifier of drying equipment some practical factors that should be considered at the stage of selection of the dryer system. Rarely, if ever, is it possible to select a dryer that meets all criteria. In most cases, however, one can modify the dryer system design or operation to meet all the essential specifications of the user.

Menon and Mujumdar [15] have indicated some energy-saving measures involved in conditioning of the feed, dryer design, and heat recovery from the exhaust stream, including the use of heat pumps. Some novel techniques, like displacement drying, steam drying, drying using superheated steam, RF drying, press drying, and combined impingement and through drying, have been proposed.

Although prior experience is a guide commonly used in specifying dryers, it is important to recognize that earlier dryers were often specified in times when energy costs were minimal and requirements of product quality and production rates were different. There is also a variation in energy costs from one geographic location to another and certainly from one country to another. It is strongly recommended that the process engineer make a selection of dryer from current conditions and geographic location while taking into account future expected trends. In many instances the most widely used dryers for a specific product have been found to be poor choices under prevailing conditions. Table 1.4 gives a

TABLE 1.4
Capacity and Energy Consumption for Selected Dryers

Dryer Type	Typical Evaporation Capacity (kg H ₂ O/h m ² or kg H ₂ O/h m ³)	Typical Energy Consumption (kJ/kg of H ₂ O Evaporated)
Tunnel dryer	—	5,500–6,000
Band dryer	—	4,000–6,000
Impingement dryer	50/m ²	5,000–7,000
Rotary dryer	30–80/m ³	4,600–9,200
Fluid bed dryer		4,000–6,000
Flash dryer	5–100/m ³ (depends on particle size)	4,500–9,000
Spray dryer	1–30/m ³	4,500–11,500
Drum dryer (for pastes)	6–20/m ²	3,200–6,500

Note: Figures are only approximate and are based on current practice. Better results can often be obtained by optimizing operating conditions and using advanced technology to modify the earlier designs.

summary of typical drying capacities and energy consumption in existing common industrial dryers.

1.7 DESIGN OF DRYERS

The engineer concerned with process design has to choose for a given dryer those conditions that enable the specified properties of the product to be obtained. The performance characteristics of alternative systems should also be assessed before the final choice of a dryer type. Almost always some small-scale tests are needed to determine the material's drying characteristics required to predict the way in which the raw material would behave in the actual unit. A flowchart illustrating the various steps needed to design a dryer is shown in Ref. [3].

For a preliminary estimate of dryer size, Toei [24] gives a simple method based on data obtained from operating industrial dryers. For convection dryers, the rate of heat transfer (kcal/h) is given by

$$q = (ha)(V)(t - t_m), \quad \text{for batch type}$$

and

$$q = (ha)(V)(t - t_m)_{\ln}, \quad \text{for continuous type}$$

For conduction dryers,

$$q = UA(t_k - t_m)$$

t_m is the product temperature (K)

t is the inlet temperature (K)

$(t - t_m)_{\ln}$ is the logarithmic mean (K) of the temperature differences between the hot air and the product at the inlet and outlet, respectively

ha is the volumetric heat transfer coefficient (kcal/s K m³)

U is the overall heat transfer coefficient (kcal/s K m²)

A is the heating area in contact with the product (m²)

t_k is the temperature of the heat source (K)

Table 1.4 is excerpted from Ref. [24]. Here the volume used to define ha includes void volumes, for example, above and below a fluid bed dryer. Thus it is the overall volume and is subject to significant variation. It is also dependent on the critical moisture of the solids. Note the units when using this table. An estimate must be made *a priori* about the drop in air temperature across the dryer and rise of product temperature in order to use Table 1.5 for a very rough sizing of a dryer.

Extensive work has been done in developing theories of drying, such as the theory of simultaneous transport, theories involving flow through porous media, and simplified models, like the wetted surface model and the receding-plane model. The characteristic drying rate curve has also found use in design. Details of these theories are available in Refs. [3,13,17,26].

In designing a dryer, basically one or more of the following sources of information are used: (1) information obtained from customers, (2) previous experience in the form of files on dryers sold and tended, or (3) pilot-plant tests and bench-scale tests. For design a computer is usually used; the input parameters are based on hard data and partly on experience. Pilot-plant tests ensure that the material can be processed in the desired manner. However, the scale-up procedures are by no means straightforward. The theory in the computer program is restricted to heat and mass balances. The design is based on drying times guessed by the designer, who may be guided by drying data obtained in bench-scale and pilot-plant tests. Design procedures for various dryers have been outlined in Refs. [7,16]. Factors affecting product quality should be borne in mind before deciding on a dryer. (For discussion of some new drying technologies, see Ref. [28].)

1.8 GUIDELINES FOR DRYER SELECTION

In view of the enormous choices of dryer types one could possibly deploy for most products, selection of the best type is a challenging task that should not be taken lightly nor should

TABLE 1.5
Approximate Values of ha (kcal/h °C m³) for Various Dryer Types

Type	h_a	$(t - t_m)_{\ln}$ (°C)	Inlet Hot Air Temperature (°C)
<i>Convection</i>			
Rotary	100–200	Countercurrent: 80–150 Cocurrent: 100–180	200–600 300–600
Flash	2000–6000	Parallel flow only: 100–180	400–600
Fluid bed	2000–6000	50–150	100–600
Spray	20–80 (large five)	Counterflow: 80–90 Cocurrent: 70–170	200–300 200–450
Tunnel	200–300	Counterflow: 30–60 Cocurrent: 50–70	100–200 100–200
Jet flow	$h = 100–150$	30–80	60–150
<i>Conduction</i>			
Drum	U (kcal/h · °C · m ²)	$t_k - t_m$ (°C)	
Agitated through rotary with steam tubes, etc.	100–200	50–80	
	60–130 (smaller for sticky solids)	50–100	

it be left entirely to dryer vendors who typically specialize in only a few types of dryers. The user must take a proactive role and employ vendors' experience and bench-scale or pilot-scale facilities to obtain data, which can be assessed for a comparative evaluation of several options. A wrong dryer for a given application is still a poor dryer, regardless of how well it is designed. Note that minor changes in composition or physical properties of a given product can influence its drying characteristics, handling properties, etc., leading to a different product and in some cases severe blockages in the dryer itself. Tests should be carried out with the "real" feed material and not a "simulated" one where feasible.

Although here we will focus only on the selection of the dryer, it is very important to note that in practice one must select and specify a drying system which includes predrying stages (e.g., mechanical dewatering, evaporation, preconditioning of feed by solids backmixing, dilution or pelletization, and feeding) as well as the postdrying stages of exhaust gas cleaning, product collection, partial recirculation of exhausts, cooling of product, coating of product, agglomeration, etc. The optimal cost-effective choice of dryer will depend, in some cases significantly, on these stages. For example, a hard pasty feedstock can be diluted to a pumpable slurry, atomized and dried in a spray dryer to produce a powder, or it may be pelletized and dried in a fluid bed or in a through circulation dryer, or dried as is in a rotary or fluid bed unit. Also, in some cases, it may be necessary to examine the entire flow sheet to see if the drying problem can be simplified or even eliminated. Typically, nonthermal dewatering is an order-of-magnitude less expensive than evaporation which, in turn, is manyfold energy efficient than thermal drying. Demands on product quality may not always permit one to select the least expensive option based solely on heat and mass transfer considerations, however. Often, product quality requirements have overriding influence on the selection process. For high-value products (e.g., pharmaceuticals and certain high-value foodstuffs) the selection of dryers depends mainly on the value of the dried products since cost of drying becomes a very small fraction of the sale price of the product. On the other hand for very low value products or even no value products (e.g., waste sludges) the choice of drying system depends entirely on the cost of drying, so the lowest cost system is selected for such applications.

As a minimum, the following quantitative information is necessary to arrive at a suitable dryer:

- Dryer throughput; mode of feedstock production (batch/continuous)
- Physical, chemical, and biochemical properties of the wet feed as well as desired product specifications; expected variability in feed characteristics
- Upstream and downstream processing operations
- Moisture content of the feed and product
- Drying kinetics; moist solid sorption isotherms
- Quality parameters (physical, chemical, biochemical)
- Safety aspects, e.g., fire hazard and explosion hazards, toxicity

- Value of the product
- Need for automatic control
- Toxicological properties of the product
- Turndown ratio, flexibility in capacity requirements
- Type and cost of fuel, cost of electricity
- Environmental regulations
- Space in plant

For high-value products like pharmaceuticals, certain foods and advanced materials, quality considerations override other considerations since the cost of drying is unimportant. Throughputs of such products are also relatively low in general. In some cases, the feed may be conditioned (e.g., size reduction, flaking, pelletizing, extrusion, backmixing with dry product) before drying, which affects the choice of dryers.

As a rule, in the interest of energy savings and reduction of dryer size, it is desirable to reduce the feed liquid content by less expensive operations such as filtration, centrifugation, and evaporation. It is also desirable to avoid overdrying, which increases the energy consumption as well as drying time.

Drying of food and biotechnological products require adherence to good manufacturing practice (GMP) and hygienic equipment design and operation. Such materials are subject to thermal as well as microbiological degradation during drying as well as in storage. If the feed rate is low (<100 kg/h), a batch-type dryer may be suited. Note that there is a limited choice of dryers that can operate in the batch mode.

In less than 1% of cases the liquid to be removed is a nonaqueous (organic) solvent or a mixture of water with a solvent. This is not uncommon in drying of pharmaceutical products, however. Special care is needed to recover the solvent and to avoid potential danger of fire and explosion. Table 1.6 presents a typical checklist of most dryer vendors use to select and quote an industrial dryer.

Drying kinetics play a significant role in the selection of dryers. Apart from simply deciding the residence time required, it limits the types of suitable dryers. Location of the moisture (whether near surface or distributed in the material), nature of moisture (free or strongly bound to solid), mechanisms of moisture transfer (rate-limiting step), physical size of product, conditions of drying medium (e.g., temperature, humidity, flow rate of hot air for a convective dryer), pressure in dryer (low for heat-sensitive products), etc., have a bearing on the type of suitable dryer as well as the operating conditions. Most often, not more than one dryer type will likely meet the specified selection criteria.

We will not focus on novel or special drying techniques here for lack of space. However, it is worth mentioning that many of the new techniques use superheated steam as the drying medium or are simply intelligent combinations of traditional drying techniques, e.g., combination of heat transfer modes, multistaging of different dryer types. Superheated steam as the convective drying medium offers several advantages, e.g., higher drying rates under certain conditions,

TABLE 1.6
Typical Checklist for Selection of Industrial Dryers

Physical form of feed	• Granular, particulate, sludge, crystalline, liquid, pasty, suspension, solution, continuous sheets, planks, odd-shapes (small/large) • Sticky, lumpy
Average throughput	• kg/h (dry/wet); continuous • kg per batch (dry/wet)
<i>Expected variation in throughput (turndown ratio)</i>	
Fuel choice	• Oil • Gas • Electricity
<i>Pre- and postdrying operations (if any)</i>	
For particulate feed products	• Mean particle size • Size distribution • Particle density • Bulk density • Rehydration properties
Inlet–outlet moisture content	• Dry basis • Wet basis
<i>Chemical/biochemical/microbiological activity</i>	
Heat sensitivity	• Melting point • Glass transition temperature
<i>Sorption isotherms (equilibrium moisture content)</i>	
Drying time	• Drying curves • Effect of process variables
Special requirements	• Material of construction • Corrosion • Toxicity • Nonaqueous solution • Flammability limits • Fire hazard • Color/texture/aroma requirements (if any)
Footprint of drying system	• Space availability for dryer and ancillaries

better quality for certain products, lower net energy consumption if the excess steam produced in the dryer is used elsewhere in the process, elimination of fire and explosion hazard. Vacuum steam drying of timber, for example, can reduce drying times by a factor of up to 4, at the same time enhancing wood quality and reducing net fuel and electricity consumption by up to 70%. The overall economics are also highly favorable.

As the product quality requirements become increasingly stringent and as the environmental legislation becomes more and more demanding it is often found that we need to switch from one drying technology to the other. The rising cost of energy as well as the differences in the cost of fossil fuels versus electrical energy can also affect the choice of a dryer. Since up to 70% of the life cycle cost of a convective dryer is due to energy it is important to choose an energy-efficient dryer where possible even at a higher initial cost. Note that energy costs will continue to rise in the future so this will

become increasingly important. Fortunately, improved efficiency also translates into better environmental implications in terms of reduced emissions of the greenhouse gas (CO_2) as well as NO_x resulting from combustion.

New dryers are developed continuously as a result of industrial demands. Over 250 U.S. patents are granted each year related to dryers (equipment) and drying (process); in the European Community about 80 patents are issued annually on dryers. Kudra and Mujumdar [44] have discussed a wide assortment of novel drying technologies, which are beyond the scope of this chapter. Suffice it to note that many of the new technologies (e.g., superheated steam, pulse combustion—newer gas-particle contactors as dryers) will eventually replace conventional dryers in the next decade or two. New technologies are inherently more risky and more difficult to scale-up. Hence there is natural reluctance to their adoption. Readers are encouraged to review the new developments in order to make sure their selection is the most appropriate one for the application at hand. Some conventional and more recent drying techniques are listed in Table 1.7.

As is evident from the brief discussion the process of dryer selection is rather complex as typically several different choices may exist and the final choice rests on numerous criteria. Indeed, several researchers have developed expert systems for computer-aided selection of dryers. It is important to note that selection of the drying system (e.g., pre- and postdrying operations as well as the dryer) is the correct goal

TABLE 1.7
Conventional vs. Innovative Drying Techniques

Feed Type	Dryer Type	New Techniques
Liquid suspension	• Drum • Spray	• Fluid/spouted beds of inert particles • Spray/fluid bed combination • Vacuum belt dryer • Pulse combustion dryers
Paste/sludge	• Spray • Drum • Paddle	• Spouted bed of inerts • Fluid bed (with solid backmixing) • Superheated steam dryers
Particles	• Rotary • Flash • Fluidized bed (hot air or combustion gas)	• Superheated steam FBD • Vibrated bed • Ring dryer • Pulsated fluid bed • Jet-zone dryer • Yamato rotary dryer
Continuous sheets (coated paper, paper, textiles)	• Multicylinder contact dryers • Impingement (air)	• Combined impingement/radiation dryers • Combined impingement and through dryers (textiles, low basis weight paper) • Impingement and MW or RF

of the exercise. Finally, even if the dryer is selected correctly it is just as important to operate it under optimal conditions to obtain desired product quality and production rate at minimum total cost.

1.9 CONCLUSIONS

Recent studies have resulted in significant advances in the understanding of the thermodynamics of drying hygroscopic materials, kinetics of drying, evaporation of multicomponent mixtures from porous bodies, behavior of particulate motion in various dryers, and so on. In general the empirical knowledge gained in the past two decades has been of considerable value in the design of industrial dryers, including modeling of dryers and control. On the other hand the understanding at the microscopic level of the drying mechanisms remain at a rudimentary level in the sense that modeling “drying” remains a complex and challenging task. Numerous textbooks [29–33] have appeared in recent years that focus on one or more aspects of drying. The interested reader is referred to the series *Advances in Drying* [26,34] and the journal *Drying Technology* [35] for more recent developments in the field of drying. In addition, the series *Drying of Solids* [36–38] contains valuable recent information on drying technology. The technical catalogs published by various manufacturers of drying and ancillary equipment are also very valuable. The proceedings of the biennial international drying symposia (IDS), published in the *Drying* series provide a useful guide to current R&D in drying [39–41].

It should be stressed that one must think in terms of the drying system and not just the dryer when examining an industrial dehydration problem. The preprocessing steps (feeding, dewatering, etc.) of the feed as well as the postprocessing steps (e.g., cooling, granulation, blending, etc.) and cleaning of the dryer emissions are often just as important as the dryer itself. In view of the increasingly stringent environmental regulations coming in force around the world it is not unusual for the dryer itself to cost only a small fraction of the total drying system cost. This edition of the handbook contains details concerning feeders as well as treatment of dryer emissions.

The selection of dryers and drying systems is another important area of practical significance. The cost of a poorly selected drying system is often underestimated since the user must pay for it over the entire life span of the system. If past experience is used as the sole guide, we automatically eliminate the potential benefits of specifying some of the newer dryers marketed recently around the world. Indeed, to cover the broad spectrum of new and special drying technologies that have appeared in the marketplace in the past decade, a new chapter (see Chapter 36) is devoted exclusively to this subject. The reader is urged to go through that chapter in conjunction with any specific drying application to become familiar with some of the newly developed technologies that may afford some advantages over the conventional drying technologies. Use of superheated steam rather than hot air as the drying medium for direct dryers has attracted

considerable attention in recent years. The topic is covered in some detail in a separate chapter (see Chapter 35) from the applications viewpoint.

Scale-up of dryers is perhaps the central issue of most significance in the design of dryers. Information on this subject is rather limited and widely scattered. Genskow [42] provides perhaps the only compilation of papers dealing with the scale-up of several dryer types. For design, analysis or scale-up of dryers, Houska et al. [43] have presented the general equations and methodology as applied to a number of industrial dryers. Unfortunately, the extreme diversity of products and dryer types precludes development of a single design package.

It is difficult to generate rules for both classification and selection of dryers because exceptions occur rather frequently. Often, minor changes in feed or product characteristics result in different dryer types as the appropriate choices. It is not uncommon to find that different dryer types are used to dry apparently the same material. The choice is dependent on production throughput, flexibility requirements, cost of fuel as well as on the subjective judgment of the individual who specified the equipment.

We have not considered novel dryers in this chapter. Kudra and Mujumdar [44] have discussed in detail most of the nonconventional and novel drying technologies. Most of them have yet to mature; a few have been commercialized successfully for certain products. It is useful to be aware of such advances so that the user can make intelligent decisions about dryer selection. Since dryer life is typically 25–40 years that effect of a poor “prescription” can have a long-term impact on the economic health of the plant. It is typically not a desirable option to depend exclusively on prior experience, reports in the literature or vendors’ recommendations. Each drying problem deserves its own independent evaluation and solution.

This chapter has not considered various mathematical models for drying kinetics. The reader is referred to Turner and Mujumdar [45] as well as publications in the journal *Drying Technology* for numerous approaches to mathematical modeling and numerical techniques applicable in drying technology. Mathematical models are increasingly used for design, optimization, and control of industrial dryers.

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NOMENCLATURE

A	heat transfer area, m^2
C_{LW}	heat capacity of liquid moisture, $J/(kg\ K)$
C_p	heat capacity at constant pressure, $J/(kg\ K)$
C_{pG}	heat capacity of dry gas, $J/(kg\ K)$
C_{pW}	heat capacity of moisture vapor, $J/(kg\ K)$
C_{pY}	humid heat, $J/(kg\ K)$

D	molecular diffusivity, m^2/s
E	efficiency of the dryer (dimensionless)
f	relative drying rate (dimensionless)
G_G	dry gas flow rate, kg/s
G_V	evaporation rate, kg/s
ha	volumetric heat transfer coefficient, $\text{kcal}/(\text{m}^3 \text{K})$
h_C	convective heat transfer coefficient, $\text{W}/(\text{m}^2 \text{K})$
h_R	radiative heat transfer coefficient, $\text{W}/(\text{m}^2 \text{K})$
H	enthalpy, J/kg
H_{GG}	dry gas enthalpy, J/kg
H_{GW}	moisture vapor enthalpy, J/kg
I_G	humid enthalpy, J/kg
I_{GS}	enthalpy of dry gas, J/kg
I_{LW}	enthalpy of added moisture, J/kg
K	mass transfer coefficient $\text{kg}/(\text{m}^2 \text{s})$
m_G	mass of dry air, kg
m_W	moisture of mass vapor, kg
M_G	molar mass of dry gas, kg/mol
M_W	molar mass of moisture, kg/mol
N	rate of drying per unit surface area, $\text{kg}/(\text{m}^2 \text{s})$
N_G	molar gas flow per unit area, $\text{mol}/(\text{m}^2 \text{s})$
P_G	partial pressure of dry gas, Pa
P_W	partial pressure of moisture vapor, Pa
P_W^0	vapor pressure, Pa
P	total pressure, Pa
Q	heat flux, W/m^2
Q	heat quality, J
R	gas constant, $\text{J}/(\text{mol K})$
T	temperature, K
T_{av}	average temperature, K
T_C	critical temperature, K
T_D	dew point temperature, K
T_G	gas temperature (dry bulb), K
T_{GS}	adiabatic saturation temperature, K
T_0	initial or reference temperature, K
T_W	wet bulb temperature, K
U	internal energy, J/kg
U	overall heat transfer coefficient, $\text{kcal}/(\text{s m}^2)$
V_H	humid volume, m^3/kg
V_W	specific molar volume, m^3/mol
V_L	specific molar volume of liquid moisture, m^3/mol
W	moisture content (wet basis) (dimensionless)
X	moisture content (dry basis) (dimensionless)
X^*	equilibrium moisture content (dimensionless)
X_{cr}	critical moisture content (dimensionless)
X_0	initial moisture content (dimensionless)
Y	humidity (mass ratio vapor/dry gas) (dimensionless)
Y_{GS}	humidity at adiabatic saturation temperature (dimensionless)
Y_S	saturation humidity (dimensionless)
Y_W	wet bulb humidity (dimensionless)

GREEK SYMBOLS

ΔH_{GM}	residual gas-mixing enthalpy, J/kg
ΔH_R	molar latent heat of reference substance, J/mol
ΔH_{VD}	latent heat of vaporization at T_D , J/kg

ΔH_{V0}	latent heat of vaporization at T_0 , J/kg
ΔH_{VS}	latent heat of vaporization at T_S , J/kg
ΔH_{VW}	latent heat of vaporization at T_W , J/kg
ΔH_W	molar latent heat of vaporization, J/mol
θ	temperature difference, K
θ^*	temperature difference on saturation, K
ρ	gas density, kg/m^3
Φ	humidity-potential coefficient (dimensionless)
Φ	characteristic moisture content (dimensionless)
Φ_E	Ackermann correction, (dimensionless)
ψ	relative humidity (dimensionless)

DIMENSIONLESS GROUPS

Lu	Luikov number
Nu	Nusselt number
Pr	Prandtl number
Sc	Schmidt number

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2 Experimental Techniques in Drying

Károly Molnár

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2.1 INTRODUCTION

The calculation of drying processes requires a knowledge of a number of characteristics of drying techniques, such as the characteristics of the material, the coefficients of conductivity and transfer, and the characteristics of shrinkage. In most cases these characteristics cannot be calculated by analysis, and it is emphasized in the description of mathematical models of the physical process that the so-called global conductivity and transfer coefficients, which reflect the total effect on the partial processes, must frequently be interpreted as experimental characteristics. Consequently, these characteristics can be determined only by adequate experiments. With experimental data it is possible to apply analytical or numerical solutions of simultaneous heat and mass transfer to practical calculations.

The general aim of drying experiments is as follows [1]:

1. Choice of adequate drying equipment
2. Establishing the data required for planning
3. Investigation of the efficiency and capacity of existing drying equipment
4. Investigation of the effect of operational conditions on the shape and quality of the product
5. Study of the mechanism of drying

When the aim of experiments is the choice of adequate drying equipment and the determination of the data required for planning, the effect of the different variables must be examined. It is practical to carry out such a series of experiments in small- or pilot-plant equipment, which operates from both thermal- and materials-handling aspects as does actual equipment, and

by means of which the effect of the parameters to be examined can be studied over a wide range of conditions.

Experiments are frequently carried out in equipment of plant scale, for example, in order to establish data for certain materials, required for planning, to select the type of adequate drying equipment, or to test the efficiency of existing equipment or its suitability for drying other materials. In general, the efficiency and, further, the heat and mass balances, can be determined from data obtained with equipment of plant size.

Owing to the diversity of the aims of experimental investigations and the large number of drying characteristics, the techniques of experimental determination are extremely diverse and in many cases very specific. Therefore, without attempting to be complete, only the methods generally applied are presented here; some special cases are mentioned with reference to the sources of the literature.

2.2 DETERMINATION OF MOISTURE CONTENT

2.2.1 DETERMINATION OF THE MOISTURE CONTENT OF SOLID MATERIALS

Although determination of the moisture content of wet materials appears simple, the results obtained are often not sufficiently accurate, namely, many materials may suffer not only moisture loss but also chemical changes (oxidation, decomposition, destructive distillation, and others) on heating, and this may change the material as well. At the same time the adsorbed water must be distinguished from the so-called water of crystallization, which is frequently a very complex problem.

On selecting the techniques of moisture determination one must take into account the desired accuracy, the case of the procedure, the length of the investigation, and the complexity of the required instruments and equipment. Possible methods of measuring the distribution of moisture content during drying are presented separately.

2.2.1.1 Direct Methods

The direct methods consist essentially of determination of the moisture content of a sample by drying carried out in a drying oven with or without blow through of air, or by drying in a vacuum chamber or in a vacuum desiccator. The sample material must be prepared in every case in the following way. The material is disintegrated into pieces of 1–2 mm³, and a sample of known mass (4–5 g) is placed into a previously dried and weighed glass container, which is put into the drying chamber and dried at 102°C–105°C. The measurement of mass is carried out at ambient temperature, previously allowing the sample to be cooled in a desiccator. The drying process may be considered complete when the difference between the values obtained for the moisture content of the material by two consecutive measurements does not exceed $\pm 0.05\%$. The literature indicates that this process is faster when drying is carried out at 130°C–150°C. However, our investigations proved that results obtained in this way may

deviate by 0.5%–1.0%. Thus, the quick method appears to be suitable only for an approximate determination of the moisture content of a material.

The drying period may be significantly shortened by blowing air through the drying chamber, provided this air has been previously heated to 102°C–105°C, generally by electricity [2].

Drying of foods and other materials sensitive to heat is carried out at 60°C in a vacuum desiccator, in order to prevent their decomposition. In this case drying requires several days.

2.2.1.2 Indirect Methods

Under industrial conditions the moisture present in material must be determined by faster methods, such as by electrical methods of which three main varieties have become widespread: moisture determination based on the change of the ohmic dc resistance, a measurement of the electrostatic capacitance (dielectric constant of the material), and a measurement of the loss in an ac field. Other quick methods are the chemical methods developed mainly for the most frequently occurring case, when the moisture is water, such as the Karl–Fischer analysis based on the chemical reaction of iodine in the presence of water [3], the distillation method, in which moisture is determined by distillation with toluene, and the extraction method, which is carried out with absolute ethanol.

For the determination of factors required for the description of the moisture transport mechanism of wet materials, the distribution of the moisture content of the dried material and even its change during the drying process must be known. These measurements are usually carried out under laboratory conditions. Two main varieties of these measurements are widespread:

1. Mechanical disintegration of the material for the rapid determination of the moisture content of the individual elements and, on compressing the specimen, its further drying [4,5]
2. Special adaptation of one of the above-mentioned electrical methods (see Section 2.2.1.2) [6]

2.2.2 DETERMINATION OF MOISTURE CONTENT OF GASES

The following methods have become widespread for the determination of the moisture content of gases: determination of the absolute value of moisture content by gravimetry or barometry (direct method), measurement of wet gas particles in air by determination of the dry and wet bulb temperatures and the dew point temperature (indirect method), and measurement of a property of the wet gas that depends on the moisture content, such as the absorptivity of the wet gas to electromagnetic waves.

Most frequently applied is the indirect method—determining the dry and wet bulb temperatures or measuring the dew point temperature, which requires a slightly more expensive instrument. However, on using these methods, the determination of the moisture content of the wet gas

cannot be carried out in every case with sufficient accuracy. When high accuracy is required, one of the absolute methods must be applied. (In case of air of 50°C temperature and 50% relative moisture content, a change of 1°C in the temperature of the wet bulb results in a 9% change of the absolute moisture content, or a change of 1°C in the dew point temperature results in a 7% change of the absolute moisture content, respectively.)

The absolute determination of the moisture content of gases can be carried out by the absorption method (allowing gas to flow through a silica-gel bed or allowing it to be absorbed by methanol and then determining the water content by Karl–Fischer titration, for example). A detailed description of these and other special methods are given in Refs. [3,7].

2.3 EXPERIMENTAL DETERMINATION OF THE SORPTION EQUILIBRIUM CHARACTERISTICS OF MATERIALS

2.3.1 INTERPRETATION OF THE EQUILIBRIUM MOISTURE CONTENT OF MATERIALS

A significant part of materials contains a multitude of capillaries, micropores and macropores, and cells and micelles of various dimensions and shapes. In these materials the potential sites of moisture are determined by structural buildup.

The moisture-binding properties of materials are affected by their interior structural buildup. Thus, the equilibrium characteristics determined by measurement may be applied only to structural materials strictly identical with the material used for the measurement.

Although the materials may approximate the equilibrium moisture content either by moisture uptake (adsorption) or by drying (desorption), the value of equilibrium vapor pressure generally depends, for capillary-porous materials, upon the direction of the approximation of the equilibrium moisture content. This phenomenon is known as *sorption hysteresis*. For this reason it is necessary to distinguish adsorption equilibrium moisture from desorption equilibrium moisture.

The equilibrium moisture content is developed as a result of an interaction between the material and the environment: $\bar{X}^* = \bar{X}^*(p_v, T)$. Changes in the moisture content of the material are due to conditions (p_v, T) prevailing on the surface of the material. After a sufficiently long time—with steady-state limit conditions—an internal moisture diffusion balance takes place until the equilibrium moisture content is attained. In an equilibrium state a steady internal moisture distribution exists. Although theoretically an infinite time is needed for its formation, a practically acceptable accurate approximation may be attained by a number of procedures and methods with a finite time.

2.3.2 INTERPRETATION OF THE EQUILIBRIUM VAPOR PRESSURE

The vapor pressure at which a material of a given $\bar{X}$ moisture content is at a given temperature T is a sorption equilibrium

(i.e., it does not lose and does not take up any moisture) is the equilibrium vapor pressure $p_v^* = p_v^*(T)_{\bar{X}}$.

Generally, in drying, a knowledge of the equilibrium vapor pressure at a constant temperature is needed. Thus, since when T is constant p_v^* is constant, the equilibrium relative vapor content was applied in drying as a characteristic of the vapor pressure:

$$\psi = \frac{p_v}{p_{ov}^*} \quad (2.1)$$

2.3.3 CHARACTERISTIC FUNCTIONS OF SORPTION EQUILIBRIUM

Since the function $\bar{X}^* = \bar{X}^*(T, p_v)$ expresses sorption equilibrium, sorption isotherms $\bar{X}^* = \bar{X}^*(p_v)_T$, sorption isobars $\bar{X}^* = \bar{X}^*(T)_{p_v}$, and sorption isosteres $p_v = p_v(T)_{\bar{X}^*}$ are applied to drying. Since sorption isotherms are most frequently applied, the method of their determination is dealt with here. Sorption isotherms are determined from point to point:

$$\bar{X}_1^*(\Psi_1)_T, \bar{X}_2^*(\Psi_2)_T \dots \bar{X}_n^*(\Psi_n)_T$$

Each pair of values determining a “point” is in general the result of a measurement.

The elements of this measurement are as follows:

1. Presentation of the pair of values to be measured on the condition that T is constant
2. Measurement of the value of p_v and ψ during the measurement or at the end of the measurement
3. At the end of the measurement the determination of equilibrium moisture content $\bar{X}^*$ of the sample of material

Widespread methods of measurement in practice differ from each other decisively in the presentation of p_v and ψ with respect to $\bar{X}^*$. Techniques of determining the moisture content of wet gases are described in detail in Section 2.2.2 and those of the materials are described in Section 2.2.1. In general, two main methods of developing the pair of values to be measured are applied.

2.3.3.1 Measuring Techniques Carried Out at a Constant Vapor Pressure

During measurement, p_v and ψ are kept at a constant value until the moisture $\bar{X}$ of the material sample attains the constant equilibrium value of $\bar{X}^*$. Practically, the sample is dried to equilibrium by means of a gas held at a constant state.

2.3.3.2 Measuring Techniques Based on Developing an Equilibrium Vapor Pressure

Equilibrium vapor pressure is developed by measurement of the material sample itself, and measurement is continued until the values of p_v and ψ become constant in the measuring space.

In preparing a drying (desorption) isotherm, the source of moisture serves as the sample.

2.3.4 MEASURING TECHNIQUES AT CONSTANT VAPOR PRESSURE

The gravimetric, the bithermal, and the method based on the interpolation of weight changes are the best known, but of these the gravimetric method is the most widespread. This method is described here in detail; references are provided for the others [8–10,29]. Constant vapor pressure is maintained by means of an acid or salt solution of a specified concentration. It is known that the moisture content of wet air, in equilibrium with an acid or salt solution of known concentration, is constant at a given temperature. Partial pressures of water vapor developed with the use of solutions of sulfuric acid can be determined, as shown in Figure 2.1. The relative vapor pressure above an aqueous solution of sulfuric acid as a function of concentration and temperature can be calculated by the correlation [2]

$$\log \frac{p_v}{p_{ov}^*} = \left(a_1 - \frac{a_2}{T} \right) + \frac{1}{\log p_{ov}^*} \quad (2.2)$$

where values for constants a_1 and a_2 are given in Table 2.1. The term p_{ov}^* in correlation Equation 2.2 is the saturation vapor pressure of water referred to the given temperature (Table 2.2).

The relative equilibrium vapor pressures developed above saturated salt solutions are summarized in Table 2.3 [2,11,12].

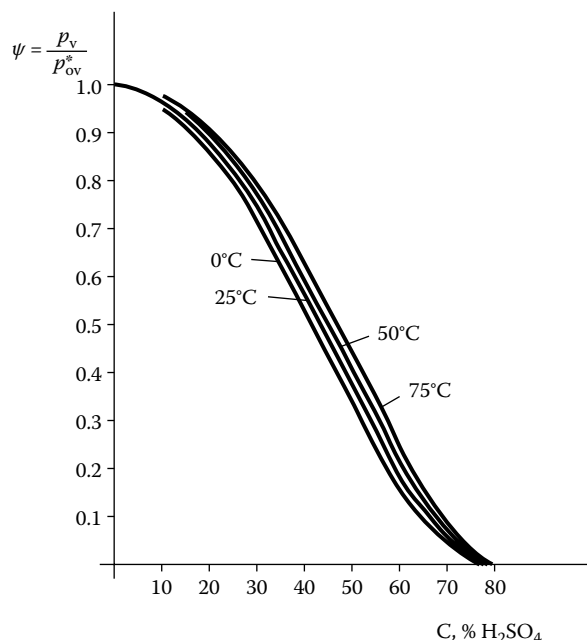


FIGURE 2.1 Partial pressures of water vapor developed with the use of solutions of sulfuric acid.

TABLE 2.1
Values of a_1 and a_2 against H_2SO_4
Concentration C

C (%)	a_1	a_2	C (%)	a_1	a_2
10	8.925	2259	60	8.841	2457
20	8.922	2268	70	9.032	2688
30	8.864	2271	80	9.293	3040
40	8.84	2299	90	9.265	3390
50	8.832	2357	95	9.79	3888

With the gravimetric method a closed measuring space must exist in the thermostat so that a constant vapor pressure is formed and maintained in the manner already described. The equilibrium moisture content of the sample placed in the equilibrium airspace serves as a source for one point of the isotherm.

The static and the dynamic methods have become widespread; their names indicate a stagnant volume of air in the measuring space and airflow, respectively. The static measuring arrangement is shown in Figure 2.2, and that of a dynamic system is shown in Figure 2.3.

A variant of the dynamic method may be, for example, an air conditioner of high accuracy that maintains the constant state of a sample placed in the airflow, which is kept at given T and ψ values [13]. Development of the equilibrium moisture content of the material is attained when the weight of the sample becomes constant. This may be checked by gravimetric measurements.

The sorption equilibrium of the system material-air-solution in principle is attained only after an infinite time. However, the constant weight is observed after a finite time.

The measurement period of the static method is, particularly with high ψ values, very long (several hundred hours). Thus this method cannot be applied to perishable materials even when the thickness δ of the material sample is small and its specific surface area is high. The mass of the sample cannot be significantly decreased because of the hazard of increasing the relative error of measurement originating in the inaccuracy of the weight measurement.

With change in the moisture content of the sample during investigation the concentration of the solution is also altered, as are p_v and ψ . This change is negligible when the mass of the solution is higher by at least two orders of magnitude than the mass of the sample. But even in this case it is advisable to repeatedly determine the concentration of the solution when weight constancy is achieved.

With the dynamic method of measurement the period of measurement is significantly shorter because over the course of evaporation the material transfer resistance decreases to a greater extent. According to the suggestion of Likov [2] the optimum rinsing rate of the measuring space is $0.2\text{--}0.3 \times 10^{-4}$ (m³/s).

The gravimetric method is simple and yields reliable results. There are also fully automated measuring systems [14] provided with computerized evaluation possibilities [15].

TABLE 2.2
Saturation Vapor Pressure of Water Referred to the Given Temperature

Temperature (T)			Temperature (T)		
°C	K	Pressure ($P \times 10^{-4}$) (Pa)	°C	K	Pressure ($P \times 10^{-4}$) (Pa)
0	273.15	0.061076	51	324.15	1.296047
1	274.15	0.065656	52	325.15	1.361163
3	276.15	0.075747	53	326.15	1.429221
4	277.15	0.081287	54	327.15	1.500123
5	278.15	0.008891	55	328.15	1.573967
6	279.15	0.093477	56	329.15	1.650950
7	280.15	0.100126	57	330.15	1.731168
8	281.15	0.107206	58	331.15	1.814623
9	282.15	0.114728	59	332.15	1.901509
10	283.15	0.122711	60	333.15	1.991731
11	284.15	0.131174	61	334.15	2.085874
12	285.15	0.140157	62	335.15	2.183941
13	286.15	0.149669	63	336.16	2.284949
14	287.15	0.159741	64	337.15	2.390861
15	288.15	0.170410	65	338.15	2.500696
16	289.15	0.181698	66	339.15	2.614453
17	290.15	0.193642	67	340.15	2.733113
18	291.15	0.206234	68	341.15	2.855696
19	292.15	0.219571	69	342.15	2.984164
20	293.15	0.233692	70	343.15	3.116553
21	294.15	0.248599	71	344.15	3.252846
22	295.15	0.264289	72	345.15	3.396043
23	296.15	0.280764	73	346.15	3.543134
24	297.15	0.298220	74	347.15	3.696126
25	298.15	0.316657	75	348.15	3.854994
26	299.15	0.335976	76	349.15	4.018765
27	300.15	0.356374	77	350.15	4.189401
28	301.15	0.377850	78	351.15	4.364940
29	302.15	0.400406	79	352.15	4.547344
30	303.15	0.424138	80	353.15	4.735631
31	304.15	0.449145	81	354.15	4.930784
32	305.15	0.375328	82	355.15	5.132801
33	306.15	0.502885	83	356.15	5.341682
34	307.15	0.531815	84	357.15	5.557429
35	308.15	0.562215	85	358.15	5.780040
36	309.15	0.593989	86	359.15	6.010496
37	310.15	0.627429	87	360.15	6.248797
38	311.15	0.662439	88	361.15	6.494944
39	312.15	0.669116	89	362.15	6.748937
40	313.15	0.737460	90	363.15	7.010774
41	314.15	0.777765	91	364.15	7.280457
42	315.15	0.819836	92	365.15	7.560927
43	316.15	0.863868	93	366.15	7.849243
44	317.15	0.909959	94	367.15	8.146384
45	318.15	0.958208	95	368.15	8.452352
46	319.15	1.008516	96	369.15	8.769106
47	320.15	1.061178	97	370.15	9.094687
48	321.15	1.116193	98	371.15	9.430075
49	322.15	1.173562	99	372.15	9.778211
50	323.15	1.233480	100	373.15	10.13223

TABLE 2.3
Relative Equilibrium Vapor Pressures Developed above Saturated Salt Solutions

Salt	T (°C)	$\Psi = \frac{P_v}{P_{ov}^*}$	Salt	T (°C)	$\Psi = \frac{P_v}{P_{ov}^*}$
BaCl ₂ · 2H ₂ O	24.5	0.88	KI	100	0.562
CaCl ₂ · 6H ₂ O	2	0.398	K ₂ S	100	0.562
	10	0.38	KNO ₂	20	0.45
	18.5	0.35	LiCl · H ₂ O	20	0.15
	20.0	0.323		73.06	0.112
	24.5	0.31		88.96	0.0994
CaHPO ₄ · H ₂ O	22.8	0.952	LiSO ₄	24.7	0.865
	30.1	0.94	MgCl ₂	41.7	0.33
	39.0	0.954		50.8	0.311
Ca(NO ₃) ₂ · 4H ₂ O	18.5	0.56		62.3	0.301
	24.5	0.51		79.5	0.287
CaSO ₄ · 5H ₂ O	20	0.98	Mg(C ₂ H ₃ O ₂) ₂ · 4H ₂ O	20	0.65
CdBR ₂	24	0.896	Mg(NO ₃) ₂ · 6H ₂ O	18.5	0.56
	31.6	0.857		24.5	0.52
	41.5	0.835		33.9	0.505
CdSO ₄	23.9	0.906		42.6	0.472
	31.4	0.865		54.8	0.428
CuCl ₂ · 2H ₂ O	28.9	0.67		76.3	0.3275
	36.1	0.669	MgSO ₄	30	0.942
CuSO ₄	22.6	0.974		40.8	0.865
	29.5	0.97		55.1	0.843
	38.7	0.964		95.4	0.776
	91.74	0.888	NH ₄ Cl	20.0	0.792
CH ₄ N ₂ O	26.6	0.77		25.0	0.793
	34.8	0.719		30.0	0.795
	46.1	0.658	NH ₄ Cl + KNO ₃	20	0.726
C ₄ H ₆ O ₆	24.25	0.877		25	0.716
	32.43	0.818		30	0.686
	43.27	0.760	NH ₄ H ₂ PO ₄	20	0.931
CrO ₃	20	0.35		25	0.93
H ₂ C ₂ O ₄ · 2H ₂ O	20	0.76		30	0.929
H ₃ PO ₅ · 5H ₂ O	24.5	0.09		40	0.9048
KCl	24.5	0.866	(NH ₄) ₂ SO ₄	25	0.811
	32.3	0.825		30	0.811
	41.8	0.833		108.2	0.75
	55.4	0.832	NH ₄ Br	95.6	0.772
	95.5	0.773	NH ₄ NO ₃	28	0.706
KBr	20	0.84		42	0.488
	10	0.692		52.9	0.469
KC ₂ H ₃ O ₂	168.0	0.13		76	0.332
	20	0.20			0.229
K ₂ CO ₃ · 2H ₂ O	18.5	0.44	NaBr	100	
	24.5	0.43	NaBr · 2H ₂ O	20	0.58
KCNS	20	0.47	NaBrO ₃	20	0.92
K ₂ CrO ₄	20	0.88	NaC ₂ H ₃ O ₂ · H ₂ O	20	0.76
KF	100	0.229	NaCl	26.82	0.758
K ₂ HPO ₄	20	0.92		34.2	0.743
KHSO ₄	20	0.86		96.8	0.738
			NaCl + KClO ₃	16.4	0.366
NaCl + KNO ₃	16.4	0.326	Na ₄ P ₂ O ₇	22.6	0.973
NaCl + KNO ₃ + NaNO ₃	16.4	0.305	Na ₂ SO ₃ · 7H ₂ O	20	0.95
				23.5	0.9215
NaC ₂ H ₃ O ₂ · 3H ₂ O	20	0.76		30.9	0.894

TABLE 2.3 (continued)
Relative Equilibrium Vapor Pressures Developed above Saturated Salt Solutions

Salt	T (°C)	$\Psi = \frac{p_v}{p_{ov}^*}$	Salt	T (°C)	$\Psi = \frac{p_v}{p_{ov}^*}$
$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	18.5	0.92		40.8	0.867
	24.5	0.87		55.4	0.83
NaClO_3	20	0.75		94.9	0.785
	100	0.54	$\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$	20	0.93
$\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$	20	0.52	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	20	0.93
NaF	100	0.966		31.3	0.8746
$\text{Na}_2\text{HPO}_4 \cdot 2\text{H}_2\text{O}$	20	0.95	$\text{Pb}(\text{NO}_3)_2$	20	0.98
$\text{NaHSO}_4 \cdot 2\text{H}_2\text{O}$	20	0.52		103.5	0.884
NaI	100	0.504	TiCl_3	100.1	0.997
NaNO_2	20	0.66	Ti_2SO_4	104.7	0.848
NaNO_3	27.4	0.7295	$\text{ZuCl}_2 \cdot \frac{1}{2}\text{H}_2\text{O}$	20	0.10
	35.1	0.708	$\text{Zu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	20	0.42
	59.0	0.702	ZuSO_4	5	0.947
	102.0	0.654	$\text{ZuSO}_4 \cdot 7\text{H}_2\text{O}$	20	0.90

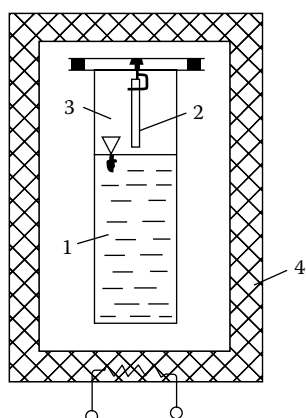


FIGURE 2.2 The measuring arrangement of a static method: (1) sulfuric acid or salt solutions, (2) sample, (3) measuring place, (4) thermostat.

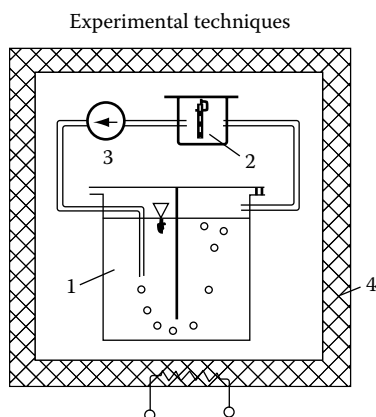


FIGURE 2.3 Schematic diagram of gravimetric dynamic system: (1) bubbling recipient with sulfuric acid or salt solution, (2) sample, (3) air pump, (4) thermostat.

2.3.5 MEASURING TECHNIQUES BASED ON DEVELOPING AN EQUILIBRIUM VAPOR PRESSURE

The sample of a wet material placed in a closed measuring space adjusts, and after a sufficiently long period, the vapor pressure of the space reaches equilibrium with the moist solid. If the change of moisture content of the solid is to be negligible, the moisture in the sample must be higher by at least three orders of magnitude than the amount of water required for saturation of the space. The saturation of the measuring space practically takes place at a constant moist solid vapor pressure, and the time required to attain equilibrium can be fairly long, with knowledge of the air mass present in the measuring space, the evaporation surface of the material sample, and the average evaporation coefficient [8]. For this measurement determination of the partial pressure and of the relative vapor content of the space are of fundamental importance.

Vapor pressure can be measured by manometer on freezing out the water vapor. After insertion of the material sample, the measuring space is evacuated at constant temperature. The pressure value can be read on an accurate micromanometer. The measuring space is removed from the thermostat, water vapor is frozen out, and the manometer is read again. The difference between the two values read on the manometer is equal to the equilibrium vapor pressure of the material.

If the medium in the measuring space is allowed to flow, the measurement will be lowered since the coefficient of evaporation increases. However, care must be taken during measurement to maintain the mass flow of the flowing air at a constant value.

The essence of the isotonic method [16] is that solutions of acids or salts of various concentration are placed in the closed space, where the mass of these solutions is

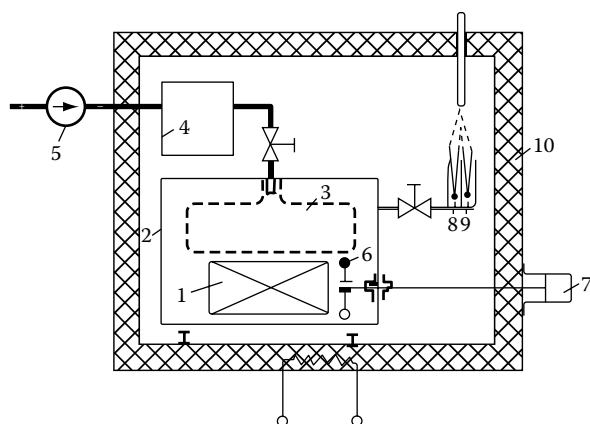


FIGURE 2.4 Schematic diagram of the arrangement based on measuring the wet bulb temperature: (1) sample, (2) measuring chamber, (3) foil balloon, (4) dashpot, (5) air pump, (6) paddle wheel with magnetic clutch, (7) motor, (8) thermometer, (9) wet bulb thermometer, (10) thermostat.

negligible in comparison with the mass of the material sample, together with the material sample itself. These solutions act as sources of absorbents of moisture, respectively, from the material sample. At a given temperature over the course of measurement their composition approaches equilibrium concentration of the solution, corresponding to the equilibrium pressure.

In a method utilizing the deliquescence of crystalline salts, the salt crystals deliquesce in a space with a vapor pressure that exceeds the vapor pressure of their saturated solution. In this way it is possible to draw from their state conclusions about whether the vapor pressure of the measuring space is actually below or above the limit value. The dry bulb temperature and the dew point temperature of air determine the state of air present in the measuring space and also the partial pressures. The method based on this consists essentially of measuring the dew point of the air. The measurement of the equilibrium vapor tension of the air can be carried out by means of hygroscopic cell.

The state of the air is unequivocally determined by the dry and wet bulb temperatures of the air. Methods based on the measurement of the wet bulb thermometer apply this principle. Figure 2.4 presents the principle of the measuring equipment of a method of this type.

2.4 TECHNIQUES AND EQUIPMENT OF THE INVESTIGATION OF DRYING KINETICS

In drying, the heat transfer coefficient h and the mass transfer coefficient K between the drying gas and the wet material, the heat diffusivity κ_h and moisture (mass) diffusivity κ_m coefficients, and the thermal conductivity λ_h and moisture conductivity λ_m coefficients are the parameters of interest. Although several methods are employed for the determination of these parameters, here only the techniques and equipments that allow their determination via a drying experiment will be described.

From the simpler drying experiments a fair amount of data may be obtained on the basis of which—in fractional drying—conclusions can be drawn concerning the time required for the prescribed change in moisture content and the transport characteristics listed above.

When the heat and mass diffusivity properties of the drying material are also needed, a technique of investigation and measurement is applied such that the distribution of temperature and moisture content can also be measured during drying, along with the thickness of the dried material. When, however, during investigation, the above-mentioned distributions are not measured and only the integral-average moisture content of the dried material and eventually the temperature of the drying surface (or the average temperature) are measured, the model used to analyze the data is the so-called “lumped parameter” type as opposed to the “distributed parameter” model applied when local values are measured.

2.4.1 DESCRIPTION OF THE MEASURING EQUIPMENT

Experimental measuring equipment developed for drying by convection is shown in Figure 2.5. The equipment is in fact a controlled climate air duct. The wet material is placed on a balance in the measuring space. The temperature and moisture content of the drying air is measured by wet and dry bulb thermometers. Other online humidity measuring devices may be used for convenience.

The flow rate can be determined by means of an orifice meter. Prior to beginning the measurement, the fan is started and the desired airflow adjusted by a throttle valve; the temperature and moisture content of the air are controlled by means of an electric heater and a steam valve. In the next section the development of an adequate value of flow rate may be ensured by means of screens. The time-varying mass of the sample to be dried is measured continuously and recorded.

2.4.2 DRYING EXPERIMENTS (LUMPED APPROACH)

To determine the drying rate, the mass of a sample placed in the airflow (constant temperature, humidity, and velocity) must be measured as a function of time. In order to obtain results that can be applied for scale-up, the following aspects must be taken into account: the sample must not be too small and the conditions of drying must if possible be identical to the conditions anticipated in the industrial unit [1,17]:

1. The sample must be placed in a similar way in the laboratory unit.
2. The ratios of the drying surface to the nondrying surface must be identical.
3. The conditions of heat transfer by radiation must be similar.
4. The temperature and the moisture content of air (drying gas) and its velocity and direction must be identical with respect to the sample.

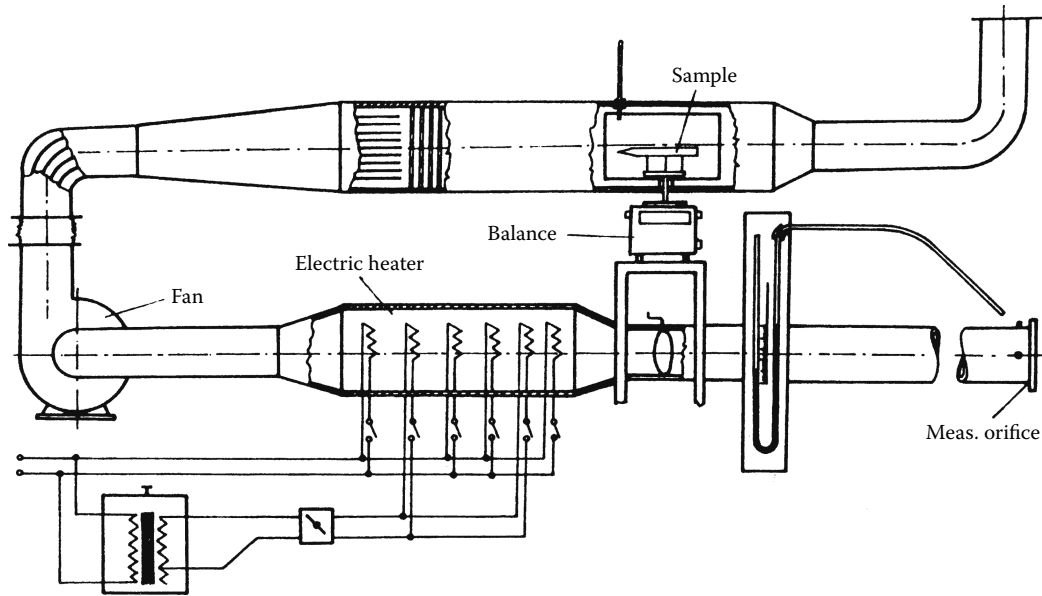


FIGURE 2.5 Drying apparatus.

It is practical to carry out several experiments with material samples of various thicknesses and to determine in every case the dry matter content of the sample in the way described in Section 2.2.

When the temperature, moisture content, and velocity of the drying air are constant, drying takes place under constant drying conditions. This way of drying is ensured by the equipment sketched in Figure 2.5.

Since the wet mass is

$$m_{sw} = m_w + m_s \quad (2.3)$$

and the moisture content is

$$X = \frac{m_w}{m_s} \quad (2.4)$$

thus

$$\frac{m_{sw}}{m_s} = 1 + X$$

That is,

$$X = \frac{m_{sw} - m_s}{m_s} \quad (2.5)$$

These correlations indicate that, with a knowledge of the drying wet mass as a function of time [$m_{sw} = m_{sw}(\tau)$] and of the bone-dry mass of the sample, it is possible to plot the moisture content of the sample as a function of time (Figure 2.6).

This curve can be directly applied for the determination of time of drying greater masses to a prescribed lower moisture content, provided the drying is carried out under identical conditions. However, better information is obtained when, on

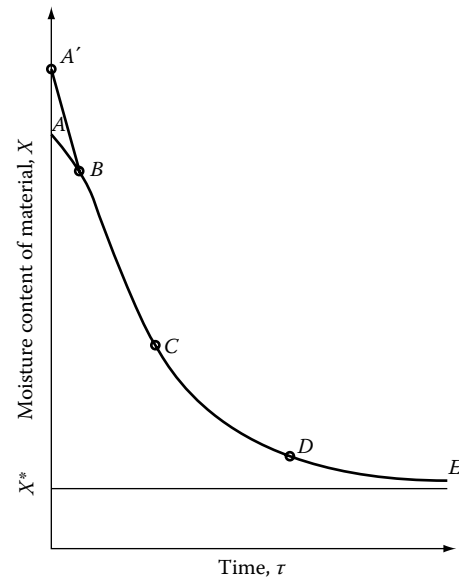


FIGURE 2.6 Batch, convection drying curve.

the basis of Figure 2.6, the drying rate is plotted against the moisture content of the material. The drying rate is defined by

$$N_w = -\frac{1}{A_s} \frac{dm_{sw}}{d\tau} = \frac{m_s}{A_s} \frac{dX}{d\tau} \quad (2.6)$$

In Equation 2.6, A_s denotes the contact surface of the drying gas and the dried material, and a cross section perpendicular to the airflow in through circulation drying.

According to Equation 2.6, with the knowledge of the weight loss curve we can plot Figure 2.7, which represents the drying rate. If the drying rate is known under one set of constant drying conditions, it can be extrapolated within limits to

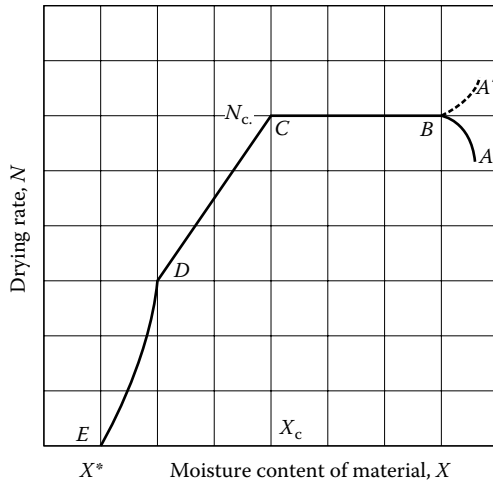


FIGURE 2.7 Drying rate curve.

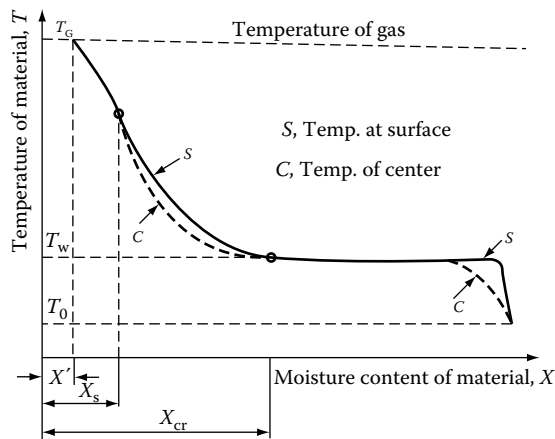


FIGURE 2.8 Temperature curves of a wet material at the surface and the middle of the sample.

other conditions [17]. Figure 2.8 shows the variation of drying surface temperature with time whereas Figure 2.9 summarizes the characteristic parameters of the drying process.

The determination of transport coefficients by a drying experiment is achieved as follows. During constant rate drying, the temperature of the drying surface is, in a purely convective drying, the so-called wet bulb temperature. With this value, the steady-state heat flux is

$$j_q = h(T_G - T_w) \quad (2.7)$$

where j_q can also be calculated using the constant drying rate

$$j_q = N_{wc} \Delta H \quad (2.8)$$

Thus,

$$h = N_{wc} \left(\frac{\Delta H}{T_G - T_w} \right) \quad (2.9)$$

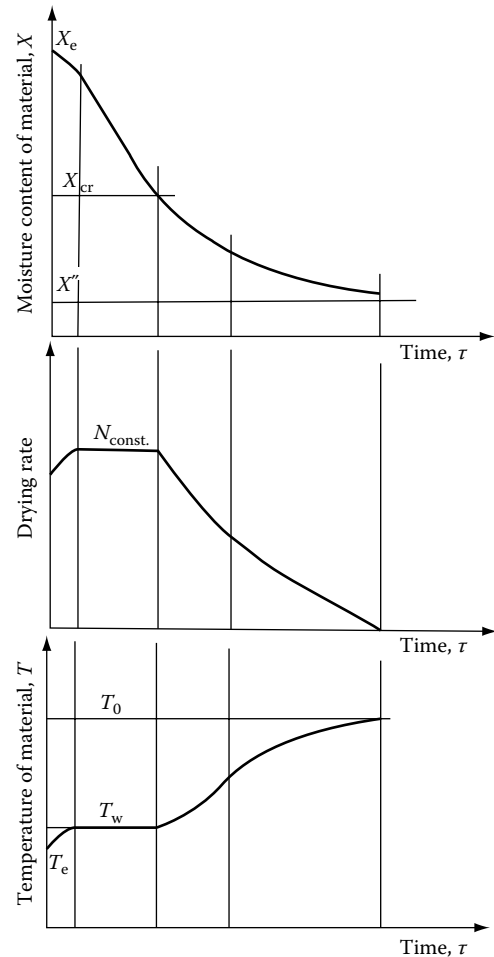


FIGURE 2.9 Characteristic functions of a convective drying.

In terms of the mass transfer coefficient,

$$N_{wc} = K(Y_w - Y_G) \quad (2.10)$$

where Y_w is the moisture content of the saturated air at temperature T_w , the coefficient of evaporation (mass transfer) is therefore

$$K = \frac{N_{wc}}{Y_w - Y_G} \quad (2.11)$$

The heat and mass transfer coefficients on the gas side, characterizing the steady state, may be considered constant during the entire drying process, under constant drying conditions. During the falling rate period, the process is significantly affected by the internal heat and moisture diffusivities. Therefore, determination of these coefficients is also necessary. It may be noted here that the transport coefficients on the gas side also change during this period owing to the continuous decrease in vapor diffusivity of the material in the surface layer [18].

2.4.3 TECHNIQUES OF INVESTIGATION WITH DISTRIBUTED PARAMETERS

In the initial constant rate drying period, a knowledge of the heat diffusivity and heat conductivity coefficients of the wet material is necessary because they are the controlling

factors for heat transport within the material. In this section of drying it is presumed that the pores of the wet material are saturated by moisture. Consequently, the above-mentioned characteristics of a heterogeneous material consisting of a solid skeleton and, within this, a capillary system (filled with moisture) of diversified sizes and shapes must be determined.

2.4.3.1 Determination of the Thermal Conductivity and Diffusivity of Wet Materials

The method for drying a plane slab is described, but a similar method may also be developed for bodies of other geometric shapes [2,8,30].

During drying of wet capillary-porous slabs, when only surface moisture is removed and the value of the so-called phase-change criterion ε is zero—that is, no evaporation takes place within the material itself—the development of the temperature of the wet material at the surface and in the plane of symmetry can be determined analytically [8,19].

Using an apparatus of the type shown in Figure 2.5, the surface temperature T_s and the slab midplane temperature T_z are monitored in addition to the change in weight of the sample. In the initial transient period, surface temperature variation is plotted as a function of the corresponding temperatures at the midplane (Figure 2.10). Figure 2.11 shows a plot of the logarithm of the absolute difference between T_s and T_z versus time. Except at very short times, the curve is linear as expected from the analytical solution of transient one-dimensional diffusion of mass or heat.

The surface temperature is expressed by the function [20]

$$\frac{T_s - T_w}{T_0 - T_w} = \sum_{k=1}^{k=\infty} 2 \left(\frac{\sin \delta_k \cos \delta_k}{\delta_k + \sin \delta_k \cos \delta_k} \right) \exp(-\delta_k^2 Fo) \quad (2.12)$$

and the temperature of the center of the slab by the function

$$\frac{T_z - T_w}{T_0 - T_w} = \sum_{k=1}^{k=\infty} 2 \left(\frac{\sin \delta_k}{\delta_k + \sin \delta_k \cos \delta_k} \right) \exp(-\delta_k^2 Fo) \quad (2.13)$$

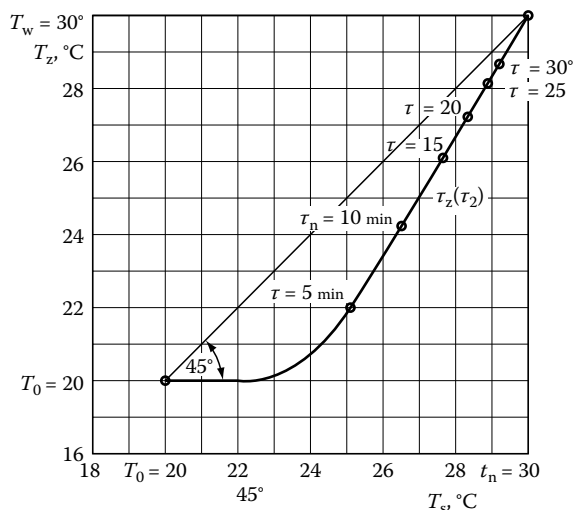


FIGURE 2.10 Temperature on the surface versus the temperature on the symmetry plane.

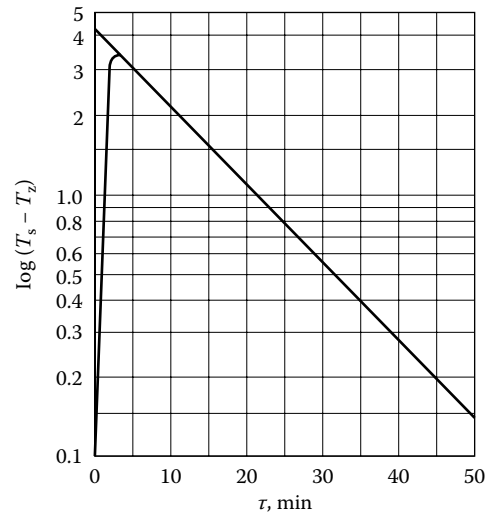


FIGURE 2.11 Plot of $\log(T_s - T_z)$ versus time.

where the eigenvalues δ_k are roots of the transcendent equation

$$\text{ctg } \delta_k = \frac{k}{Bi_{hm}} \quad (2.14)$$

In the above equations the Biot and Fourier numbers are

$$Bi_{hm} = \frac{h}{\lambda} Z \left(1 + \frac{h_m}{h} \right) \quad (2.15)$$

and

$$Fo = \frac{\kappa_r}{Z^2} \quad (2.16)$$

The coefficient h_m in Equation 2.15 is given by

$$h_m = \Delta HKS^* \quad (2.17)$$

where

$$S^* = \frac{Y_w - Y_G}{T_w - T_d} \cong \frac{Y_s - Y_G}{T_s - T_d} = \text{constant} \quad (2.18)$$

is the equation of the equilibrium curve, considered a straight line. It appears from Equations 2.12 and 2.13 that after certain time (Fourier number) only the first term of the series is predominant. Neglecting further terms we obtain

$$\frac{T_z - T_w}{T_s - T_w} = \frac{dT_z}{dT_s} = \frac{1}{\cos \delta_1} \quad (2.19)$$

and

$$\frac{d[\ln(T_s - T_z)]}{d\tau} = -\frac{\delta_1^2 \kappa_h}{Z^2} \quad (2.20)$$

In Equation 2.19, δ_1 is an eigenvalue, with which the thermal diffusivity κ_h can be determined from Equation 2.20.

The function $T_z(T_s)$ shown in Figure 2.10 becomes nonlinear after a certain time, when the superficial moisture disappears. However, on lengthening the straight section, the point of intersection of unit steepness (45°), starting from the origin, indicates the limiting temperature T_w that the temperature of the material approaches, over the course of drying of the surface moisture.

Knowing κ , the thermal conductivity coefficient λ of the wet material can be calculated from Equation 2.21 when c_{sw} , the specific heat of the wet material, and $\bar{\rho}_{os}$, the density of the dry material, are known:

$$\kappa = \frac{\lambda}{\bar{\rho}_{os} c_{sw}} \quad (2.21)$$

In an example of the determination of the thermal diffusivity and thermal conductivity, the wet material is a suspension of water and powdered chalk. Its characteristics are given as $T_0 = 14.3^\circ\text{C}$, $Z = 40 \text{ mm} = 4 \times 10^{-2} \text{ m}$, $c_{sw} = 2.51 \text{ kJ/kg}^\circ\text{C}$, and $\rho_{os} = 1500 \text{ kg/m}^3$. The drying gas is air. Its characteristics are $T_G = 52.5^\circ\text{C}$, $T_d = 10.8^\circ\text{C}$, $P = 1 \text{ bar}$, and $c_{PG} = 1.05 \text{ kJ/kg}^\circ\text{C}$. Values of temperatures observed during drying are given in Figure 2.12.

T_w extrapolated from Figure 2.12 is $T_w = 25.2^\circ\text{C}$. According to Equation 2.19,

$$\frac{dT_z}{dT_s} = \frac{1}{\cos \delta_1} = \frac{T_{z81'} - T_{z24'}}{T_{s81'} - T_{s24'}} = \frac{20.73 - 16.13}{24.18 - 23.17} = 4.55$$

where

$$\cos \delta_1 = 0.22$$

that is, the first eigenvalue is $\delta_1 = 1.35$.

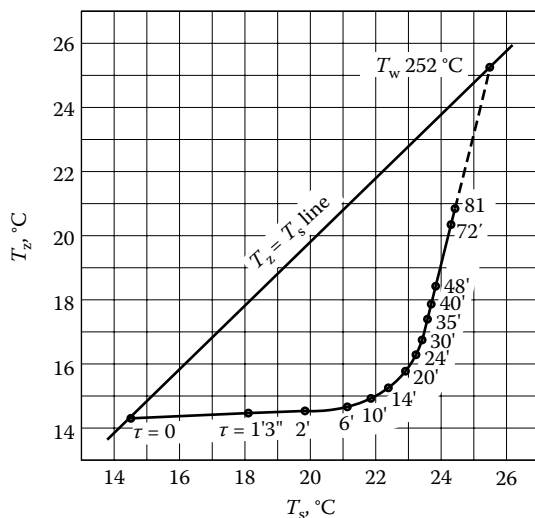


FIGURE 2.12 Experimental $T_z(T_s)$ curve for suspension of water and powdered chalk.

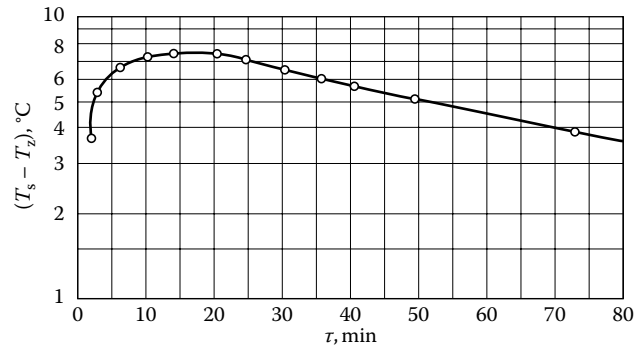


FIGURE 2.13 Experimental $\log(T_s - T_z) - \tau$ functions for suspension of water and powdered chalk.

Figure 2.13 shows the logarithm of the difference between the temperatures at the surface and in the center plane plotted against time. Using the data of Figure 2.13, it follows according to Equation 2.20 that

$$\begin{aligned} -\frac{d[\ln(T_s - T_z)]}{d\tau} &= \frac{\delta_1^2}{Z^2} \\ &= \frac{\ln(T_{s24'} - T_{z24'}) - \ln(T_{s81'} - T_{z81'})}{81 - 24} \\ &= 0.7551/\text{h} \end{aligned}$$

Thus the heat diffusivity of the given wet material is

$$\begin{aligned} &= \frac{\lambda}{c_{sw} \bar{\rho}_{os}} = 0.755 \frac{Z^2}{\delta_1^2} = 0.755 \frac{16 \times 10^{-4}}{1.35^2} \\ &= 0.655 \times 10^{-3} \text{ m}^2/\text{h} \end{aligned}$$

and the thermal conductivity is

$$\begin{aligned} \lambda &= c_{sw} \bar{\rho}_{os} = (0.655 \times 10^{-3})(2.51)(1500) \\ &= 2.47 \text{ kJ/m h}^\circ\text{C} \end{aligned}$$

2.4.3.2 Determination of the Mass Diffusivity and Moisture Conductivity Coefficient of Wet Materials

The mass diffusivity coefficient can be determined from the drying curve already discussed. It has been shown for drying of a plane slab [20], that in the falling rate period of drying as the equilibrium moisture content is approached, the time rate of change to the weight of the slab as a function of its weight is a linear function, given by

$$\frac{-(dm/dr)}{m - m^*} = v_1^2 \frac{m}{Z^2} = v_1^2 \frac{\lambda_m}{c_m \bar{\rho}_{os} Z^2} \quad (2.22)$$

The eigenvalue v_1 in Equation 2.22 is defined by

$$\text{ctg } v_1 = \frac{v_1}{Bi_m} = \frac{v_1 \lambda_m}{S^{**} K Z c_m} \quad (2.23)$$

where S^{**} in Equation 2.23 is given by

$$S^{**} = \frac{P_{vs} - P_{vw}}{X_s - X_w} \quad (2.24)$$

therefore the slope of the sorption isotherm (assuming that the isotherms are not strongly dependent on temperature).

On the basis of Equations 2.22 and 2.23, we obtain

$$\nu_1 \operatorname{ctg} \nu_1 = \frac{(-dm/d\tau) \bar{\rho}_{0s} Z}{(m - m^*) K S^{**}} \quad (2.25)$$

The right side of Equation 2.25 contains known data when the weight of the wet material is measured as a function of time and when we have already determined, according to the previous section, the mass transfer coefficient K from data obtained in the constant rate drying period. The first eigenvalue can be determined numerically (by trial and error or by iteration) from the transcendental equation. (The value of the ν ranges between 0 and π , Ref. [2].) The mass diffusivity coefficient κ_m can be calculated directly from Equation 2.22 if ν_1 is known. When the moisture capacity coefficient c_m is determined separately [2,19], then the moisture conductivity coefficient λ_m can also be determined from Equation 2.23.

2.4.3.3 Determination of the Thermal Conductivity and Effective Diffusivity Coefficient of the Dry Material

In the course of drying of macroporous and capillary-porous materials—in the falling rate period of drying—the moisture from the pores of the material is removed, and an evaporation front penetrates from the surface of the material into the interior. At a given time the primarily wet material can be divided into two parts separated by an evaporation front: a part already dried and a part that is still wet. The moisture evaporated at the plane of phase change must pass through the pores of the already dried layer in order to leave the material. At the same time the heat required for drying has to pass through the already dried layer to carry out the evaporation. This model of drying is known as the *penetrating or receding front model* and is described in the literature [2,4,21,22].

Besides drying with a penetrating front, other drying phenomena also occur in which, in the course of the drying, development of a layer inhibits the drying process (in drying two-layer materials, such as salami varieties, seeds coated with an earth layer, and so on). The mathematical model of the drying process of such two-layer materials of constant thickness is described in detail in the literature [23,24]. For a description of drying by convection of double-layer materials, in the drying section with a penetrating front, and also of the macroporous and capillary-porous bodies—besides the material characteristics and transport coefficients—knowledge of the thermal diffusivity of the dry layer and the effective diffusivity coefficient value is also necessary. When the mean free path of the molecules removed is shorter than the characteristic dimension of the capillary, the vapor

diffusion process through the pores of the dry layer is the so-called molecular diffusion process. However, the length of the diffusion path and the cross section available for vapor flow may change the direction of the diffusion. For this case, Krischer and Kröll [25] defined an effective diffusivity coefficient

$$D_a = \frac{D_{VG}}{\mu_D} \quad (2.26)$$

where

D_{VG} is the conventional diffusivity of the vapor in free air
 μ_D is the coefficient of resistance due to diffusion

Determination of the effective diffusivity coefficient by diffusion experiments is widely known in the literature [2,19]. Since these experimental measurements are relatively complex, we describe here the determination of this coefficient by means of a simple series of drying experiments. For drying double layers the rate of drying can be defined as [22]

$$j_m = K_G (p_{v\xi} - p_{VG}) \quad (2.27)$$

where

$$K_G = \frac{1}{1/k_G M_V + \xi (RT/D_a M_V)} \quad (2.28)$$

is the overall mass transfer coefficient, p_{VG} the partial pressure of vapor of the bulk gas phase, and $p_{v\xi}$ the vapor pressure at the evaporation plane (see Figure 2.14).

It can be shown that a steady state exists in which the rate of drying is

$$j_{me} = \frac{1}{1/k_G M_V + \xi (RT/D_a M_V)} (p_{v\xi e} - p_{VG}) \quad (2.29)$$

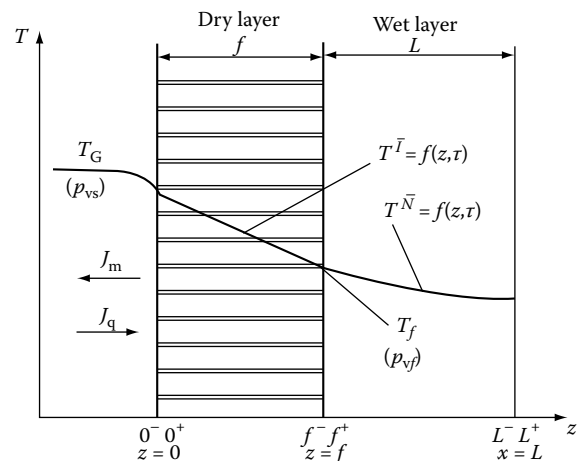


FIGURE 2.14 The double-layer model investigated.

where $p_{v_{\xi e}}$ is the equilibrium partial vapor pressure corresponding to the equilibrium temperature of the site evaporation front. The equilibrium temperature is given by

$$T_{\xi e} = \frac{T_d + T_G (K_H / \Delta H S K_G)}{1 + (K_H / \Delta H S K_G)} \quad (2.30)$$

In Equation 2.30,

$$S = \frac{P_{v_{\xi e}} - P_{VG}}{T_{\xi e} - T_d} \quad (2.31)$$

and the overall heat transfer coefficient is

$$K_H = \frac{1}{1/h + \xi/\lambda_l} \quad (2.32)$$

On rearranging Equation 2.29, we obtain

$$\frac{P_{v_{\xi e}} - P_{VG}}{j_{me}} = \frac{1}{k_G M_V} + \xi \frac{RT}{D_a M_V} \quad (2.33)$$

Equation 2.33 can be used as the basis for determining D_a if the mass transfer coefficient k_G is known. Other necessary parameters are measurable from the weight loss curve.

For the determination of the thermal conductivity of the dry layer, a steady state yields

$$j_{qe} = j_{me} \Delta H \quad (2.34)$$

or

$$K_H (T_G - T_{\xi e}) = j_{me} \Delta H \quad (2.35)$$

Substituting K_H given by Equation 2.32 into Equation 2.35 and rearranging we obtain

$$\frac{T_G - T_{\xi e}}{\Delta H j_{me}} = \frac{1}{h} + \xi \frac{1}{\lambda_l} \quad (2.36)$$

This equation can be used to determine λ_l if all other parameter values are known.

In an experimental apparatus utilizing constant (and known) drying conditions, change in weight of the two-layer slab as well as the temperature of the evaporation front must be measured as functions of time. Appropriate plots of Equations 2.33 and 2.36 using this data yield the effective and thermal conductivity of the dry layer. An example of the determination of the thermal conductivity and effective diffusivity coefficients of dry layers follows.

A two-layer slab consisting of a sand layer and a layer of gypsum is used in this illustration. Five different thicknesses of the sand layer were employed. The flow rate of the drying air was the same in all experiments. The measured data at steady state are summarized in Table 2.4. Table 2.5 gives

TABLE 2.4
Data Measured on the Stabilized State

No. Experiments	$\frac{\xi}{Z}$	ξ (mm)	$j_{me} \cdot 10^5$ (kg/m ² s)	$T_{\xi e}$ (°C)	T_d (°C)	T_G (°C)
1	0.2	4	8.14	40.0	10.0	48.0
2	0.3	6	6.44	40.9	6.5	48.6
3	0.5	10	4.53	41.9	11.5	48.0
4	0.7	14	3.25	42.0	12.0	47.2
5	0.8	16	3.11	42.2	11.5	47.7

TABLE 2.5
Calculated Results (from Table 2.4)

No. Experiments	ξ (mm)	$p_{v_{\xi e}}$ (bar)	p_{VG} (bar)	$\frac{P_{v_{\xi e}} - P_{VG}}{j_{me}}$ (bar · m ² · s/kg)	$\frac{T_G - T_{\xi e}}{\Delta H j_{me}}$ (°C · m ² · s/kJ)
1	4	0.07374	0.01251	752.21	40.91
2	6	0.07889	0.009871	1071.72	47.2
3	10	0.08317	0.01383	1519.2	55.8
4	14	0.0836	0.01429	2131.2	66.75
5	16	0.08449	0.01383	2271.6	73.72

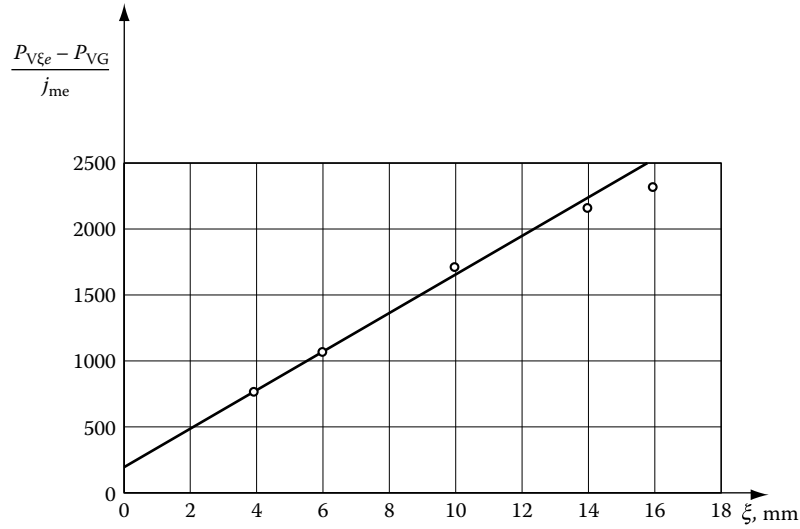


FIGURE 2.15 Determination of the mass transfer coefficient and the effective diffusivity.

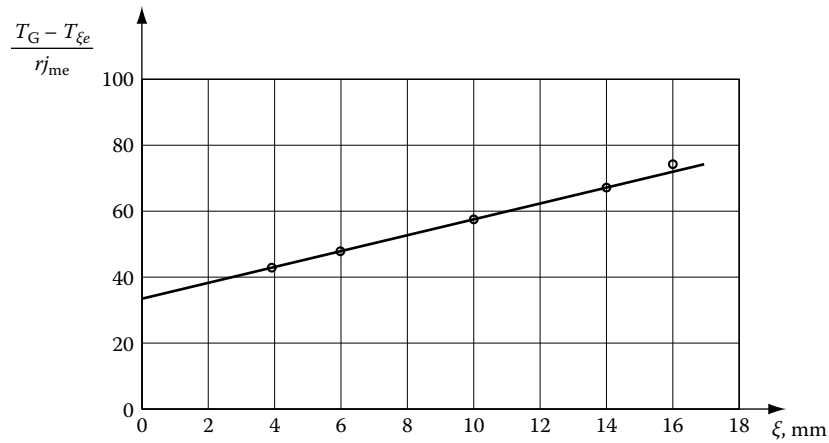


FIGURE 2.16 Determination of the heat transfer coefficient and the thermal conductivity.

the calculated results using Equations 2.33 and 2.36. Plots corresponding to Equations 2.33 and 2.36 are presented in Figures 2.15 and 2.16. The transfer coefficients are determined from the slopes of the linear regions of the curves. The results are

$$\begin{aligned} k_G &= 5.56 \times 10^{-4} && \text{kmol}/(\text{m}^2 \text{ bar s}) \\ D_a &= 9.1 \times 10^{-6} && \text{m}^2/\text{s} \\ h &= 0.031 && \text{kJ}/(\text{m}^2 \text{ s K}) \\ \lambda_1 &= 0.4 && \text{J}/(\text{m s K}) \end{aligned}$$

appropriate to define volumetric transfer coefficients for heat and mass transfer, h_v and K_v , respectively, as follows:

$$h_v = h \frac{dA_s}{dV} \approx h \frac{A_s}{V} \quad (2.37)$$

and

$$K_v = K \frac{dA_s}{dV} \approx K \frac{A_s}{V} \quad (2.38)$$

2.5 DRYING OF FIXED AND MOVING BEDS

In through circulation fixed bed dryers and in moving or suspended bed dryers (spouted beds, fluidized beds, rotary dryers, and so on), the actual interfacial area participating in heat and mass transfer is unknown. For such cases it is

where

dA_s and A_s are the differential and total interfacial areas, respectively, for heat and mass transfer whereas dV and V are the corresponding differential and total bed volumes, respectively.

2.5.1 DETERMINATION OF THE VOLUMETRIC HEAT TRANSFER COEFFICIENT

The method described here is for a plant-size, direct rotary dryer although the technique is applicable to other dryers of granular media as well. In a plant scale, effects of various parameters cannot be studied in depth as one could in a laboratory or pilot-plant size equipment. Hence, such data are not generally applicable.

Figure 2.17 is a schematic of the dryer employed for this investigation. It consists of a cylindrical shell (1 in Figure 2.17) covered with a heating jacket (2) and fitted with an agitator (3) within the shell. Heat for drying can be supplied directly by hot air or indirectly by circulating a heating fluid through the jacket. In direct drying the drying air is heated.

The stirrer carries out agitation and disintegration of the dried grains driving rings (4 in Figure 2.17) and plates (5) are attached in a staggered fashion on the agitator in order to increase the turbulence of the drying medium and the residence time of the material to be dried. The scraper-stirrer plates (8), which stir the material and feed it into the flow of the drying gas, are fastened to these rings and plates, respectively. The material to be dried is fed through the hopper (6) to the feeder (7). Exit air leaves through the dust separator (9).

The parameters measured include flow rate of the solids feed, mass flow rate of air, solids temperature at entry and exit, and the humidity of the drying air entering and leaving the dryer. The volumetric heat transfer coefficient can then be calculated from the definition

$$Q_H = hA_s \Delta T_m \quad (2.39)$$

where the average temperature difference for cocurrent flow is the logarithmic average value

$$\begin{aligned} \Delta T_m &= \Delta T_{\log} \\ &= \frac{(T_{G \text{ in}} - T_{SW \text{ in}}) - (T_{G \text{ out}} - T_{SW \text{ out}})}{\ln \frac{T_{G \text{ in}} - T_{SW \text{ in}}}{T_{G \text{ out}} - T_{SW \text{ out}}}} \end{aligned} \quad (2.40)$$

Dividing both sides of Equation 2.39 by the empty volume of the equipment V results in

$$\frac{Q_H}{V} = h \frac{A_s}{V} \Delta T_m \quad (2.41)$$

that is,

$$\frac{Q_H}{V} = h_v \Delta T_m \quad (2.42)$$

The transfer rate is given by

$$\begin{aligned} Q_H &= G_T c_{pG} (T_{G \text{ in}} - T_{G \text{ out}}) \\ &= L_{SW} c_{SW} (T_{SW \text{ out}} - T_{SW \text{ in}}) \end{aligned} \quad (2.43)$$

On the basis of Equations 2.42 and 2.43, the volumetric heat transfer coefficient can also be determined as

$$\begin{aligned} h_v &= \frac{G_T c_{pG} (T_{G \text{ in}} - T_{G \text{ out}})}{V \Delta T_{\log}} \\ &= \frac{L_{SW} c_{SW} (T_{SW \text{ out}} - T_{SW \text{ in}})}{V \Delta T_{\log}} \end{aligned} \quad (2.44)$$

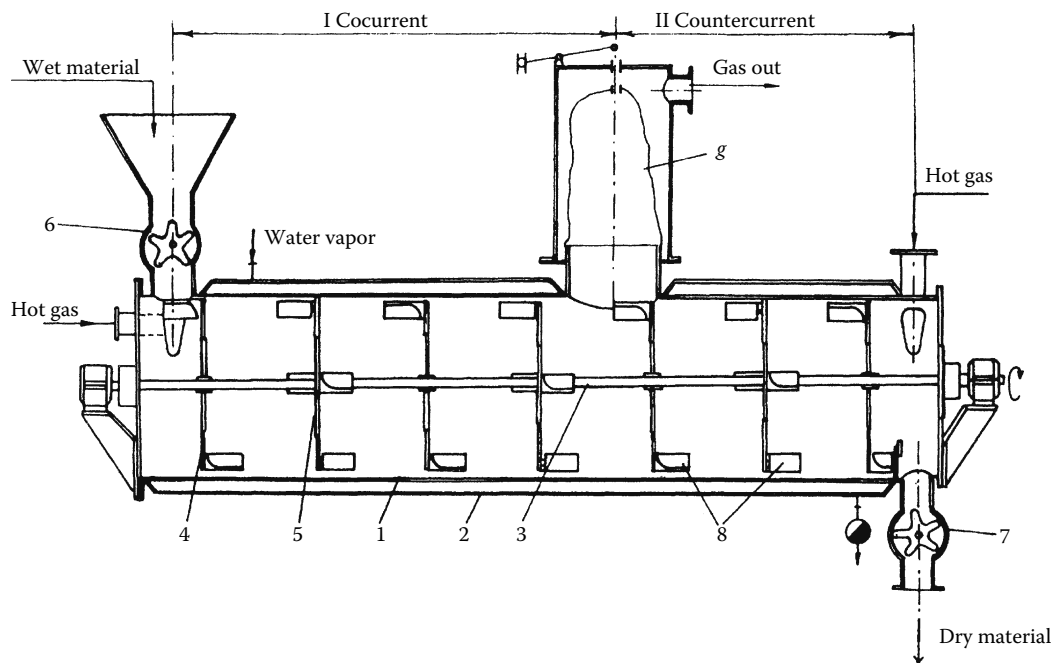


FIGURE 2.17 The contact-fluid dryer (see text for discussion).

In these experiments already dried granular material must be used to produce a process of pure sensible heat transfer.

For a direct-heated rotary dryer, the volumetric heat transfer coefficient can be estimated from the following empirical correlation

$$h_V = 2950 n \phi^{0.173} G_0^{0.9} \quad (2.45)$$

where

ϕ is the fractional holdup (volume of granular material as a fraction of the total volume of the dryer)

n is the rpm of the agitator

G_0 is the mass flow rate of the drying air

2.5.2 DETERMINATION OF THE HEAT AND MASS TRANSFER COEFFICIENTS IN THROUGH CIRCULATION DRYING

In general one needs only the volumetric heat and mass transfer coefficients for design purposes. In some special cases (e.g., drying alfalfa plants), the material to be dried may consist of a mixture of materials with distinctly different characteristics (the stems and the leaves in the case of alfalfa). The stems have a well-defined smaller surface area whereas the leaves have large areas which may, however, be covered, therefore not all areas are available for heat and mass transfer. The drying rates are quite dissimilar for the two components, therefore care must be exercised in drying whole alfalfa plants. Refs. [26,27] give the necessary details for the determination of the volumetric as well as real heat and mass transfer coefficients. These references give a methodology for evaluation of the interfacial area for heat and mass transfer on the basis of certain assumptions.

The volumetric mass and heat transfer coefficients can be determined simply from the definitions (Equations 2.46 and 2.47) for an adiabatic operation. (See Ref. [28] for details.)

$$K_V = \frac{G_T}{AH} \frac{Y_{G \text{ out}} - Y_{G \text{ in}}}{\Delta Y_{\log}} \quad (2.46)$$

and

$$h_v = \frac{G_{TCWG}}{AH} \frac{T_{Gin} - T_{Gout}}{\Delta T_{log}} \quad (2.47)$$

where

$$\Delta Y_{\log} = \frac{(Y_W - Y_{G \text{ in}}) - (Y_W - Y_{G \text{ out}})}{\ln \frac{Y_W - Y_{G \text{ in}}}{Y_W - Y_{G \text{ out}}}} \quad (2.48)$$

and

$$\Delta T_{\log} = \frac{(T_{G \text{ in}} - T_W) - (T_{G \text{ out}} - T_W)}{\ln \frac{T_{G \text{ in}} - T_W}{T_{G \text{ out}} - T_W}} \quad (2.49)$$

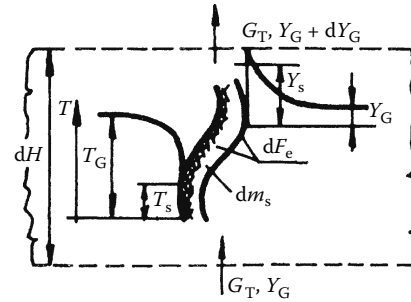


FIGURE 2.18 Elementary part of alfalfa bed.

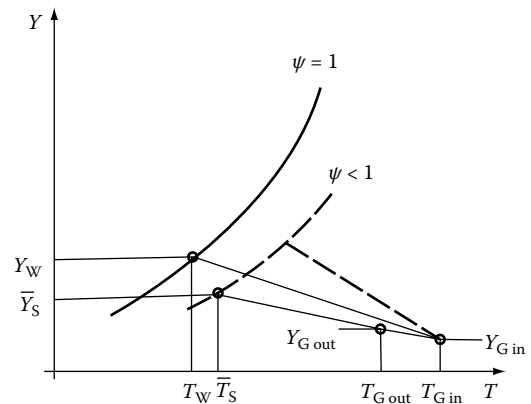


FIGURE 2.19 Saturation of air during circulation through bed.

Further parameters can be seen in Figures 2.18 and 2.19. When the saturation is nonadiabatic and the bed height is sufficiently small, it is evident according to Ref. [28] that

$$\frac{1}{T_s - T_G} \cong \frac{1}{\bar{T}_s - \bar{T}_G} \quad (2.50)$$

and

$$\left(\overline{\frac{1}{Y_s - Y_G}}\right) \cong \frac{1}{\bar{Y}_s - \bar{Y}_G} \quad (2.51)$$

The volumetric transport coefficients are in this case

$$K_V(\tau) = \frac{G_T}{A\Delta H} \frac{\Delta Y_G}{\bar{Y}_s - \bar{Y}_G} \quad (2.52)$$

and

$$h_v(\tau) = \frac{G_T c_{wG}}{A \Delta H} \frac{\Delta T_G}{\bar{T}_s - \bar{T}_G} \quad (2.53)$$

In the above correlations,

$$\Delta Y_G = Y_{G\text{out}} - Y_{G\text{in}} \quad (2.54)$$

and

$$\Delta T_G = T_{G \text{ out}} - T_{G \text{ in}} \quad (2.55)$$

$\bar{T}_s$ can be calculated by iteration from the correlation

$$\frac{\Delta Y_G}{\Delta T_G} = \frac{\frac{M_V}{M_G} \frac{p_{Ov}^*}{P} \Psi(\bar{T}_s, \bar{X}) - \bar{Y}_G}{\bar{T}_s - \bar{T}_G} \quad (2.56)$$

and with $\bar{T}_s$ known, $\bar{Y}_s$ can also be determined:

$$\bar{Y}_s = \frac{M_V}{M_G} \frac{p_{Ov}^*}{P} \Psi(\bar{T}_s, \bar{X}) \quad (2.57)$$

2.6 CONCLUSION

Simple experimental methods are discussed in this chapter for determination of the key equilibrium and heat and mass transfer parameters need in engineering calculations for common dryers. The coverage is by no means all-inclusive. The reader is referred to the literature cited for details.

NOMENCLATURE

A	surface, m ²
Bi	Biot number
c_m	moisture capacity, kg/kg
c_{sw}	specific heat capacity of wet material, J/(kg K)
D_a	effective diffusivity coefficient, m ² /s
D_{VG}	molecular diffusivity coefficient between vapor and gas, m ² /s
$ Fo$	Fourier number
G_T	total gas flow rate, kg/s
G_0	total gas flow, kg/(m ² s)
h	heat transfer coefficient, W/(m ² K)
h_m	modified material transfer coefficient, Equation 2.17, W/(m ² K)
h_v	volumetric heat transfer coefficient, W/(m ³ K)
ΔH	sum of heat of evaporation plus heat of moisture, J/kg
j_m	mass flow density, kg/(m ² s)
j_q	heat flow density, W/m ²
k_G	mass transfer coefficient, kmol/(m ² s Pa)
K	mass transfer coefficient, kg/(m ² s Pa)
K_G	overall mass transfer coefficient, kg/(m ² s Pa)
K_H	overall heat transfer coefficient, W/(m ² K)
K_V	volumetric mass transfer coefficient, kg/(m ² s)
L_{sw}	mass flow of wet material, kg/s
m	mass, kg
n	revolutions per minute, 1/s
N_w	drying rate, kg/(m ² s)
N_{ew}	constant drying rate, kg/(m ² s)
p	partial pressure, Pa

p_o	partial vapor pressure of pure component, Pa
Q	heat rate, W
R	gas constant, J/(kg K)
S	slope of equilibrium curve defined by Equation 2.31, Pa/K
S^*	slope of equilibrium curve, per K
S^{**}	slope of equilibrium curve defined by Equation 2.24, Pa
T	temperature, K
T_m	average temperature, K
X	moisture content of material, dry basis
Y	moisture content of air, dry basis
Z	half thickness of the plane, m
V	volume, m ³

SUBSCRIPTS

d	belonging to the dew point
e	equilibrium value
G	gas
h	heat transfer
in	entering
log	logarithmic mean value
m	mass transfer
0	initial value
out	leaving
s	surface
S	solid
V	vapor
W	wet, wet bulb
Z	in the symmetry plane of a plate
ξ	at the border of the dry and wet layer

SUPERSCRIPTS

$-$	Average value
$*$	equilibrium

GREEK SYMBOLS

δ	thickness, m
δ_k	eigenvalue, defined by Equation 2.14
ϕ	fractional holdup of solids
ε	phase-change criterion
k_h	heat diffusivity, m ² /s
k_m	moisture diffusivity coefficient, m ² /s
λ_h	thermal conductivity, W/(m K)
λ_m	moisture conductivity, kg/(ms K)
ν	eigenvalue, defined by Equation 2.23
ψ	relative moisture content
μ_D	diffusion-resistance coefficient
$\bar{\rho}_{Os}$	bulk density of solids, kg/m ³
τ	time, s, h
ξ	thickness of dry zone, m
Δ	symbol of difference

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3 Basic Process Calculations and Simulations in Drying

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3.1 INTRODUCTION

Since the publication of the first and second editions of this handbook, we have been witnessing a revolution in methods of engineering calculations. Computer tools have become easily available and have replaced the old graphical methods. An entirely new discipline of computer-aided process design (CAPD) has emerged. Today even simple problems are solved using dedicated computer software. The same is not necessarily true for drying calculations; dedicated software for this process is still scarce. However, general computing tools including Excel, Mathcad, MATLAB®, and Mathematica are easily available in any engineering company. Bearing this in mind, we have decided to present here a more computer-oriented calculation methodology and simulation methods than to rely on old graphical and shortcut methods. This does not mean that the computer will relieve one from thinking. In this respect, the old simple methods and rules of thumb are still valid and provide a simple commonsense tool for verifying computer-generated results.

3.2 OBJECTIVES

Before going into details of process calculations we need to determine when such calculations are necessary in industrial practice. The following typical cases can be distinguished:

- Design—(a) selection of a suitable dryer type and size for a given product to optimize the capital and operating costs within the range of limits imposed—this case is often termed *process synthesis* in CAPD; (b) specification of all process parameters and dimensioning of a selected dryer type so the set of design parameters or assumptions is fulfilled—this is the common *design problem*.
- Simulation—for a given dryer, calculation of dryer performance including all inputs and outputs, internal distributions, and their time dependence.
- Optimization—in design and simulation an optimum for the specified set of parameters is sought. The objective function can be formulated in terms of economic, quality, or other factors, and restrictions may be imposed on ranges of parameters allowed.
- Process control—for a given dryer and a specified vector of input and control parameters the output parameters at a given instance are sought. This is a special case when not only the accuracy of the obtained results but the required computation time

is equally important. Although drying is not always a rapid process, in general for real-time control, calculations need to provide an answer almost instantly. This usually requires a dedicated set of computational tools like neural network models.

In all of the above methods we need a *model* of the process as the core of our computational problem. A model is a set of equations connecting all process parameters and a set of constraints in the form of inequalities describing adequately the behavior of the system. When all process parameters are determined with a probability equal to 1 we have a *deterministic* model, otherwise the model is a *stochastic* one.

In the following sections we show how to construct a suitable model of the process and how to solve it for a given case. We will show only deterministic models of convective drying. Models beyond this range are important but relatively less frequent in practice.

In our analysis we will consider each phase as a continuum unless stated otherwise. In fact, elaborate models exist describing aerodynamics of flow of gas and granular solid mixture where phases are considered noncontinuous (e.g., bubbling bed model of fluid bed, two-phase model for pneumatic conveying, etc.).

3.3 BASIC CLASSES OF MODELS AND GENERIC DRYER TYPES

Two classes of processes are encountered in practice: steady state and unsteady state (batch). The difference can easily be seen in the form of general balance equation of a given entity for a specific volume of space (e.g., the dryer or a single phase contained in it):

$$\text{Inputs} - \text{outputs} = \text{accumulation} \quad (3.1)$$

For instance, for mass flow of moisture in a solid phase being dried (in kg/s) this equation reads:

$$W_S X_1 - W_S X_2 - w_D A = m_s \frac{dX}{d\tau} \quad (3.2)$$

In steady-state processes, as in all continuously operated dryers, the accumulation term vanishes and the balance equation assumes the form of an algebraic equation. When the process is of batch type or when a continuous process is being started up or shut down, the accumulation term is nonzero and the balance equation becomes an ordinary differential equation (ODE) with respect to time.

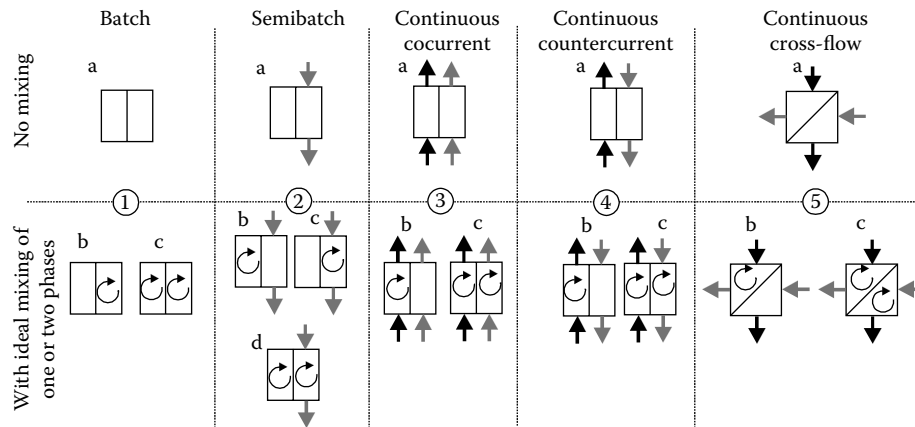


FIGURE 3.1 Generic types of dryers.

In writing Equation 3.1, we have assumed that only the input and output parameters count. Indeed, when the volume under consideration is perfectly mixed, all phases inside this volume will have the same property as that at the output. This is the principle of a lumped parameter model (LPM).

If a property varies continuously along the flow direction (in one dimension for simplicity), the balance equation can only be written for a differential space element. Here Equation 3.2 will now read

$$W_S X - W_S \left(X + \frac{\partial X}{\partial l} dl \right) - w_D dA = dm_S \frac{\partial X}{\partial \tau} \quad (3.3)$$

or, after substituting $dA = a_V S dl$ and $dm_S = (1 - \varepsilon) \rho_S S dl$, we obtain

$$-W_S \frac{\partial X}{\partial l} - w_D a_V S = (1 - \varepsilon) \rho_S S \frac{\partial X}{\partial \tau} \quad (3.4)$$

As we can see for this case, which we call a distributed parameter model (DPM), in steady state (in the one-dimensional case) the model becomes an ODE with respect to space coordinate, and in unsteady state it becomes a partial differential equation (PDE). This has a far-reaching influence on methods of solving the model. A corresponding equation will have to be written for yet another phase (gaseous), and the equations will be coupled by the drying rate expression.

Before starting with constructing and solving a specific dryer model it is recommended to classify the methods, so typical cases can easily be identified. We will classify typical cases when a solid is contacted with a heat carrier. Three factors will be considered:

1. Operation type—we will consider either batch or continuous process with respect to given phase.
2. Flow geometry type—we will consider only parallel flow, cocurrent, countercurrent, and cross-flow cases.
3. Flow type—we will consider two limiting cases, either plug flow or perfectly mixed flow.

These 3 assumptions for 2 phases present result in 16 generic cases as shown in Figure 3.1. Before constructing a model it is desirable to identify the class to which it belongs so that writing appropriate model equations is facilitated.

Dryers of type 1 do not exist in industry; therefore, dryers of type 2 are usually called *batch dryers* as is done in this text. An additional term—*semicontinuous*—will be used for dryers described in Section 3.9. Their principle of operation is different from any of the types shown in Figure 3.1.

3.4 GENERAL RULES FOR A DRYER MODEL FORMULATION

When trying to derive a model of a dryer we first have to identify a volume of space that will represent a dryer. If a dryer or a whole system is composed of many such volumes, a separate submodel will have to be built for each volume and the models connected together by streams exchanged between them. Each stream entering the volume must be identified with parameters. Basically for systems under constant pressure it is enough to describe each stream by the name of the component (humid gas, wet solid, condensate, etc.), its flowrate, moisture content, and temperature. All heat and other energy fluxes must also be identified.

The following five parts of a deterministic model can usually be distinguished:

1. Balance equations—they represent Nature's laws of conservation and can be written in the form of Equation 3.1 (e.g., for mass and energy).
2. Constitutive equations (also called kinetic equations)—they connect fluxes in the system to respective driving forces.
3. Equilibrium relationships—necessary if a phase boundary exists somewhere in the system.
4. Property equations—some properties can be considered constant but, for example, saturated water vapor pressure is strongly dependent on temperature even in a narrow temperature range.

5. Geometric relationships—they are usually necessary to convert flowrates present in balance equations to fluxes present in constitutive equations. Basically they include flow cross-section, specific area of phase contact, etc.

Typical formulation of basic model equations will be summarized later.

3.4.1 MASS AND ENERGY BALANCES

Input–output balance equations for a typical case of convective drying and LPM assume the following form:

3.4.1.1 Mass Balances

Solid phase:

$$W_S X_1 - W_S X_2 - w_{Dm} A = m_S \frac{dX}{d\tau} \quad (3.5)$$

Gas phase:

$$W_B Y_1 - W_B Y_2 + w_{Dm} A = m_B \frac{dY}{d\tau} \quad (3.6)$$

3.4.1.2 Energy Balances

Solid phase:

$$W_S i_{m1} - W_S i_{m2} + (\Sigma q_m - w_{Dm} h_A) A = m_S \frac{di_m}{d\tau} \quad (3.7)$$

Gas phase:

$$W_B i_{g1} - W_B i_{g2} - (\Sigma q_m - w_{Dm} h_A) A = m_B \frac{di_g}{d\tau} \quad (3.8)$$

In the above equations Σq_m and w_{Dm} are a sum of mean interfacial heat fluxes and a drying rate, respectively. Accumulation in the gas phase can almost always be neglected even in a batch process as small compared to accumulation in the solid phase. In a continuous process the accumulation in solid phase will also be neglected.

In the case of DPMs for a given phase the balance equation for property Γ reads:

$$\begin{aligned} \text{div}[\Gamma \cdot u] - \text{div}[D \cdot \text{grad} \Gamma] \pm \beta a_v \Delta \Gamma \pm G \\ - \frac{\partial \Gamma}{\partial \tau} = 0 \end{aligned} \quad (3.9)$$

where the LHS terms are, respectively (from the left): convective term, diffusion (or axial dispersion) term, interfacial term, source or sink (production or destruction) term, and accumulation term.

This equation can now be written for a single phase for the case of mass and energy transfer in the following way:

$$\text{div}[\rho X \cdot u] - \text{div}[D \cdot \text{grad}(\rho X)] \pm k_X a_v \Delta X - \frac{\partial \rho X}{\partial \tau} = 0 \quad (3.10)$$

$$\begin{aligned} \text{div}[\rho c_m T \cdot u] - \text{div}\left[\frac{\lambda}{\rho c_m} \cdot \text{grad}(\rho c_m T)\right] \\ \pm \alpha a_v \Delta T + q_{ex} - \frac{\partial \rho c_m T}{\partial \tau} = 0 \end{aligned} \quad (3.11)$$

Note that density here is related to the whole volume of the phase: e.g., for solid phase composed of granular material it will be equal to $\rho_m(1 - \epsilon)$. Moreover, the interfacial term is expressed here as $k_X a_v \Delta X$ for consistency, although it is expressed as $k_Y a_v \Delta Y$ elsewhere (see Equation 3.27).

Now, consider a one-dimensional parallel flow of two phases either in co- or countercurrent flow, exchanging mass and heat with each other. Neglecting diffusional (or dispersion) terms, in steady state the balance equations become

$$W_S \frac{dX}{dl} = -w_D a_v S \quad (3.12)$$

$$\pm W_B \frac{dY}{dl} = w_D a_v S \quad (3.13)$$

$$W_S \frac{di_m}{dl} = (q - w_D h_{Av}) a_v S \quad (3.14)$$

$$\pm W_B \frac{di_g}{dl} = -(q - w_D h_{Av}) a_v S \quad (3.15)$$

where the LHSs of Equations 3.13 and 3.15 carry the positive sign for cocurrent and the negative sign for countercurrent operation. Both heat and mass fluxes, q and w_D , are calculated from the constitutive equations as explained in the following section. Having in mind that

$$\frac{di_g}{dl} = (c_B + c_A Y) \frac{dt_g}{dl} + (c_A t_g + \Delta h_{v0}) \frac{dY}{dl} \quad (3.16)$$

and that enthalpy of steam emanating from the solid is

$$h_{Av} = c_A t_m + \Delta h_{v0} \quad (3.17)$$

we can now rewrite Equations 3.12 through 3.15 in a more convenient working version

$$\frac{dX}{dl} = -\frac{S}{W_S} w_D a_v \quad (3.18)$$

$$\frac{di_m}{dl} = \frac{S}{W_S} \frac{a_v}{c_S + c_{Al} X} \left[q + w_D \left((c_{Al} - c_A) t_m - \Delta h_{v0} \right) \right] \quad (3.19)$$

$$\frac{dY}{dl} = \frac{1}{\chi} \frac{S}{W_B} w_D a_V \quad (3.20)$$

$$\frac{dt_g}{dl} = -\frac{1}{\chi} \frac{S}{W_B} \frac{a_V}{c_B + c_A Y} [q + w_D c_A (t_g - t_m)] \quad (3.21)$$

where χ is 1 for cocurrent and -1 for countercurrent operation.

For a monolithic solid phase convective and interfacial terms disappear and in unsteady state, for the one-dimensional case, the equations become

$$D_{\text{eff}} \frac{\partial^2 X}{\partial x^2} = \frac{\partial X}{\partial \tau} \quad (3.22)$$

$$\lambda \frac{\partial^2 t_m}{\partial x^2} = c_p \rho_m \frac{\partial t}{\partial \tau} \quad (3.23)$$

These equations are named Fick's law and Fourier's law, respectively, and can be solved with suitable boundary and initial conditions. Literature on solving these equations is abundant, and for diffusion a classic work is that of Crank (1975). It is worth mentioning that, in view of irreversible thermodynamics, mass flux is also due to thermodiffusion and barodiffusion. Formulation of Equations 3.22 and 3.23 containing terms of thermodiffusion was favored by Luikov (1966).

3.4.2 CONSTITUTIVE EQUATIONS

They are necessary to estimate either the local nonconvective fluxes caused by conduction of heat or diffusion of moisture or the interfacial fluxes exchanged either between two phases or through system boundaries (e.g., heat losses through a wall). The first are usually expressed as

$$q = -\lambda \frac{dt}{dl} \quad (3.24)$$

$$j = -\rho D_{\text{eff}} \frac{dX}{dl} \quad (3.25)$$

and they are already incorporated in the balance equations (3.22 and 3.23). The interfacial flux equations assume the following form:

$$q = \alpha (t_g - t_m) \quad (3.26)$$

$$w_D = k_Y \phi (Y^* - Y) \quad (3.27)$$

where ϕ is

$$\phi = \frac{M_A/M_B}{Y^* - Y} \ln \left(1 + \frac{Y^* - Y}{M_A/M_B + Y} \right) \quad (3.28)$$

While the convective heat flux expression is straightforward, the expression for drying rate needs explanation. The drying rate can be calculated from this formula, when drying is controlled by gas-side resistance. The driving force is then the difference between absolute humidity at equilibrium with solid surface and that of bulk gas. When solid surface is saturated with moisture, the expression for Y^* is identical to Equation 3.48; when solid surface contains bound moisture, Y^* will result from Equation 3.46 and a sorption isotherm. This is in essence the so-called equilibrium method of drying rate calculation.

When the drying rate is controlled by diffusion in the solid phase (i.e., in the falling drying rate period), the conditions at solid surface are difficult to find, unless we are solving the DPM (Fick's law or equivalent) for the solid itself. Therefore, if the solid itself has lumped parameters, its drying rate must be represented by an empirical expression. Two forms are commonly used.

3.4.2.1 Characteristic Drying Curve

In this approach the measured drying rate is represented as a function of the actual moisture content (normalized) and the drying rate in the constant drying rate period:

$$w_D = w_{Dl} f(\Phi) \quad (3.29)$$

The f function can be represented in various forms to fit the behavior of typical solids. The form proposed by Langrish et al. (1991) is particularly useful. They split the falling rate periods into two segments (as it often occurs in practice) separated by Φ_B . The equations are:

$$\begin{aligned} f &= \Phi_B^{a-c} & \text{for } \Phi \leq \Phi_B \\ f &= \Phi^a & \text{for } \Phi > \Phi_B \end{aligned} \quad (3.30)$$

Figure 3.2 shows the form of a possible drying rate curve using Equation 3.30.

Other such equations also exist in the literature (e.g., Hallström and Wimmerstedt, 1983; Nijdam and Keey, 2000).

3.4.2.2 Kinetic Equation (e.g., Thin-Layer Equations)

In agricultural sciences it is common to present drying kinetics in the form of the following equation:

$$\Phi = f(\tau, \text{process parameters}) \quad (3.31)$$

The function f is often established theoretically, for example, when using the drying model formulated by Lewis (1921)

$$\frac{dX}{d\tau} = k(X - X^*) \quad (3.32)$$

After integration one obtains

$$\Phi = \exp(-k\tau) \quad (3.33)$$

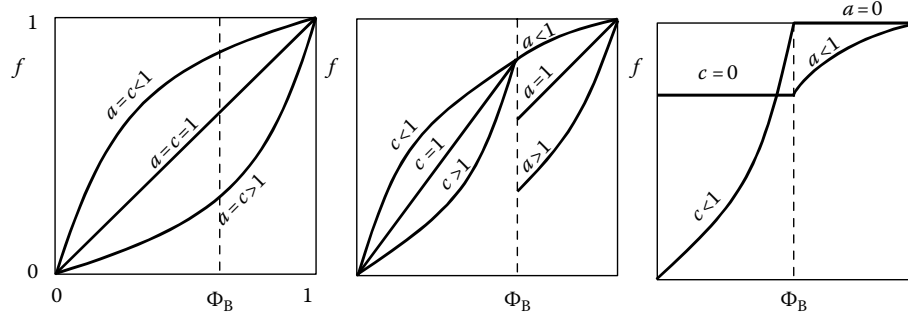


FIGURE 3.2 The influence of parameters a and c of Equation 3.30 on CDC shape.

A similar equation can be obtained by solving Fick's equation in spherical geometry:

$$\Phi = \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \exp\left(-n^2 \pi^2 \frac{D_{\text{eff}}}{R^2} \tau\right) \quad (3.34)$$

By truncating the RHS side one obtains

$$\Phi = \frac{6}{\pi^2} \exp\left(-\pi^2 \frac{D_{\text{eff}}}{R^2} \tau\right) = a \exp(-k\tau) \quad (3.35)$$

This equation was empirically modified by Page (1949), and is now known as the Page equation:

$$\Phi = \exp(-k\tau^n) \quad (3.36)$$

A collection of such equations for popular agricultural products is contained in Jayas et al. (1991). Other process parameters such as air velocity, temperature, and humidity are often incorporated into these equations.

The volumetric drying rate, which is necessary in balance equations, can be derived from the TLE in the following way:

$$\begin{aligned} w_D a_V &= -\frac{m_s}{A} a_V \frac{d\Phi}{d\tau} (X_c - X^*) \\ &= -\frac{m_s}{V} \frac{d\Phi}{d\tau} (X_c - X^*) \end{aligned} \quad (3.37)$$

while

$$m_s = V(1 - \varepsilon) \rho_s \quad (3.38)$$

and

$$w_D a_V = -(1 - \varepsilon) \rho_s (X_c - X^*) \frac{d\Phi}{d\tau} \quad (3.39)$$

The drying rate ratio of CDC is then calculated as

$$f = -\frac{(1 - \varepsilon) \rho_s (X_c - X^*)}{k_Y \phi (Y^* - Y) a_V} \frac{d\Phi}{d\tau} \quad (3.40)$$

To be able to calculate the volumetric drying rate from TLE, one needs to know the voidage ε and specific contact area a_V in the dryer.

When dried solids are monolithic or grain size is overly large, the above lumped parameter approximations of drying rate would be unacceptable, in which case a DPM represents the entire solid phase. Such models are shown in Section 3.7.

3.4.3 AUXILIARY RELATIONSHIPS

3.4.3.1 Humid Gas Properties and Psychrometric Calculations

The ability to perform psychrometric calculations forms a basis on which all drying models are built. One principal problem is how to determine the solid temperature in the constant drying rate conditions.

In psychrometric calculations we consider thermodynamics of three phases: inert gas phase, moisture vapor phase, and moisture liquid phase. Two gaseous phases form a solution (mixture) called humid gas. To determine the degree of complexity of our approach we will make the following assumptions:

- Inert gas component is insoluble in the liquid phase
- Gaseous phase behavior is close to ideal gas; this limits our total pressure range to less than 2 bar
- Liquid phase is incompressible
- Components of both phases do not chemically react with themselves

Before writing the psychrometric relationships we will first present the necessary approximating equations to describe physical properties of system components.

Dependence of saturated vapor pressure on temperature (e.g., Antoine equation):

$$\ln p_s = A - \frac{B}{C + t} \quad (3.41)$$

Dependence of latent heat of vaporization on temperature (e.g., Watson equation):

$$\Delta h_v = H(t - t_{\text{ref}})^n \quad (3.42)$$

Dependence of specific heat on temperature for vapor phase—polynomial form:

$$c_A = c_{A0} + c_{A1}t + c_{A2}t^2 + c_{A3}t^3 \quad (3.43)$$

Dependence of specific heat on temperature for liquid phase—polynomial form:

$$c_{A1} = c_{A10} + c_{A11}t + c_{A12}t^2 + c_{A13}t^3 \quad (3.44)$$

Table 3.1 contains coefficients of the above listed property equations for selected liquids and Table 3.2 for gases. These data can be found in specialized books (e.g., Reid et al., 1987; Yaws, 1999) and computerized data banks for other liquids and gases.

3.4.3.2 Relations between Absolute Humidity, Relative Humidity, Temperature, and Enthalpy of Humid Gas

With the above assumptions and property equations we can use Equations 3.45 through 3.47 for calculating these basic relationships (note that moisture is described as component A and inert gas as component B).

Definition of relative humidity φ (we will use here φ defined as decimal fraction instead of RH given in percentage points):

$$\varphi(t) = \frac{p}{p_s(t)} \quad (3.45)$$

Relation between absolute and relative humidities:

$$Y = \frac{M_A}{M_B} \frac{\varphi p_s(t)}{P_0 - \varphi p_s(t)} \quad (3.46)$$

Definition of enthalpy of humid gas (per unit mass of dry gas):

$$i_g = (c_A Y + c_B)t + \Delta h_{v0} Y \quad (3.47)$$

Equations 3.46 and 3.47 are sufficient to find any two missing humid gas parameters from Y , φ , t , i_g , if the other two are given. These calculations were traditionally done graphically using a psychrometric chart, but they are easy to perform numerically. When solving these equations one must remember that resulting Y for a given t must be lower than that at saturation, otherwise the point will represent a fog (super-saturated condition), not humid gas.

TABLE 3.1
Coefficients of Approximating Equations for Properties of Selected Liquids

Property		Water	Ethanol	Isopropanol	Toluene
Molar mass, kg/kmol	M_A	18.01	46.069	60.096	92.141
Saturated vapor pressure, kPa	A	16.376953	16.664044	18.428032	13.998714
	B	3878.8223	3667.7049	4628.9558	3096.52
	C	229.861	226.1864	252.636	219.48
Heat of vaporization, kJ/kg	H	352.58	110.17	104.358	47.409
	t_{ref}	374.14	243.1	235.14	318.8
	n	0.33052	0.4	0.371331	0.38
Specific heat of vapor, kJ/(kg · K)	c_{A0}	1.883	0.02174	0.04636	−0.4244
	$c_{A1} \times 10^3$	−0.16737	5.662	5.95837	6.2933
	$c_{A2} \times 10^6$	0.84386	−3.4616	−3.54923	−3.9623
	$c_{A3} \times 10^9$	−0.26966	0.8613	−16.3354	0.93604
Specific heat of liquid, kJ/(kg · K)	c_{A10}	2.822232	−1.4661	5.58272	−0.61169
	$c_{A11} \times 10^2$	1.182771	4.0052	−4.6261	1.9192
	$c_{A12} \times 10^4$	−0.350477	−1.5863	1.701	−0.56354
	$c_{A13} \times 10^8$	3.60107	22.873	−16.3354	5.9661

TABLE 3.2
Coefficients of Approximating Equations for Properties of Selected Gases

Property		Air	Nitrogen	CO ₂
Molar mass, kg/kmol	M_B	28.9645	28.013	44.010
Specific heat of gas, kJ/(kg · K)	c_{B0}	1.02287	1.0566764	0.48898
	$c_{B1} \times 10^3$	−0.5512	−0.197286	1.46505
	$c_{B2} \times 10^6$	0.181871	0.49471	−0.94562
	$c_{B3} \times 10^9$	−0.05122	−0.18832	0.23022

3.4.3.3 Calculations Involving Dew-Point Temperature, Adiabatic-Saturation Temperature, and Wet-Bulb Temperature

Dew-point temperature (DPT) is the temperature reached by humid gas when it is cooled until it becomes saturated (i.e., $\phi = 1$). From Equation 3.46 we obtain

$$Y_s = \frac{M_A}{M_B} \frac{p_s(t)}{P_0 - p_s(t)} \quad (3.48)$$

To find DPT when Y is known this equation must be solved numerically. On the other hand, the inverse problem is trivial and requires substituting DPT into Equation 3.48.

Adiabatic-saturation temperature (AST) is the temperature reached when adiabatically contacting limited amounts of gas and liquid until equilibrium. The suitable equation is

$$\frac{i_g - i_{gs,AST}}{Y - Y_{s,AST}} = c_{A1} t_{AS} \quad (3.49)$$

Wet-bulb temperature (WBT) is the one reached by a small amount of liquid exposed to an infinite amount of humid gas in steady state. The following are the governing equations.

- For water–air system, approximately

$$\frac{t - t_{WB}}{Y - Y_{s,WBT}} = - \frac{\Delta h_{v,WBT}}{c_H} \quad (3.50)$$

where

$$c_H(t) = c_A(t)Y + c_B(t) \quad (3.51)$$

Incidentally, this equation is equivalent to Equation 3.49 (see Treybal [1980]) for air and water vapor system.

- For other systems with higher Lewis numbers the deviation of WBT from AST is noticeable and can reach several degrees Celsius, thus causing serious errors in drying rate estimation. For such systems the following equation is recommended (Keey, 1978):

$$\frac{t - t_{WB}}{Y - Y_{s,WBT}} = - \frac{\Delta h_{v,WBT}}{c_H} Le^{-2/3} \phi \quad (3.52)$$

Typically in the wet-bulb calculations the following two situations are common:

- One searches for humidity of gas of which both dry- and wet-bulb temperatures are known: it is enough to substitute relationships for Y_s , Δh_v , and c_H into Equation 3.52 and solve it for Y .
- One searches for WBT once dry-bulb temperature and humidity are known: the same substitutions are necessary but now one solves the resulting equation for WBT.

The Lewis number

$$Le = \frac{\lambda_g}{c_p \rho_g D_{AB}} \quad (3.53)$$

is defined usually for conditions midway of the convective boundary layer. Recent investigations (Berg et al., 2002) indicate that Equation 3.52 needs corrections to become applicable to systems of high WBT approaching boiling point of liquid. However, for common engineering applications it is usually sufficiently accurate.

Over a narrow temperature range, e.g., for water–air system between 0°C and 100°C, to simplify calculations one can take constant specific heats equal to $c_A = 1.91$ and $c_B = 1.02$ kJ/(kg·K). In all calculations involving enthalpy balances specific heats are averaged between the reference and actual temperature.

3.4.3.4 Construction of Psychrometric Charts

Construction of psychrometric charts by computer methods is common. Three types of charts are most popular: Grosvenor chart, Grosvenor (1907) (or the psychrometric chart), Mollier chart, Mollier (1923) (or enthalpy-humidity chart), and Salin chart (or deformed enthalpy-humidity chart); these are shown schematically in Figure 3.3.

Since the Grosvenor chart is plotted in undistorted Cartesian coordinates, plotting procedures are simple. Plotting methods are presented and charts of high accuracy produced as explained in Shallcross (1994). Procedures for the Mollier chart plotting are explained in Pakowski (1986) and Pakowski and Mujumdar (1987), and those for the Salin chart in Soininen (1986).

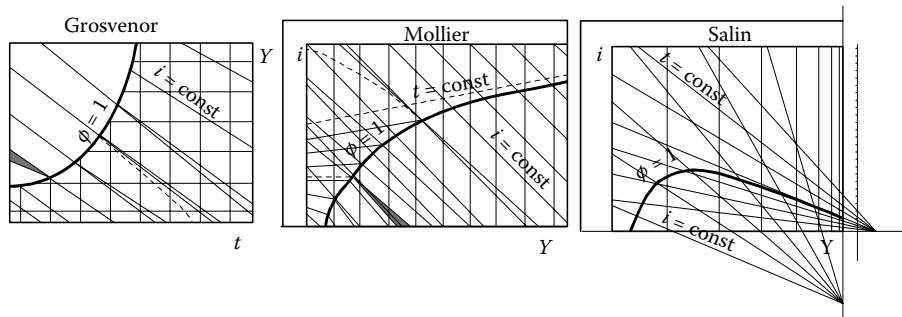


FIGURE 3.3 Schematics of the Grosvenor, Mollier, and Salin charts.

It is worth stressing that computer-generated psychrometric charts are used mainly as illustration material for presenting computed results or experimental data. They are now seldom used for graphical calculation of dryers.

3.4.3.5 Wet Solid Properties

Humid gas properties have been described together with humid gas psychrometry. The pertinent data for wet solid are presented below.

Sorption isotherms of the wet solid are, from the point of view of model structure, equilibrium relationships, and are a property of the solid–liquid–gas system. For the most common air–water system, sorption isotherms are, however, traditionally considered as a solid property. Two forms of sorption isotherm equations exist—explicit and implicit:

$$\varphi^* = f(t, X) \quad (3.54)$$

$$X^* = f(t, a_w) \quad (3.55)$$

where a_w is the water activity and is practically equivalent to φ . The implicit equation, favored by food and agricultural sciences, is of little use in dryer calculations unless it can be converted to the explicit form. In numerous cases it can be done analytically. For example, the GAB equation

$$X^* = \frac{a\varphi}{(1-b\varphi)(1+c\varphi)} \quad (3.56)$$

can be solved analytically for φ , and when the wrong root is rejected, the only solution is

$$\varphi^* = \frac{-((a/X) + b - c) + \sqrt{((a/X) + b - c)^2 + 4bc}}{2bc} \quad (3.57)$$

Numerous sorption isotherm equations (of approximately 80 available) cannot be analytically converted to the explicit form. In this case they have to be solved numerically for φ^* each time Y^* is computed, i.e., at every drying rate calculation. This slows down computations considerably.

Sorptional capacity varies with temperature, and the thermal effect associated with this phenomenon is isosteric heat of sorption, which can be numerically calculated using the Clausius–Clapeyron equation

$$\Delta h_s = -\frac{R}{M_A} \left[\frac{d \ln \varphi}{d(1/T)} \right]_{X=\text{const}} \quad (3.58)$$

If the sorption isotherm is temperature-independent the heat of sorption is zero; therefore a number of sorption isotherm equations used in agricultural sciences are useless from the point of view of dryer calculations unless drying is isothermal. It is noteworthy that in the model equations derived in this section the heat of sorption is neglected, but it can easily be added by introducing Equation 3.59 for the solid enthalpy in energy balances of the solid phase.

Wet solid enthalpy (per unit mass of dry solid) can now be defined as

$$i_m = (c_s + c_{A1}X)t_m - \Delta h_s X \quad (3.59)$$

The specific heat of dry solid c_s is usually presented as a polynomial dependence of temperature.

Diffusivity of moisture in the solid phase due to various governing mechanisms will be here termed as an effective diffusivity. It is often presented in the Arrhenius form of dependence on temperature

$$D_{\text{eff}} = D_0 \exp\left(-\frac{E_a}{RT}\right) \quad (3.60)$$

However, it also depends on moisture content. Various forms of dependence of D_{eff} on t and X are available (e.g., Marinos-Kouris and Maroulis [1995]).

3.4.4 PROPERTY DATABASES

As in all process calculations, reliable property data are essential (but not a guarantee) for obtaining sound results. For drying, three separate databases are necessary: for liquids (moisture), for gases, and for solids. Data for gases and liquids are widespread and are easily available in printed form (e.g., Yaws [1999]) or in electronic version. Relatively good property prediction methods exist (Reid et al., 1987). However, when it comes to solids, we are almost always confronted with a problem of availability of property data. Only a few source books exist with data for various products (Nikitina, 1968; Ginzburg and Savina, 1982; Iglesias and Chirife, 1984). Some data are available in this handbook also. However, numerous data are spread over technical literature and require a thorough search. Finally, since solids are not identical even if they represent the same product, it is always recommended to measure all the required properties and fit them with necessary empirical equations.

The following solid property data are necessary for an advanced dryer design:

- Specific heat of bone-dry solid
- Sorption isotherm
- Diffusivity of water in solid phase
- Shrinkage data
- Particle size distribution for granular solids

3.5 GENERAL REMARKS ON SOLVING MODELS

Whenever an attempt to solve a model is made, it is necessary to calculate the degrees of freedom of the model. It is defined as

$$N_D = N_V - N_E \quad (3.61)$$

where

N_V is the number of variables

N_E is the number of independent equations

It applies also to models that consist of algebraic, differential, integral, or other forms of equations. Typically the number of variables far exceeds the number of available equations. In this case several selected variables must be made constants; these selected variables are then called *process variables*. The model can be solved only when its degrees of freedom are zero. It must be borne in mind that not all vectors of process variables are valid or allow for a successful solution of the model.

To solve models one needs appropriate tools. They are either specialized for the specific dryer design or may have a form of universal mathematical tools. In the second case, certain experience in handling these tools is necessary.

3.6 BASIC MODELS OF DRYERS IN STEADY STATE

3.6.1 INPUT-OUTPUT MODELS

Input-output models are suitable for the case when both phases are perfectly mixed (cases 3c, 4c, and 5c in Figure 3.1), which almost never happens. On the other hand, this model is very often used to represent a case of unmixed flows when there is lack of a DPM. Input-output modeling consists basically of balancing all inputs and outputs of a dryer and is often performed to identify, for example, heat losses to the surroundings, calculate performance, and for dryer audits in general.

For a steady-state dryer balancing can be made for the whole dryer only, so the system of Equations 3.5 through 3.8 now consists of only two equations

$$W_S(X_1 - X_2) = W_B(Y_2 - Y_1) \quad (3.62)$$

$$W_S(i_{m2} - i_{m1}) = W_B(i_{g1} - i_{g2}) + q_c - q_l + \Delta q_t + q_m \quad (3.63)$$

where subscripts on heat fluxes indicate: c, indirect heat input; l, heat losses; t, net heat carried in by transport devices; and m, mechanical energy input. Let us assume that all q , W_S , W_B , X_1 , i_{m1} , Y_1 , i_{g1} are known as in a typical design case. The remaining variables are X_2 , Y_2 , i_{m2} , and i_{g2} . Since we have two equations, the system has two degrees of freedom and

cannot be solved unless two other variables are set as process parameters. In design we can assume X_2 since it is a design specification, but then one extra parameter must be assumed. This of course cannot be done rationally, unless we are sure that the process runs in constant drying rate period—then i_{m2} can be calculated from WBT. Otherwise, we must look for other equations, which could be the following:

$$W_S(X_1 - X_2) = Va_Y k_Y \Delta Y_m \quad (3.64)$$

$$W_S(i_{m2} - i_{m1}) = (Va_Y \alpha \Delta t_m + \Sigma q - k_Y \Delta Y_m h_A) \quad (3.65)$$

Provided that we know all kinetic data, a_Y , k_Y , and α , these two equations carry only one new variable V since temperatures can be derived from suitable enthalpies. Provided that we know how to calculate the averaged driving forces, the model now can be solved and exit stream parameters and volume of the dryer calculated. The success, however, depends on how well we can estimate the averaged driving forces.

3.6.2 DISTRIBUTED PARAMETER MODELS

3.6.2.1 Cocurrent Flow

For cocurrent operation (case 3a in Figure 3.1) both the case design and simulation are simple. The four balance equations (3.18 through 3.21) supplemented by a suitable drying rate and heat flux equations are solved starting at inlet end of the dryer, where all boundary conditions (i.e., all parameters of incoming streams) are defined. This situation is shown in Figure 3.4.

In the case of design the calculations are terminated when the design parameter, usually final moisture content, is reached. Distance at this point is the required dryer length. In the case simulation the calculations are terminated once the dryer length is reached.

Parameters of both gas and solid phase (represented by gas in equilibrium with the solid surface) can be plotted in a psychrometric chart as process paths. These phase diagrams (no timescale is available there) show schematically how the process goes on.

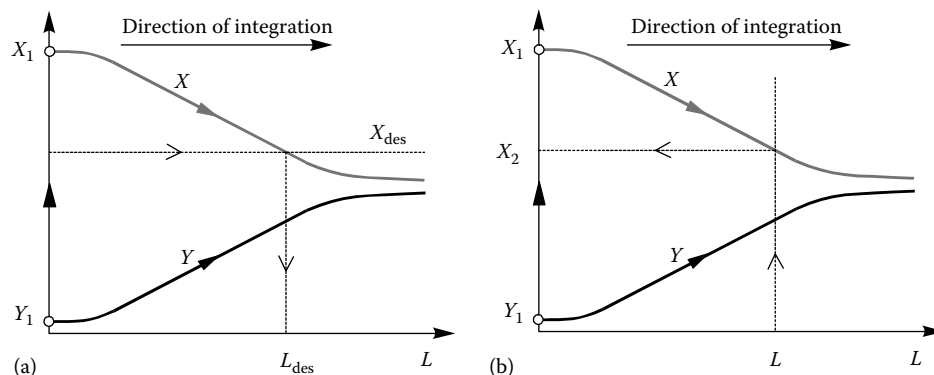


FIGURE 3.4 Schematic of design and simulation in cocurrent case: (a) design; (b) simulation. X_{des} is the design value of final moisture content.

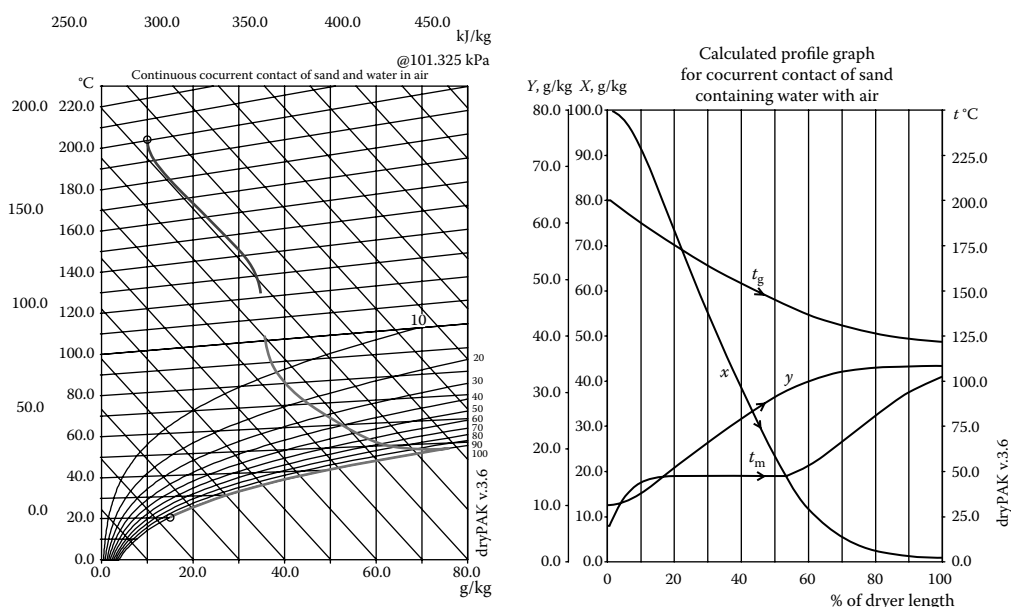


FIGURE 3.5 Process paths and longitudinal distribution of parameters for cocurrent drying of sand in air.

To illustrate the case the model composed of Equations 3.18 through 3.21, 3.26, and 3.27 is solved for a set of typical conditions and the results are shown in Figure 3.5.

3.6.2.2 Countercurrent Flow

The situation in countercurrent case (case 4a in Figure 3.1) design and simulation is shown in Figure 3.6. In both cases we see that boundary conditions are defined at opposite ends of the integration domain. It leads to the *split boundary value* problem.

In design this problem can be avoided by using the design parameters for the solid specified at the exit end. Then, by writing input–output balances over the whole dryer, inlet parameters of gas can easily be found (unless local heat losses or other distributed parameter phenomena need also be considered). However, in simulation the split boundary value problem exists and must be solved by a suitable numerical method, e.g., the *shooting method*. Basically

the method consists of assuming certain parameters for the exiting gas stream and performing integration starting at the solid inlet end. If the gas parameters at the other end converges to the known inlet gas parameters, the assumption is satisfactory; otherwise, a new assumption is made. The process is repeated under control of a suitable convergence control method, e.g., Wegstein. Figure 3.7 contains a sample countercurrent case calculation for the same material as that used in Figure 3.5.

3.6.2.3 Cross-Flow

3.6.2.3.1 Solid Phase is One-Dimensional

This is a simple case corresponding to case 5b of Figure 3.1. By assuming that the solid phase is perfectly mixed in the direction of gas flow, the solid phase becomes one-dimensional. This situation occurs with a continuous plug-flow fluid bed dryer. Schematic of an element of the dryer length with finite thickness Δl is shown in Figure 3.8.

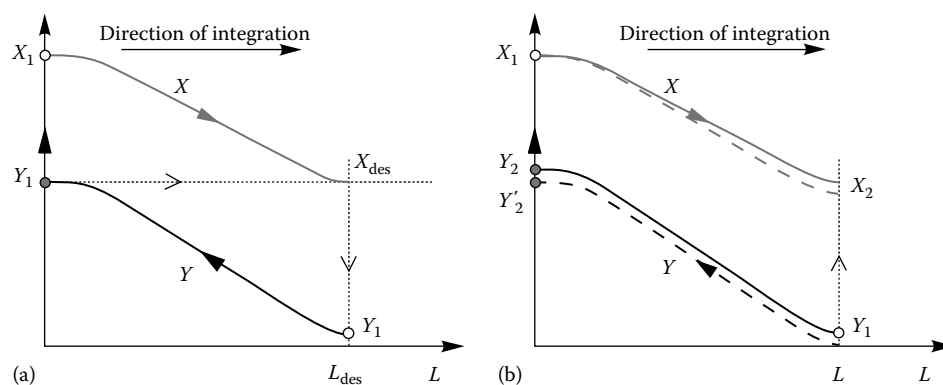


FIGURE 3.6 Schematic of design and simulation in cocurrent case: (a) design—split boundary value problem is avoided by calculating Y_1 from the overall mass balance; (b) simulation—split boundary value problem cannot be avoided, broken line shows an unsuccessful iteration, solid line shows a successful iteration—with Y_2 assumed the Y profile converged to Y_1 .

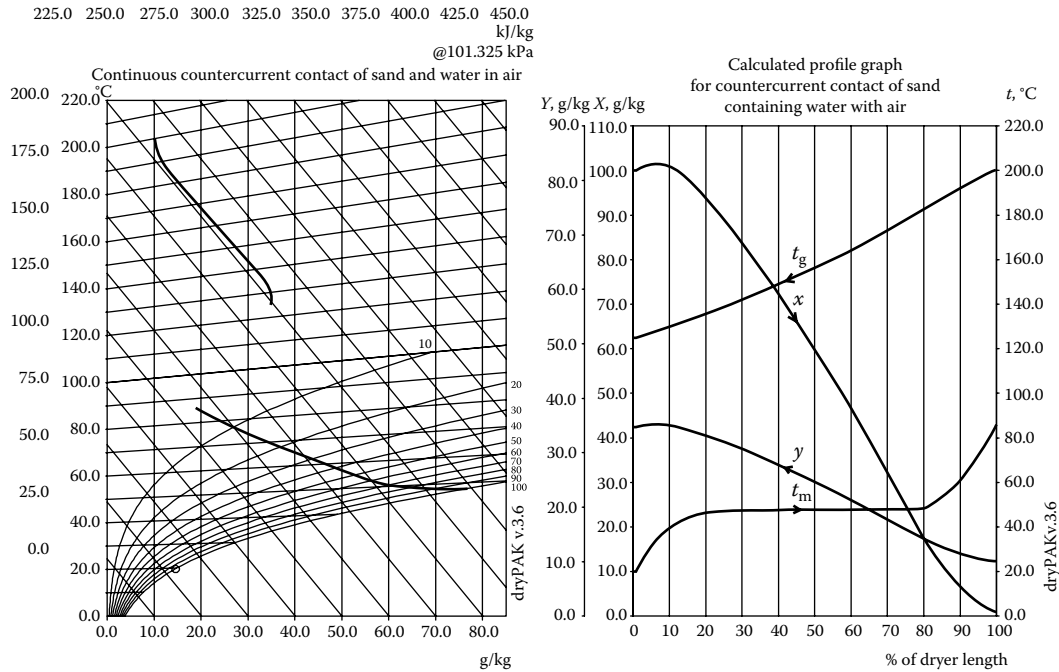


FIGURE 3.7 Process paths and longitudinal distribution of parameters for countercurrent drying of sand in air.

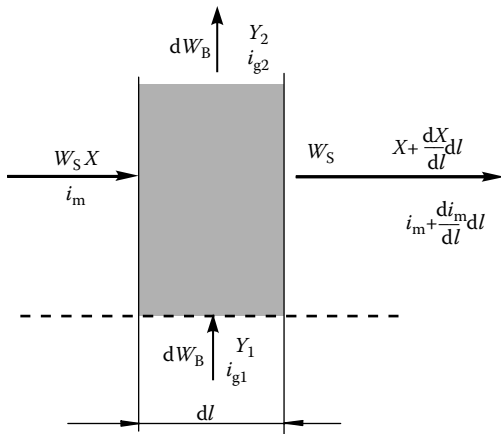


FIGURE 3.8 Element of a cross-flow dryer.

The balance equations for the solid phase can be derived from Equations 3.12 and 3.14 of the parallel flow:

$$\frac{W_S}{S} \frac{dX}{dl} = -w_D a_V \quad (3.66) \quad \text{or}$$

$$\frac{W_S}{S} \frac{di_m}{dl} = (q - w_D h_{Av}) a_V \quad (3.67)$$

The analogous equations for the gas phase are:
Mass balance

$$\frac{1}{S} \frac{dW_B}{dl} (Y_2 - Y_1) = w_D a_V \quad (3.68)$$

Energy balance

$$\frac{1}{S} \frac{dW_B}{dl} (i_{g2} - i_{g1}) = -(q - w_D h_{Av}) a_V \quad (3.69)$$

In the case of an equilibrium method of calculation of the drying rate the kinetic equations are:

$$w_D = k_Y \Delta Y_m \quad (3.70)$$

$$q = \alpha \Delta t_m \quad (3.71)$$

In other models (CDC and TLE) the drying rate will be modified as shown in Section 3.4.2.

Since the heat and mass coefficients can be defined on the basis of either the inlet driving force or the mean logarithmic driving force, ΔY_m and Δt_m are calculated respectively as

$$\Delta Y_m = (Y^* - Y_1) \quad (3.72)$$

$$\Delta Y_m = \frac{Y_2 - Y_1}{\ln((Y^* - Y_1)/(Y^* - Y_2))} \quad (3.73)$$

$$\Delta t_m = (t_m - t_{g1}) \quad (3.74)$$

or

$$\Delta t_m = \frac{t_{g2} - t_{g1}}{\ln((t_m - t_{g2})/(t_m - t_{g1}))} \quad (3.75)$$

To solve Equations 3.68 and 3.69 one needs to assume a uniform distribution of gas over the whole length of the dryer, and therefore

$$\frac{dW_B}{dl} = \frac{W_B}{L} \quad (3.76)$$

When the algebraic Equations 3.68 and 3.69 are solved to obtain the exiting gas parameters Y_2 and i_{g2} , one can plug the LHS of these equations into Equations 3.66 and 3.67 to obtain

$$\frac{dX}{dl} = \frac{1}{W_S} \frac{W_B}{L} (Y_2 - Y_1) \quad (3.77)$$

$$\frac{di_m}{dl} = \frac{1}{W_S} \frac{W_B}{L} (i_{g2} - i_{g1}) \quad (3.78)$$

The following equations can easily be integrated starting at the solids inlet. In Figure 3.9 sample process parameter profiles along the dryer are shown.

Cross-flow drying in a plug-flow, continuous fluid bed is a case when axial dispersion of flow is often considered. Let us briefly present a method of solving this case. First, the governing balance equations for the solid phase will have the following form derived from Equations 3.10 and 3.11

$$u_m \frac{dX}{dl} = E_m \frac{d^2 X}{dl^2} - \frac{a_v w_D}{\rho_s(1-\varepsilon)} \quad (3.79)$$

$$u_m \frac{di_m}{dl} = E_h \frac{d^2 i_m}{dl^2} + \frac{a_v (q - w_D h_{Av})}{\rho_s(1-\varepsilon)} \quad (3.80)$$

or

$$u_m \frac{dt_m}{dl} = E_h \frac{d^2 t_m}{dl^2} + \frac{a_v}{\rho_s(1-\varepsilon)} \frac{1}{c_s + c_{A1} X} \times [q + ((c_{A1} - c_A)t_m - \Delta h_{v0})w_D] \quad (3.81)$$

where

$$u_m = \frac{W_s}{\rho_s(1-\varepsilon)} \quad (3.82)$$

These equations are supplemented by equations for w_D and q according to Equations 3.70 and 3.71. It is a common assumption that $E_m = E_h$, because in fluid beds they result from longitudinal mixing by rising bubbles. Boundary conditions (BCs) assume the following form:

At $l = 0$

$$X = X_0 \quad \text{and} \quad i_m = i_{m0} \quad (3.83)$$

At $l = L$

$$\frac{dX}{dl} = 0 \quad \text{and} \quad \frac{di_m}{dl} = 0 \quad (3.84)$$

The second BC is due to Danckwerts and has been used for chemical reactor models. This leads, of course, to a split boundary value problem, which needs to be solved by an appropriate numerical technique. The resulting longitudinal profiles of solid moisture content and temperature in a dryer for various Peclet numbers ($Pe = u_m L/E$) are presented in Figure 3.10.

As one can see, only at low Pe numbers, profiles differ significantly. When $Pe > 0.5$, the flow may be considered a plug-flow.

3.6.2.3.2 Solid Phase is Two-Dimensional

This case happens when solid phase is not mixed but moves as a block. This situation happens in certain dryers for wet grains. The model must be derived for differential bed element as shown in Figure 3.11.

The model equations are now:

$$\frac{dX}{dl} = -\frac{w_D a_v}{\sigma_H} \quad (3.85)$$

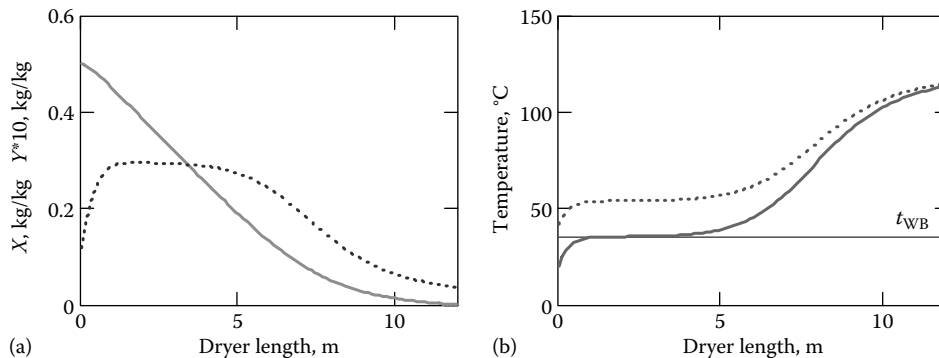


FIGURE 3.9 Longitudinal parameter distribution for a cross-flow dryer with one-dimensional solid flow. Drying of a moderately hygroscopic solid: (a) material moisture content (solid line) and local exit air humidity (broken line); (b) material temperature (solid line) and local exit air temperature (broken line). t_{WB} is wet bulb temperature of the incoming air.

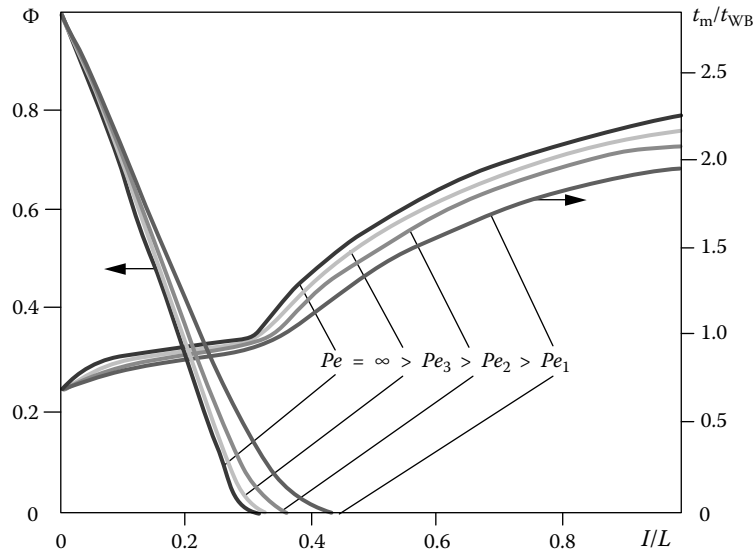


FIGURE 3.10 Sample profiles of material moisture content and temperature for various Pe numbers.

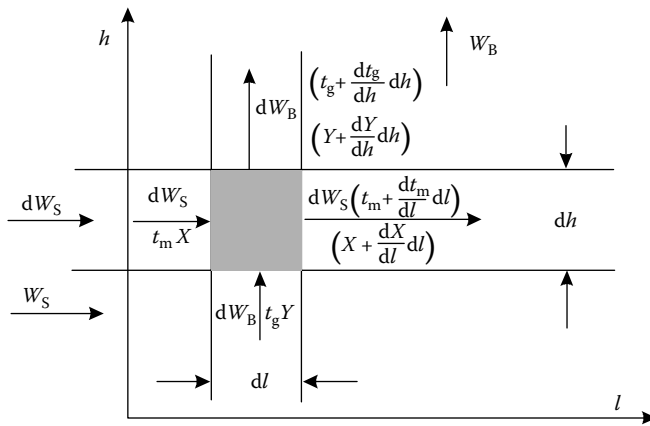


FIGURE 3.11 Schematic of a two dimensional cross-flow dryer.

$$\frac{dY}{dh} = \frac{w_D a_V}{\sigma_L} \quad (3.86)$$

$$\frac{dt_m}{dl} = \frac{a_V}{\sigma_H} \frac{1}{c_S + c_{A1} X} \times \left[q - ((c_A - c_{A1}) t_m + \Delta h_{v0}) w_D \right] \quad (3.87)$$

$$\frac{dt_g}{dh} = -\frac{a_V}{\sigma_L} \frac{1}{c_B + c_A Y} \left[q + c_A (t_g - t_m) w_D \right] \quad (3.88)$$

The symbols σ_H and σ_L are flow densities per 1 m for solid and gas mass flowrates, respectively, and are defined as follows:

$$\sigma_H = \frac{dW_S}{dh} = \frac{W_S}{H} \quad (3.89)$$

$$\sigma_L = \frac{dW_B}{dl} = \frac{W_B}{L} \quad (3.90)$$

The third term in these formulations applies when distribution of flow is uniform, otherwise an adequate distribution

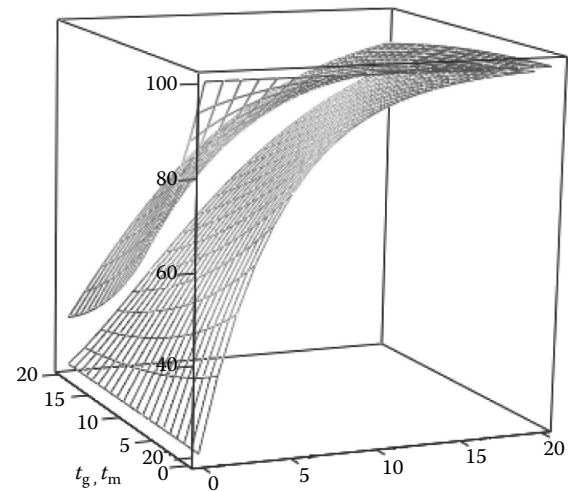


FIGURE 3.12 Solution of a two-dimensional cross-flow dryer model for cooling of granular solid with hot air. Solid flow enters through the front face of the cube, gas flows from left to right. Upper surface, solid temperature; lower surface, gas temperature.

function must be used. An exemplary model solution is shown in Figure 3.12. The solution only presents the heat transfer case (cooling of granular solid with air), so mass transfer equations are neglected.

3.7 DISTRIBUTED PARAMETER MODELS FOR THE SOLID

This case occurs when dried solids are monolithic or have large grain size so that LPM for the drying rate would be an unacceptable approximation. To answer the question as to whether this case applies one has to calculate the Biot number for mass transfer. It is recommended to calculate it from Equation 3.100 since various definitions are found in

the literature. When $Bi < 1$, the case is externally controlled and no DPM for the solid is required.

3.7.1 ONE-DIMENSIONAL MODELS

3.7.1.1 Nonshrinking Solids

Assuming that moisture diffusion takes place in one direction only, i.e., in the direction normal to surface for plate and in radial direction for cylinder and sphere, and that no other way of moisture transport exists but diffusion, the following second Fick's law may be derived

$$\frac{\partial X}{\partial \tau} = \frac{1}{r^n} \frac{\partial}{\partial r} \left[r^n D_{\text{eff}}(t_m, X) \frac{\partial X}{\partial r} \right] \quad (3.91)$$

where

$n = 0$ for plate, 1 for cylinder, 2 for sphere

r is current distance (radius) measured from the solid center

This parameter reaches a maximum value of R , i.e., plate is $2R$ thick if dried at both sides.

Initially we assume that moisture content is uniformly distributed and the initial solid moisture content is X_0 . To solve Equation 3.91 one requires a set of BCs. For high Bi numbers ($Bi > 100$) BC is called BC of the first kind and assumes the following form at the solid surface:

At $r = R$

$$X = X^*(\tau, Y) \quad (3.92)$$

For moderate Bi numbers ($1 < Bi < 100$) it is known as BC of the third kind and assumes the following form:

At $r = R$

$$-D_{\text{eff}} \rho_m \left(\frac{\partial X}{\partial r} \right)_i = k_Y [Y^*(X, t)_i - Y] \quad (3.93)$$

where subscript i denotes the solid–gas interface. BC of the second kind as known from calculus (constant flux at the surface)

At $r = R$

$$w_{Di} = \text{const.} \quad (3.94)$$

has little practical interest and can be incorporated in BC of the third kind. Quite often (here as well), therefore, BC of the third kind is named BC of the second kind. Additionally, at the symmetry plane we have

At $r = 0$

$$\frac{\partial X}{\partial r} = 0 \quad (3.95)$$

When solving the Fick's equation with constant diffusivity it is recommended to convert it to a dimensionless form. The following dimensionless variables are introduced for this purpose:

$$\Phi = \frac{X - X^*}{X_c - X^*}, \quad Fo = \frac{D_{\text{eff}0} \tau}{R^2}, \quad \zeta = \frac{r}{R} \quad (3.96)$$

In the nondimensional form Fick's equation becomes

$$\frac{\partial \Phi}{\partial Fo} = \frac{1}{\zeta^n} \frac{\partial}{\partial \zeta} \left[\zeta^n \frac{D_{\text{eff}}}{D_{\text{eff}0}} \frac{\partial \Phi}{\partial \zeta} \right] \quad (3.97)$$

and the BCs assume the following form:

BC I BC II

$$\text{at } \zeta = 1, \quad \Phi = 0 \left(\frac{\partial \Phi}{\partial \zeta} \right)_i + Bi_D^* \Phi = 0 \quad (3.98)$$

$$\text{at } \zeta = 0, \quad \frac{\partial \Phi}{\partial \zeta} = 0 \quad \frac{\partial \Phi}{\partial r} = 0 \quad (3.99)$$

where

$$Bi_D^* = m_{XY} \frac{k_Y \phi R}{D_{\text{eff}} \rho_m} \quad (3.100)$$

is the modified Biot number in which m_{XY} is a local slope of equilibrium curve given by the following expression:

$$m_{XY} = \frac{Y^*(X, t_m)_i - Y}{X - X^*} \quad (3.101)$$

The diffusional Biot number modified by the m_{XY} factor should be used for classification of the cases instead of $Bi_D = k_Y R / (D_{\text{eff}} \rho_m)$ encountered in several texts. Note that due to dependence of D_{eff} on X Biot number can vary during the course of drying, thus changing classification of the problem.

Since drying usually proceeds with varying external conditions and variable diffusivity, analytical solutions will be of little interest. Instead we suggest using a general-purpose tool for solving parabolic (Equation 3.97) and elliptic PDE in one-dimensional geometry like the *pdepe* solver of MATLAB. The result for $Bi_D^* = 5$ obtained with this tool is shown in Figure 3.13. The results were obtained for isothermal conditions. When conditions are nonisothermal, a question arises as to whether it is necessary to simultaneously solve Equations 3.22 and 3.23. Since Biot numbers for mass transfer far exceed those for heat transfer, usually the problem of heat transfer is purely external, and internal profiles of temperature are almost flat. This allows one to use LPM for the energy balance. Therefore, to monitor the solid temperature it is enough to supplement Equation 3.22 with the following energy balance equation:

$$\frac{dt_m}{d\tau} = \frac{A}{m_s} \frac{1}{c_s + c_{A1} \bar{X}} \left[q + ((c_A - c_{A1}) t_m + \Delta h_{v0}) w_D \right] \quad (3.102)$$

If Equations 3.22 and 3.23 must be solved simultaneously, the problem becomes stiff and requires specialized solvers.

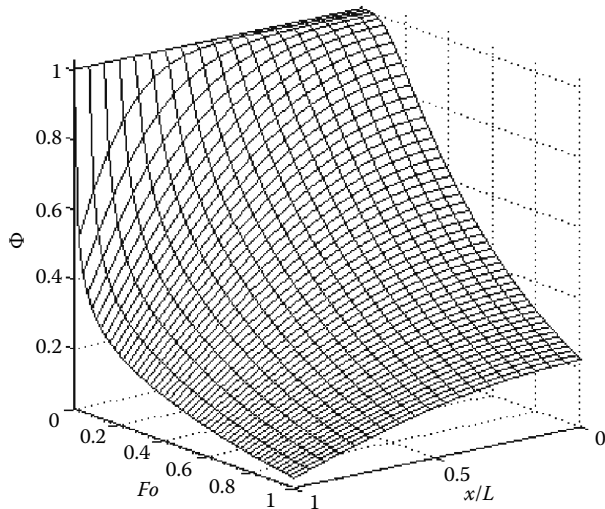


FIGURE 3.13 Solution of the DPM isothermal drying model of one-dimensional plate by *pdepe* solver of MATLAB®. Finite difference discretization by uniform mesh both for space and time, $Bi_D^* = 5$. Fo is dimensionless time, x/L is dimensionless distance.

3.7.1.2 Shrinking Solids

3.7.1.2.1 Unrestrained Shrinkage

When solids shrink volumetrically (majority of food products does), their volume is usually related to moisture content by the following empirical law:

$$V = V_s(1 + sX) \quad (3.103)$$

If one assumes that, for instance, a plate shrinks only in the direction of its thickness, the following relationship may be deduced from the above equation:

$$R = R_s(1 + sX) \quad (3.104)$$

where

R is the actual plate thickness

R_s is the thickness of absolutely dry plate

In Eulerian coordinates, shrinking causes an advective mass flux, which is difficult to handle. By changing the coordinate system to Lagrangian, i.e., the one connected with dry mass basis, it is possible to eliminate this flux. This is the principle of a method proposed by Kechaou and Roques (1990). In Lagrangian coordinates Equation 3.91 for one-dimensional shrinkage of an infinite plate becomes:

$$\frac{\partial X}{\partial \tau} = \frac{\partial}{\partial \zeta} \left[\frac{D_{\text{eff}}}{(1 + sX)^2} \frac{\partial X}{\partial \zeta} \right] \quad (3.105)$$

All boundary and initial conditions remain but the BC of Equation 3.94 now becomes

$$\left(\frac{\partial X}{\partial \zeta} \right)_{\zeta=R_s} = - \frac{(1 + sX)^2}{\rho_s D_{\text{eff}}} k_Y (Y^* - Y) \quad (3.106)$$

In Equations 3.105 and 3.106, ζ is the Lagrangian space coordinate, and it changes from 0 to R_s . For the above case of one-dimensional shrinkage the relationship between r and ζ is identical to that in Equation 3.104:

$$r = \zeta(1 + sX) \quad (3.107)$$

The model was proved to work well for solids with $s > 1$ (gelatin, polyacrylamide gel). An exemplary solution of this model for a shrinking gelatin film is shown in Figure 3.14.

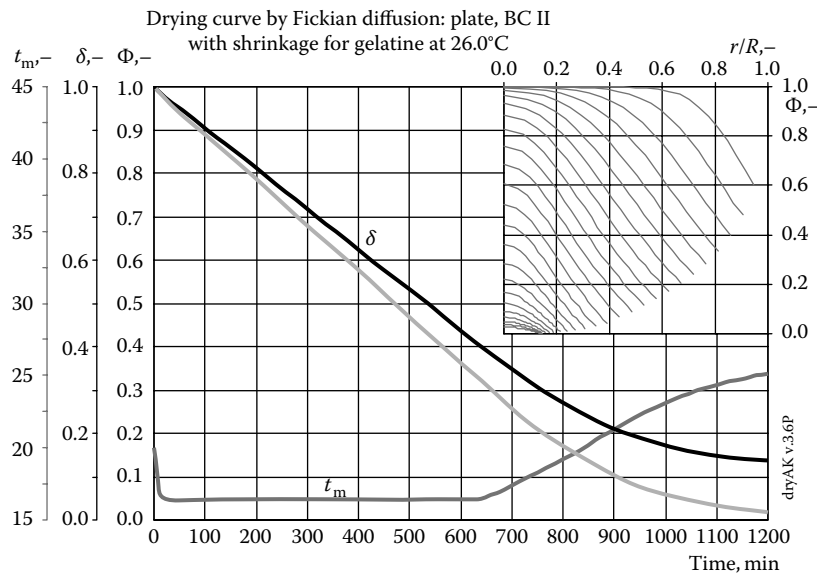


FIGURE 3.14 Solution of a model of drying for a shrinking solid. Gelatin plate 3 mm thick, initial moisture content 6.55 kg/kg. Shrinkage coefficient $s = 1.36$. Main plot shows dimensionless moisture content Φ , dimensionless thickness $\delta = R/R_0$, solid temperature t_m . Insert shows evolution of the internal profiles of Φ .

3.7.1.2.2 Restrained Shrinkage

For many materials shrinkage accompanying the drying process may be opposed by the rigidity of the solid skeleton or by viscous forces in liquid phase as it is compressed by shrinking external layers. This results in development of stress within the solid. The development of stress is interesting from the point of view of possible damage of dried product by deformation or cracking. In order to account for this, new equations have to be added to Equations 3.10 and 3.11. These are the balance of force equation and liquid moisture flow equation written as

$$G\nabla^2 \mathbf{U} + \frac{G}{1-2\nu} \nabla e - \alpha \nabla p = 0 \quad (3.108)$$

$$\frac{k}{\mu_{A1}} \nabla^2 p = \frac{1}{Q} \frac{\partial p}{\partial \tau} + \alpha \frac{\partial e}{\partial \tau} \quad (3.109)$$

where

$\mathbf{U}$ is the deformation matrix

e is strain tensor element

p is internal pressure (Q and α are constants)

The equations were developed by Biot and are explained in detail by Hasatani and Itaya (1996). Equations 3.108 and 3.109 can be solved together with Equations 3.10 and 3.11 provided that a suitable rheological model of the solid is known. The solution is almost always obtained by the finite element method due to inevitable deformation of geometry. Solution of such problems is complex and requires much more computational power than any other problem in this section.

3.7.2 TWO- AND THREE-DIMENSIONAL MODELS

In fact some supposedly three-dimensional cases can be converted to one-dimensional by transformation of the coordinate system. This allows one to use a finite difference method, which is easy to program. Lima et al. (2001) show how ovoid solids (e.g., cereal grains, silkworm cocoons) can be modeled by a one-dimensional model. This even allows for uniform shrinkage to be considered in the model. However, in the

case of two- and three-dimensional models when shrinkage is not negligible, the finite difference method can no longer be used. This is due to unavoidable deformation of corner elements, as shown in Figure 3.15.

The finite element methods have been used instead for two- and three-dimensional shrinking solids (see Perre and Turner [1999, 2000]). So far no commercial software was proven to be able to handle drying problems in this case and all reported simulations were performed by programs individually written for the purpose.

3.7.3 SIMULTANEOUS SOLVING DPM OF SOLIDS AND GAS PHASE

Usually in texts the DPM for solids (e.g., Fick's law) is solved for constant external conditions of gas. This is especially the case when analytical solutions are used. As the drying progresses, the external conditions change. At present with powerful ODE integrators there is essentially only computer power limit for simultaneously solving PDEs for the solid and ODEs for the gas phase. Let us discuss the case when spherical solid particles flow in parallel to gas stream exchanging mass and heat.

The internal mass transfer in the solid phase described by Equation 3.91 will be discretized by a finite difference method into the following set of equations

$$\frac{dX_i}{d\tau} = f(X_{i-1}, X_i, X_{i+1}, v) \quad \text{for } i = 1, \dots, \text{number of nodes} \quad (3.110)$$

where

X_i is the moisture content at a given node

v is the vector of process parameters

We will add Equations 3.19 through 3.21 to this set. In the last three equations the space increment dl can be converted to time increment by

$$dl = \frac{S(1-\epsilon)\rho_m}{W_s} d\tau \quad (3.111)$$

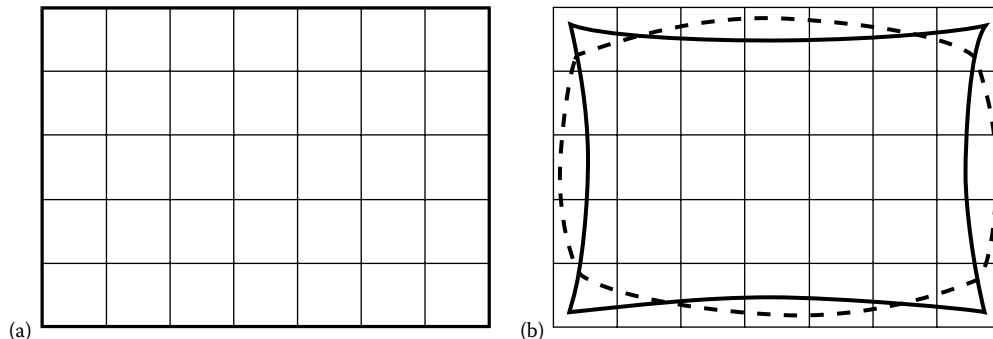


FIGURE 3.15 Finite difference mesh in the case two-dimensional drying with shrinkage: (a) before deformation; (b) after deformation. Broken line—for unrestrained shrinkage, solid line—for restrained shrinkage.

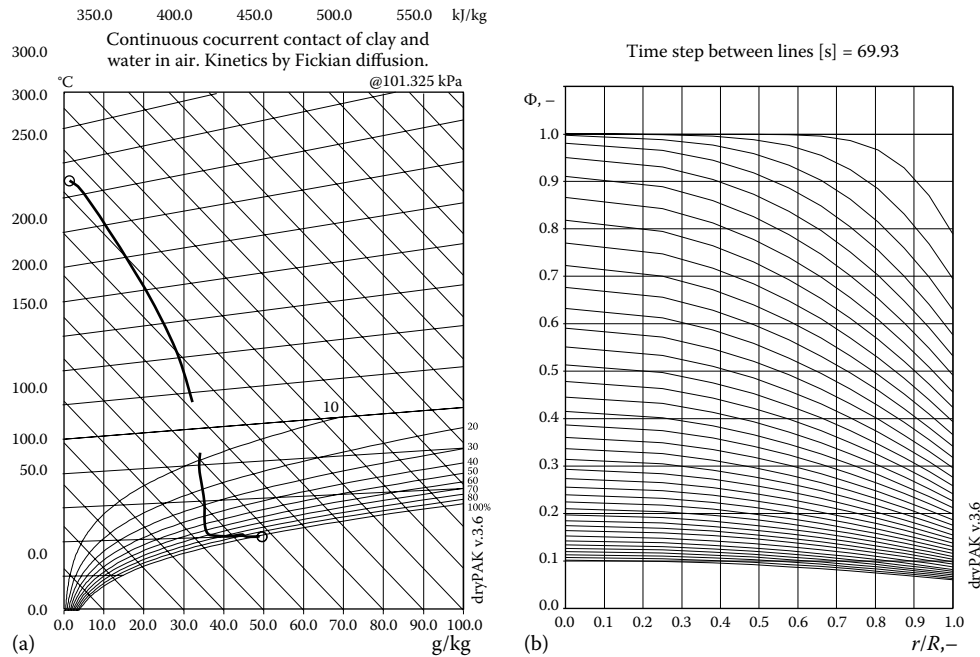


FIGURE 3.16 Cocurrent drying of clay spheres $d = 10$ mm in air at $t_g = 250^\circ\text{C}$. Solid throughput 0.1 kg/s, air throughput 0.06 kg/s. Simultaneous solution for gas phase and solid phase: (a) process trajectories—solid is represented by air in equilibrium with surface; (b) internal moisture distribution profiles.

The resulting set of ODEs can be solved by any ODE solver. The drying rate can be calculated between time steps (Equation 3.112) from temporal change of space-averaged moisture content. As a result one obtains simultaneously spatial profiles of moisture content in the solid as well as longitudinal distribution of parameters in the gas phase. Exemplary results are shown for cocurrent flash drying of spherical particles in Figure 3.16.

3.8 MODELS FOR BATCH DRYERS

We will not discuss here cases pertinent to startup or shut-down of typically continuous dryers but concentrate on three common cases of batch dryers. In batch drying the definition of drying rate, i.e.,

$$w_D = -\frac{m_S}{A} \frac{dX}{d\tau} \quad (3.112)$$

provides a basis for drying time computation.

3.8.1 BATCH-DRYING OVEN

The simplest batch dryer is a tray dryer shown in Figure 3.17. Here wet solid is placed in thin layers on trays and on a truck, which is then loaded into the dryer.

The fan is started and a heater power turned on. A certain air ventilation rate is also determined. Let us assume that the solid layer can be described by an LPM. The same applies to the air inside the dryer; because of internal fan, the air is well mixed and the case corresponds to case 2d in Figure 3.1. Here, the air humidity and temperature inside the dryer will

change in time as well as solid moisture content and temperature. The resulting model equations are therefore

$$m_S \frac{dX}{d\tau} = -w_D A \quad (3.113)$$

$$W_B Y_0 - W_B Y = m_S \frac{dX}{d\tau} + m_B \frac{dY}{d\tau} \quad (3.114)$$

$$m_S \frac{di_m}{d\tau} = (q - w_D h_{Av}) A \quad (3.115)$$

$$W_B i_{g0} - W_B i_g + \Sigma q = m_S \frac{di_m}{d\tau} + m_B \frac{di_g}{d\tau} \quad (3.116)$$

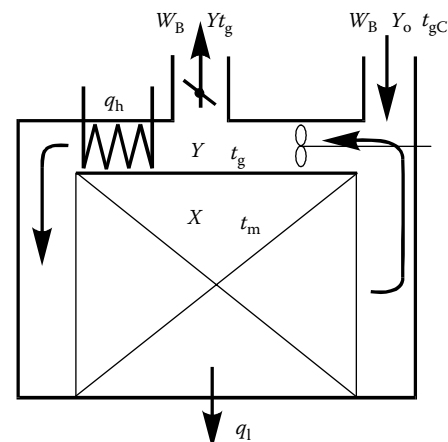


FIGURE 3.17 Schematic of a batch drying oven.

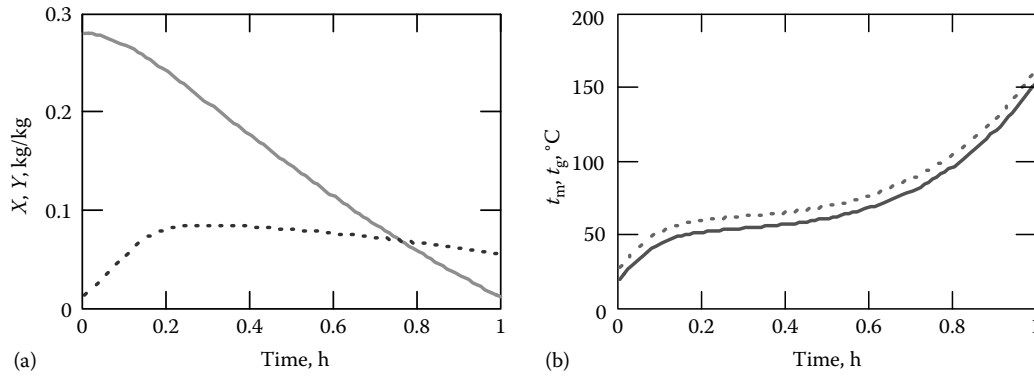


FIGURE 3.18 Solution of a batch oven dryer model—solid dry mass is 90 kg, internal heater power is 20 kW and air ventilation rate is 0.1 kg/s (dry basis); external air humidity is 2 g/kg and temperature 20°C: (a) moisture content X (solid line) and air humidity Y (broken line); (b) material temperature t_m (solid line) and air temperature t_g (broken line).

Note that Equation 3.113 is in fact the drying rate definition (Equation 112). In writing these equations we assume that the stream of air exiting the dryer has the same parameters as the air inside—this is a result of assuming perfect mixing of the air.

This system of equations is mathematically *stiff* because changes of gas parameters are much faster than changes in solid due to the small mass of gas in the dryer. It is advisable to neglect accumulation in the gas phase and assume that gas phase instantly follows changes of other parameters. Equations 3.114 and 3.116 will now have an asymptotic form of algebraic equations. Equations 3.113 through 3.116 can now be converted to the following working form:

$$\frac{dX}{d\tau} = -w_D \frac{A}{m_S} \quad (3.117)$$

$$W_B (Y_0 - Y) + w_D A = 0 \quad (3.118)$$

$$\frac{dt_m}{d\tau} = \frac{1}{c_S + c_A X} \frac{A}{m_S} \left[q + w_D \left((c_{A1} - c_A) t_m - \Delta h_{v0} \right) \right] \quad (3.119)$$

$$W_B \left[(c_B + c_A Y_0) t_{g0} - (c_B + c_A Y) t_{g0} + (Y - Y_0) c_A t_g \right] - A \left[q + w_D c_A (t_g - t_m) \right] + \Sigma q = 0 \quad (3.120)$$

The system of equations (Equations 3.117 and 3.119) is then solved by an ODE solver for a given set of data and initial conditions. For each time step air parameters Y and t_g are found by solving Equations 3.118 and 3.120. Sample simulation results for this case are plotted in Figure 3.18. Note that at the end of drying, the temperature in the dryer increases excessively due to constant power being supplied to the internal heater. The model may serve as a tool to control the process, e.g., increase the ventilation rate W_B when drying becomes too slow or reduce the heater power when temperature becomes too high as in this case.

3.8.2 BATCH FLUID BED DRYING

In this case the solid phase may be considered as perfectly mixed, so it will be described by an input–output model with accumulation term. On the other hand, the gas phase changes its parameters progressively as it travels through the bed. This situation is shown in Figure 3.19.

Therefore, gas phase will be described by a DPM with no accumulation and the solid phase will be described by an LPM with an accumulation term. The resulting equations are:

$$\frac{dX}{d\tau} = -\frac{a_v}{(1-\epsilon)\rho_s} \frac{1}{H} \int_0^H w_D dh \quad (3.121)$$

$$\frac{dY}{dh} = \frac{S}{W_B} w_D a_v \quad (3.122)$$

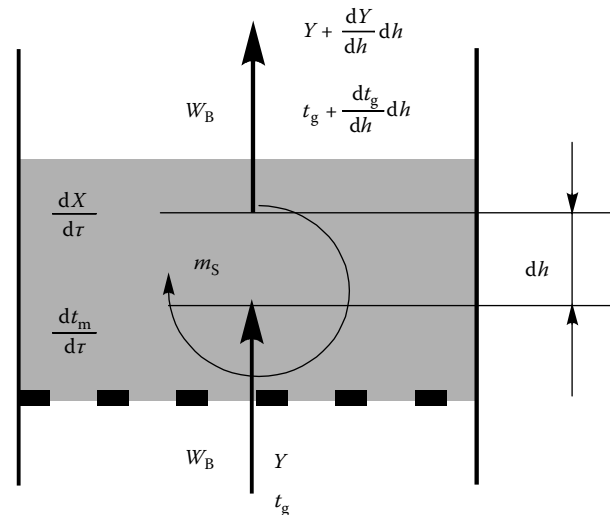


FIGURE 3.19 Schematic of a batch fluid bed dryer.

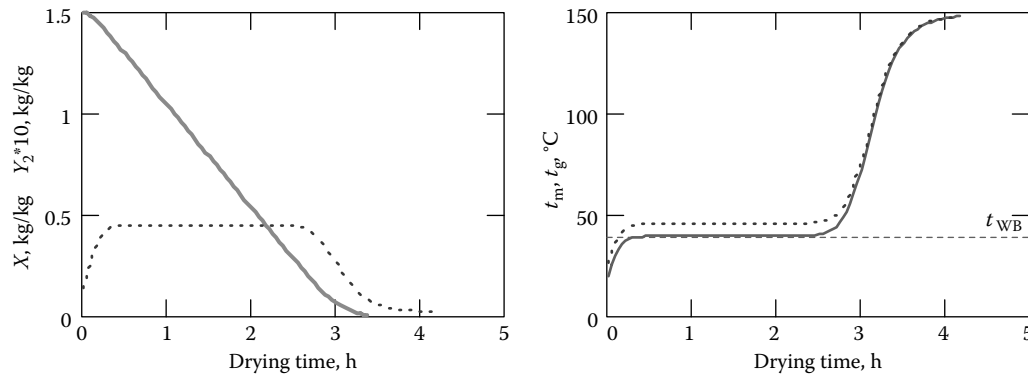


FIGURE 3.20 Temporal changes of solid moisture content and temperature and exit air humidity and temperature in a sample batch fluid bed dryer. Bed diameter 0.6 m, bed height 1.2 m, particle diameter 3 mm, particle density 1200 kg/m³, air temperature 150°C, and humidity 1 g/kg.

$$\frac{dt_m}{d\tau} = \frac{a_v}{(1-\epsilon)\rho_s} \frac{1}{c_s + c_{Al}X} \frac{1}{H} \times \int_0^H \left[q - ((c_A - c_{Al})t_m + \Delta h_{v0})w_D \right] dh \quad (3.123)$$

$$\frac{dt_g}{dh} = -\frac{S}{W_B} a_v \frac{1}{c_B + c_A Y} \left[q + c_A (t_g - t_m) w_D \right] \quad (3.124)$$

Equations 3.122 and 3.124 for the gas phase serve only to compute distributions of Y and t_g along bed height, which is necessary to calculate q and w_D . They can easily be integrated numerically, e.g., by the Euler method, at each time step. The integrals in Equations 3.121 and 3.123 can be numerically calculated, e.g., by the trapezoidal rule. This allows Equations 3.121 and 3.123 to be solved by any ODE solver. The model has been solved for a sample case and the results are shown in Figure 3.20.

3.8.3 DEEP BED DRYING

In deep bed drying solid phase is stationary and remains in the dryer for a certain time while gas phase flows through it continuously (case 2a of Figure 3.1). Drying begins at the inlet end of gas and progresses through the entire bed.

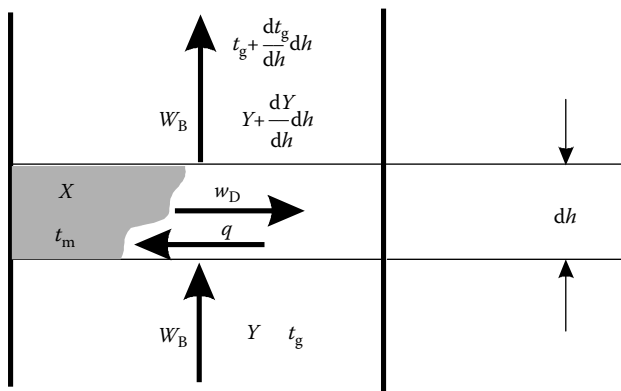


FIGURE 3.21 Schematic of batch drying in a deep layer.

A typical desorption wave travels through the bed. The situation is shown schematically in Figure 3.21.

The above situation is described by the following set of equations:

$$\frac{dX}{d\tau} = -\frac{w_D a_v}{(1-\epsilon)\rho_s} \quad (3.125)$$

$$\frac{dY}{dh} = \frac{S}{W_B} w_D a_v \quad (3.126)$$

$$\frac{dt_m}{d\tau} = \frac{a_v}{(1-\epsilon)\rho_s} \frac{1}{c_s + c_{Al}X} = \left[q - ((c_A - c_{Al})t_m + \Delta h_{v0})w_D \right] \quad (3.127)$$

$$\frac{dt_g}{dh} = -\frac{S}{W_B} a_v \frac{1}{c_B + c_A Y} \left[q + c_A (t_g - t_m) w_D \right] \quad (3.128)$$

The equations can be solved by finite difference discretization and a suitable numerical technique. Figure 3.22 presents the results of a simulation of drying cereal grains in a thick bed using Mathcad. Note how a desorption wave is formed, and also that the solid in deeper regions of the bed initially takes up moisture from the air humidified during its passage through the entry region.

Given a model together with its method of solution it is relatively easy to vary BCs, e.g., change air temperature in time or switch the gas flow from top to bottom intermittently, and observe the behavior of the system.

3.9 MODELS FOR SEMICONTINUOUS DRYERS

In some cases the dryers are operated in such a way that a batch of solids is loaded into the dryer and it progressively moves through the dryer. New batches are loaded at specified time intervals and at the same moment dry batches are removed at the other end. Therefore, the material is not moving continuously but by step increments. This is a typical situation in a tunnel dryer where trucks are loaded at one end of a tunnel and unloaded at the other, as shown in Figure 3.23.

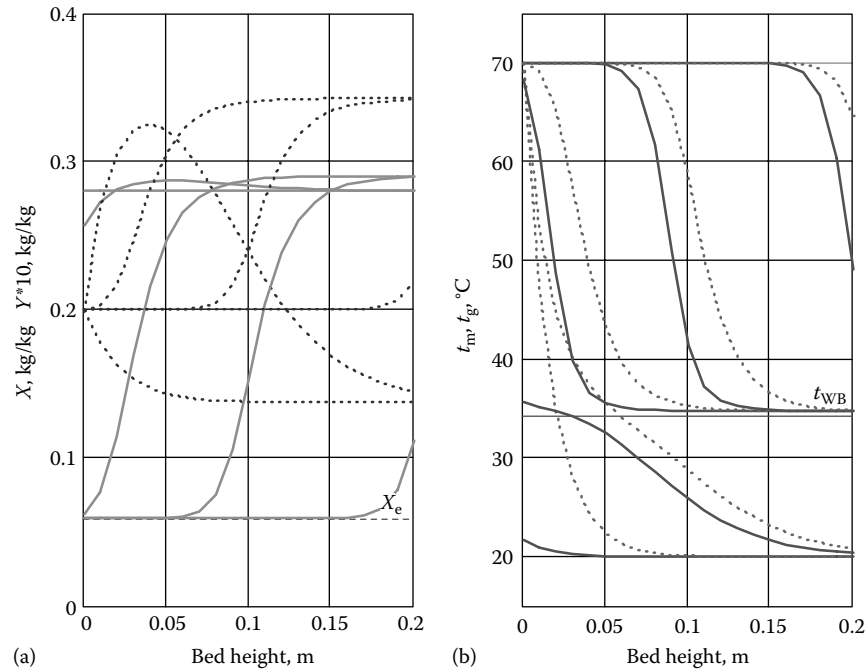


FIGURE 3.22 Simulation of deep bed drying of cereal grains: (a) moisture content profiles (solid lines) and gas humidity profiles (broken lines); (b) material temperature (solid lines) and air temperature (broken lines). Initial solid temperature 20°C and gas inlet temperature 70°C . Profiles are calculated at 0.33, 1.67, 3.33, 6.67, and 11.67 min of elapsed time. X_e is equilibrium moisture content and t_{WB} is wet-bulb temperature.

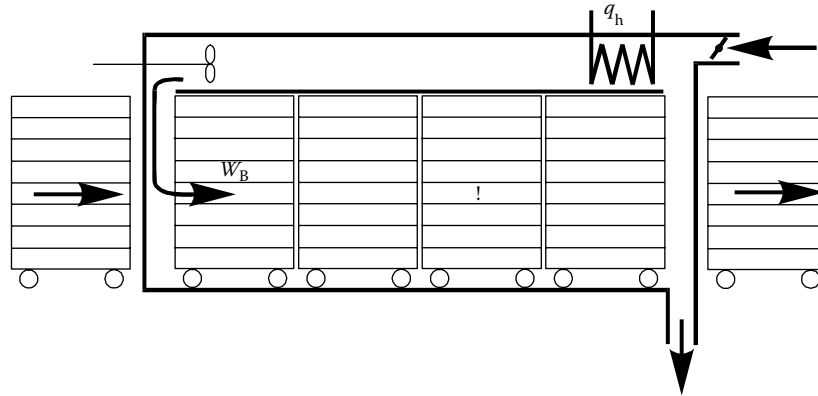


FIGURE 3.23 Schematic of a semicontinuous tunnel dryer.

To simplify the case one can take an LPM for each truck and a DPM for circulating air. As before, we will neglect accumulation in the gas phase but of course consider it in the solid phase. The resulting set of equations is

$$\frac{dX^i}{d\tau} = -\frac{w_{Di}A_i}{m_{Si}} \quad (3.129)$$

$$\frac{dY}{dl} = \frac{S}{W_B} w_D a_V \quad (3.130)$$

$$\frac{dt_m^i}{d\tau} = \frac{A_i}{m_{Si}} \frac{1}{c_S + c_{A1}X^i} \times \left[q_i - \left((c_A - c_{A1})t_m^i + \Delta h_{v0} \right) w_{Di} \right] \quad (3.131)$$

$$\frac{dt_g}{dl} = -\frac{S}{W_B} a_V \frac{1}{c_B + c_{A1}Y} \left[q + c_A (t_g - t_m) w_D \right] \quad (3.132)$$

where i is the number of a current truck. Additionally, a balance equation for mixing of airstreams at fresh air entry point is required. The semi-steady-state solution is when a new cycle of temporal change of X^i and t_m^i will be identical to the old cycle. In order to converge to a semi-steady state the initial profiles of X^i and t_m^i must be assumed. Usually a linear distribution between the initial and the final values is enough. The profiles are adjusted with each iteration until a cyclic solution is found.

The solution of this system of equations is schematically shown in Figure 3.24 for semi-steady-state operation and four trucks in the dryer. In each truck moisture content drops

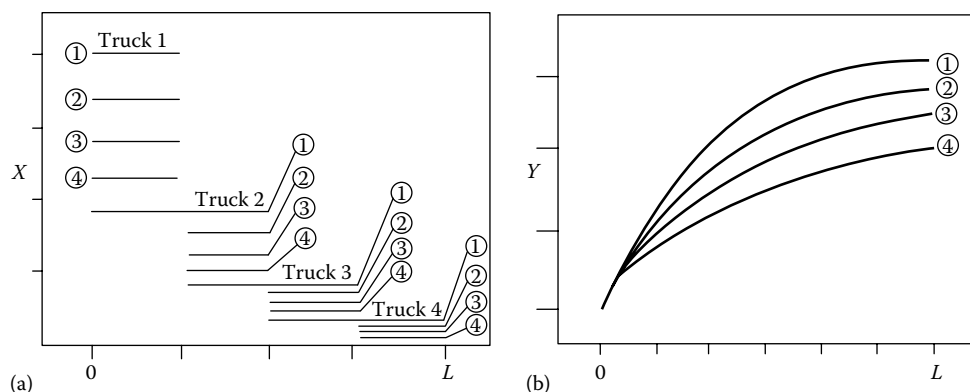


FIGURE 3.24 Schematic of the model solution for semicontinuous tunnel dryer for cocurrent flow of air vs. truck direction—mass transfer only: (a) moisture content in trucks at specified equal time intervals; (b) humidity profiles at specified time intervals. 1, 2, 3, 4—elapsed times.

in time until the load–unload time interval. Then the truck is moved one position forward so the last moisture content for this truck at former position becomes its initial moisture content at the new position. A practical application of this model for drying of grapes is presented by Caceres-Huambo and Menegalli (2002).

3.10 SHORTCUT METHODS FOR DRYER CALCULATION

When no data on sorptional properties, water diffusivity, shrinkage, etc., are available, dryer design can only be approximate, nevertheless useful, as a first approach. We will identify here two such situations.

3.10.1 DRYING RATE FROM PREDICTED KINETICS

3.10.1.1 Free Moisture

This case exists when drying of the product entirely takes place in the constant drying rate period. It is almost always possible when the solid contains unbound moisture. Textiles, minerals, and inorganic chemicals are examples of such solids.

Let us investigate a continuous dryer calculation. In this case solid temperature will reach, depending on a number of transfer units in the dryer, a value between AST and WBT, which can easily be calculated from Equations 3.49 and 3.50. Now mass and energy balances can be closed over the whole dryer and exit parameters of air and material obtained. Having these, the averaged solid and gas temperatures and moisture contents in the dryer can be calculated. Finally the drying rate can be calculated from Equation 3.27, which in turn allows one to calculate solid area in the dryer. Various aspect ratios of the dryer chamber can be designed; one should use judgment to calculate dryer cross-section in such a way that air velocity will not cause solid entrainment, etc.

3.10.1.2 Bound Moisture

In this case we can predict drying rate by assuming that it is linear, and at $X = X^*$ drying rate is zero, whereas at $X = X_{cr}$ drying rate is w_{DI} . The equation of drying rate then becomes

$$w_D = w_{DI} \frac{X - X^*}{X_c - X^*} = w_{DI} \Phi \quad (3.133)$$

This equation can be used for calculation of drying time in batch drying. Substituting this equation into Equation 3.112 and integration from the initial X_0 to final moisture content X_p , the drying time is obtained

$$\tau = \frac{m_s}{Aw_{DI}} (X_c - X^*) \ln \frac{X_0 - X^*}{X_p - X^*} \quad (3.134)$$

Similarly, Equation 3.133 can be used in a model of a continuous dryer.

3.10.2 DRYING RATE FROM EXPERIMENTAL KINETICS

Another simple case is when the drying curve has been obtained experimentally. We will discuss both batch and continuous drying.

3.10.2.1 Batch Drying

We may assume that if the solid size and drying conditions in the industrial dryer are the same, the drying time will also be the same as obtained experimentally. Other simple scaling rules apply, e.g., if a batch fluid bed thickness is double of the experimental one, the drying time will also double.

3.10.2.2 Continuous Drying

Here the experimental drying kinetics can only be used if material flow in the dryer is of plug type. In other words, it is as if the dryer served as a transporter of a batch container where drying is identical to that in the experiment. However, when a certain degree of mixing of the solid phase occurs, particles of the solid phase exiting the dryer will have

various residence times and will therefore differ in moisture content. In this case we can only talk of average final moisture content. To calculate this value we will use methods of residence time distribution (RTD) analysis. If the empirical drying kinetics curve can be represented by the following relationship:

$$X = f(X_0, \tau) \quad (3.135)$$

and mean residence time by

$$\tau_r = \frac{m_S}{W_S} \quad (3.136)$$

the average exit solid moisture content can be calculated using the external RTD function E as

$$\bar{X} = \int_0^{\infty} E(\tau) X(X_0, \tau) d\tau \quad (3.137)$$

Formulas for E function are presented in Table 3.3 for the most common flow models.

Figure 3.25 is an exemplary comparison of a batch and real drying curves. As can be seen, drying time in real flow conditions is approximately 50% longer here.

TABLE 3.3
External RTD Function for Selected Models of Flow

Model of Flow	E Function	
Plug flow	$E(\tau) = \delta(\tau - \tau_r)$	(3.138)
Perfectly mixed flow	$E(\tau) = \frac{1}{\tau_r} e^{-\tau/\tau_r}$	(3.139)
Plug flow with axial dispersion	$E(\tau) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tau - \tau_r)^2}{2\sigma^2}\right)$	(3.140)
	where for $Pe \geq 10$, $\sigma^2/\tau_r^2 = 2/Pe$	
n -Perfectly mixed uniform beds	$E(\tau) = \frac{n}{\tau_r} \frac{(n(\tau/\tau_r))^{n-1}}{(n-1)!} \exp\left(-n\frac{\tau}{\tau_r}\right)$	(3.141)

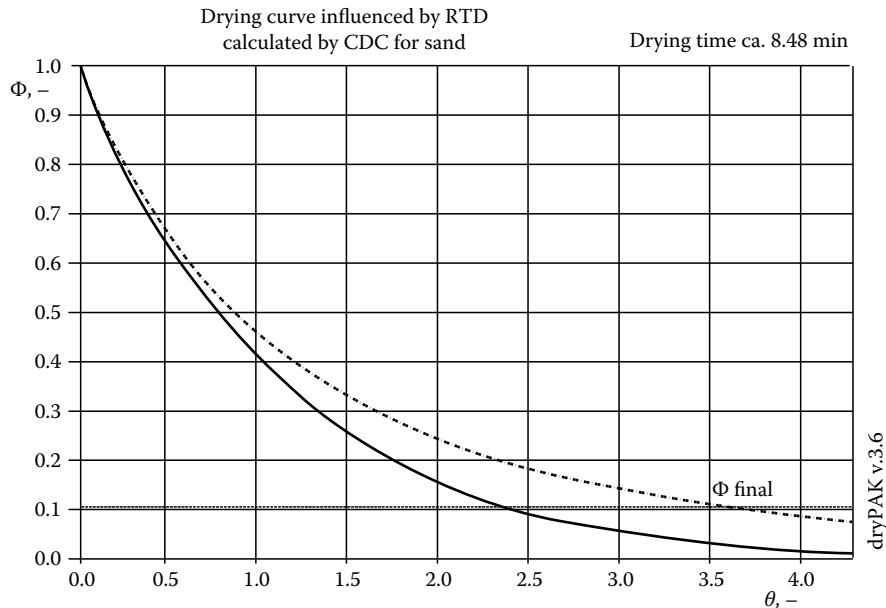


FIGURE 3.25 Experimental (solid line) and actual (broken line) drying kinetic curves for three tanks in series model of flow. θ is the ratio of the actual to mean residence times, Φ is dimensionless moisture content.

3.11 SOFTWARE TOOLS FOR DRYER CALCULATIONS

Menshutina and Kudra (2001) present 17 commercial and semicommercial programs for drying calculations that they were able to identify on the market. Only a few of them perform process calculations of dryers including dryer dimensioning, usually for fluid bed dryers. Typically a program for dryer calculations performs balancing of heat and mass and, if dimensioning is possible, the program requires empirical coefficients, which the user has to supply. Similarly, the drying process is designed in commercial process simulators used in chemical and process engineering. A program that does all calculations presented in this chapter does not exist. However, with present-day computer technology, construction of such software is possible; dryPAK (Silva and Correa, 1998; Pakowski, 1999) is a program that evolves in this direction. The main concept in dryPAK is that all models share the same database of humid gas, moist material properties, methods for calculation of drying rate, etc. The results are also visualized in the same way. Figures 3.5, 3.7, 3.14, and 3.16 were in fact produced with dryPAK.

General-purpose mathematical software can greatly simplify solving new models of not-too-complex structure. Calculations shown in Figures 3.9, 3.12, 3.18, 3.20, and 3.22 were produced with Mathcad. Mathcad or MATLAB can significantly reduce the effort involved with numerical solutions of equation systems as they contain a multitude of solvers for both algebraic and differential equations. Problems that would require several days of work can now be solved within hours. To let the reader get acquainted with this tool several Mathcad files containing selected solutions of problems presented in this section will be made available at <http://chemeng.p.lodz.pl/books/HID/>. Both MATLAB and Mathcad offer associated tools for visual modeling of dynamic systems (Simulink® and VisiSim, respectively) that make simulation of batch system even easier.

3.12 CONCLUSION

In this chapter we have illustrated how dryer calculations can be made by constructing a model of a dryer and solving it using appropriate numerical methods. Using general-purpose mathematical software solving models is a task that can be handled by any engineer. The results can be obtained in a short time and provide a sound basis for more detailed dryer calculations. For more advanced and specialized dryer design dedicated software should be sought. However, the question of how to obtain the necessary property data of dried materials remains. This question is as important now as it was before since very little has been done in the area of materials databases. The data are spread over the literature and, in the case of unsuccessful search, an experimental determination of the missing data is necessary.

NOMENCLATURE

A	interfacial area of phase contact, m^2
a, b, c	constants of GAB equation
a_v	characteristic interfacial area per unit volume of dryer, $1/m$
c	specific heat, $kJ/(kg \cdot K)$
D	diffusivity, m^2/s
E	axial dispersion coefficient, m^2/s
E	external RTD function
f	ratio of drying rates in CDC equation
G	shear modulus, Pa
h	specific enthalpy per unit mass of species, kJ/kg
Δh_s	latent heat of sorption, kJ/kg
Δh_v	latent heat of vaporization, kJ/kg
i	specific enthalpy per dry basis, kJ/kg
k	permeability, m^2
k_Y	mass transfer coefficient, $kg/(m^2 \cdot s)$
L	total length, m
l	running length, m
M	molar mass, $kg/kmol$
m	mass holdup, kg
p	vapor pressure, Pa
P_0	total pressure, Pa
q	heat flux, kW/m^2
R	maximum radius, m
R	universal gas constant, $kJ/(kmol \cdot K)$
r	actual radius, m
S	cross-sectional area normal to flow direction, m^2
s	shrinkage coefficient
t	temperature, $^{\circ}C$
T	absolute temperature, K
W	mass flowrate, kg/s
w_D	drying rate, $kg/(m^2 \cdot s)$
X	moisture content per dry basis, kg/kg
x	coordinate in Cartesian system, m
Y	absolute humidity per dry basis, kg/kg
V	total volume, m^3
α	heat transfer coefficient, $kW/(m^2 \cdot K)$
δ	Dirac delta function
ε	voidage
Φ	dimensionless moisture content $= (X - X^*)/(X_c - X^*)$
ϕ	correcting coefficient in Equation 3.27
φ	relative humidity
λ	thermal conductivity, $kW/(m \cdot K)$
μ	viscosity, $Pa \cdot s$
ν	Poisson's ratio
ρ	density, kg/m^3
τ	time, s

SUBSCRIPTS AND SUPERSSCRIPTS

A	moisture
AS	adiabatic saturation
B	dry gas
c	critical (for moisture content)
g	humid gas

i	at interface
m	wet solid
m	mean value
s	at saturation
S	dry solid
WB	wet bulb
v	vapor phase
*	in equilibrium
—	space averaged

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4 Transport Properties in the Drying of Solids

Dimitris Marinos-Kouris and Z.B. Maroulis

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4.1 INTRODUCTION

Drying is a complicated process involving simultaneous heat, mass, and momentum transfer phenomena, and effective models are necessary for process design, optimization, energy integration, and control. The development of mathematical models to describe drying processes has been a topic of many research studies for several decades. Undoubtedly, the observed progress has limited empiricism to a large extent. However, the design of dryers is still a mixture of science and practical experience. Thus the prediction of Luikov that by 1985 “would obviate the need for empiricism in selecting optimum drying conditions,” represented an optimistic perspective, which, however, shows that the efforts must be increased [1]. Presently, more and more sophisticated drying models are becoming available, but a major question that still remains is the measurement or determination of the parameters used in the models. The measurement or estimation of the necessary parameters should be feasible and practical for general applicability of a drying model.

In the early 1970s, Nonhebel and Moss stated that “the choice of drying plant, or design of special plant to meet unprecedented conditions” would require use of 34 parameters [2]. Regardless of the truth of such a statement, that is, of the actual number of parameters necessary for the design of a dryer, there is an obvious need for a large amount of data. Nowadays, the completeness and accuracy of such data reflect to a large extent our ability to perform effective process design. It should be noted that in spite of the intense activities in the drying literature (*Drying Technology Journal*, *Advances in Drying*, *Drying, International Drying Symposium*, etc.), the problem of property data still remains an important one because such data are widely scattered and not systematically evaluated. Moreover, whereas the need “for accurate design data is increasing, the rate of accumulation of new data is not increasing fast enough” [3]. The lack of data is expected to continue and, as noted by Keey, “it is probably unrealistic to expect complete hygrothermal data for materials of commercial interest” [4].

Out of the full set of thermophysical properties necessary for the analysis of drying of a material, this chapter examines only those that are critical. As such, we consider the thermodynamic

and transport properties, which are usually incorporated in a drying model as model parameters, and which are:

- Effective moisture diffusivity
- Effective thermal conductivity
- Air boundary heat and mass transfer coefficients
- Drying constant
- Equilibrium material moisture content

Effective thermal conductivity and effective moisture diffusivity are related to internal heat and mass transfer, respectively, while air boundary heat and mass transfer coefficients are related to external heat and mass transfer, respectively. The above transport properties are usually coefficients in the corresponding flow rate and driving force relationship. The equilibrium material moisture content, on the other hand, is usually related to the mass transfer driving force.

The above transport properties in conjunction with a transport phenomena mechanistic model can adequately describe the drying kinetics, but sometimes an additional property, the drying constant, is also used. The drying constant is essentially a combination of the above transport properties and it must be used in conjunction with the so-called thin-layer model.

Effective moisture diffusivity and effective thermal conductivity are in general functions of material moisture content and temperature, as well as of the material structure. Air boundary coefficients are functions of the conditions of the drying air, that is humidity, temperature, and velocity, as well as system geometry. Equilibrium moisture content of a given material is a function of air humidity and temperature. The drying constant is a function of material moisture content, temperature, and thickness, as well as air humidity, temperature, and velocity.

The required accuracy of the above properties depends on the controlling resistance to heat and mass transfer. If, for example, drying is controlled by the internal moisture diffusion, then the effective moisture diffusivity must be known with high accuracy. This situation is valid when large particles are drying with air of high velocity. Drying of small particles with low velocity of air is controlled by the external mass transfer, and the corresponding coefficient should be known with high accuracy. But there are situations in which

heat transfer is the controlling resistance. This happens, for example, in drying of solids with high porosity, in which high mass and low heat transfer rates are obtained.

The purpose of this chapter is to examine the above properties related to drying processes, particularly drying kinetics. Most of the following topics are discussed for each property:

- Definition
- Methods of experimental measurement
- Data compilation
- Effect of various factors
- Theoretical estimation

The statement of Poersch (quoted in Ref. [4]) that it is possible for someone to dry a product based on experience and without theoretical knowledge but not the reverse is worth repeating here. To this we may add the comment that it is impossible to efficiently dry a product without complete and precise thermophysical data.

4.2 MOISTURE DIFFUSIVITY

4.2.1 DEFINITION

Diffusion in solids during drying is a complex process that may involve molecular diffusion, capillary flow, Knudsen flow, hydrodynamic flow, or surface diffusion. If we combine all these phenomena into one, the effective diffusivity can be defined from Fick's second law

$$\frac{\partial X}{\partial t} = D \nabla^2 X. \quad (4.1)$$

where

- D (m^2/s) is the effective diffusivity
- X ($\text{kg}/\text{kg db}$) is the material moisture content
- t (s) is the time

The moisture transfer in heterogeneous media can be conveniently analyzed by using Fick's law for homogeneous materials, in which the heterogeneity of the material is accounted for by the use of an effective diffusivity.

Equation 4.1 shows the time change of the material moisture distribution, that is, it describes the movement of moisture within the solid. The previous equation can be used for design purposes in cases in which the controlling mechanism of drying is the diffusion of moisture.

Pakowski and Mujumdar [5] describe the use of Equation 4.1 for the calculation of the drying rate, whereas Strumillo and Kudra [6] describe its use in calculating the drying time. Solutions of the Fickian equation for a variety of initial and boundary conditions are exhaustively described by Crank [7].

4.2.2 METHODS OF EXPERIMENTAL MEASUREMENT

There is no standard method for the experimental determination of diffusivity. The diffusivity in solids can be

TABLE 4.1
Methods for the Experimental Measurement of Moisture Diffusivity

Method	Refs.
Sorption kinetics	[8]
Permeation methods	[8]
Concentration–distance curves	[10–12]
<i>Other methods</i>	
Radiotracer methods	[8]
Nuclear magnetic resonance (NMR)	[8,13,14]
Electron spin resonance (ESR)	[8,15]
<i>Drying technique</i>	
Simplified methods	[16]
Regular regime method	[17–19]
Numerical solution—regression analysis	See Section 4.7

determined using the methods presented in Table 4.1. These methods have been developed primarily for polymeric materials [7–9]. Table 4.1 also includes the relevant entries in the “References” section for the application of the methods in food systems.

4.2.2.1 Sorption Kinetics

The sorption (adsorption or desorption) rate is measured with a sorption balance (spring or electrical) whereas the solid sample is kept in a controlled environment. Assuming negligible surface resistance to mass transfer, the method is based on Fick's diffusion equation.

4.2.2.2 Permeation Method

The permeation method is a steady-state method applied to a film of material. According to this method, the permeation rate of a diffusant through a material of known thickness is measured under constant, well-defined, surface concentrations. The analysis is also based on Fick's diffusion equation.

4.2.2.3 Concentration–Distance Curves

The concentration–distance curves method is based on the measurement of the distribution of the diffusant concentration as a function of time. Light interference methods, as well as radiation adsorption or simply gravimetric methods, can be used for concentration measurements. Various sample geometries can be used, for example semiinfinite solid, two joint cylinders with the same or different material, and so on. The analysis is based on the solution of Fick's equation.

4.2.2.4 Other Methods

Modern methods for the measurement of moisture profiles lead to diffusivity measurement methods. Such methods discussed in the literature are radiotracer methods, nuclear magnetic resonance (NMR), electron spin resonance (ESR), and the like.

4.2.2.5 Drying Methods

The simplified, regular regime, and regression analysis methods are particularly relevant for drying processes. In them, the samples are placed in a dryer and moisture diffusivity is estimated from drying data. All the drying methods are based on Fick's equation of diffusion, and they differ with respect to the solution methodology. The following analysis is considered.

4.2.2.5.1 Simplified Methods

Fick's equation is solved analytically for certain sample geometries under the following assumptions:

- Surface mass transfer coefficient is high enough so that the material moisture content at the surface is in equilibrium with the air drying conditions.
- Air drying conditions are constant.
- Moisture diffusivity is constant, independent of material moisture content and temperature.

The analytical solution for slab, spherical, or cylindrical samples is used in the analysis. Several alternatives exist concerning the methodology of estimation of diffusivity using the above equations. They are discussed in the COST 90 bis project of European Economic Community (EEC) [16]. These alternatives differ essentially on the variable on which a regression analysis is applied.

4.2.2.5.2 Regular Regime Method

The regular regime method is based on the experimental measurement of the *regular regime curve*, which is the drying curve when it becomes independent of the initial concentration profile. Using this method, the concentration-dependent diffusivity can be calculated from one experiment.

4.2.2.5.3 Numerical Solution—Regression Analysis Method

The regression analysis method can be considered as a generalization of the other two types of methods. It can estimate simultaneously some additional transport properties; it is analyzed in detail in Section 4.7.

4.2.3 DATA COMPILATION

Effective diffusivities, reported in the literature, have been usually estimated from drying or sorption rate data. Experimental data are scarce because of the effect of the experimental method, the method of analysis, the variations in composition and structure of the examined materials, and so on. Data of effective diffusion coefficients are available for inorganic materials [20], polymers [8], and foods [21,22].

Table 4.2 gives some literature values of the effective diffusivity of moisture in various materials. A number of data from the above-mentioned bibliographic entries are also included in Table 4.2. New data up to 1992 are also incorporated. Foods are the most investigated materials in the literature, and they are presented separately. Table 4.2 was prepared for the needs of this chapter, that is, to show the

range of variation of diffusivity for various materials and not to present some experimental values. That is why most of the data are presented as ranges.

The data of Table 4.2 are further displayed in Figures 4.1 through 4.4. The moisture diffusivity is plotted versus the number of material for food and other materials in Figure 4.1. Diffusivities in foods have values in the range 10^{-13} to 10^{-6} m²/s, and most of them (82%) are accumulated in the region 10^{-11} to 10^{-8} . Diffusivities of other materials have values in the range 10^{-12} to 10^{-5} , whereas most of them (58%) are accumulated in the region 10^{-9} to 10^{-7} . These results are also clarified in the histograms of Figure 4.2. Diffusivities in foods are less than those in other materials. This is because of the complicated biopolymer structure of food and, probably, the stronger binding of water in them.

The influence of material moisture content and temperature from the statistical point of view is shown in Figures 4.3 and 4.4. Figure 4.3 shows the diffusivities versus the material moisture content for all the materials. The positive effect of material moisture content on diffusivity is evident. The same trend is noted in Figure 4.4 with regard to the temperature. It should be noted that the observed trends in the previous figures are the result of examining different materials at various temperatures and moistures and from various sources. The influence of material moisture content and temperature for each material is discussed in the next section.

In general, comparison among diffusivities reported in the literature is difficult because of the different methods of estimation and the variation of composition, especially for foods. However, on the basis of Figures 4.3 and 4.4, it is concluded that the differences in diffusivity among materials are less than that between temperature or material moisture content of the same material. Diffusivities of other solutes in various materials are also presented in the literature (e.g., see Ref. [68]).

4.2.4 FACTORS AFFECTING DIFFUSIVITY

Moisture diffusivity depends strongly on temperature and, often, very strongly on the moisture content, but there are few reliable figures. In porous materials the void fraction affects diffusivity significantly, and the pore structure and distribution do so even more.

The temperature dependence of the diffusivity can generally be described by the Arrhenius equation, which takes the form

$$D = D_0 \exp\left(\frac{-E}{RT}\right) \quad (4.2)$$

where

- D_0 (m²/s) is the Arrhenius factor
- E (kJ/kmol) is the activation energy for diffusion
- R (kJ/(kmol K)) is the gas constant
- T (K) is the temperature

The moisture content dependence of the diffusivity can be introduced in the Arrhenius equation by considering either the activation energy or the Arrhenius factor as an empirical

TABLE 4.2
Effective Moisture Diffusivity in Some Materials

Classification ^a	Material	Water Content (kg/kg db)	Temperature (°C)	Diffusivity (m ² /s)	Refs.
<i>Food</i>					
1	Alfalfa stems	<3.70	26	2.6E – 12–2.6E – 09	[23]
2	Apple	0.12	60	6.5E – 12–1.2E – 10	[24]
		0.15–7.00	30–76	1.2E – 10–2.6E – 10	[25]
3	Avocado		31–56	1.1E – 10–3.3E – 10	[26]
4	Beet		65	1.5E – 09	[26]
5	Biscuit	0.10–0.65	20–100	9.4E – 10–9.7E – 08	[27]
6	Bread	0.10–0.70	20–100	2.5E – 09–5.5E – 07	[27]
7	Carrot	0.03–11.6	42–80	9.0E – 10–3.3E – 09	[28]
8	Corn	0.05–0.23	40	1.0E – 12–1.0E – 10	[29]
		0.19–0.27	36–62	7.2E – 11–3.3E – 10	[30]
9	Fish muscle	0.05–0.30	30	8.1E – 11–3.4E – 10	[31]
10	Garlic	0.20–1.60	22–58	1.1E – 11–2.0E – 10	[32]
11	Milk foam	0.20	40	1.1E – 09	[33]
	Milk skim	0.25–0.80	30–70	1.5E – 11–2.5E – 10	[34]
12	Muffin	0.10–0.65	20–100	8.4E – 10–1.5E – 07	[27]
13	Onion	0.05–18.7	47–81	7.0E – 10–4.9E – 09	[35]
14	Pasta, semolina	0.01–0.25	40–125	3.0E – 13–1.5E – 10	[36]
	Pasta, corn based	0.10–0.40	40–80	5.0E – 11–1.3E – 10	[37]
	Pasta, durum wheat	0.16–0.35	50–90	2.5E – 12–5.6E – 11	[38]
15	Pepper, green	0.04–16.2	47–81	5.0E – 10–9.2E – 09	[35]
16	Pepperoni	0.19	12	4.7E – 11–5.7E – 11	[39]
17	Potato	0.60	54	2.6E – 10	[40]
		<4.00	65	4.0E – 10	[41]
		0.15–3.50	65	1.7E – 09	[42]
		0.01–7.20	39–82	5.0E – 11–2.7E – 09	[43]
18	Rice	0.18–0.36	60	1.3E – 11–2.3E – 11	[44]
		0.28–0.64	40–56	1.0E – 11–6.9E – 11	[45]
19	Soybeans, defatted	0.05	30	2.0E – 12–5.4E – 12	[46]
20	Starch, gel	0.10–0.30	25	1.0E – 12–2.3E – 11	[47]
		0.20–3.00	30–50	1.0E – 10–1.2E – 09	[48]
		0.75	25–140	1.0E – 10–1.5E – 09	[49]
	Starch granular	0.10–0.50	25–140	5.0E – 10–3.0E – 09	[49]
21	Sugar beet	2.50–3.60	40–80	4.0E – 10–1.3E – 09	[50,51]
22	Tapioca root	0.16–1.95	97	9.0E – 10	[52,53]
23	Turkey	0.04	22	8.0E – 15	[54]
24	Wheat	0.12–0.30	21–80	6.9E – 12–2.8E – 10	[55]
		0.13–0.20	20	3.3E – 10–3.7E – 09	[56]
<i>Other materials</i>					
1	Asbestos cement	0.10–0.60	20	2.0E – 09–5.0E – 09	[20]
2	Avicel (FMC Corp.)		37	5.0E – 09–5.0E – 08	[57]
3	Brick powder	0.08–0.16	60	2.5E – 08–2.5E – 06	[58]
4	Carbon, activated		25	1.6E – 05	[59]
5	Cellulose acetate	0.05–0.12	25	2.0E – 12–3.2E – 12	[60]
6	Clay brick	0.20	25	1.3E – 08–1.4E – 08	[61]
7	Concrete	0.10–0.40	20	5.0E – 10–1.2E – 08	[20]
	Concrete, pumice	0.20	25	1.8E – 08	[61]
8	Diatomite	0.05–0.50	20	3.0E – 09–5.0E – 09	[20]
9	Glass wool	0.10–1.80	20	2.0E – 09–1.5E – 08	[20]
	Glass spheres, 10 µm	0.01–0.22	60	1.84E – 8 ± 0.94E – 8	[16]
10	Hyde clay	0.10–0.40		5.0E – 09–1.0E – 08	[62]
11	Kaolin clay	<0.50	45	1.5E – 08–1.5E – 07	[20]
12	Model system		68	3.1E – 09	[63]

(continued)

TABLE 4.2 (continued)
Effective Moisture Diffusivity in Some Materials

Classification ^a	Material	Water Content (kg/kg db)	Temperature (°C)	Diffusivity (m ² /s)	Refs.
13	Peat	0.30–2.50	45	4.0E – 08–5.0E – 08	[20]
14	Sand	<0.15	45	8.0E – 08–1.5E – 07	[20]
	Sand, sea	0.07–0.13	60	2.5E – 08–2.5E – 06	[58]
	Sand	0.05–0.10		1.0E – 07–1.0E – 06	[64]
15	Silica alumina	0.59–1.18	60	2.5E – 08–2.5E – 06	[58]
16	Silica gel		25	3.0E – 06–5.6E – 06	[59]
17	Tobacco leaf		30–50	3.2E – 11–8.1E – 11	[65]
18	Wood, soft		40–90	5.0E – 10–2.5E – 09	[66]
	Wood, yellow poplar	1.00	100–150	1.0E – 08–2.5E – 08	[67]

^a Classification number for each material used in Figure 4.1.

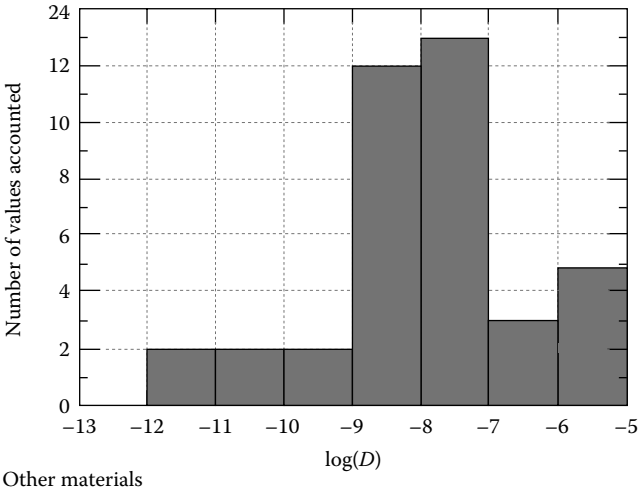
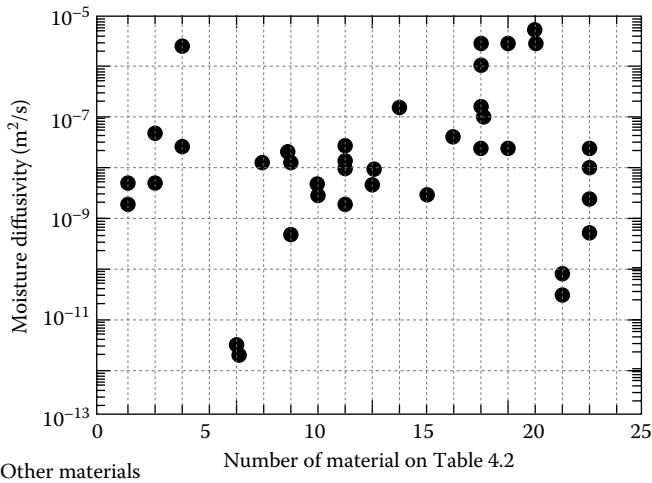
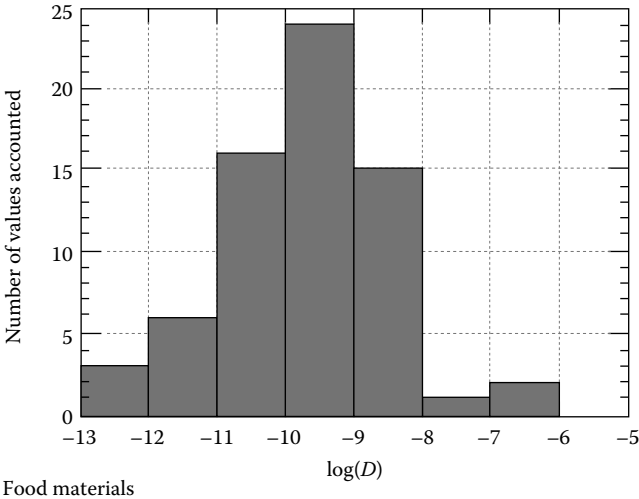
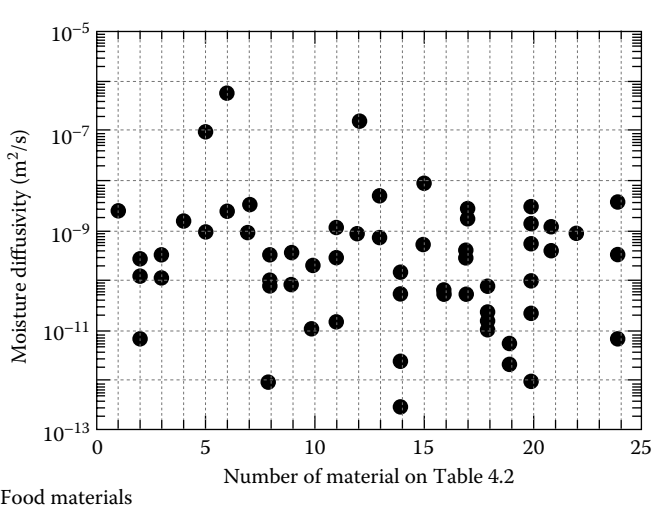


FIGURE 4.1 Moisture diffusivity in various materials (data from Table 4.2).

FIGURE 4.2 Histograms of diffusivities in various materials (data from Table 4.2).

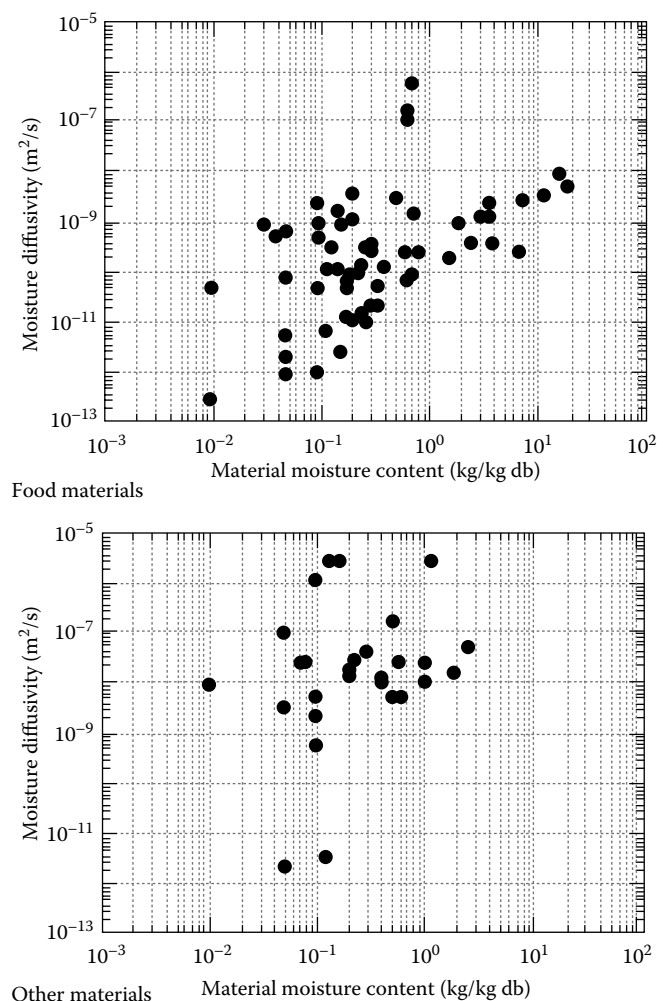


FIGURE 4.3 Moisture diffusivity versus material moisture content (data from Table 4.2).

function of moisture. Both modifications can be considered simultaneously. Other empirical equations not based on the Arrhenius equation can be used.

The moisture diffusivity is an increasing function of the temperature and moisture of the material. Yet, in certain categories of polymers, deviation from this kind of behavior has been observed. For instance, for several of the less hydrophilic polymers (e.g., polymethacrylates and polycrylates) the moisture diffusivity decreases with increasing water content. On the other hand, the moisture diffusivity appears to be independent of the concentration—and hence constant—for some hydrophobic polyolefins.

Table 4.3 gives some relationships that describe simultaneous dependence of the diffusivity upon temperature and moisture. Some rearrangement of the equations proposed has been done in order to present them in a uniform format. Table 4.4 lists parameter values for typical equations of Table 4.3.

Equations T3.1 through T3.4 in Table 4.3 suggest that the material moisture content can be taken into account by considering the preexponential factor of the Arrhenius equation

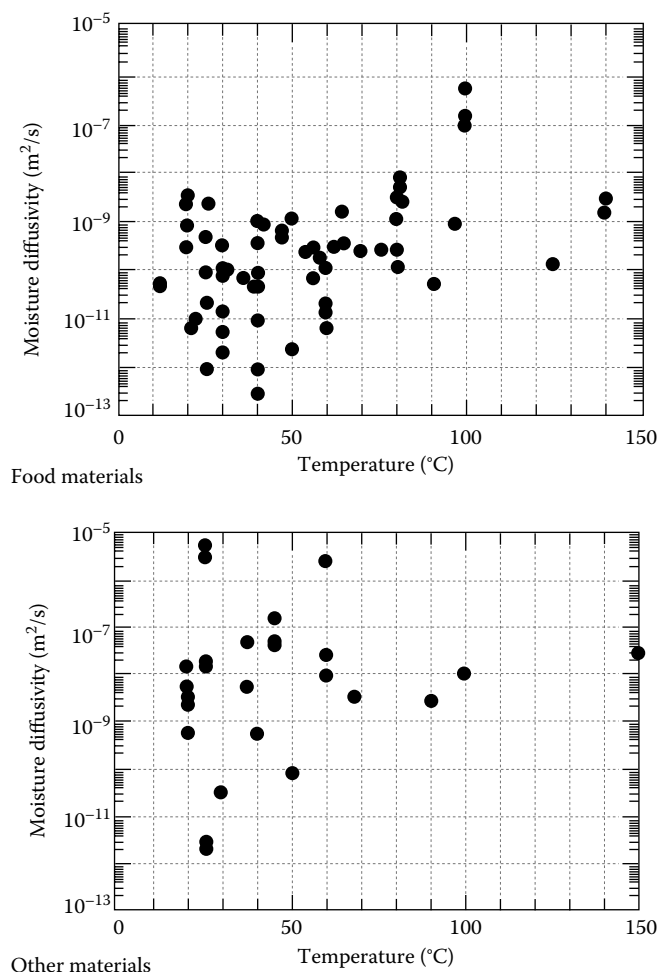


FIGURE 4.4 Moisture diffusivity versus material temperature (data from Table 4.2).

as a function of material moisture content. Polynomial functions of first order can be considered (Equation T3.1), as well as of higher order (Equation T3.2 or T3.3). The exponential function can also be used (Equation T3.4).

Equations T3.5 and T3.6 in Table 4.3 are obtained by considering the activation energy for diffusion as a function of material moisture content. Equations T3.7 through T3.10 are not based on the Arrhenius form. They are empirical and they use complicated functions concerning the discrimination of the moisture and temperature effects (except, of course, Equation T3.7). Equation T3.11 is more sophisticated as it considers different diffusivities of bound and free water and introduces the functional dependence of material moisture content on the binding energy of desorption. Equation T3.12 introduces the effect of porosity on moisture diffusivity.

With regard to the number of parameters involved (a significant measure concerning the regression analysis), it is concluded that at least three parameters are needed (Equations T3.1, T3.5, and T3.7).

Equations T3.5 and T3.7 in Table 4.3 were applied to potato and clay brick, respectively, and the results are

TABLE 4.3
Effect of Material Moisture Content and Temperature on Diffusivity

Equation No.	Materials of Application	Equation	No. of Parameters	Refs.
T3.1	Apple, carrot, starch	$D(X,T) = a_0 \exp(a_1 X) \exp(-a_2/T)$	3	[49,69,70]
T3.2	Bread, biscuit, muffin	$D(X,T) = a_0 \exp\left(\sum_{i=1}^3 a_i X^i\right) \exp\left(\frac{-a_2}{T}\right)$	5	[27]
T3.3	Polyvinylalcohol	$D(X,T) = a_0 \exp\left(\sum_{i=1}^{10} a_i X^i\right) \exp\left(\frac{-a_2}{T}\right)$	12	[71]
T3.4	Vegetables	$D(X,T) = a_0 \exp(-a_1/X) \exp(-a_2/T)$	3	[72]
T3.5	Glucose, coffee extract, skim milk, apple, potato, animal feed	$D(X,T) = a_0 \exp[-a_1(1/T - 1/a_2)] \exp(-a_{12}X)$ $a_1 = a_{10} + a_{11}$	5	[18]
T3.6	Silica gel	$D(X,T) = a_0 \exp(-a_1/T)$ $a_1 = a_{10} + a_{11}X$	3	[73]
T3.7	Clay brick, burned clay, pumice concrete	$D(X,T) = a_0 X^{a_1} T^{a_2}$	3	[61]
T3.8	Corn	$D(X,T) = a_0 \exp(a_1 X) \exp(-a_2/T)$ $a_1 = a_{11}T + a_{10}$	4	[30]
T3.9	Rough rice	$D(X,T) = a_1 \exp(a_2 X)$ $a_1 = a_{10} \exp(a_{11}T)$ $a_2 = a_{20} \exp(a_{21}T + a_{22}T^2)$	5	[74,75]
T3.10	Wheat	$D(X,T) = a_0 + a_1 X + a_2 X^2$ $a_0 = a_{01} \exp(a_{02}T)$ $a_1 = a_{11} \exp(a_{12}T)$ $a_2 = a_{21} \exp(a_{22}T)$	6	[76]
T3.11	Semolina, extruded	$D(X,T) = a_0 \exp\left(\frac{-a_2}{T}\right) \frac{a_2 \exp(-a_3/T)}{1 + a_2 \exp(-a_3/T)}$	4	[77]
T3.12	Porous starch	$D(X,T) = (a_0 + a_1 X^a) \exp\left(\frac{-a_3}{T}\right)$ $a_0 = F(\epsilon)$	>5	[78]

D , moisture diffusivity; X , material moisture content; T , temperature; a_i , constants; ϵ , porosity.

TABLE 4.4
Application Examples

Material	Equation	Constants	Refs.
Clay brick, burned clay	$D = D_0 \left(\frac{T}{T_0}\right)^{a_T} \left(\frac{X}{X_0}\right)^{a_X}$	$D_0 = 7.36 \times 10^{-9} \text{ m}^2/\text{s}$, $T_0 = 273 \text{ K}$, $a_T = 9.5$, $X_0 = 0.35 \text{ kg/kg db}$, $a_X = 0.5$ for clay brick; $D_0 = 1.11 \times 10^{-9} \text{ m}^2/\text{s}$, $T_0 = 273 \text{ K}$, $a_T = 6.5$, $X_0 = 0.40 \text{ kg/kg db}$, $a_X = 0.5$ for burned clay	[61]
Polyvinylalcohol	$D = D_0 \exp[-E/R(1/T - 1/T_0)]$, $D_0 = \sum a_i X^i$	$T_0 = 298 \text{ K}$, $E = 3.05 \times 10^4 \text{ J/mol}$, $R = 8.314 \text{ J/(mol K)}$, $a_0 = -0.104015 \times 10^2$ $a_1 = 0.363457 \times 10^2$, $a_2 = -0.469291 \times 10^3$ $a_3 = 0.634869 \times 10^4$, $a_4 = -0.517559 \times 10^5$ $a_5 = 0.250188 \times 10^6$, $a_6 = -0.747613 \times 10^6$ $a_7 = 0.139929 \times 10^7$, $a_8 = -0.159715 \times 10^7$ $a_9 = 0.101503 \times 10^7$, $a_{10} = -0.274672 \times 10^6$	[71]
Potato, carrot	$D = D_0 \exp(-X_0/X) \exp(-T_0/T)$	$D_0 = 2.41 \times 10^{-7} \text{ m}^2/\text{s}$, $X_0 = 7.62 \times 10^{-2} \text{ kg/kg db}$ $T_0 = 1.49 \times 10^{+30} \text{ }^\circ\text{C}$ for potato; $D_0 = 2.68 \times 10^{-4} \text{ m}^2/\text{s}$ $X_0 = 8.92 \times 10^{-2} \text{ kg/kg db}$, $T_0 = 3.68 \times 10^{+30} \text{ }^\circ\text{C}$ for carrot	[72]
Silica gel	$D = D_0 \exp((E_0 - E_1 X)/T)$	$D_0 = 5.71 \times 10^{-7} \text{ m}^2/\text{s}$, $E_0 = 2450 \text{ K}$, $E_1 = 1400 \text{ K/(kg/kg db)}$	[73]

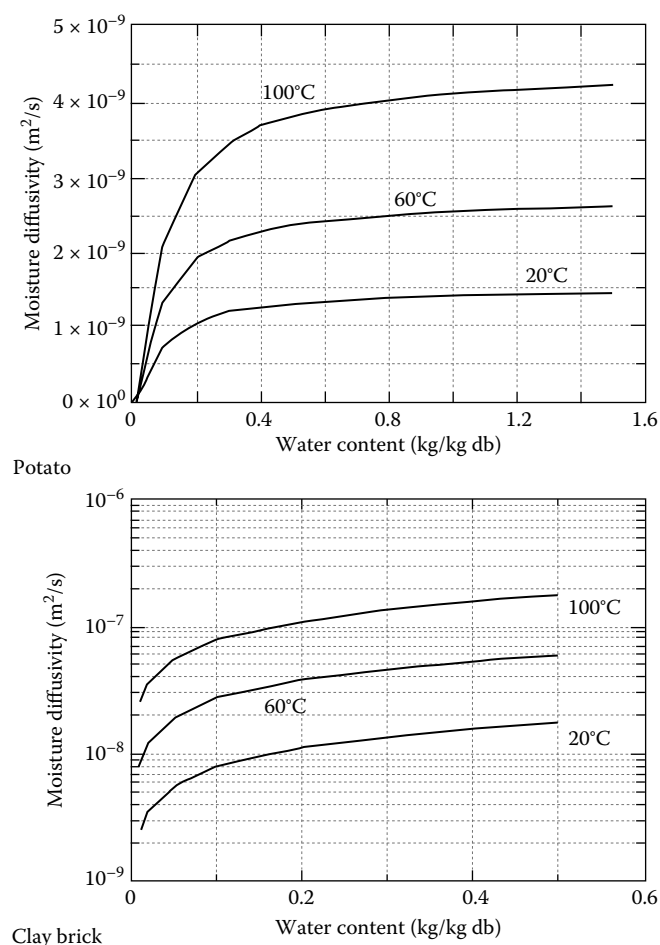


FIGURE 4.5 Effect of material moisture content and temperature on moisture diffusivity. (Data for potato are from Kiranoudis, C.T. et al., *Drying Technol.*, 10(4), 1097, 1992 and data for clay brick are from Haertling, M., in *Drying'80*, Vol. 1, Mujumdar, A.S. (ed.), Hemisphere Publishing, New York, 1980, pp. 88–98.)

presented in Figure 4.5. Both materials exhibit typical behavior. Diffusivity at low moisture content shows a steep descent when the moisture content decreases.

The equations listed in Table 4.3 resulted from fitting to experimental data. The reason for the success of this procedure is the apparent simple dependence of diffusivity upon the material moisture content and temperature, which, as stated above, can be described even by three parameters only. The equations of Table 4.3 have been chosen by the respective researchers as the most appropriate for the material listed.

A single relation for the dependence of diffusivity upon the material moisture content and temperature general enough so as to apply to all the materials would be especially useful. It is expected that such a relation will be proposed soon.

The effect of pore structure and distribution on moisture diffusion can be examined by considering the material as a two-(or multi-) phase (dry material, water, air in voids, etc.) system and by considering some structural models to express

the system geometry. Although a lot of work has been done in the analogous case of thermal conductivity, little attention has been given to the case of moisture diffusivity, and even less experimental validation of the structural models has been obtained. The similarity, however, of the relevant transport phenomena (i.e., heat and mass transfer) permits, under certain restrictions, the use of conclusions derived from one area in the other. Thus, the literature correlations for the estimation of the effective diffusion coefficient, in many cases, had been initially developed for the thermal conductivity in porous media [79].

4.2.5 THEORETICAL ESTIMATION

The prediction of the diffusion coefficients of gases from basic thermophysical and molecular properties is possible with great accuracy using the Chapman–Enskog kinetic theory. Diffusivities in liquids, on the other hand, in spite of the absence of a rigorous theory, can be estimated within an order of magnitude from the well-known equations of Stokes and Einstein (for large spherical molecules) and Wilke (for dilute solutions).

Diffusion of gases, vapors, and liquids in solids, however, is a more complex process than the diffusion in fluids because of the heterogeneous structure of the solid and its interactions with the diffusing components. As a result, it has not yet been possible to develop an effective theory for the diffusion in solids. Usually, diffusion in solids is handled by the researchers in a manner analogous to heat conduction. In the following paragraphs typical methods are described for the development of semiempirical correlations for diffusivity.

For the estimation of the diffusion coefficient in isotropic macroporous media, the relation

$$D = \left(\frac{\delta \varepsilon}{\tau^2} \right) D_A \quad (4.3)$$

has been proposed [79]. In this equation, ε is the porosity, τ is the tortuosity, δ is the constrictivity, and D_A is the vapor diffusivity in air in the absence of porous media. In spite of its simplicity, Equation 4.3 will not attain practical utility unless it is validated with additional pore space models, its parameters (ε , τ , δ) determined for a large number of systems, and the effect of the solid's moisture properly accounted for.

An equation has been derived relating the effective diffusivity of porous foodstuffs to various physical properties such as molecular weight, bulk density, vapor space permeability, water activity as a function of material moisture content, water vapor pressure, thermal conductivity, heat of sorption, and temperature [80]. A predictive model has been proposed to obtain effective diffusivities in cellular foods. The method requires data for composition, binary molecular diffusivities, densities, membrane and cell wall permeabilities, molecular weights, and water viscosity and molar volume [81]. The effect of moisture upon the effective diffusivity is taken into account via the binding energy of sorption in an equation suggested in Ref. [77].

4.3 THERMAL CONDUCTIVITY

4.3.1 DEFINITION

The *thermal conductivity* of a material is a measure of its ability to conduct heat. It can be defined using Fourier's law for homogeneous materials:

$$\frac{\partial T}{\partial t} = \left(\frac{k}{\rho c_p} \right) \nabla^2 T \quad (4.4)$$

where

k is the thermal conductivity (kW/(m K))

ρ is the density (kg/m³)

c_p is the specific heat of the material (kJ/(kg K))

T is the temperature (K)

t is the time (s)

The quantity $(k/\rho c_p)$ is the thermal diffusivity. For heterogeneous materials, the effective thermal conductivity is used in conjunction with Fourier's law.

Equation 4.4 is used in cases in which heat transfer during drying takes place through conduction (internally controlled drying). This, for example, is the situation when drying large particles, relatively immobile, that are immersed in the heat transfer medium.

As far as heat and mass transfer is concerned, the drying process is internally controlled whenever the respective Biot number (Bi_H , Bi_M) is greater than 1 [5].

4.3.2 METHODS OF EXPERIMENTAL MEASUREMENT

The effective thermal conductivity can be determined using the methods presented in Table 4.5, which includes the relevant references. Measurement techniques for thermal conductivity can be grouped into steady-state and transient-state methods. Transient methods are more popular because they can be run for as short as 10 s, during which time the moisture migration and other property changes are kept minimal.

4.3.2.1 Steady-State Methods

In steady-state methods, the temperature distribution of the sample is measured at steady state, with the sample

placed between a heat source and a heat sink. Different geometries can be used, those for longitudinal heat flow and radial heat flow.

4.3.2.2 Longitudinal Heat Flow (Guarded Hot Plate)

The longitudinal heat flow (guarded hot plate) method is regarded as the most accurate and most widely used apparatus for the measurement of thermal conductivity of poor conductors of heat. This method is most suitable for dry homogeneous specimens in slab forms. The details of the technique are given by the American Society for Testing and Materials (ASTM) Standard C-177 [82].

4.3.2.3 Radial Heat Flow

Whereas the longitudinal heat flow methods are most suitable for slab specimens, the radial heat flow techniques are used for loose, unconsolidated powder or granular materials. The methods can be classified as follows:

- Cylinder with or without end guards
- Sphere with central heating source
- Concentric cylinder comparative method

4.3.2.4 Unsteady State Methods

Transient-state or unsteady-state methods make use of either a line source of heat or plane sources of heat. In both cases, the usual procedure is to apply a steady heat flux to the specimen, which must be initially in thermal equilibrium, and to measure the temperature rise at some point in the specimen, resulting from this applied flux [83]. The Fitch method is one of the most common transient methods for measuring the thermal conductivity of poor conductors. This method was developed in 1935 and was described in the National Bureau of Standards Research Report No. 561. Experimental apparatus is commercially available.

4.3.2.5 Probe Method

The probe method is one of the most common transient methods using a line heat source. This method is simple and quick. The probe is a needle of good thermal conductivity that is provided with a heater wire over its length and some means of measuring the temperature at the center of its length. Having the probe embedded in the sample, the temperature response of the probe is measured in a step change of heat source and the thermal conductivity is estimated using the transient solution of Fourier's law. Detailed descriptions as well as the necessary modifications for the application of the above-mentioned methods in food systems are given in Refs. [83,89,90].

4.3.3 DATA COMPILATION

Despite the limited data of effective moisture diffusivity, a lot of data are reported in the literature for thermal conductivity. Data for mainly homogeneous materials are available in handbooks such as the *Handbook of Chemistry*

TABLE 4.5
Methods for the Experimental
Measurement of Thermal Conductivity

Method	Refs.
<i>Steady-state method</i>	
Longitudinal heat flow (guarded hot plate)	[82]
Radial heat flow	[83]
<i>Unsteady-state method</i>	
Fitch	[84,85]
Plane heat source	[86]
<i>Probe method</i>	[87,88]

TABLE 4.6
Effective Thermal Conductivity in Some Materials

Material	Temperature (°C)	Thermal Conductivity (W/(m K))	Refs.
Aerogel, silica	38	0.022	[94]
Asbestos	427	0.225	[94]
Bakelite	20	0.232	[94]
Beef, 69.5% water	−18	0.622	[99]
Beef fat, 9% water	−10	0.311	[100]
Brick, common	20	0.173–0.346	[94]
Brick, fire clay	800	1.37	[94]
Carrots	−15 to −19	0.622	[101]
Concrete	20	0.813–1.40	[94]
Corkboard	38	0.043	[94]
Diatomaceous earth	38	0.052	[94]
Fiber-insulating board	38	0.042	[94]
Fish	−20	1.50	[100]
Fish, cod, and haddock	−20	1.83	[102]
Fish muscle	−23	1.82	[103]
Glass, window	20	0.882	[94]
Glass wool, fine	38	0.054	[94]
Glass wool, packed	38	0.038	[94]
Ice	0	2.21	[94]
Magnesia	38	0.067	[94]
Marble	20	2.77	[94]
Paper		0.130	[94]
Peach	18–27	1.12	[104]
Peas	18–27	1.05	[104]
Peas	−12 to −20	0.501	[101]
Plums	−13 to −17	0.294	[101]
Potato	−10 to −15	1.09	[101]
Potato flesh	18–27	1.05	[104]
Rock wool	38	0.040	[94]
Rubber, hard	0	0.150	[94]
Strawberries	18–27	1.35	[104]
Turkey breast	−25	0.167	[100]
Turkey leg	−25	1.51	[100]
Wood, oak	21	0.207	[94]

and Physics [91], the *Chemical Engineers' Handbook* [92], *ASHRAE Handbook of Fundamentals* [93], Rohsenow and Choi [94], and many others. For foods and agricultural products, data are available in Refs. [83,88,95–97]. For selected pharmaceutical materials, data are presented by Pakowski and Mujumdar [98].

Some data for thermal conductivity are presented in Table 4.6. These values are distributed as shown in Figure 4.6. The distribution is different from that of moisture diffusivity (Figure 4.2), which is normal. For thermal conductivity, the values are uniformly distributed in the range 0.25–2.25 W/(m K), whereas a lot of data are accumulated below 0.25 W/(m K).

4.3.4 FACTORS AFFECTING THERMAL CONDUCTIVITY

The thermal conductivity of homogeneous materials depends on temperature and composition, and empirical equations

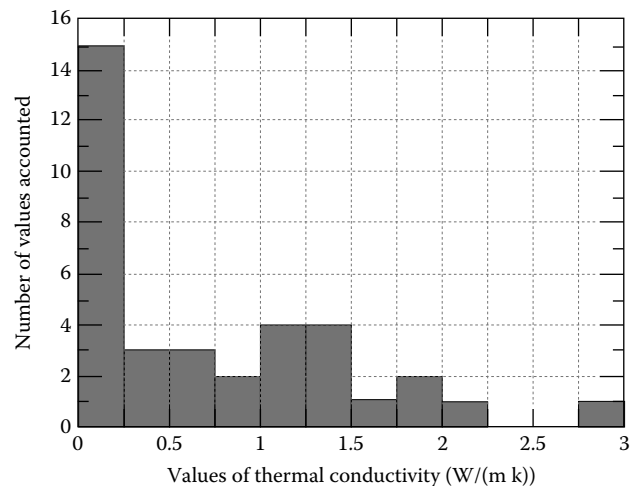


FIGURE 4.6 Distribution of thermal conductivity values (data from Table 4.5).

are used for its estimation. For each material, polynomial functions of first or higher order are used to express the temperature effect. A large number of empirical equations for the calculation of thermal conductivity as a function of temperature and humidity are available in the literature [83,92].

For heterogeneous materials, the effect of geometry must be considered using structural models. Utilizing Maxwell's and Eucken's work in the field of electricity, Luikov et al. [105] initially used the idea of an elementary cell, as representative of the model structure of materials, to calculate the effective thermal conductivity of powdered systems and solid porous materials. In the same paper, a method is proposed for the estimation of the effective thermal conductivity of mixtures of powdered and solid porous materials.

Since then, a number of structural models have been proposed, some of which are given in Table 4.7. The perpendicular model assumes that heat conduction is perpendicular to alternate layers of the two phases, whereas the parallel model assumes that the two phases are parallel to heat conduction. In the mixed model, heat conduction is assumed to take place by a combination of parallel and perpendicular heat flow. In the random model, the two phases are assumed to be mixed randomly. The Maxwell model assumes that one phase is continuous, whereas the other phase is dispersed as uniform spheres. Several other models have been reviewed in Refs. [107,110,111], among others.

The use of some of these structural models to calculate the thermal conductivity of a hypothetical porous material is presented in Figure 4.7. The parallel model gives the larger value for the effective thermal conductivity, whereas the perpendicular model gives the lower value. All other models predict values in between. The use of structural

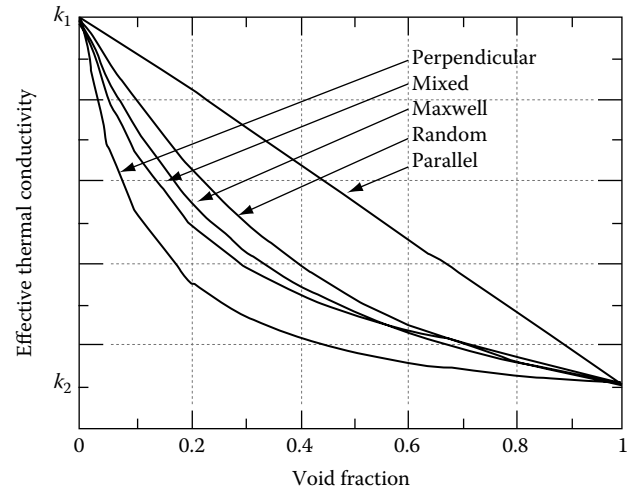


FIGURE 4.7 Effect of geometry on the thermal conductivity of heterogeneous materials using structural models.

models has been successfully extended to foods [108,112], which exhibit a more complex structure than that of other materials, whereas this structure often changes during the heat conduction.

A systematic general procedure for selecting suitable structural models, even in multiphase systems, has been proposed in Ref. [113]. This method is based on a model discrimination procedure. If a component has unknown thermal conductivity, the method estimates the dependence of the temperature on the unknown thermal conductivity, and the suitable structural models simultaneously.

An excellent example of applicability of the above is in the case of starch, a useful material in extrusion. The granular starch consists of two phases, the wet granules and the air-vapor mixture in the intergranular space. The starch granule also consists of two phases, the dry starch and the water. Consequently, the thermal conductivity of the granular starch depends on the thermal conductivities of pure materials (i.e., dry pure starch, water, air, and vapor, all functions of temperature) and the structures of granular starch and the starch granule. It has been shown that the parallel model is the best model for both the granular starch and the starch granule [113]. These results led to simultaneous experimental determination of the thermal conductivity of dry pure starch versus temperature. Dry pure starch is a material that cannot be isolated for direct measurement.

4.3.5 THEORETICAL ESTIMATION

As in the case of the diffusion coefficient, the thermal conductivity in fluids can be predicted with satisfactory accuracy using theoretical expressions, such as the formulas of Chapman and Enskog for monoatomic gases, of Eucken for polyatomic ones, or of Bridgman for pure liquids. The thermal conductivity of solids, however, has not yet been predicted using basic thermophysical or molecular properties, just like the analogous diffusion coefficient. Usually, the

TABLE 4.7
Structural Models for Thermal Conductivity
in Heterogeneous Materials

Model	Equation	Refs.
Perpendicular (series)	$\frac{1}{k} = \frac{(1-\varepsilon)}{k_1} + \frac{\varepsilon}{k_2}$	[106,107]
Parallel	$k = (1-\varepsilon)k_1 + \varepsilon k_2$	[106,107]
Mixed	$\frac{1}{k} = \frac{1-F}{(1-\varepsilon)k_1 + \varepsilon k_2} + F \left(\frac{1-\varepsilon}{k_1} + \frac{\varepsilon}{k_2} \right)$	[106,107]
Random	$k = k_1^{(1-\varepsilon)} k_2^\varepsilon$	[106,107]
Effective medium theory	$k = k_1 [b + (b^2 + 2(k_1/k_2)(Z-2))^{1/2}]$ $b = \frac{Z(1-\varepsilon)/2 - 1 + (k_2/k_1)(\varepsilon Z/2 - 1)}{(Z-2)}$	[108]
Maxwell	$k = \frac{k_2 [k_1 + 2k_2 - 2(1-\varepsilon)(k_2 - k_1)]}{k_1 + 2k_2 + (1-\varepsilon)(k_2 - k_1)}$	[109]

k , effective thermal conductivity; k_1 , thermal conductivities of phase i ;
 ε , void fraction of phase 2; F , Z , parameters.

thermal conductivities of solids must be established experimentally since they depend upon a large number of factors that cannot be easily measured or predicted.

A large number of correlations are listed in the literature for the estimation of thermal conductivity as a function of characteristic properties of the material. Such relations, however, have limited practical utility since the values of the necessary properties are not readily available.

A method has been developed for the prediction of thermal conductivity as a function of temperature, porosity, material skeleton thermal conductivity, thermal conductivity of the gas in the porous, mechanical load on the porous material, radiation, and optical and surface properties of the material's particles [105]. The method produced satisfactory results for a wide range of materials (quartz sand, powdered Plexiglas, perlite, silica gel, etc.).

It has been proposed that the thermal conductivity of wet beads of granular material be estimated as a function of material content and the thermal conductivity of each of the three phases [114]. The results of the method were validated in a small number of materials such as crushed marble, slate, glass, and quartz sand.

Empirical equations for estimating the thermal conductivity of foods as a function of their composition have been proposed in the literature. In particular, it has been suggested that the thermal conductivity of foods is a first-degree function of the concentrations of the constituents (water, protein, fat, carbohydrate, etc.) [97].

4.4 INTERPHASE HEAT AND MASS TRANSFER COEFFICIENTS

4.4.1 DEFINITION

The interphase heat transfer coefficient is related to heat transfer through a relative stagnant layer of the flowing air, which is assumed to adhere to the surface of the solid during drying (generally heating or cooling). It may be defined as the proportionality factor in the equation (Newton's law)

$$Q = h_H A (T_A - T) \quad (4.5)$$

where

h_H (kW/(m² K)) is the surface heat transfer coefficient at the material–air interface

Q (kW) is the rate of heat transfer

A (m²) is the effective surface area

T (K) is the solid temperature at the interface

T_A (K) is the bulk air temperature

By analogy, a surface mass transfer coefficient can be defined using the following equation:

$$J = h_M A (X_A - X_{AS}) \quad (4.6)$$

where

h_M (kg/(m² s)) is the surface mass transfer coefficient at the material–air interface

J (kg/s) is the rate of mass transfer

A (m²) is the effective surface area

X_{AS} (kg/kg) and X_A (kg/kg) are the air humidities at the solid interface and the bulk air

Equations 4.5 and 4.6 are used in cases in which the drying is externally controlled. This occurs when the Biot number (Bi_H , Bi_M) for heat and mass transfer is less than 0.1 [5].

Volumetric heat and mass transfer coefficients are often used instead of surface heat and mass transfer coefficients. They can be defined using the equations

$$h_{VH} = \alpha h_H \quad (4.7)$$

$$h_{VM} = \alpha h_M \quad (4.8)$$

where α is the specific surface defined as follows:

$$\alpha = \frac{A}{V} \quad (4.9)$$

where

A (m²) is the effective surface area

V (m³) is the total volume of the material

Different coefficients can be defined using different driving forces.

4.4.2 METHODS OF EXPERIMENTAL MEASUREMENT

The methods of experimental measurement of heat and mass transfer coefficients are summarized in Table 4.8, and resulted mainly from heat and mass transfer investigations in packed beds. Heat transfer techniques are either steady or unsteady state. In steady-state methods, the heat flow is

TABLE 4.8
Methods for the Experimental Measurement of Heat and Mass Transfer Coefficients

Method	Refs.
<i>Steady-state heating methods</i>	
Material heating	[115]
Wall heating	[116]
Microwave heating	[117]
<i>Unsteady-state heating methods</i>	
Step change of input air temperature	[118,119]
Pulse change of input air temperature	[120,121]
Cyclic temperature variation of input air	[122,123]
Constant rate drying experiments	[124,125]
Simultaneous estimation of transport properties using drying experiments	See Section 4.7

measured together with the temperatures, and the heat transfer coefficient is obtained using Newton's law. Three different methods for heating are presented in Table 4.8. In unsteady-state techniques, the temperature of the outlet air is measured as a response to variations of the inlet air temperature. A transient model incorporating the heat transfer coefficient is used for analysis. Step, pulse, or cyclic temperature variations of the input air temperature have been used. Drying experiments during the constant drying rate period have also been used for estimating heat and mass transfer coefficients. A generalization of this method for simultaneous estimation of transport properties using drying experiments is presented in Section 4.7.

4.4.3 DATA COMPILATION

All the data available in the literature are in the form of empirical equations, and they are examined in the next section.

4.4.4 FACTORS AFFECTING THE HEAT AND MASS TRANSFER COEFFICIENTS

Both heat and mass transfer coefficients are influenced by thermal and flow properties of the air and, of course, by the geometry of the system. Empirical equations for various geometries have been proposed in the literature. Table 4.9 summarizes the most popular equations used for drying.

TABLE 4.9
Equations for Estimating Heat and Mass Transfer Coefficients

Equation No.	Geometry	Equation	Refs.
T9.1	Packed beds (heat transfer)	$j_H = 1.06Re^{-0.41}$ $350 < Re < 4,000$	[126]
T9.2	Packed beds (mass transfer)	$j_M = 1.82Re^{-0.51}$ $40 < Re < 350$	[127]
T9.3	Flat plate (heat transfer, parallel flow)	$j_H = 0.036Re^{-0.2}$ $500,000 < Re$	[128]
T9.4	Flat plate (heat transfer, parallel flow)	$h_H = 0.0204G^{0.8}$ $0.68 < G < 8.1$; $45 < T < 150^\circ\text{C}$	[129]
T9.5	Flat plate (heat transfer, perpendicular flow)	$h_H = 1.17 G^{0.37}$ $1.1 < G < 5.4$	[129]
T9.6	Flat plate (heat transfer, parallel flow)	$Nu = 0.036(Re^{0.8} - 9,200)Pr^{0.43}$ $1.0 \times 10^5 < Re < 5.5 \times 10^6$	[130]
T9.7	Packed beds (heat transfer)	$Nu' = (0.5Re'^{1/2} + 0.2Re'^{2/3})Pr^{1/3}$ $2 \times 10^3 < Re' < 8 \times 10^3$	[130]
T9.8	Rotary dryer (heat transfer)	$j_H = 1.0Re^{-0.5}Pr^{1/3}$	[131]
T9.9	Rotary dryer (heat transfer)	$Nu = 0.33Re^{0.6}$	[131]
T9.10	Rotary dryer (heat transfer)	$h_{vH} = 0.52G^{0.8}$	[131]
T9.11a	Fluidized beds (heat transfer)	$Nu = 0.0133Re^{1.6}$ $0 < Re < 80$	[6]
T9.11b	Fluidized beds (heat transfer)	$Nu = 0.316Re^{0.8}$ $80 < Re < 500$	[6]
T9.12a	Fluidized beds (mass transfer)	$Sh = 0.374Re^{1.18}$ $0.1 < Re < 15$	[6]
T9.12b	Fluidized beds (mass transfer)	$Sh = 2.01Re^{0.5}$ $15 < Re < 250$	[6]
T9.13	Droplets in spray dryer (heat transfer)	$Nu = 2 + 0.6Re^{1/2}Pr^{1/3}$ $2 < Re < 200$	[132]
T9.14	Droplets in spray dryer (mass transfer)	$Sh = 2 + 0.6Re^{1/2}Sc^{1/3}$ $2 < Re < 200$	[132]
T9.15	Spouted beds (heat transfer)	$Nu = 5.0 \times 10^{-4} Re_s^{1.46} (u/u_s)^{1/3}$	[6]
T9.16	Spouted beds (mass transfer)	$Sh = 2.2 \times 10^{-4} Re^{1.45} (D/H_0)^{1/3}$	[6]
T9.17	Pneumatic dryers (heat transfer)	$Nu = 2 + 1.05Re^{1/2}Pr^{1/3}Gu^{0.175}$ $Re < 1,000$	[6]
T9.18	Pneumatic dryers (mass transfer)	$Sh = 2 + 1.05Re^{1/2}Pr^{1/3}Gu^{0.175}$ $Re < 1,000$	[6]
T9.19	Impingement drying	Several equations for various configurations	[133–135]

For nomenclature, see Table 4.10.

TABLE 4.10
Dimensionless Groups of Physical Properties

Name	Definition
Biot for heat transfer	$Bi_H = h_H d / 2k$
Biot for mass transfer	$Bi_M = h_M d / 2\rho_A D_A$
Gukhman number	$Gu = (T_A - T) / T_A$
Heat transfer factor	$j_H = St Pr^{2/3}$
Mass transfer factor	$j_M = (h_M / u_A \rho_A) Sc^{2/3}$
Nusselt number	$Nu = h_H d / k_A$
Prandtl number	$Pr = c_p \mu / k_A$
Reynolds number	$Re = u_A \rho_A d / \mu$
Schmidt number	$Sc = \mu / \rho_A D_A$
Sherwood number	$Sh = h_M d / \rho_A D_A$
Stanton number	$St = h_H / u_A \rho_A c_p$

c_p , specific heat (kJ/(kg K)); d , particle diameter (m); D_A , diffusivity in solid (m²/s); D_A , vapor diffusivity in air (m² s); ϵ , void fraction in packed bed; G , mass flow rate of air (kg/(m² s)); h_H , heat transfer coefficient (kW/(m² K)); h_M , mass transfer coefficient (kg/(m² s)); h_{vH} , volumetric heat transfer coefficient (kW/(m³ K)); h_{vM} , volumetric mass transfer coefficient (kg/(m³ s)); k , thermal conductivity of solid (kW/(m·K)); k_A , thermal conductivity of air (kW/(m·K)); μ , dynamic viscosity of air (kg/(m·s)); Nu' , $Nu' = Nu \cdot \epsilon / (1 - \epsilon)$; Q_A , density of air (kg/m³); Re' , $Re' = Re (1 - \epsilon)$; Re_s , Re based on u_s instead of u ; T_A , air temperature (°C); T , material temperature (°C); u_A , air velocity (m/s); u_s , air velocity for incipient spouting (m/s).

The empirical equations incorporate dimensionless groups, which are defined in Table 4.10. Some nomenclature needed for understanding Table 4.9 is also included in Table 4.10.

Equations T9.1 through T9.5 in Table 4.9 are the most widely used equations in estimating heat and mass transfer coefficients for simple geometries (packed beds, flat plates).

For packed beds, the literature contains many references. In 1965, Barker reviewed 244 relevant papers [136]. The equation suggested by Whitaker [130] is selected and presented in Table 4.9 as Equation T9.7. It has been obtained by fitting to data of several investigators (see Refs. [126,127]). Equation T9.6 for flat plates comes from the same investigation [130], and it is also included in Table 4.9. In drying of granular materials, the equations reviewed in Ref. [137] should be examined.

Rotary dryers are usually controlled by heat transfer. Thus, Equations T9.8 through T9.10 in Table 4.9 are proposed in Ref. [131] for the estimation of the corresponding heat transfer coefficients.

Heat and mass transfer in fluidized beds have been discussed in Refs. [6,138–141]. The latter reviewed the most important correlations and proposed Equations T9.11 and T9.12 of Table 4.9 for the calculation of heat and mass transfer coefficients, respectively. Further information for fluidized bed drying can be found in Ref. [142].

Vibration can intensify heat and mass transfer between the particles and gas. The following correction has been

suggested for the heat and mass transfer coefficients when vibration occurs [6]

$$h_{H'} = h_H \left(\frac{A' f'}{u_A} \right)^{0.65} \quad (4.10)$$

$$h_{M'} = h_M \left(\frac{A' f'}{u_A} \right)^{0.65} \quad (4.11)$$

where

u (m/s) is the air velocity

A (m) is the vibration amplitude

f (s⁻¹) is the frequency of vibration

Further information on vibrated bed dryers can be found in Ref. [143].

For spray dryers, the popular equation of Ranz and Marshall [132] is presented in Table 4.9 (Equations T9.13 and T9.14). They correlated data obtained for suspended drops evaporating in air.

Heat and mass transfer in a spouted bed has not been fully investigated yet because of the complex character of the flow path of the particles in a bed with zones under different aerodynamic conditions [6]. However, Equations T9.15 and T9.16 of Table 4.9 can be used.

Heat transfer coefficients for pneumatic dryers have been reviewed in Ref. [6]. The majority of authors examined and use an equation similar to Equations T9.13 and T9.14 of Table 4.9 for spray dryers. For immobile particles, the exponent of the Re number is close to 0.5 and for free-falling particles, it is 0.8. Equation T9.17 of Table 4.9 is proposed. The mass transfer coefficient could be estimated by the analogy $Sh = Nu$ [6]. In extensive reviews [133–135], correlations for estimating heat and mass transfer coefficients in impingement drying under various configurations are discussed.

The calculated heat and mass transfer coefficients using some of the equations presented in Table 4.9 are plotted versus air velocity with some simplifications in Figures 4.8 and 4.9.

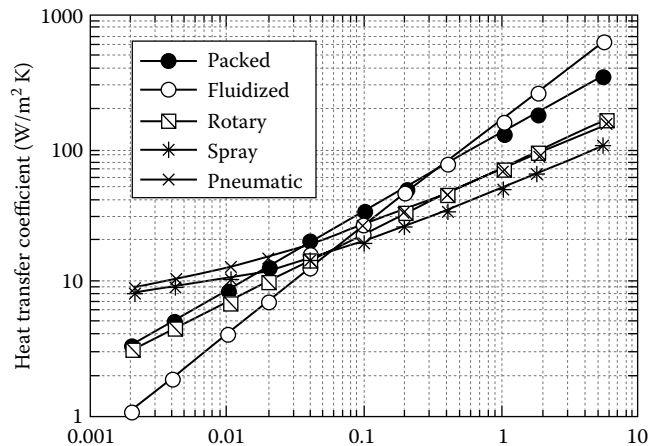


FIGURE 4.8 Heat transfer coefficients versus air velocity for some dryers (particle size 10 mm; drying conditions $T_A = 80^\circ\text{C}$, $X_A = 10$ g/kg db).

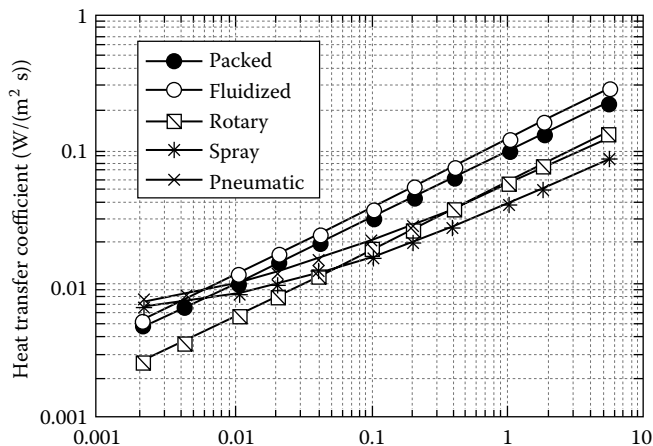


FIGURE 4.9 Mass transfer coefficients versus air velocity for some dryers (particle size 10 mm; drying conditions $T_A = 80^\circ\text{C}$, $X_A = 10 \text{ g/kg db}$).

These figures can be used to estimate approximately the heat and mass transfer coefficients for various dryers. The simplifications made for the construction of these figures concern the drying air and material conditions. For instance, the air temperature is taken as 80°C , the air humidity as 0.010 kg/kg db , and the particle size as 10 mm (typical drying conditions). For other conditions, the equations of Table 4.9 should be used.

4.4.5 THEORETICAL ESTIMATION

No theory is available for estimating the heat and mass transfer coefficients using basic thermophysical properties. The analogy of heat and mass transfer can be used to obtain mass transfer data from heat transfer data and vice versa. For this purpose, the Chilton–Colburn analogies can be used [129]

$$j_M = j_H = \frac{f}{2} \quad (4.12)$$

where

f is the well-known Fanning friction factor for the fluid
 j_H and j_M are the heat and mass transfer factors defined in Table 4.10

Discrepancies of the above classical analogy have been discussed in Ref. [144].

In air conditioning processes, the heat and mass transfer analogy is usually expressed using the Lewis relationship

$$\frac{h_H}{h_M} = c_p \quad (4.13)$$

where c_p (kJ/(kg K)) is the specific heat of air.

4.5 DRYING CONSTANT

4.5.1 DEFINITION

The transport properties discussed above (moisture diffusivity, thermal conductivity, interface heat, and mass transfer coefficients) describe completely the drying kinetics. However, in the literature sometimes (mainly in foods, especially in cereals) instead of the above transport properties, the drying constant K is used. The drying constant is a combination of these transport properties.

The drying constant can be defined using the so-called thin-layer equation. Lewis suggested that during the drying of porous hygroscopic materials, in the falling rate period, the rate of change in material moisture content is proportional to the instantaneous difference between material moisture content and the expected material moisture content when it comes into equilibrium with the drying air [145]. It is assumed that the material layer is thin enough or the air velocity is high so that the conditions of the drying air (humidity, temperature) are kept constant throughout the material. The thin-layer equation has the following form:

$$\frac{-dX}{dt} = K(X - X_e) \quad (4.14)$$

where

X (kg/kg db) is the material moisture content

X_e (kg/kg db) is the material moisture content in equilibrium with the drying air

t (s) is the time

A review of several other thin-layer equations can be found in Refs. [76,146].

Equation 4.14 constitutes an effort toward a unified description of the drying phenomena regardless of the controlling mechanism. The use of similar equations in the drying literature is ever increasing. It is claimed, for example, that they can be used to estimate the drying time as well as for the generalization of the drying curves [6].

The drying constant K is the most suitable quantity for purposes of design, optimization, and any situation in which a large number of iterative model calculations are needed. This stems from the fact that the drying constant embodies all the transport properties into a simple exponential function, which is the solution of Equation 4.14 under constant air conditions. On the other hand, the classical partial differential equations, which analytically describe the four prevailing transport phenomena during drying (internal–external, heat–mass transfer), require a lot of time for their numerical solution and thus are not attractive for iterative calculations.

4.5.2 METHODS OF EXPERIMENTAL MEASUREMENT

The measurement of the drying constant is obtained from drying experiments. In a drying apparatus, the air temperature, humidity, and velocity are controlled and kept constant,

whereas the material moisture content is monitored versus time. The drying constant is estimated by fitting the thin-layer equation to experimental data.

4.5.3 FACTORS AFFECTING THE DRYING CONSTANT

The drying constant depends on both material and air properties as it is a phenomenological property representative of several transport phenomena. So, it is a function of material moisture content, temperature, and thickness, as well as air humidity, temperature, and velocity.

Some relationships describing the effect of the above factors on the drying constant are presented in Table 4.11. Equations T11.1 and T11.2 are Arrhenius-type equations, which take into account the temperature effect only. The effect of water activity can be considered by modifying the activation energy (Equation T11.1) on the preexponential factor (Equation T11.2). Equations T11.1 and T11.2 consider the same factors in a different form. Equation T11.4 takes into account only the air velocity effect, whereas Equation T11.5 considers all the factors affecting the drying constant. Table 4.12 lists parameter values for typical equations of Table 4.11.

Equations T11.2 and T11.5 were applied to shelled corn [151] and to green pepper [35], respectively, and the results are presented in Figure 4.10. The effects of air temperature and velocity, as well as particle dimensions, are shown for green pepper drying, whereas the air temperature and the

small air–water activity effects are shown for the low air temperature drying of wheat.

4.5.4 THEORETICAL ESTIMATION

It is impossible to estimate an empirical constant using theoretical arguments. The estimation of an empirical constant using theoretical arguments has little, if any, meaning. Nevertheless, if we assume that for some drying conditions the controlling mechanism is the moisture diffusion in the material, then the drying constant can be expressed as a function of moisture diffusivity. For slabs, for example, the following equation is valid:

$$K = \frac{\pi^2 D}{L^2} \quad (4.15)$$

where

D (m²/s) is the effective diffusivity

L (m) is the thickness of the slab

4.6 EQUILIBRIUM MOISTURE CONTENT

4.6.1 DEFINITION

A knowledge of the state of thermodynamic equilibrium between the surrounding air and the solid is a basic prerequisite for drying, as it is for any similar mass transfer situation.

TABLE 4.11
Effect of Various Factors on the Drying Constant

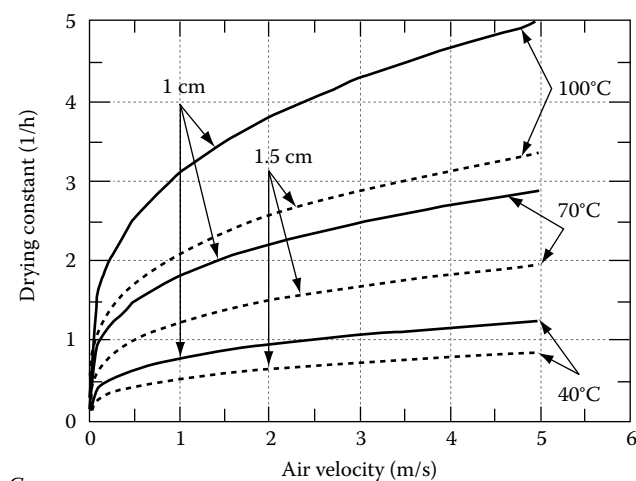
Equation No.	Materials of Application	Equation	Refs.
T11.1a	Grains, barley, various tropical agricultural products	$K(T_A) = b_0 \exp[-b_1/T_A]$	[75,147,148]
T11.1b	Barley, wheat	$K(T_A) = b_0 \exp[-b_1/(b_2 + b_3 T_A)]$	[149]
T11.2a	Melon	$K(a_w, T_A) = b_0 \exp[-(b_1 + b_2 a_w)/T_A]$	[150]
T11.2b	Corn, shelled	$K(a_w, T_A) = b_0 \exp(-b_1 a_w) \exp[-b_2/(b_3 + b_4 T_A)]$	[151]
T11.3a	Rice	$K(a_w, T_A) = b_0 + b_1 T_A - b_2 a_w$	[152]
T11.3b	Wheat	$K(a_w, T_A) = b_0 + b_1 T_A^2 - b_2 a_w$	[153]
T11.4	Carrot	$K(u_A) = \exp(-b_1 + b_2 \ln u_A)$	[154]
T11.5	Potato, onion, carrot, pepper	$K(a_w, T_A, d, u_A) = b_0 a_w^{b_1} T_A^{b_2} d^{b_3} u_A^{b_4}$	[35]

K , drying constant; T_A , temperature; u_A , air velocity; a_w , water activity; d , particle diameter; b_i , parameters.

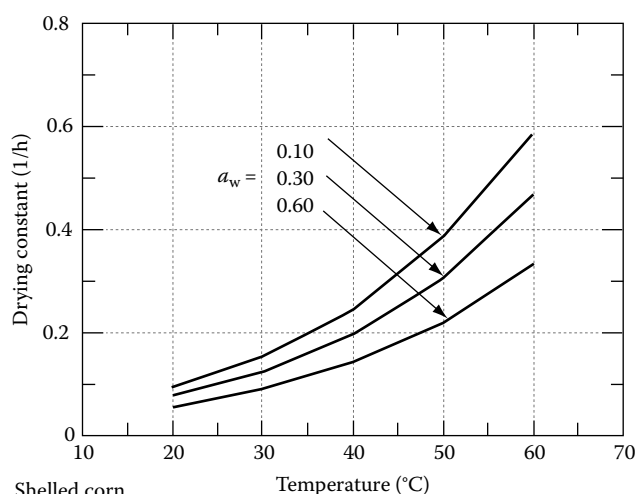
TABLE 4.12
Application Examples

Material	Equation	Constants	Refs.
Shelled corn	$K = b_0 \exp(-b_1 a_w) \exp[-b_2/(b_3 + b_4 T_A)]$ $0.1 < a_w < 0.6, 23.5 < T_A < 56.9^\circ\text{C}$	$b_0 = 170/\text{s}, b_1 = 1.15, b_2 = 8259$ $b_3 = 492, b_4 = 1.8/^\circ\text{C}$	[151]
Green pepper	$K = b_0 X_A^{b_1} T_A^{b_2} d^{b_3} u_A^{b_4}$ $0.006 < X_A < 0.022 \text{ kg/kg db}$ $60 < T_A < 90^\circ\text{C}, 0.005 < d < 0.015 \text{ m}, 3 < u_A < 5 \text{ m/s}$	$b_0 = 1.11 \times 10^{-8}/\text{s}, b_1 = 9.03 \times 10^{-2},$ $b_2 = 1.54, b_3 = -0.982, b_4 = 0.293$	[35]

Source: Brunauer, S. et al., *Am. Chem. Soc. J.*, 62, 1723, 1940. With permission.



Green pepper



Shelled corn

FIGURE 4.10 Effect of various factors on the drying constant. (Data for green pepper are from Kiranoudis, C.T. et al., *Drying Technol.*, 10(4), 995, 1992 and data for shelled corn are from Westerman, P.W. et al., *Trans. ASAE*, 16, 1136, 1973.)

The moisture content of the material when it comes into equilibrium with drying air is a useful property included in most drying models. The relation between equilibrium material moisture content and the corresponding water activity for a given temperature is known as the *sorption isotherm*. The water activity a_w at the pressures and temperatures that usually prevail during drying is equal to the relative humidity of air.

The equilibrium moisture of a material can be attained either by adsorption or by desorption, as expressed by the respective isotherms of Figure 4.11. The usually observed deviation of the two curves is due to the phenomenon of hysteresis, which has not yet been quantitatively described. Many explanations for the phenomenon have been put forth that converge in that there are more active sites during the desorption than during adsorption. It is clear from Figure 4.11 that the desorption isotherm is the curve to use for the process of drying.

In essence, the sorption isotherms express the minimum value of material moisture content that can be reached by a solid

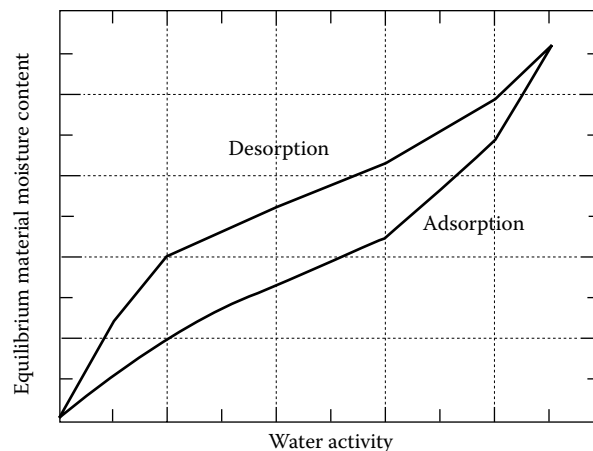


FIGURE 4.11 Hysteresis between adsorption and desorption isotherms.

during drying in relation to the relative humidity of the drying air. On the basis of such isotherms, the equilibrium material moisture content can be calculated. Such equilibrium values are necessary for the formulation of the mass transfer driving forces.

Moreover, the isotherms determine the proper storage environment and the packaging conditions, especially for foods. Through the isotherms, the isosteric heat of sorption can be determined and, hence an accurate prediction can be made of the energy requirements for the drying of a solid. The utility of the isotherm is extended to the determination of the moisture sorption mechanism as well as to the degree of bound water.

Brunauer et al. [155] classified the sorption isotherms into five different types (see Figure 4.12). The sorption isotherms of the hydrophilic polymers, such as natural fibers and foods, are of type II. The isotherms of the less hydrophilic rubbers, plastics, synthetic fibers, and foods rich in soluble components are of type III. The isotherms of certain inorganic materials (such as aluminum oxides) are of type IV. For many materials, however, the sorption isotherms cannot be properly classified since they belong to more than one type.

4.6.2 METHODS OF EXPERIMENTAL MEASUREMENT

A comprehensive review of existing experimental measuring methods is given in Refs. [156,157]. Sorption isotherms can be determined according to two basic principles, gravimetric and hygrometric.

4.6.2.1 Gravimetric Methods

During the measurement, the air temperature and the water activity are kept constant until the moisture content of the sample attains the constant equilibrium value. The air may be circulated (dynamic methods) or stagnant (static). The material weight may be registered continuously (continuous methods) or discontinuously (discontinuous methods).

4.6.2.2 Hygrometric Methods

During the measurement, the material moisture content is kept constant until the surrounding air attains the constant

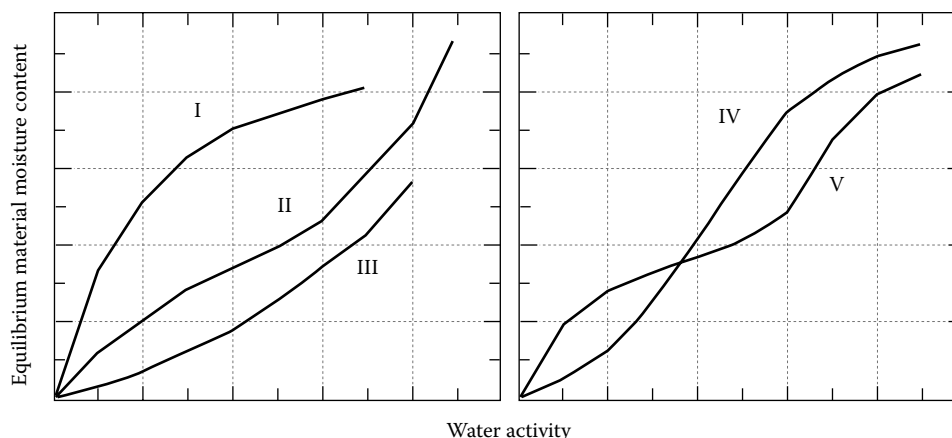


FIGURE 4.12 The five types of isotherms. (From Brunauer, S. et al., *Am. Chem. Soc. J.*, 62, 1723, 1940.)

equilibrium value. The air–water activity is measured via hygrometer or manometer.

The working group in the COST 90 bis Project has developed a reference material (microcrystalline cellulose, MCC) and a reference method for measuring water sorption isotherms, and conducted a collaborative study to determine the precision (repeatability and reproducibility) with which the sorption isotherm of the reference material may be determined by the reference method. A detailed procedure for the resulting standardized method was presented, and the factors influencing the results of the method were discussed [158–160].

4.6.3 DATA COMPILATION

A large volume of data of equilibrium moisture content appears in the literature. Data for more than 35 polymeric materials, such as natural fibers, proteins, plastics, and synthetic fibers, are given in Ref. [8]. Isotherms for 32 materials (organic and inorganic) are also given in Ref. [92]. The literature is especially rich in sorption isotherms of foods due to the fact that the value of water activity is a critical parameter for food preservation safety and quality.

A bibliography on sorption isotherms of food materials is presented in Ref. [161]. The collection comprises 2200 references, including about 900 papers with information on equilibrium moisture content of foods in defined environments. The papers are listed alphabetically according to the names of the first author, but they are also grouped according to product.

Additional bibliographies should also be mentioned. The *Handbook of Food Isotherms* contains more than 1000 isotherms, with a mathematical description of over 800 [162]. About 460 isotherms were obtained from the monograph of Ref. [163]. Data on sorption properties of selected pharmaceutical materials are presented in Ref. [98].

4.6.4 FACTORS AFFECTING THE EQUILIBRIUM MOISTURE CONTENT

Equilibrium material moisture content depends upon many factors, among which are the chemical composition, the

physical structure, and the surrounding air conditions. A large number of equations (theoretical, semiempirical, empirical) have been proposed, none of which, however, can describe the phenomenon of hysteresis. Another basic handicap of the equations is that their applicability is not satisfactory over the entire range of water activity ($0 \leq a_w \leq 1$).

Table 4.13 lists the best-known isotherm equations. The Langmuir equation can be applied in type I isotherm behavior. The Brunauer–Emmet–Tetter (BET) equation has

TABLE 4.13
Effect of Water Activity and Temperature
on the Equilibrium Moisture Content

Equation Name	Equation	Refs.
Langmuir	$a_w \left(\frac{1}{X} - \frac{1}{b_0} \right) = \frac{1}{b_0 b_1}$	[164]
Brunauer–Emmet–Tetter (BET)	$\frac{a_w}{(1 - a_w)X} = \frac{1}{b_0 b_1} + \frac{b_1 - 1}{b_0 b_1} a_w$	[165]
Halsey	$a_w = \exp \left[-\frac{b_1}{RT} \left(\frac{X}{b_2} \right)^{b_3} \right]$	[166]
Henderson	$1 - a_w = \exp \left[-b_1 T X^{b_2} \right]$	[167]
Chung and Pfof	$\ln a_w = -\frac{b_1}{RT} \exp(-b_2 X)$	[168]
Chen and Clayton	$\ln a_w = -b_1 T^{b_2} \exp(-b_3 T^{b_2} X)$	[169]
Iglesias and Chirife	$\ln a_w = -\exp \left[(-b_1 T + b_2) X^{b_3} \right]$	[170]
Guggenheim–Anderson–de Boer (GAB)	$X = \frac{b_0 b_1 b_2 a_w}{(1 - b_1 a_w)(1 - b_1 a_w + b_1 b_2 a_w)}$ $b_1 = b_{10} \exp \left(\frac{b_{11}}{RT} \right), b_2 = b_{20} \exp \left(\frac{b_{21}}{RT} \right)$	[171,172]

X , equilibrium material moisture content; a_w , water activity; T , temperature; b_1 , parameters.

been successfully applied to almost all kinds of materials, but especially to hydrophilic polymers for $a_w < 0.5$. The Halsey equation is suitable for materials of types I–III. The Henderson equation is less versatile than that of Halsey. For cereal and other field crops, the Chung and Pfof equation is considered suitable, whereas that of Iglesias and Chirife has been successfully applied on isotherms of type III (i.e., foods rich in soluble components).

The Guggenheim–Anderson–de Boer (GAB) equation is considered as the most versatile model, capable of application to situations over a wide range of water activities ($0.1 < a_w < 0.9$) and to various materials (inorganic, foods, etc.). The GAB equation is probably the most suitable for process analysis and design of drying because of its reliability, its simple mathematical form, and its wide use (with materials and water activity ranges). Table 4.14 lists parameter values of the GAB equation for some foods.

Two selected food materials are presented as an example in Figure 4.13. Potatoes exhibit a typical behavior. Equilibrium material moisture content is increased [173]. Raisins, on the other hand, exhibit an inverse temperature effect at large water activities [174]. As shown in Figure 4.13, potatoes and raisins exhibit sorption isotherms of types II and III, respectively.

The isotherms at 25°C for some organic and inorganic materials are presented in Figure 4.14 [92]. In Figure 4.14, one can observe the various isotherm types, like type I for activated charcoal and silica gel, type II for leather, type III for soap, and so on.

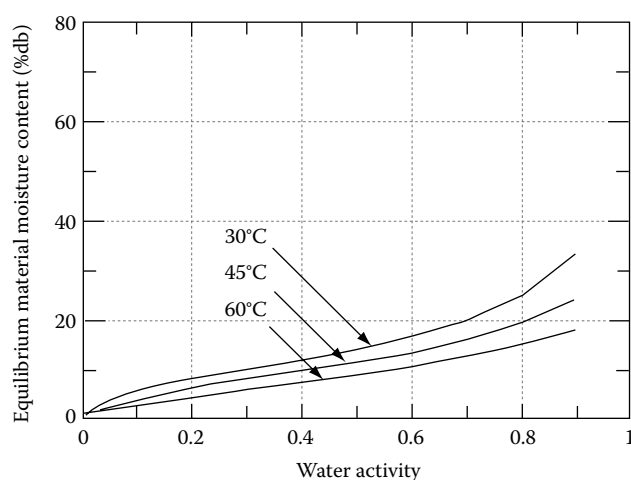
Various regression analysis methods for fitting the above equations to experimental data have been discussed in the literature. The direct nonlinear regression exhibits several advantages over indirect nonlinear regression [174]. Linear regression, on the other hand, can give highly erroneous results and should be avoided [175]. When there exist differences in the variance of the data, the direct nonlinear weighted regression method should be used [176].

TABLE 4.14

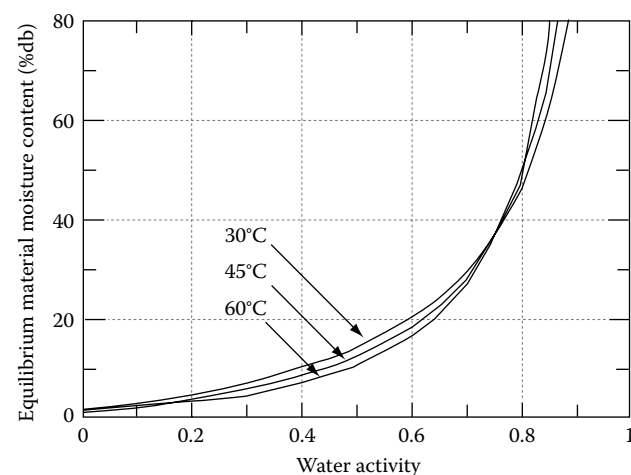
Application of the Guggenheim–Anderson–de Boer Model to Some Fruits and Vegetables

Material	b_0	$b_{10} \times 10^5$	b_{11}	b_{20}	b_{21}
Potato	8.7	1.86	34.1	5.68	6.75
Carrot	21.2	5.94	28.9	8.03	5.49
Tomato	18.2	1.99	34.5	5.52	6.70
Pepper	21.1	1.46	33.4	5.56	6.56
Onion	20.2	2.30	32.5	5.79	6.43
Raisin	12.5	0.17	22.4	1.77	-1.53
Fig	11.7	0.05	25.2	1.77	-1.55
Prune	13.3	0.07	23.9	1.82	-1.65
Apricot	15.1	0.11	21.1	2.13	-2.05

Sources: Kiranoudis, C.T. et al., *J. Food Eng.*, 20(1), 55, 1992; Maroulis, Z.B. et al., *J. Food Eng.*, 7(1), 63, 1988.



Potatoes



Sultana raisins

FIGURE 4.13 Effect of air–water activity and temperature on equilibrium material moisture content for two foods. (Data for potatoes from Kiranoudis, C.T. et al., *J. Food Eng.*, 20(1), 55, 1992 and data for sultana raisins from Maroulis, Z.B. et al., *J. Food Eng.*, 7(1), 63, 1988.)

4.7 SIMULTANEOUS ESTIMATION OF HEAT AND MASS TRANSPORT PROPERTIES FROM DRYING EXPERIMENTS

4.7.1 PRINCIPLES OF ESTIMATION

In the previous sections, methods of experimental determination of heat and mass transport properties have been discussed. These methods use special apparatus and are based on the equation of definition of the corresponding property. This section discusses the experimental determination of these properties from drying experiments. Some relevant techniques have been already discussed by Molnar [125]. However, a generalized method based on model-building techniques is presented here. The method uses a drying experimental apparatus and estimates the heat and mass transport properties as parameters of a drying model that incorporates these properties [28,43,177–181]. An outline of the method is described below.

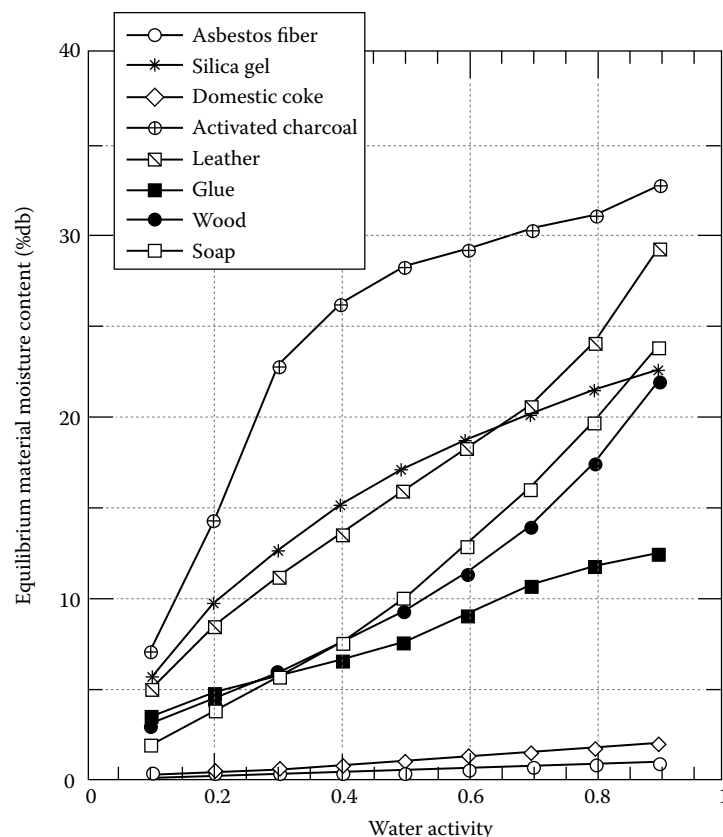


FIGURE 4.14 Equilibrium material moisture content for some organic and inorganic materials. (Data from Perry, R.H. and Chilton, C.H., *Chemical Engineers' Handbook*, 4th and 5th edns., McGraw-Hill, New York, 1963, 1973.)

First, an experimental drying apparatus is used. In such an apparatus, the air passes through the drying material and the air humidity, temperature, and velocity are controlled, whereas the material moisture content and, eventually, the material temperature are monitored versus time. Second, a mathematical model that takes into account the controlling mechanisms of heat and mass transfer is considered. This model includes the heat and mass transport properties as model parameters or, even more, includes the functional dependence of the relevant factors on the transport properties. Third, a regression analysis procedure is used to obtain the transport properties as model parameters by fitting the model to experimental data of material moisture content and temperature.

Theoretically, all the properties describing the drying kinetics could be estimated simultaneously. We can define the drying kinetics (in an analogous manner to reaction kinetics) as the dependence of factors affecting the drying on the drying rate. Drying is not a chemical reaction, but it involves simultaneous heat and mass transfer phenomena. Consequently, the properties describing these phenomena describe the drying process as well.

If, for example, the phenomena considered are

The moisture diffusion in the solid toward its external surface

The vaporization and convective transfer of the vapor into the airstream

The conductive heat transfer within the solid mass
The convective heat transfer from the air to the solid's surface

then the following properties describe the drying kinetics:

Effective moisture diffusivity
Air boundary mass transfer coefficient
Effective thermal conductivity
Air boundary heat transfer coefficient

and consequently they can be estimated.

Alternatively, if the drying constant is assumed to describe the drying kinetics by the thin-layer equation, then the drying constant can be estimated using this method.

4.7.2 EXPERIMENTAL DRYING APPARATUS

A typical drying apparatus is shown in Figure 4.15. The apparatus consists of two parts, the air conditioning section and the measuring section. The air conditioning section includes the heater, the humidifier, and the fan, which are handled via a temperature, a humidity, and a flow controller, respectively. In the measuring section, the air properties, that is, temperature, humidity, and velocity, as well as the material properties (weight and temperature) are continuously recorded. The use of a computer for online measurement and control is preferable.

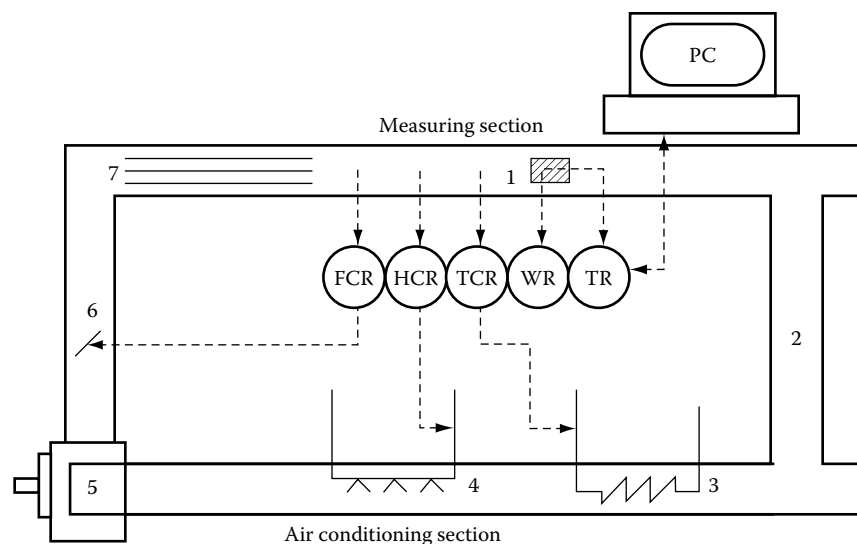


FIGURE 4.15 Typical experimental drying apparatus: (1) sample; (2) air recirculating duct; (3) heater; (4) humidifier; (5) fan; (6) valve; (7) straighteners; FCR, airflow control and recording; HCR, air humidity control and recording; TCR, air temperature control and recording; WR, sample weight recording; TR, sample temperature recording; PC, personal computer, for on-line measurement and control.

4.7.3 THE DRYING MODEL

An information flow diagram for a drying model appropriate for this method is shown in Figure 4.16. This model can calculate the material moisture content and temperature as a function of position and time whenever the air humidity, temperature, and velocity are known as a function of time, together with the model parameters. If the model takes into account the controlling mechanisms of heat and mass transfer, then the transport properties (moisture diffusivity, thermal conductivity, boundary heat and mass transfer coefficients) are included in the model as parameters. If the dependence of drying conditions (material moisture content, temperature, and thickness, as well as air humidity, temperature, and velocity) on transport properties is also considered, then the constants of the relative empirical equations are considered as model parameters. In Figure 4.16 the part of the model that contains equations

for the heat and mass transfer phenomena is termed the *process model*, whereas the equations describing the dependence of drying conditions on transport properties form the *properties model*.

In the process model, each mechanism of heat and mass transfer is expressed using a driving force and a transport property as a coefficient of proportionality between the rate and the corresponding driving force. In the properties model, several formulas can be considered. Some assumptions have been suggested in the previous sections.

4.7.4 REGRESSION ANALYSIS

The parameters of a model can be estimated by fitting the model to experimental data [182,183]. Using the model of Section 4.7.3, two regression analysis procedures can be applied [43]: transport properties estimation and transport properties equations estimation.

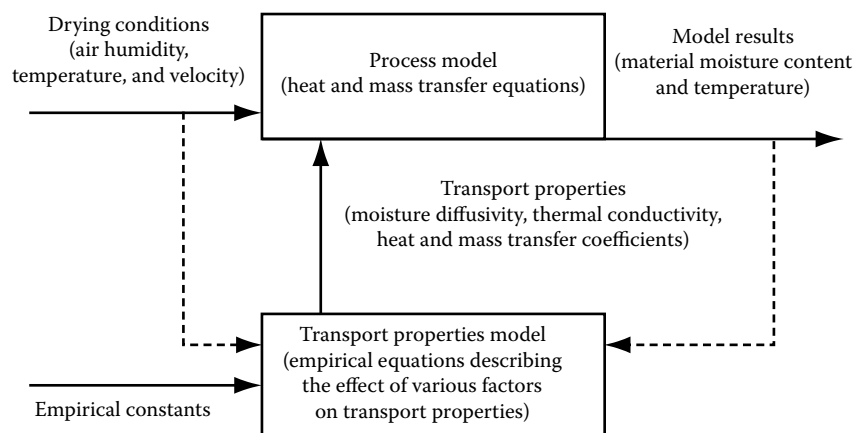


FIGURE 4.16 Model information flow diagram.

4.7.4.1 Transport Properties Estimation

It is assumed that during the drying experiments the drying conditions are not varying very much with time, and the transport properties can be considered constant (not functions of the drying conditions). The transport properties are estimated as parameters of the process model by fitting it to experimental data. Only the properties of the controlling mechanisms can be obtained. Consequently, the precision and correlations of the estimates should be examined. A model discrimination procedure is suggested to discard the noncontrolling mechanisms.

4.7.4.2 Transport Properties Equations Estimation

Several empirical equations describing the dependence of transport properties on various factors are tested using a model discrimination procedure. The constants of the empirical equations are estimated as parameters of the total model (process model plus properties model) by fitting it to experimental data.

The information flow diagram for the regression analysis proposed is shown in Figure 4.17.

4.7.5 APPLICATION EXAMPLE

The method described above is applied to a wide set of experimental data in potato drying [43].

4.7.5.1 Experimental Drying Apparatus

An experimental drying apparatus similar to that shown in Figure 4.15 was used [35]. In each experiment, the air–water activity, temperature, and velocity were controlled, and the material moisture content and temperature were monitored versus time. A total number of 100 experiments were performed for three different particle dimensions (5, 10, and 15 mm) at five air temperature (60°C, 65°C, 70°C, 75°C, and 80°C), three air velocities (3, 4, and 5 m/s) and at air humidity ranging from 6 to 22 g/kg db.

4.7.5.2 Drying Model

A mathematical drying model involving simultaneous heat and mass transfer is considered for the analysis [43]. The model considered has the following form:

Moisture diffusion into the solid

$$\frac{\partial(\rho X)}{\partial t} = \nabla(\rho D \nabla X) \quad (4.16)$$

$$D = a_0 \exp\left(\frac{-a_1}{X}\right) \exp\left(\frac{-a_2}{T}\right) \quad (4.17)$$

Boundary layer vapor transfer

$$-(\rho D \nabla X) = h_M (a_{we} - a_w) \quad (4.18)$$

$$h_M = c_0 X_A^{c_1} T_A^{c_2} u_A^{c_3} \quad (4.19)$$

Heat conduction in the solid

$$\frac{\partial(\rho h_s)}{\partial t} = \nabla(k \nabla T) \quad (4.20)$$

$$k = b_0 \exp\left(\frac{-b_1}{X}\right) \exp\left(\frac{-b_2}{T}\right) \quad (4.21)$$

Boundary layer heat transfer

$$-(k \nabla T) = h_H (T - T_A) - \Delta H_s h_M (a_{we} - a_w) \quad (4.22)$$

$$h_H = d_0 X_A^{d_1} T_A^{d_2} u_A^{d_3} \quad (4.23)$$

where

X and T are the material moisture content and temperature, respectively

u_A , a_w , and T_A are the air velocity, water activity, and temperature, respectively

The thermophysical and thermodynamic properties, material density ρ , material specific enthalpy h_s , heat of vaporization of

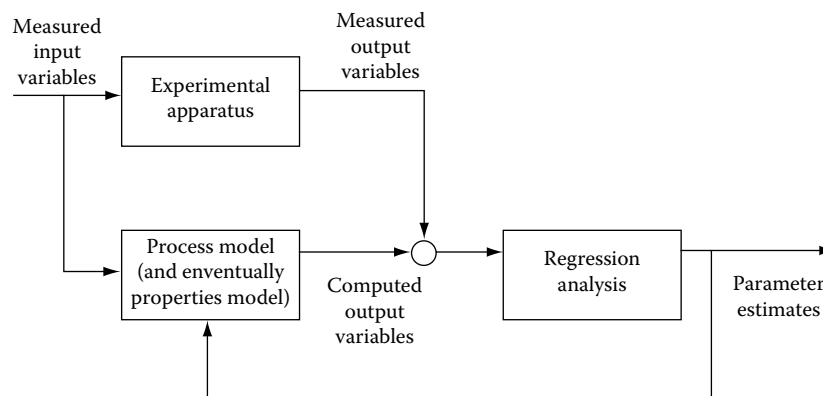


FIGURE 4.17 Regression analysis information flow diagram.

water ΔH_s , and equilibrium air–water activity a_{we} are known functions of material moisture content and temperature.

The transport properties, moisture diffusivity D , and thermal conductivity k are functions of material moisture content and temperature, whereas heat and mass transfer coefficients, h_H , h_M , are functions of air velocity, water activity, and temperature.

The following adjustable constants are introduced to the relevant properties model: a_i , b_i , c_i , d_i .

4.7.5.3 Regression Analysis

If X_i and T_i are the experimental values of material moisture content and temperature and $X_{i,c}$ and $T_{i,c}$ are the corresponding calculated values using the mathematical model, then the relative deviations between experimental and calculated values (relative residuals) can be defined as follows:

$$e_{X_i} = \frac{(X_{i,c} - X_i)}{X_i} \quad (4.24)$$

$$e_{T_i} = \frac{(T_{i,c} - T_i)}{T_i} \quad (4.25)$$

The relative standard deviations between experimental and calculated values of material moisture content S_X and temperature S_T are defined as follows:

$$S_{X^2} = \frac{\sum e_{X_i}^2}{N} \quad (4.26)$$

$$S_{T^2} = \frac{\sum e_{T_i}^2}{N} \quad (4.27)$$

where N is the number of experimental points (including different measurements and different experiments).

A linear combination of S_X and S_T is used for parameter estimation and the resulting S_X , S_T are used for model validation [181,183]. The regression analysis is performed simultaneously on all experiments.

4.7.5.4 Results

The application of the proposed method proved that:

The moisture diffusivity is a function of material moisture content and temperature.

The thermal conductivity is high and cannot be estimated from these experiments.

Heat and mass transfer coefficients are constant in the region of experimentation.

More specifically, the results obtained are as follows:

$$D = 2.94 \times 10^{-7} \text{ m}^2/\text{s} \exp(-1.58 \times 10^3 K/T) \quad (4.28)$$

$$\exp(-6.72 \times 10^{-2} \text{ kg/kg}/X) \quad (4.29)$$

$$h_M = 5.84 \times 10^{-7} \text{ kg/m}^2/\text{s} \quad (4.29)$$

$$h_H = 1.64 \times 10^{-1} \text{ W/m}^2 \quad (4.30)$$

The resulting model calculates the material moisture content and temperature close to experimental values and it is considered satisfactory.

4.8 TRANSPORT PROPERTIES OF FOODS

The transport properties of foods received much attention in the literature [184–188]. The main results presented by Saravacos and Maroulis [188] are summarized in this section. The results refer to moisture diffusivity and thermal conductivity. Recently published values of moisture diffusivity and thermal conductivity in various foods were retrieved from the literature and were classified and analyzed statistically to reveal the influence of material moisture content and temperature. Empirical models relating moisture diffusivity and thermal conductivity to material moisture content and temperature were fitted to all examined data for each material. The data were screened carefully using residual analysis techniques. A promising model was proposed based on an Arrhenius-type effect of temperature, which uses a parallel structural model to take into account the effect of material moisture content.

4.8.1 MOISTURE DIFFUSIVITY

A total of 175 papers were retrieved from the literature from which 1773 data were obtained. These data refer to more than 100 food materials classified into 11 food categories. Among the available data only 19 materials have more than 10 data, which come from more than 3 publications. The resulting model is summarized in Table 4.15 and the results of parameter estimation are presented in Table 4.16. Figures 4.18 through 4.21 present the model-calculated values for selected food materials as a function of moisture content and temperature.

TABLE 4.15
Mathematical Model for Calculating Moisture Diffusivity in Foods as a Function of Moisture Content and Temperature

Proposed mathematical model

$$D = \frac{1}{1+X} D_0 \exp \left[-\frac{E_0}{R} \left(\frac{1}{T} - \frac{1}{T_r} \right) \right] + \frac{X}{1+X} D_i \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_r} \right) \right]$$

where

D is the moisture diffusivity (m^2/s)

X is the material moisture content (kg/kg db)

T is the material temperature ($^\circ\text{C}$)

T_r is the reference temperature (60°C)

$R = 0.0083143$ is the ideal gas constant $\text{kJ}/(\text{mol K})$

Adjustable model parameters

D_0 (m^2/s) Diffusivity at moisture $X = 0$ and temperature $T = T_r$

D_i (m^2/s) Diffusivity at moisture $X = \infty$ and temperature $T = T_r$

E_0 (kJ/mol) Activation energy for diffusion in dry material at $X = 0$

E_i (kJ/mol) Activation energy for diffusion in wet material at $X = \infty$

TABLE 4.16
Parameter Estimates of the Proposed Mathematical Model

Material	No. of Papers	No. of Data	D_i (m ² /s)	D_o (m ² /s)	E_i (kJ/mol)	E_o (kJ/mol)	s.d. (m ² /s)
<i>Cereal products</i>							
<i>Corn</i>							
Dent	4	26	4.40E-09	0.00E+00	0.0	10.4	1.48E-10
Grains	3	15	1.19E-08	0.00E+00	49.4	73.1	3.30E-10
Kernel	3	28	1.15E-09	6.66E-11	10.2	57.8	3.17E-10
Pericarp	4	25	5.87E-10	5.32E-10	0.0	33.8	1.88E-11
<i>Pasta</i>	3	13	1.13E-09	0.00E+00	10.0	5.0	2.34E-11
<i>Rice</i>							
Kernel	3	21	1.39E-09	0.00E+00	16.2	2.0	7.71E-12
Rough rice	3	12	9.75E-09	0.00E+00	12.5	2.0	5.52E-11
Wheat	7	35	2.27E-09	0.00E+00	12.7	0.7	3.66E-11
<i>Fruits</i>							
Apple	6	22	1.94E-09	1.30E-09	0.0	46.3	9.53E-11
Banana	8	39	7.97E-10	1.16E-10	16.7	56.6	1.92E-10
Banana	4	34	2.03E-09	4.66E-10	9.9	4.6	1.77E-10
<i>Grapes</i>							
Seedless	3	32	5.35E-09	0.00E+00	34.0	10.4	1.45E-10
Raisins	3	10	8.11E-10	1.05E-10	21.4	50.1	6.88E-11
<i>Model foods</i>							
Amioca	4	49	1.52E-08	1.52E-08	0.0	33.3	1.02E-09
Hylon-7	5	48	1.96E-08	1.96E-08	0.0	24.2	3.87E-09
<i>Vegetables</i>							
Carrot	9	90	2.47E-09	1.54E-09	13.9	11.3	1.69E-09
Garlic	4	22	5.33E-10	1.68E-11	15.4	7.1	7.43E-11
Onion	4	31	1.45E-08	0.00E+00	70.2	10.4	1.58E-09
Potato	16	106	1.57E-09	4.31E-10	44.7	76.9	4.02E-10

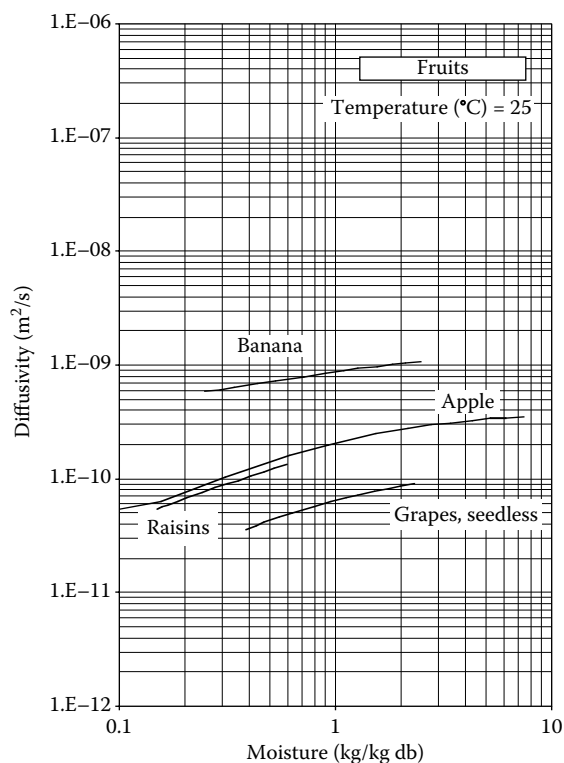


FIGURE 4.18 Predicted values of moisture diffusivity of fruits at 25°C.

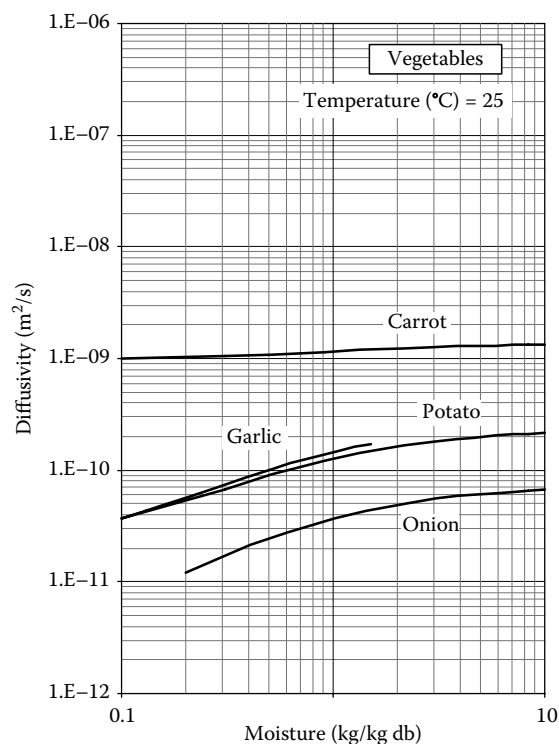


FIGURE 4.19 Predicted values of moisture diffusivity of vegetables at 25°C.

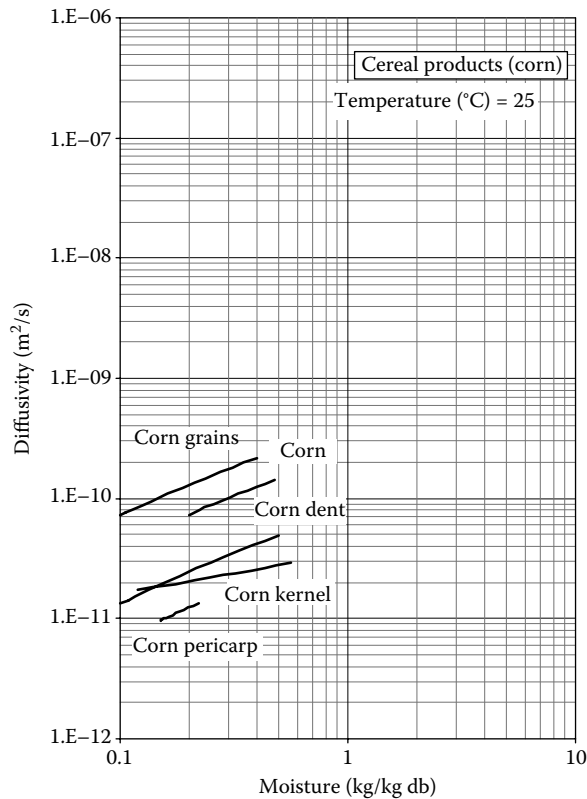


FIGURE 4.20 Predicted values of moisture diffusivity of corn at 25°C.

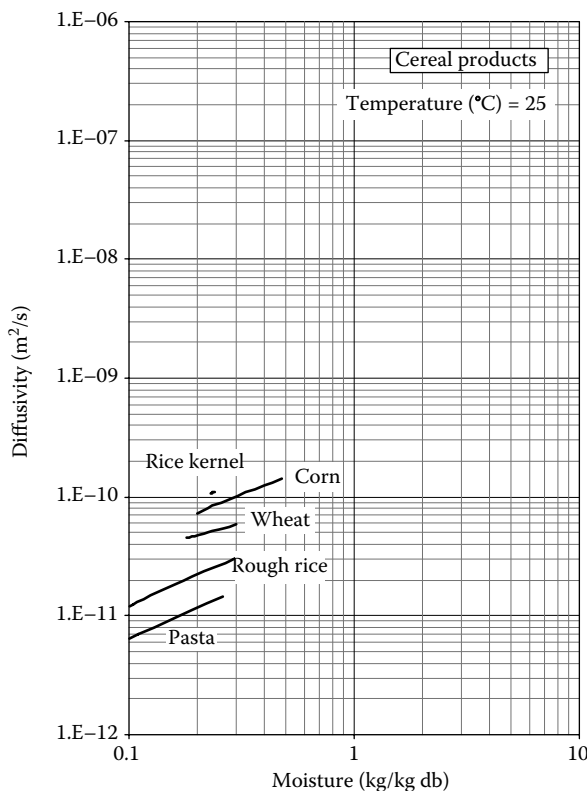


FIGURE 4.21 Predicted values of moisture diffusivity of cereal products at 25°C.

The regression procedure was applied simultaneously to all the data of each material, regardless of the data sources. Thus, the results are not based on the data of only one author and, consequently, they are of higher accuracy and general applicability.

The diffusivity parameters D_0 and D_i of the proposed model vary in the range of 10^{-10} to 10^{-8} m²/s. It should be noted that the self-diffusivity of water is approximately 10^{-9} m²/s, and the moisture diffusivity in bone-dry food material should be lower (in our analysis, by a factor of 100).

Low moisture diffusivities are found in nonporous and sugar-containing foods, whereas higher values of moisture diffusivity characterize porous food materials. Diffusivities higher than the self-diffusivity of water are indicative of vapor diffusion in porous solids.

The moisture diffusivity increases, in general, with increasing moisture content. Temperature has a positive effect, which depends strongly on the food material. The energy of activation for diffusion E of water is, in general, higher in the dry food materials. Some observed exceptions may be explained by the prevailing type of diffusion. Thus, lower values of activation energy for diffusion are expected for porous foods, where vapor diffusion is important. In general, temperature has a stronger effect on diffusivity in liquids and solids than in the gaseous state.

4.8.2 THERMAL CONDUCTIVITY

A total of 146 papers were retrieved from the literature from which 1210 data were obtained. These data refer to more than 100 food materials classified into 11 food categories. Among the available data only 13 materials have more than 10 data, which come from more than 3 publications. This procedure is applied to these data and the results of parameter estimation are presented in Tables 4.17 and 4.18. Figures 4.22 through 4.24 present

TABLE 4.17
Mathematical Model for Calculating Thermal Conductivity in Foods as a Function of Moisture Content and Temperature

Proposed mathematical model

$$\lambda = \frac{1}{1+X} \lambda_0 \exp \left[-\frac{E_0}{R} \left(\frac{1}{T} - \frac{1}{T_r} \right) \right] + \frac{X}{1+X} \lambda_i \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_r} \right) \right]$$

where

λ is the thermal conductivity (W/(m K))

X is the material moisture content (kg/kg db)

T is the material temperature (°C)

$T_r = 60^\circ\text{C}$ a reference temperature

$R = 0.0083143$ is the ideal gas constant kJ/(mol K)

Adjustable model parameters

λ_0 (W/(m K)) thermal conductivity at moisture $X = 0$ and temperature

$T = T_r$

λ_i (W/(m K)) thermal conductivity at moisture $X = \infty$ and temperature $T = T_r$

E_0 (kJ/mol) activation energy for heat conduction in dry material at $X = 0$

E_i (kJ/mol) activation energy for heat conduction in wet material at $X = \infty$

TABLE 4.18
Parameter Estimates of the Proposed Mathematical Model

Material	No. of Papers	No. of Data	λ_i (W/(m K))	λ_0 (W/(m K))	E_i (kJ/mol)	E_0 (kJ/mol)	s.d. (W/(m K))
<i>Cereal products</i>							
Corn	3	15	1.580	0.070	7.2	5.0	0.047
<i>Fruits</i>							
Apple	12	68	0.589	0.287	2.4	11.7	0.114
Orange	4	13	0.642	0.106	1.3	0.0	0.007
Pear	5	15	0.658	0.270	2.4	1.9	0.016
<i>Model foods</i>							
Amioca	5	29	0.718	0.120	3.2	14.4	0.037
Starch	4	24	0.623	0.243	0.3	0.4	0.006
Hylon	3	21	0.800	0.180	9.9		0.072
<i>Vegetables</i>							
Potato	12	37	0.611	0.049	0.0	47.0	0.059
Tomato	5	28	0.680	0.220	0.2	5.0	0.047
<i>Dairy</i>							
Milk	5	33	0.665	0.212	1.7	1.9	0.005
<i>Meat</i>							
Beef	6	37	0.568	0.280	2.2	3.2	0.017
<i>Other</i>							
Rapeseed	3	35	0.239	0.088	3.6	0.6	0.023
<i>Baked products</i>							
Dough	3	15	0.800	0.273	2.7	0.0	0.183

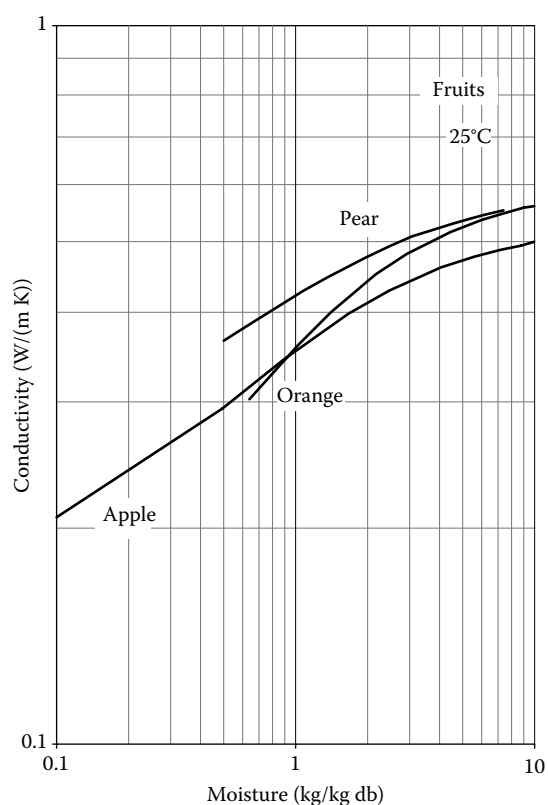


FIGURE 4.22 Predicted values of thermal conductivity of fruits at 25°C.

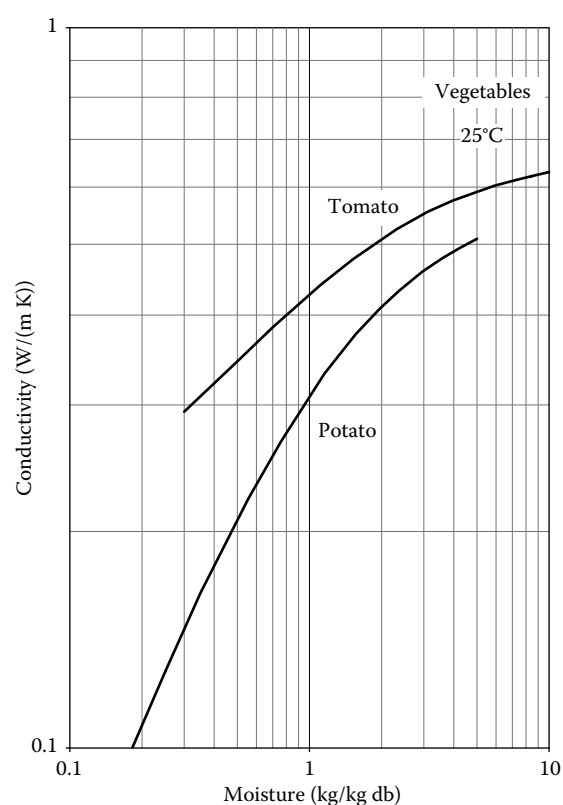


FIGURE 4.23 Predicted values of thermal conductivity of vegetables at 25°C.